

Service Parts Catalog - Parte 1

HP LaserJet M501 / M506 / M507 / M527 / M528 / E52645

TechGuide IA — Consulta Inteligente de Manuais HP

Service Manual

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HP LaserJet M501, M506, M507, E50045,

E50145, MFP M527, M528, E52545, E52645

- Service Manual

SUMMARY

This guide provides theory of operation, troubleshooting, and repair information.

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Revision history

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4 3/2025 Deleted table entry for 320 GB HP

Secure High-Performance Hard Disk

Drive in Supplies and accessories on

page 1309. 3 10/2024 Updated special instructions for Paper

Delivery assembly

2 9/2024 Updated DC controller part numbers in the following sections: Removal and replacement: DC controller on page 967 Table 1 1
12/2023 New combined manual created.

Repair and replace Manual Revision history

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11 6/2023 Updated DC controller part number in the following section: Removal and replacement: DC controller on page 967 10 7/2021
Added: Assembly-Intermediate Paper Feed as Not Shown in Table 2-10 Parts and diagrams: Internal assemblies (3 of 4) on page 1325 9
6/2021 Updated Customer self-repair kits for Tray 2 rollers Customer self-repair kits on page 1312 8 10/2020 Added new remove and replace
procedure for paper delivery assembly. See Removal and replacement: Paper delivery assembly on page 1186

Added one new step to Removal and

replacement fuser power supply. 7 5/2020 Added all M507 Control panel part numbers, new formatter numbers, updated SCB number 6
1/2020 Updated two photos 5 11/2019 Corrected part numbers for Formatters iii

Repair and replace Manual Revision history (continued)

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4 1/2020 Updated document feeder roller removal and replacement topic. See Removal and replacement: Document feeder rollers (MFP) on page 1258. 3 5/2019 Updated or added repair topics to include the M507/E50145/M528/E52645: ● See Removal and replacement: Control panel (M506x/E50045x/ M507x/E50145x) on page 504 ● See Removal and replacement: White backing (MFP) on page 513 ● See Removal and replacement: Control panel (M501, 507n/dn, and E50145n/dn) on page 530 ● See Removal and replacement: Document feeder (MFP) on page 547 ● See Removal and replacement: Top cover (SFP) on page 633 ● See Removal and replacement: Wireless printed-circuit board (WiFi) (M507x/E50145x) on page 751 ● See Install accessory: Internal USB ports (M506/E50045) on page 1285 ● See Install accessory: Trusted platform module (TPM) on page 1297 iv Revision history

Repair and replace Manual Revision history (continued)

Revision number Revision date Revision notes

3 5/2019 Updated or added parts topics to include the M507/E50145/M528/E52645: ● See Supplies and accessories on page 1309 ● See Customer self-repair kits on page 1312 ● See Parts and diagrams: Document feeder and scanner whole units on page 1314 ● See Parts and diagrams: Covers (M501/M506/E50045/ M507/E50145) on page 1316 ● See Parts and diagrams: Covers (M527/E52545/M528/E52645) on page 1319 ● See Parts and diagrams: Internal assemblies (1 of 4) on page 1321 ● See Parts and diagrams: Internal assemblies (2 of 4) on page 1323 ● See Parts and diagrams: Internal assemblies (3 of 4) on page 1325 ● See Parts and diagrams: Internal assemblies (4 of 4) on page 1327 2 4/2016 LaserJet Pro M501 content added. 1 10/2015 HP LaserJet Enterprise M506/M527 Repair Manual initial release.

Troubleshooting Manual Revision history

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5 4/2021 Updated the M527/E52545/M528/E52645 formatter connector locations figure and legend. See Formatter PCA (M527/E52545/M528/E52645) on page 163. Corrected callouts on diagrams motor and fans. see: Motor and fans on page 170 v

Troubleshooting Manual Revision history (continued)

Revision number Revision date Revision notes

3 5/2019 Added FutureSmart 4 steps to the following procedures: ● See Use the solve problems checklist on page 79 ● See Print the Settings menu map on page 82 ● See Print current settings pages on page 83 ● See Print the event log on page 102 ● See HP Jetdirect LEDs on page 139 ● See Disable cartridge check (all except M501) on page 142 ● See Paper path test (all except M501) on page 143 ● See Paper path sensors test (all except M501) on page 144 ● See Manual sensor tests (all except M501) on page 146 ● See Tray/bin manual sensor test (all except M501) on page 148 ● See Print/stop test (all except M501) on page 150 ● See Individual component diagnostics (special mode test; all except M501) on page 152 ● See Use the scanner tests on page 154 ● See Print a configuration page on page 175 ● See Print a fuser test page (all except M501) on page 181 ● See Control panel menus (all except M501) on page 194 ● See Check toner-cartridge status on page 282 ● See Print a cleaning page on page 282 ● See Step three: Set the individual tray alignment on page 284 ● See Print quality troubleshooting guide on page 293 ● See Change jam recovery (all except M501) on page 416 ● See Service menu on page 428 ● See Restore factory-set defaults (all except M501) on page 432 ● See Determine the installed revision of firmware (all except M501) on page 441 ● See USB flash drive (control panel menu) (all except M501) on page 445 vi Revision history

Troubleshooting Manual Revision history (continued)

Revision number Revision date Revision notes

3 5/2019 Added a procedure to access the touchscreen diagnostic mode for the M507x ad E50145dn printers. ● See Touchscreen diagnostic mode (M507x and E50145dn) on page 116 3 5/2019 Updated model list for control panel diagnostic keyboard test. ● See Keyboard test (M527c/z, E52545c, M528c/z, E52645c only) on page 128 2 4/2016 LaserJet Pro M501 content added. 1 10/2015 HP LaserJet Enterprise M506/M527 Repair Manual initial release. vii

Table of contents

1 Assembly locations ... 1 Printer front view (SFP) ... 1 Printer front view (MFP) ... 2 Printer back view (SFP) ... 3 Printer back view (MFP) ... 3
2 Printer specifications ... 5 Printer dimensions ... 6 HP LaserJet Pro M501 dimensions ... 6 HP LaserJet Enterprise
M506/E50045/M507/E50145 dimensions ... 8 HP LaserJet Enterprise MFP M527/E52545/M528/E52645 dimensions ... 10 Printer space
requirements ... 13 Power consumption, electrical specifications, and acoustic emissions ... 14 Operating-environment range ... 14 3 Theory of
operations ... 15 Conventions used in this guide ... 15 HP service and support ... 15 Basic operation ... 18 Sequence of operation ... 19
Formatter-control system ... 21 Sleep mode (all except M501) ... 21 Sleep delay (M501) ... 22 Auto On / Auto Off mode (M501) ... 22 Printer
job language (PJM) ... 22 Printer management language (PML) ... 23 Control panel ... 23 Easy-access USB port (all except M501) ... 23
Wireless (wireless models only) ... 24 Low end data model (LED) overview (M501) ... 24 Advanced control language (ACL) overview (M501)
... 24 Near field communication (NFC models only) ... 24

CPU ... 24

Input/output (I/O) ... 24 Memory ... 25 viii

Firmware ... 25 Nonvolatile random access memory (NVRAM) ... 25 Random access memory (RAM) ... 25 HP Memory Enhancement technology (MEt) ... 25 Engine-control system ... 26 DC controller ... 26 Motors ... 27 Fans ... 28 Solenoids ... 28 Clutches ... 28 Switches ... 29 Photo interrupter sensors ... 29 Sensors ... 29 LEDs ... 29 Low-voltage power supply (LVPS) ... 30 Low-voltage power supply voltages description ... 31 Over-current/over-voltage protection ... 32 Sleep mode operation (all except M501) ... 32 Low-voltage power supply failure detection (all except M501) ... 33 Safety ... 33 Low-voltage power supply functions ... 33 High-voltage power supply ... 33 Fuser bias ... 34 High-voltage power supply circuits ... 34 Fuser control ... 34 Fuser circuits ... 35 Fuser control functions ... 35 Fuser heater protection ... 37 Fuser unit life detection (all except M501) ... 37 Fuser identification (all except M501) ... 37 Engine laser/scanner system ... 38 Laser/scanner failure detection ... 38 Safety ... 39 Image-formation process ... 39 Step 1: Primary charging ... 43 Step 2: Laser-beam exposure ... 43 Step 3: Development ... 44 Step 4: Transfer ... 44 Step 5: Separation ... 45 Step 6: Fusing ... 45 Step 7: Drum cleaning ... 46 Toner cartridge ... 46 Design ... 46 Memory chip ... 47 Dynamic security enabled printers in this guide ... 48 Dynamic security ... 48 Toner seal ... 48 Toner level and cartridge life detection ... 48 ix

JetIntelligence ... 48 Authentication ... 49 Anti-theft ... 49 Paper handling system ... 50 Photo sensors and switches ... 52 Motors, clutches, and solenoids ... 53 Tray 1 (multipurpose) / Tray 2 (base printer) ... 54 Tray 1 paper pickup and feed ... 54 Tray 2 paper presence detection ... 56 Tray 2 lift operation ... 56 Tray 2 paper pickup ... 56 Tray 2 multiple feed prevention ... 57 Tray 2 presence detection ... 57 Tray 2 skew feed prevention ... 57 Feed speed control ... 58 Duplexing unit ... 59 Duplexing reverse and duplex feed control ... 59 Jam detection/prevention ... 60 Input accessories ... 62 Tray 3-5 ... 62 Driver PCA ... 64 Electrical components ... 64 Paper pickup ... 65 Multiple feed prevention ... 65 Tray presence detection ... 65 Tray lift operation ... 65 Jam detection ... 66 Scanning and image capture system (M527/E52545/M528/E52645) ... 68 Document feeder system (M527/E52545/M528/E52645) ... 69 Document feed system ... 69 Sensors in the document feeder ... 69 Document feeder paper path ... 70 Document feeder simplex operation ... 71 Document feeder e-duplex operation ... 72 Deskew operation ... 72 Document feeder hinges ... 73 4 Solve problems ... 75 HP service and support ... 75 Solve problems checklist (M501) ... 77 Use the solve problems checklist (M501) ... 77 Print the menu map (M501) ... 78 Print the service page (includes the event log) (M501) ... 78 Solve problems checklist (all except M501) ... 79 Use the solve problems checklist ... 79 Print the Settings menu map ... 82 Print the settings menu map from a touchscreen control panel ... 82 x

Print the settings menu map from a LCD control panel ... 83 Print current settings pages ... 83 Print the current settings page from a touchscreen control panel ... 83 Print the current settings page from a LCD control panel ... 84 Pre-boot menu options ... 84 Open the Pre-boot menu from a touchscreen control panel ... 85 Open the Pre-boot menu from a LCD control panel ... 86 Cold reset using the Pre-boot menu from a touchscreen control panel ... 86 Cold reset using the Pre-boot menu from a LCD control panel ... 87 Remote Admin (all except M501) ... 94 Required software and network connection ... 95 Connect a remote connection ... 97 Disconnect a remote connection ... 101 Print the event log ... 102 Print the event log from the Administration menu (FutureSmart 3) or Support Tools menu (FutureSmart 4) from a touchscreen control panel ... 102 Print the event log from the Administration menu (FutureSmart 3) or Support Tools menu (FutureSmart 4) from a LCD control panel ... 103 Print the event log from the Service menu from a touchscreen control panel ... 103 Print the event log from the Service menu from a LCD control panel ... 104 Clear the event log from a touchscreen control panel ... 104 Clear the event log from a LCD control panel ... 105 Troubleshooting process ... 106 Determine the problem source ... 106 Pre-troubleshooting checklist ... 106 Troubleshooting flowchart ... 107 Power subsystem ... 109 Power-on checks ... 109 Power-on troubleshooting overview ... 109 Control panel checks (M501) ... 112 Control panel checks (all except M501) ... 112 Control panel diagnostics (M506x/M507x and M527/E52545/M528/E52645) ... 112 Touchscreen diagnostic mode (M506x, M527/E52545/M528/E52645) ... 112 Touchscreen diagnostic mode (M507x and E50145dn) ... 116 Control panel system diagnostics (M506x/M507x, M527/E52545/M528/E52645) ... 118 Control panel diagnostic flowcharts (M506x/M507x, M527/E52545/M528/E52645) ... 130 Tools for troubleshooting ... 136 Individual component diagnostics ... 136 LED diagnostics (M501) ... 136 Network port LEDs (M501) ... 136 M501 control panel LEDs ... 137 LED diagnostics (all except M501) ... 137 Understand lights on the formatter ... 137 Heartbeat LED ... 138 HP Jetdirect LEDs ... 139 Engine diagnostics ... 140 Engine test button ... 140 Defeating interlocks ... 140 xi

Disable cartridge check (all except M501) ... 142 Paper path and sensor diagnostic tests (all except M501) ... 143 Paper path test (all except M501) ... 143 Paper path sensors test (all except M501) ... 144 Manual sensor tests (all except M501) ... 146 Tray/bin manual sensor test (all except M501) ... 148 Print/stop test (all except M501) ... 150 Access the print/stop test from a touchscreen control panel ... 151 Access the print/stop test from a LCD control panel ... 151 Common print stop test timing millisecond (ms) stops ... 151 Component tests (all except M501) ... 152 Individual component diagnostics (special-mode test; all except M501) ... 152 Scanner tests (M527/E52545/M528/E52645) ... 153 Scanner tests (M527/E52545/M528/E52645) ... 154 Diagrams ... 155 Diagrams: Block diagrams ... 155 Sensors and switches ... 155 Cross section diagrams ... 158 Diagrams: Printed circuit assembly (PCA) connector locations ... 160 Diagrams: External plug and port locations ... 165 Diagrams: Locations of major assemblies ... 168 Main assemblies (printer base) ... 168 Rollers and pads (printer base) ... 169 Motor and fans ... 170 Printed circuit assemblies (PCAs; printer base) ... 171 Diagrams: General timing chart ... 172 Diagrams: General circuit diagrams ... 172 Internal test and information pages ... 174 Print a configuration page ... 175 Print the configuration page from a LCD control panel (M501) ... 175 Print the configuration page from a LCD control panel (M506/E50045/M507/E50145) ... 176 Print the configuration page from a touchscreen control panel (M527/E52545/M528/

E52645) ... 176

HP embedded Jetdirect page (all except M501) ... 177 GUID-A0173DC1-D63F-445A-B60E-61FC78C5DCC0-high.eps ... 0 Wireless page (all except M501) ... 178 Finding important information on the configuration pages ... 180 Print a fuser test page (all except M501) ... 181 Print a fuser test page from a touchscreen control panel ... 181 Print a fuser test page from a LCD control panel ... 181 Advanced configuration with HP Embedded Web Server (EWS) and HP Device Toolbox (Windows) (M501) ... 182 Open the HP Embedded Web Server (EWS) from the Start menu ... 183 Open the HP Embedded Web Server (EWS) from a Web browser ... 183 Control panel menus (M501) ... 185 Setup menu ... 185 HP Web Services menu (M501) ... 185 Reports menu (M501) ... 185 System Setup menu (M501) ... 186 xii

Service menu (M501) ... 191 Network Setup menu (M501) ... 192 Quick Forms menu (M501) ... 193 Function specific menus (M501) ... 194 Apps (M501) ... 194 Jobs (M501) ... 194 Supplies Status (M501) ... 194 Control panel menus (all except M501) ... 194 Reports menu ... 194 Settings menu ... 196 General menu (all except M501) ... 196 Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/

E52645) ... 203

Scan/Digital Send Settings menu (M527/E52545/M528/E52645) ... 225 Fax menu (M527/E52545/M528/E52645) ... 235 Manage Supplies menu ... 246 Networking menu ... 249 Support Tools menu ... 261 Maintenance menu (all except M501) ... 261 Troubleshooting menu ... 263 Error-code and control-panel-message troubleshooting overview ... 268 Error codes (types and structure) ... 268 How to search for printer documentation ... 270 How to search WISE for printer information and documentation ... 270 Accessing WISE for HP channel partners (CSDP) ... 276 Accessing WISE for HP channel partners (HP Partner Portal) ... 277 Resolving mono print quality problems ... 280 Introduction ... 280 Troubleshoot print quality ... 280 Print from a different software program ... 280 Check the paper-type setting for the print job ... 280 Check the paper type setting on the control panel ... 280 Check the paper type setting (Windows) ... 281 Check the paper type setting (macOS) ... 281 Check toner-cartridge status ... 282 Step one: Print the Supplies Status Page ... 282 Step two: Check supplies status ... 282 Print a cleaning page ... 282 Visually inspect the toner cartridge or cartridges ... 283 Check paper and the printing environment ... 283 Step one: Use paper that meets HP specifications ... 283 Step two: Check the environment ... 283 Step three: Set the individual tray alignment ... 284 Try a different print driver ... 284 Troubleshoot image defects ... 284 Print quality troubleshooting guide ... 293 Image defects table ... 293 Product specific image defects ... 296 xiii

Repetitive image defect ruler ... 296 Use a ruler to measure between repetitive defects ... 297 Print-quality troubleshooting ... 300 Image defect events ... 300 Image defect (developmental) events ... 321 Other events ... 358 Clean the printer ... 372 Clean the paper path ... 372 Print a cleaning page ... 372 Print the cleaning page from a LCD control panel (M501) ... 372 Print the cleaning page from a LCD control panel (M506/E50045/M507/E50145) ... 372 Print the cleaning page from a touchscreen control panel (M527/E52545/M528/E52645) ... 373 Enable and configure auto cleaning (M527/E52545/M528/E52645) ... 373 Check the scanner glass for dirt and smudges (M527/E52545/M528/E52645) ... 373 Clean the pickup rollers and separation pad in the document feeder (M527/E52545/M528/E52645) ... 375 Clean the Tray 1 roller and separation pad ... 376 Step 1: Remove the roller ... 376 Step 2: Remove the separation pad assembly ... 377 Step 3: Clean the Tray 1 roller and separation pad ... 379 Step 4: Install the separation pad assembly ... 379 Step 5: Install the roller ... 381 Clean the Tray 2-X rollers ... 383 Step 1: Remove the tray ... 383 Step 2: Remove the roller assembly ... 384 Step 3: Clean the Tray 2-X rollers ... 385 Step 4: Install the roller assembly ... 386 Step 5: Install the tray ... 388 Solve paper handling problems ... 390 Printer feeds incorrect page size ... 390 Printer pulls from incorrect tray ... 390 Printer will not duplex or duplexes incorrectly (duplex models) ... 390 Paper does not feed from Tray 2-X ... 391 Output is curled or wrinkled ... 391 Printer does not pick up paper or misfeeds ... 392 The printer does not pick up paper ... 392 The printer picks up multiple sheets of paper ... 393 The document feeder jams, skews, or picks up multiple sheets of paper (MFP) ... 393 Paper does not feed automatically ... 394 Clear paper jams ... 395 Paper path jam sensor locations ... 395 Auto-navigation for clearing jams ... 396 Experiencing frequent or recurring paper jams? ... 396 Clear paper jams in the document feeder ... 396 Clear paper jams in Tray 1 ... 400 Clear paper jams in Tray 2 and the 550-sheet trays ... 403 Clear paper jams in the toner-cartridge area ... 407 Clear paper jams in the rear door and the fuser area ... 409 Clear paper jams in the output bin ... 411 xiv

Clear paper jams in the duplexer ... 411 Change jam recovery (all except M501) ... 416 Change jam recovery from a touchscreen control panel ... 416 Change jam recovery from a LCD control panel ... 417 Solve performance problems ... 418 Factors affecting print performance ... 418 Print speeds ... 419 The printer does not print or it prints slowly ... 420 The printer does not print ... 420 The printer prints slowly ... 420 Solve connectivity problems ... 422 Solve USB connection problems ... 422 Solve wired network problems ... 422 Introduction ... 422 Poor physical connection ... 422 The computer is using the incorrect IP address for the printer ... 422 The computer is unable to communicate with the printer ... 423 The printer is using incorrect link and duplex settings for the network ... 423 New software programs might be causing compatibility problems ... 423 The computer or workstation might be set up incorrectly ... 423 The printer is disabled, or other network settings are incorrect ... 423 Service mode functions (M501) ... 425 Service menu (M501) ... 425 Service menu settings ... 425 Restore the factory-set defaults ... 425 Restore the factory-set defaults from a LCD control panel (M501) ... 425 Secondary service menu (M501) ... 426 Open the secondary service menu (M501) ... 426 Open the secondary service menu from a LCD control panel (M501) ... 426 Secondary service menu structure (M501) ... 426 Printer resets (M501) ... 427 NVRAM initialization (M501) ... 427 Service mode functions ... 428 Service menu ... 428 Open the service menu from a touchscreen control panel ... 428 Open the service menu from a LCD control panel ... 428 Service menu items ... 429 Printer resets (all except M501) ... 432 Restore factory-set defaults (all except M501) ... 432 Restore factory-set defaults from a touchscreen control panel ... 432 Restore factory-set defaults from a LCD control panel ... 432 Restore the service ID (all except M501) ... 433 Restore the service ID ... 433 Convert the service ID to an actual date ... 433 Printer cold reset (all except M501) ... 434 Cold reset using the Pre-boot menu from a touchscreen control panel ... 434 Cold reset using the Pre-boot menu from a LCD control panel ... 434 Format Disk and Partial Clean functions (all except M501) ... 435 xv

Active and repository firmware locations ... 435 Partial Clean ... 435 Execute a Partial Clean ... 436 Format Disk ... 437 Execute a Format Disk ... 438 Firmware upgrades (M501) ... 439 Determine the installed revision of firmware (M501) ... 439 Method one: Update the firmware using the control panel (M501) ... 439 Method two: Update the firmware using the Firmware Update Utility (M501) ... 440 Firmware upgrades (all except M501) ... 441 Determine the installed revision of firmware (all except M501) ... 441 Print the configuration page from a touchscreen control panel ... 441 Print the configuration page from a LCD control panel ... 442 Perform a firmware upgrade (all except M501) ... 443 HP Embedded Web Server (all except M501) ... 443 USB flash drive (Pre-boot menu) (all except M501) ... 444 USB flash drive firmware (Pre-boot menu) update from a touchscreen control panel ... 444 USB flash drive firmware (Pre-boot menu) update from a LCD control panel ... 445 USB flash drive (control panel menu) (all except M501) ... 445 USB flash drive firmware (control panel menu) update from a touchscreen control panel ... 445 USB flash drive firmware (control panel menu) update from a LCD control panel ... 446 Solve fax or email problems ... 447 5 Removal and replacement ... 448 Conventions used in this guide ... 448 HP service and support ... 448 Removal and replacement strategy

... 451 Introduction ... 451 Considerations during removal and replacement ... 451 Electrostatic discharge ... 452 Required tools ... 453 Fasteners used in this printer ... 453 Service approach ... 455 Before performing service ... 455 After performing service ... 455 Print quality test ... 455 Removal and replacement procedures ... 456 Customer self-repair (CSR) A parts and accessories ... 456 Removal and replacement: Toner cartridge ... 456 Removal and replacement: Staple cartridge (MFP) ... 460 Removal and replacement: embedded Multi-Media Card (eMMC) (all except M501) ... 464 Removal and replacement: Hard-disk drive (HDD) ... 469 Removal and replacement: Dual in-line memory module (DIMM) ... 476 Removal and replacement: Transfer roller ... 485 Removal and replacement: Control panel (MFP) ... 492 Removal and replacement: Control panel (M506x/E50045x/M507x/E50145x) ... 504 Removal and replacement: White backing (MFP) ... 513 xvi

Removal and replacement: Keyboard assembly (M527c/z, E52545c/z, M528c/z, and E52645c/z) ... 519 Removal and replacement: Internal parts and assemblies (base printer) ... 527 Removal and replacement: External panels, covers, and doors (base printer) ... 528 Removal and replacement: Formatter cover ... 528 Removal and replacement: Control panel (M501, 507n/dn, and E50145n/dn) ... 530 Removal and replacement: Control-panel cover (MFP) ... 534 Removal and replacement: Top-left cover (MFP) ... 537 Removal and replacement: Top-rear cover (MFP) ... 540 Removal and replacement: Document feeder (MFP) ... 547 Removal and replacement: Cartridge door assembly ... 561 Removal and replacement: Rear door assembly ... 567 Removal and replacement: Right cover ... 570 Removal and replacement: Left cover ... 582 Removal and replacement: Stapler door or stapler blank cover (M527/E52545/M528/

E52645) ... 592

Removal and replacement: Stapler stationary (inner) cover (M527/E52545/M528/E52645) ... 596 Removal and replacement: Top cover (MFP) ... 603 Removal and replacement: Top cover (SFP) ... 633 Removal and replacement: Paper delivery tray (output bin) ... 657 Removal and replacement: Sub-scanner assembly (SSA) (MFP) ... 694 Field replaceable units (FRUs) ... 724 Removal and replacement: Scanner control board (MFP) ... 724 Removal and replacement: Near field communication printed-circuit assembly (M527z/ E52545z) ... 734 Removal and replacement: Wireless printed-circuit board (WiFi) (M506x/E50045x) ... 738 Removal and replacement: Wireless printed-circuit board (WiFi) (M507x/E50145x) ... 751 Removal and replacement: Wireless printed-circuit board (M527z/E52545z/M528z/ E52645z) ... 754 Removal and replacement: Fuser ... 757 Removal and replacement: Stapler unit (M527c/f/z, E52545c/z/f, M528c/f/z, and E52645c/f/z) ... 763 Removal and replacement: Formatter ... 771 Removal and replacement: Laser/scanner assembly ... 780 Removal and replacement: Formatter case ... 818 Removal and replacement: Fan (FM1) ... 862 Removal and replacement: Fan (FM2) ... 874 Removal and replacement: Fuser power supply (FPS) ... 920 Removal and replacement: DC controller ... 967 Removal and replacement: High-voltage power supply ... 1012 Removal and replacement: Low-voltage power supply ... 1025 Removal and replacement: Main Motor M1 ... 1073 Removal and replacement: Paper feed drive assembly ... 1130 Removal and replacement: Paper delivery assembly ... 1186 Remove and replacement: Trays ... 1247 Removal and replacement: Tray 2 ... 1247 Removal and replacement: Tray 3 ... 1249 Remove and replacement: Accessories ... 1252 Removal and replacement: 550-sheet paper feeder ... 1252 Removal and replacement: Document feeder rollers (MFP) ... 1258 xvii

Removal and replacement: Fax printed-circuit board (M527c/f/z, E52545c/f/z, M528c/f/z, and E52645c/f/z) ... 1267 Install accessory: Internal USB ports (MFP) ... 1274 Install accessory: Internal USB ports (M506/E50045) ... 1285 Install accessory: Internal USB ports (M507/E50145) ... 1291 Install accessory: Trusted platform module (TPM) ... 1297 6 Parts and diagrams ... 1307 HP service and support ... 1307 Order parts by authorized service providers ... 1309 Ordering ... 1309 Orderable parts ... 1309 Supplies and accessories ... 1309 Customer self-repair kits ... 1312 How to use the parts lists and diagrams ... 1313 Parts and diagrams: Document feeder and scanner whole units ... 1314 Parts and diagrams: Covers (M501/M506/E50045/M507/E50145) ... 1316 Parts and diagrams: Covers (M527/E52545/M528/E52645) ... 1319 Parts and diagrams: Internal assemblies (1 of 4) ... 1321 Parts and diagrams: Internal assemblies (2 of 4) ... 1323 Parts and diagrams: Internal assemblies (3 of 4) ... 1325 Parts and diagrams: Internal assemblies (4 of 4) ... 1327 Parts and diagrams: 1x550-sheet paper feeder ... 1330 Alphabetical parts list ... 1331 HP service and support ... 1307 Order parts by authorized service providers ... 1309 Ordering ... 1309 Orderable parts ... 1309 Supplies and accessories ... 1309 Customer self-repair kits ... 1312 How to use the parts lists and diagrams ... 1313 Parts and diagrams: Document feeder and scanner whole units ... 1314 Parts and diagrams: Covers (M501/M506/E50045/M507/E50145) ... 1316 Parts and diagrams: Covers (M527/E52545/M528/E52645) ... 1319 Parts and diagrams: Internal assemblies (1 of 4) ... 1321 Parts and diagrams: Internal assemblies (2 of 4) ... 1323 Parts and diagrams: Internal assemblies (3 of 4) ... 1325 Parts and diagrams: Internal assemblies (4 of 4) ... 1327 Parts and diagrams: 1x550-sheet paper feeder ... 1330 Numerical parts list ... 1341 HP service and support ... 1307 xviii

Order parts by authorized service providers ... 1309 Ordering ... 1309 Orderable parts ... 1309 Supplies and accessories ... 1309 Customer self-repair kits ... 1312 How to use the parts lists and diagrams ... 1313 Parts and diagrams: Document feeder and scanner whole units ... 1314 Parts and diagrams: Covers (M501/M506/E50045/M507/E50145) ... 1316 Parts and diagrams: Covers (M527/E52545/M528/E52645) ... 1319 Parts and diagrams: Internal assemblies (1 of 4) ... 1321 Parts and diagrams: Internal assemblies (2 of 4) ... 1323 Parts and diagrams: Internal assemblies (3 of 4) ... 1325 Parts and diagrams: Internal assemblies (4 of 4) ... 1327 Parts and diagrams: 1x550-sheet paper feeder ... 1330 Appendix A Certificate of Volatility: HP LaserJet Pro M501, HP LaserJet Enterprise M506, M507, M527, M528, HP LaserJet Managd Enterprise E50145, E52645 ... 1351 Index ... 1361 xix

List of videos

View a video of how to use the HP Partner First Portal to access WISE ... 15 View a video of how to access WISE for internal HP users ... 16 View a video of how to use WISE to find technical support videos (model number search) ... 16 View a video of how to use WISE to find technical support videos (Product detail page [PDP] search) ... 17 View a video of how to use the HP Partner First Portal to access WISE ... 75 View a video of how to access WISE for internal HP users ... 76 View a video of how to use WISE to find technical support videos (model number search) ... 76 View a video of how to use WISE to find technical support videos (Product detail page [PDP] search) ... 76 View a video of how to access WISE for HP channel partners ... 277 View a video of how to use the HP Partner First Portal to access WISE ... 448 View a video of how to access WISE for internal HP users ... 449 View a video of how to use WISE to find technical support videos (model number

search) ... 449 View a video of how to use WISE to find technical support videos (Product detail page [PDP] search) ... 450 View a video of how to remove and replace the toner cartridge ... 456 View a video of how to remove and replace the staple cartridge ... 460 View a video of how to remove and replace the eMMC ... 464 View a video of how to remove and replace the HDD ... 469 View a video of how to remove and replace the DIMM ... 476 View a video of how to remove and replace the transfer roller ... 485 View a video of how to remove and replace the control panel ... 492 View a video of how to remove and replace the control panel ... 504 View a video of how to remove and replace the white backing ... 513 View a video of how to remove and replace the keyboard assembly ... 519 View a video of how to remove and replace the formatter cover (M501) ... 528 View a video of how to remove and replace the formatter cover (M506/E50045/M507/E50145) ... 528 xx

View a video of how to remove and replace the formatter cover (M527/E52545/M528/E52645) ... 528 View a video of how to remove and replace the control panel ... 530 View a video of how to remove and replace the control panel cover ... 534 View a video of how to remove and replace the top-left cover ... 537 View a video of how to remove and replace the top-rear cover ... 541 View a video of how to remove and replace the document feeder ... 547 View a video of how to remove and replace the cartridge door ... 561 View a video of how to remove and replace the rear door assembly ... 567 View a video of how to remove and replace the right cover ... 570 View a video of how to remove and replace the left cover ... 582 View a video of how to remove and replace the stapler door or stapler blank cover ... 592 View a video of how to remove and replace the stapler stationary (inner) cover ... 597 View a video of how to remove and replace the top cover ... 603 View a video of how to remove and replace the top cover (M501) ... 633 View a video of how to remove and replace the top cover (M506/E50045) ... 633 View a video of how to remove and replace the top cover (M507/E50145) ... 633 View a video of how to remove and replace the paper delivery tray (output bin) ... 657 View a video of how to remove and replace the sub-scanner assembly (SSA) ... 694 View a video of how to remove and replace the SCB ... 724 View a video of how to remove and replace the NFC ... 734 View a video of how to remove and replace the wireless PCA ... 738 View a video of how to remove and replace the wireless PCA ... 751 View a video of how to remove and replace the wireless PCA ... 754 View a video of how to remove and replace the fuser ... 757 View a video of how to remove and replace the stapler unit ... 763 View a video of how to remove and replace the formatter (M501) ... 771 View a video of how to remove and replace the formatter (M506/E50045) ... 771 View a video of how to remove and replace the formatter (M507/E50145) ... 771 View a video of how to remove and replace the formatter (M527/E52545/M528/E52645) ... 771 View a video of how to remove and replace the laser/scanner (M501) ... 780 View a video of how to remove and replace the laser/scanner (M506/E50045) ... 780 View a video of how to remove and replace the laser/scanner (M507/E50145) ... 780 xxi

View a video of how to remove and replace the laser/scanner (M527/E52545/M528/E52645) ... 780 View a video of how to remove and replace the formatter case (M501) ... 819 View a video of how to remove and replace the formatter case (M506/E50045) ... 819 View a video of how to remove and replace the formatter case (M507/E50145) ... 819 View a video of how to remove and replace the formatter case (M527/E52545/M528/E52645) ... 819 View a video of how to remove and replace the fan (FM1) ... 863 View a video of how to remove and replace the fan (FM2) (M501) ... 875 View a video of how to remove and replace the fan (FM2) (M506/E50045) ... 875 View a video of how to remove and replace the fan (FM2) (M507/E50145) ... 875 View a video of how to remove and replace the fan (FM2) (M527/E52545/M528/E52645) ... 875 View a video of how to remove and replace the FPS (M501) ... 920 View a video of how to remove and replace the FPS (M506/E50045) ... 920 View a video of how to remove and replace the FPS (M507/E50145) ... 920 View a video of how to remove and replace the FPS (M527/E52545/M528/E52645) ... 920 View a video of how to remove and replace the DCC (M501) ... 967 View a video of how to remove and replace the DCC (M506/M506/E5004) ... 967 View a video of how to remove and replace the DCC (M507/E50145) ... 967 View a video of how to remove and replace the DCC (M527/E52545/M528/E52645) ... 967 View a video of how to remove and replace the HVPS ... 1013 View a video of how to remove and replace the LVPS (M051) ... 1025 View a video of how to remove and replace the LVPS (M506/E50045) ... 1025 View a video of how to remove and replace the LVPS (M507/E50145) ... 1025 View a video of how to remove and replace the LVPS (M527/E52545/M528/E52645) ... 1025 View a video of how to remove and replace Tray 2 ... 1247 View a video of how to remove and replace Tray 3 ... 1250 View a video of how to remove and replace the 550-sheet feeder ... 1252 View a video of how to remove and replace the document feeder pickup and feed roller assembly and the separation roller assembly ... 1258 View a video of how to remove and replace the FAX PCA ... 1268 View a video of how to remove and replace the internal USB ports ... 1275 View a video of how to remove and replace the internal USB ports ... 1285 View a video of how to remove and replace the internal USB ports ... 1291 View a video of how to remove and replace the TPM ... 1297 xxii

View a video of how to use the HP Partner First Portal to access WISE ... 1307 View a video of how to access WISE for internal HP users ... 1308 View a video of how to use WISE to find technical support videos (model number search) ... 1308 View a video of how to use WISE to find technical support videos (Product detail page [PDP] search) ... 1308 xxiii

1 Assembly locations

Learn about major printer assembly locations. Printer front view (SFP) Use the diagram to locate important components on the printer. 1 Top cover (access to the toner cartridge) 2 Easy-access USB port Insert a USB flash drive for printing without a computer or to update the printer firmware. NOTE: An administrator must enable this port before use.

3 Output bin

4 On/off button 5 2.7-in QVGA LCD color display with 24-key pad (n and dn models) 6 Hardware integration pocket for connecting accessory and third-party devices NOTE: M506/E50045/M527/E52545 only: To use the hardware integration pocket (HIP), install the HP internal USB ports accessory (B5L28A). The USB port inside the HIP is not functional otherwise.

7 Formatter cover

8 Tray 2

9 Model name

10 Tray 1 11 Control panel with color touchscreen display (x models only) Assembly locations 1

12 1 x 550-sheet paper feeder, Tray 3 (included with the x model, optional for the other models) NOTE: Each printer model accepts up to three optional 1 x 550-sheet feeders (Trays 3, 4, and 5). Tray 3 is included with the x model. Printer front view (MFP) Locate features on the front of the printer. 15 1 2 3 4 5 7 6 8 16 14 13 12 11 9 10 1 Document-feeder cover (access for clearing jams) 2 Document-feeder input tray 3 Document-feeder output bin NOTE: When feeding long paper through the document feeder, extend the paper stop at the right side of the output bin. 4 Control panel with color touchscreen display (tilts up for easier viewing) 5 Physical keyboard (c and z models only). Pull the keyboard straight out to use it. 6 Convenience stapler (c, f, and z models only) 7 On/off button

8 Formatter cover

9 Optional 1 x 550-sheet paper feeder (Tray 3) NOTE: Each printer model accepts up to three optional 1 x 550-sheet feeders (Trays 3, 4, and 5). 10 Tray 2

11 Model name

2 Chapter 1 Assembly locations

12 Tray 1 13 Top cover (access to the toner cartridge) NOTE: The release button for the top cover is on the left side of the printer.

14 Standard output bin

15 Easy-access USB port (on the left side of the control panel support) Insert a USB flash drive for printing or scanning without a computer or to update the printer firmware. NOTE: An administrator must enable this port before use. 16 Hardware integration pocket (for connecting accessory and third-party devices) NOTE: M506/E50045/M527/E52545 only: To use the hardware integration pocket (HIP), install the HP internal USB ports accessory (B5L28A). The USB port inside the HIP is not functional otherwise. Printer back view (SFP) Use the diagram to locate important components on the printer. 1 Rear door (access for clearing jams)

2 Serial number and product number label

3 Top-cover-release button 4 Dust cover for Tray 2 (flips up when legal-size paper is loaded)

5 Power connection

6 Formatter (contains the interface ports) Printer back view (MFP) Locate features on the back of the printer. Printer back view (SFP) 3

1 2 6 5 4 3 1 Rear door (access for clearing jams) 2 Top-cover-release button

3 Serial number and product number label

4 Dust cover for Tray 2 (flips up when legal-size paper is loaded)

5 Power connection

6 Formatter (contains the interface ports) 4 Chapter 1 Assembly locations

2 Printer specifications

Review the following specifications for the printers. Printer specifications 5

Printer dimensions

Review the dimensions for the printers. HP LaserJet Pro M501 dimensions Review the following dimensions for the M501 printer. NOTE: An optional 550-sheet paper feeder is available for this printer. The M501 printer supports one paper feeder. Figure 2-1 Dimensions for the base printer 1 2 2 3 3 1 Table 2-1 Dimensions for the base printer Measurement Printer fully closed Printer fully opened 1. Height 289 mm (11.4 in) 289 mm (11.4 in) 2. Depth Tray 2 dust cover closed: 376 mm (14.8 in) Tray 2 dust cover open: 444 mm (17.5 in) 569 mm (22.4 in) 3. Width 410 mm (16.1 in) 410 mm (16.1 in) Weight 11.6 kg (25.6 lb) 6 Chapter 2 Printer specifications

Figure 2-2 Dimensions for the printer with the 1 x 550-sheet paper feeder 1 1 2 2 3 3 Table 2-2 Dimensions for the printer with the 1 x 550-sheet paper feeder Measurement Printer and accessories fully closed Printer and accessories fully opened1 1. Height 419 mm (16.5 in) 419 mm (16.5 in) 2. Depth Tray 2 dust cover closed: 376 mm (14.8 in) Tray 2 dust cover open: 444 mm (17.5 in) 569 mm (22.4 in) 3. Width 410 mm (16.1 in) 410 mm (16.1 in) Weight 15.4 kg (34 lb) 1 These values are subject to change. For current information, go to <http://www.hp.com/support/ljM501> or <http://www.hp.com/support/ljM506> or <http://www.hp.com/support/ljE50045> or <http://www.hp.com/support/ljM507> or <http://www.hp.com/support/ljE50145> or <http://www.hp.com/support/ljM527mfp> or <http://www.hp.com/support/ljE52545mfp> or <http://www.hp.com/support/ljM528mfp> or <http://www.hp.com/support/E52645mfp>. Figure 2-3 Dimensions for the 1 x 550-sheet paper feeder 1 2 2 3 3 1 Table 2-3 Dimensions for the 1 x 550-sheet paper feeder

Measurement Value

1. Height 130 mm (5.1 in) 2. Depth Tray closed: 376 mm (14.8 in) Tray opened: 569 mm (22.4 in) 3. Width 410 mm (16.1 in) Weight 3.8 kg (8.4 lb) HP LaserJet Pro M501 dimensions 7

HP LaserJet Enterprise M506/E50045/M507/E50145 dimensions Review the following dimensions for the M506/E50045/M507/E50145 printer. NOTE: An optional 550-sheet paper feeder is available for this printer. The M506/M507/E50145 printer supports up to three of these paper feeders at a time. Figure 2-4 M506/M507 dimensions for the n and dn models; E50045/E50145 dimensions for the dn model 1 2 2 3 3 1 Table 2-4 M506/M507 dimensions for the n and dn models; E50045/E50145 dimensions for the dn model Measurement Printer fully closed Printer fully opened 1. Height 296 mm (11.7 in) 296 mm (11.7 in) 2. Depth Tray 2 dust cover closed: 376 mm (14.8 in) Tray 2 dust cover open: 444 mm (17.5 in) 569 mm (22.4 in) 3. Width 410 mm (16.1 in) 410 mm (16.1 in) Weight 12 kg (26.5 lb) Figure 2-5 M506/M507 dimensions for the x model 1 1 2 2 3 3 Table 2-5 M506/M507 dimensions for the x model Measurement Printer fully closed Printer fully opened 1. Height 415 mm (16.3 in) 487 mm (19.2 in) 2. Depth Tray 2 dust cover closed: 376 mm (14.8 in) Tray 2 dust cover open: 444 mm (17.5 in) 569 mm (22.4 in) 8 Chapter 2 Printer specifications

Table 2-5 M506/M507 dimensions for the x model (continued) Measurement Printer fully closed Printer fully opened 3. Width 410 mm (16.1 in) 410 mm (16.1 in) Weight 13.4 kg (29.5 lb) Figure 2-6 Dimensions for the 1 x 550-sheet paper feeder 1 2 2 3 3 1 Table 2-6 Dimensions for the 1 x 550-sheet paper feeder

Measurement Value

1. Height 130 mm (5.1 in) 2. Depth Tray closed: 376 mm (14.8 in) Tray opened: 569 mm (22.4 in) 3. Width 410 mm (16.1 in) Weight 1.4 kg (3 lb) Figure 2-7 Dimensions for the cabinet/stand 1 3 2 3 2 1 Table 2-7 Dimensions for the cabinet/stand

Measurement Value

1. Height 381 mm (15.0 in) 2. Depth Door closed: 632 mm (24.9 in) Door opened and rear castors rotated: 865 mm (34.0 in) HP LaserJet Enterprise M506/E50045/M507/E50145 dimensions 9

Table 2-7 Dimensions for the cabinet/stand (continued)

Measurement Value

3. Width Door closed: 600 mm (23.6 in) Door opened and rear castors rotated: 630 mm (24.8 in) Weight 9.0 kg (20 lb) Figure 2-8 M506/E50045/M507/E50145 dimensions for the printer with three 1 x 550-sheet paper feeders and the cabinet/stand 1 3 2 1 3 2 Table 2-8 M506/E50045/M507/E50145 dimensions for the printer with three 1 x 550-sheet paper feeders and the cabinet/stand Measurement Printer and accessories fully closed Printer and accessories fully opened1 1. Height 1067 mm (42.0 in) 1067 mm (42.0 in) 2. Depth 632 mm (24.9 in) 865 mm (34.0 in) 3. Width 600 mm (23.6 in) 630 mm (24.8 in) Weight 25.2 kg (55.5 lb) 1 Do not extend more than one paper tray at a time. HP LaserJet Enterprise MFP M527/E52545/M528/E52645 dimensions Review the following dimensions for the M527/E52545/M528/E52645 printer. NOTE: An optional 550-sheet paper feeder is available for this printer. The M527/E52545/M528/E52645 printer supports up to three of these paper feeders at a time. 10 Chapter 2 Printer specifications

Figure 2-9 M527/M528 dimensions for the dn, f, c, and z models; E52545/E52645 dimensions for the dn and c models 1 1 2 2 3 3 Table 2-9 M527/M528 dimensions for the dn, f, c, and z models; E52545/E52645 dimensions for the dn and c models Measurement Printer fully closed Printer fully opened 1. Height 497 mm (19.6 in) 750 mm (29.5 in) 2. Depth Tray 2 dust cover closed: 496 mm (19.5 in) Tray 2 dust cover open: 559 mm (22.0 in) 674 mm (26.5 in) 3. Width 482 mm (19.0 in) 482 mm (19.0 in) Weight 23 kg (50.8 lb) Figure 2-10 Dimensions for the 1 x 550-sheet paper feeder 1 2 2 3 3 1 Table 2-10 Dimensions for the 1 x 550-sheet paper feeder

Measurement Value

1. Height 130 mm (5.1 in) 2. Depth Tray closed: 376 mm (14.8 in) Tray opened: 569 mm (22.4 in) 3. Width 410 mm (16.1 in) Weight 1.4 kg (3 lb) HP LaserJet Enterprise MFP M527/E52545/M528/E52645 dimensions 11

Figure 2-11 Dimensions for the cabinet/stand 1 3 2 3 2 1 Table 2-11 Dimensions for the cabinet/stand

Measurement Value

1. Height 381 mm (15.0 in) 2. Depth Door closed: 632 mm (24.9 in) Door opened and rear castors rotated: 865 mm (34.0 in) 3. Width Door closed: 600 mm (23.6 in) Door opened and rear castors rotated: 630 mm (24.8 in) Weight 9.0 kg (20 lb) 12 Chapter 2 Printer specifications

Figure 2-12 M527/E52545/M528/E52645 dimensions for the printer with three 1 x 550-sheet paper feeders and the cabinet/stand 1 3 2 1 3 2 Table 2-12 M527/E52545/M528/E52645 dimensions for the printer with three 1 x 550-sheet paper feeders and the cabinet/stand Measurement Printer and accessories fully closed Printer and accessories fully opened1 1. Height 1268 mm (49.9 in) 1521 mm (59.9 in) 2.

Depth 632 mm (24.9 in) 865 mm (34.0 in) 3. Width 600 mm (23.6 in) 630 mm (24.8 in) Weight 36.2 kg (79.8 lb) 1 Do not extend more than one paper tray at a time.

Printer space requirements

HP recommends that the following distances be added to the printer dimensions provided in this chapter to make sure there is sufficient space to open doors and covers, and to provide proper ventilation. See "HP LaserJet Enterprise M506/M507/E50145 dimensions" or "HP LaserJet Enterprise MFP M527/ M528/E52645 dimensions" in the Troubleshooting Manual. ● From the left-side or right-side of the printer to an obstruction, add: Printer space requirements 13

– 430 mm (17 in) ● From the front-side of the printer to an obstruction, add: – 610 mm (24 in) ● From the back-side of the printer to an obstruction, add: – 460 mm (18 in) Power consumption, electrical specifications, and acoustic emissions Review the following information about power consumption, electrical specifications, and acoustic emissions. See <http://www.hp.com/support/ljm501> or <http://www.hp.com/support/ljm506> or <http://www.hp.com/support/lje50045> or <http://www.hp.com/support/ljm507> or <http://www.hp.com/support/lje50145> or <http://www.hp.com/support/ljm527mfp> or <http://www.hp.com/support/lje52545mfp> or <http://www.hp.com/support/ljm528mfp> or <http://www.hp.com/support/E52645mfp> for current information. CAUTION: Power requirements are based on the country/region where the printer is sold. Do not convert operating voltages. This will damage the printer and void the printer warranty.

Operating-environment range

Review the operating-environment range. Table 2-13 Operating-environment specifications¹

Environment Recommended Allowed

Temperature 17° to 25°C (62.6° to 77°F) 15° to 30°C (59° to 86°F) Relative humidity 30% to 70% relative humidity (RH) 10% to 80% RH 1 These values are subject to change. For current information, go to <http://www.hp.com/support/ljm501> or <http://www.hp.com/support/ljm506> or <http://www.hp.com/support/lje50045> or <http://www.hp.com/support/ljm507> or <http://www.hp.com/support/lje50145> or <http://www.hp.com/support/ljm527mfp> or <http://www.hp.com/support/lje52545mfp> or <http://www.hp.com/support/ljm528mfp> or <http://www.hp.com/support/E52645mfp>. 14 Chapter 2 Printer specifications

3 Theory of operations

Learn about the printer theory of operations.

Conventions used in this guide

Learn about the conventions used in this publication. TIP: Tips provide helpful hints or shortcuts. NOTE: Notes provide important information to explain a concept or to complete a task. CAUTION: Cautions indicate procedures that you should follow to avoid losing data or damaging the product. WARNING! Warnings alert you to specific procedures that you should follow to avoid personal injury, catastrophic loss of data, or extensive damage to the product.

HP service and support

Learn about access to additional service and support information.

Additional service and support for channel partners

Channel partners can use the HP Partner Portal or the Channel Services Delivery Platform (CSDP) to access the HP Web-based Interactive Search Engine (WISE). Find information about the following topics. ● Service manuals, service cost data, and service advisories ● The latest control panel message (CPMD) troubleshooting information ● Remove and replace part instructions and videos ● Solutions for printer issues and emerging issues ● Printer specifications, warranty, and regulatory information ● Install, configure, and how to information View a video of how to use the HP Partner First Portal to access WISE. Access WISE for Channel partners (HP Partner Portal) Theory of operations 15

1. Sign-in to the HP Partner Portal (click <https://partner.hp.com> to access the sign-in page). 2. Select the Services & Support item (near the top of the screen). 3. Select the Technical Support (WISE) item. 4. Select the Technical Documentation item. 5. Use the WISE portal to search for printer information. Access WISE for Channel partners (CSDP) 1. Sign-in to the CSDP portal (click csdp.hp.com to access the sign-in page). 2. Select the Knowledge and Training item. 3. Select the HP Technical Documentation item. 4. Use the WISE portal to search for printer information. Additional service and support for HP internal personnel HP internal personnel, go to one of the following Web-based Interactive Search Engine (WISE) sites: View a video of how to access WISE for internal HP users. Americas (AMS) ● WISE - English ● WISE - Spanish ● WISE - Portuguese ● WISE - French Asia Pacific / Japan (APJ) ● WISE - English ● WISE - Japanese ● WISE - Korean ● WISE - Chinese (simplified) ● WISE - Chinese (traditional) ● WISE - Thai Europe / Middle East / Africa (EMEA) ● WISE - English

Additional technical support WISE videos

The videos below provide additional ways to access printer information using WISE. View a video of how to use WISE to find technical support videos (model number search). 16 Chapter 3 Theory of operations

View a video of how to use WISE to find technical support videos (Product detail page [PDP] search). HP service and support 17

Basic operation

The printer routes all high-level processes through the formatter, which stores font information, processes the print image, and communicates with the host computer. The basic printer operation comprises the following systems: ● The engine-control system, which includes the high-voltage and low-voltage power supplies (HVPS and LVPS), fuser control circuits, and the DC controller printed circuit assembly (PCA) ● The laser/scanner system, which forms the latent image on the photosensitive drum ● The image-formation system, which transfers a toner image onto the paper ● The pickup, feed and delivery system, which uses a system of rollers and belts to transport the paper through the printer ● Accessory (optional paper feeder) Figure 3-1 Relationship between the main printer systems

Engine-control system

Laser scanner system

Image-formation system

Pickup, feed, and delivery system

Accessory

18 Chapter 3 Theory of operations

Figure 3-2 System block diagram

Sequence of operation

Review the following information about how the DC controller PCA controls the printer operating sequence. Table 3-1 Sequence of operation

Period Duration Description

Waiting From the time the power is turned on, the door is closed, or when the printer exits Sleep mode until the printer is ready for printing. ● Heats the fuser film in the fuser ● Detects the toner cartridge ● Rotates and stops each motor ● Rotates and stops each fan ● Cleans the transfer roller Standby From the end of the waiting sequence, the last rotation until the formatter receives a print command, or until the printer is turned off. ● Is in the Ready state ● Enters Sleep mode if the formatter sends the sleep command ● Rotates and stops each fan Sequence of operation 19

Table 3-1 Sequence of operation (continued)

Period Duration Description

Initial rotation From the time the formatter receives a print command until the paper enters the paper path. ● Rotates each motor ● Rotates each fan ● Activates the high-voltage power supply (high voltage bias) ● Prepares the laser/scanner unit ● Warms the fuser to the correct temperature Printing From the time the first sheet of paper enters the paper path until the last sheet has passed through the fuser. ● Forms the image on the photosensitive drums ● Transfers the toner to the paper ● Fuses the toner image onto the paper Last rotation From the time the last sheet of paper exits the fuser until the motors stop rotating. ● Stops each motor ● Stops each fan ● Stops the high-voltage power supply (high voltage bias) ● Stops the laser/scanner unit ● Turns the fuser heater off ● If another print command is received, the printer enters the initial rotation period when the last rotation is complete. 20 Chapter 3 Theory of operations

Formatter-control system

Review the following information about how the formatter operates. The formatter performs the following functions: ● Controls sleep mode ● Receives and processes print data from the various printer interfaces ● Monitors control panel functions and relaying printer status information through the control panel and the network or bi-directional interface ● Develops and coordinates data placement and timing with the DC controller PCA ● Stores font information ● Communicates with the host computer through the network or the bidirectional interface The formatter receives a print job from the network or bidirectional interface and separates it into image information and instructions that control the printing process. The DC controller PCA synchronizes the image formation system with the paper input and output systems, and then signals the formatter to send the print image data. Sleep mode (all except M501) This feature conserves power after the printer has been idle for an adjustable period of time. When the printer is in Sleep mode, the printer retains all settings, downloaded fonts, and macros. The default setting is for Sleep mode to be enabled, and the printer enters Sleep mode after a 30-second idle time. NOTE: In the General Settings menu (a submenu of the Administration menu), this item is termed Sleep Timer Settings. The printer firmware uses a combination of timers and Sleep settings to control when the printer enters a different state, as well as what states the printer will enter. The available states are listed below, in descending order, from using the most power to using the least power: ● Active: The printer control panel is fully illuminated. The power button light is illuminated. ● Shallow sleep: The printer control panel is dim and the content is grayed out, but is still readable. The power button light is illuminated. ● Sleep: The printer control panel is off (blackened out). The power button light blinks once every three seconds. ● Deep sleep: The printer control panel is off (blackened out). The power button light blinks once every three seconds. The control panel and power button appearance is the same in this state as the sleep state. However, the printer is drawing less than 1 watt of power in the deep sleep state (as opposed to 6 watts of power in the sleep state). ● Off: This state is entered by pressing the power button or removing power from the printer. The power button light is not illuminated. The printer exits Sleep mode and enters the warm-up cycle when any of the following events occur: ● The printer receives a print job, valid data, or a PML or PJJ command. ● A control-panel button is pressed or the touchscreen is touched. Formatter-control system 21

- A cover or door is opened.
- The engine-test switch is pressed.
- A paper tray, other than Tray 1, is opened. NOTE: If the printer is in the deep sleep state, opening a paper tray will not cause the printer to exit Sleep mode. NOTE: Printer error messages override the Sleep message. The printer enters Sleep mode at the appropriate time, but the error message continues to appear. TIP: When the printer is in Sleep mode, the sub-power supply is off and the low-voltage power supply is on. Sleep delay (M501) When the printer is in sleep delay mode, the control-panel backlight is turned off, but the printer retains all printer settings, downloaded fonts, and macros. The default setting is a 5-minute idle time. This setting can be changed by using the control panel menus or the HP Embedded Web Server (EWS). The printer exits sleep delay mode and enters the warm-up cycle when any of the following occurs:
 - A print job, valid data, or a PML or PDL command is received at the serial port.
 - The control panel is touched (button press or touchscreen touch depending on model).
 - A tray is opened. TIP: Error messages override the sleep delay message. The printer enters sleep mode at the appropriate time, but the error message continues to appear.

Auto On / Auto Off mode (M501) Use the following procedure to select the Sleep/Auto Off delay time. 1. On the printer control panel, press the OK button. 2. Open the following menus: ● System Setup ● Energy Settings ● Sleep/Auto Off After 3. Use the arrow keys to select the time for the Sleep/Auto Off delay, and then press the OK button. Printer job language (PDL) PDL is an integral part of printer configuration, in addition to the standard printer command language (PCL) and PostScript (PS). With standard cabling, the printer can use PDL to perform a variety of functions, such as these: 22 Chapter 3 Theory of operations

- Two-way communication with the host computer through a network connection or a USB device port: The printer can inform the host about the control-panel settings, and the control-panel settings can be changed from the host.
- Dynamic I/O switching: The printer uses this switching to be configured with a host on each I/O. The printer can receive data from more than one I/O simultaneously, until the I/O buffer is full. This can occur even when the printer is offline.
- Context-sensitive switching: The printer can automatically recognize the personality (PS or PCL) of each job and configure itself to serve that personality.
- Isolation of print environment settings from one print job to the next: For example, if a print job is sent to the printer in landscape mode, the subsequent print jobs print in landscape only if they are formatted for landscape printing. Printer management language (PML) PML allows remote configuration of the printer and status read-back from the printer through the I/O ports.

Control panel

Review the following information about the printer control panels. The M501n and M501dn control panel is a 2-line backlit control panel display with numeric keypad and additional buttons for navigating control panel menus. The M506n and M506dn control panel is a 4-line backlit control panel display with numeric keypad and additional buttons for navigating control panel menus. The M506x, M507x, and E50145dn control panel is a 10.9 cm (4.3 in) full-color SVGA with infrared touchscreen and adjustable viewing angle. The M507n and M507dn control panel is a 6.9 cm (2.7 in) QVGA LCD color control panel with numeric keypad. The control panel for all of the M527, M528, and E52645 models is a 20.3 cm (8 in) full-color SVGA with infrared touchscreen and adjustable viewing angle. Easy-access USB port (all except M501) All models feature easy-access USB printing, for quickly printing files without sending them from a computer. The printer accepts standard USB flash drives in the USB port (usually found near the control panel). It supports the following types of files: ● .pdf ● .jpg ● .prn and .PRN ● .cvt and .Cvt ● .pdl ● .pcl and .PCL ● .ps and .PS Printer management language (PML) 23

- .doc and .docx ● .ppt and .pptx The USB port is disabled by default. Follow the instructions in the printer user guide to enable the USB port and print USB documents. Wireless (wireless models only) The M506x/M507x and M527z/M528z models contain a wireless card to enable wireless direct printing over an 802.11b/g/n wireless connection. NOTE: This card does not enable the printer to connect to the network. Low end data model (LED) overview (M501) The low-end data model (LED) provides one consistent data representation method and defines the dynamic and capabilities tickets shared between clients and devices, as well as the access protocol, event, security, and discovery methods. Advanced control language (ACL) overview (M501) The advanced control language (ACL) is a language that supports printer control and firmware downloads in printers that support both PDL/PCL and host-based printing. Each sequence of ACL commands must be preceded by a unified exit command (UEL) and an @PDL ENTER LANGUAGE=ACL command. The ACL sequence is always followed by a UEL. Any number of commands can be placed between the UELs. The only exception to these rules is the download command. If a firmware download is done, the download command must be the last command in the sequence. It will not be followed by a UEL. The firmware searches for the UEL sequence when parsing commands. However, while downloading binary data such as host-based code or NVRAM data the firmware suspends UEL parsing. To handle hosts that “disappear” during binary sequences, the firmware times out all ACL command sessions. If a timeout occurs during a non-download command sequence, it is treated as the receipt of a UEL. If a timeout occurs during firmware download, the printer resets. Near field communication (NFC) (M506x and M527z models only) The M506x and M527z models support NFC capabilities. NFC enables a connection between the printer and a mobile device, such as a smartphone or tablet, by touching the device to the NFC icon on the bottom of the control panel. Documents and images from the mobile device can then be printed through the wireless card on the printer. NOTE: The customer can purchase a NFC accessory and add this functionality to other M506/M507/ E50145 and M527/M528/E52645 models. CPU The formatter incorporates a 1.2 GHz processor. Input/output (I/O) The printer supports the following interfaces: ● Hi-Speed USB 2.0 24 Chapter 3 Theory of operations

- USB hosts ● 10/100/1000 Ethernet LAN connection with IPv4 and IPv6 ● Fax PCA (fax models only) NOTE: The M527/M528/E52645 printers include a fax phone line connection. ● Easy-access USB printing (no computer required; M506/M507/E50145 and M527/M528/E52645) ● HP near field communication (NFC) for printing from mobile devices (M506x and M527z models only) ● Wi-Fi Direct for printing from mobile devices (M506x/M507x and M527z/M528z models only) Memory The formatter incorporates different types of memory and storage to store the printer firmware as well as print-job data and user settings. NOTE: M501 only: If the printer encounters a problem when managing available memory, a clearable warning message displays on the control panel. Firmware For the M501, Memory on the formatter stores the firmware. A remote firmware upgrade process is used to overwrite and upgrade the firmware. For the M506n/M507n, M506dn/M507dn/E50145dn, M506x/M507x, and M527dn/M528dn/E52645dn models, the embedded Multi-Media Card (eMMC) on the formatter stores the firmware. For the M527c/ M528c/E52645c, M527f/M528f, and M527z/M528z models, the high-performance hard disk (HDD) stores the firmware. A firmware upgrade process is used to overwrite and upgrade the firmware. The upgrade can use a network connection (remote upgrade) or be accomplished by using a USB flash drive. Nonvolatile random access memory (NVRAM) The printer uses

NVRAM to store printer and user configuration settings. The contents of NVRAM are retained when the printer is turned off or disconnected.

Random access memory (RAM) The RAM on the formatter serves as a temporary storage area for printing and system operation. HP Memory Enhancement technology (MEt) MEt effectively doubles the amount of standard memory through a variety of font- and data compression methods. NOTE: MEt is available only when printing in printer command language (PCL) mode. It is not functional when printing in PostScript (PS) mode. Memory 25

Engine-control system

The engine-control system receives commands from the formatter and interacts with the other main systems to coordinate all printer functions.

Figure 3-3 Engine-control system

Formatter

Engine-control system

DC controller

Low-voltage power supply

Laser scanner system

Image-formation system

Pickup, feed, and delivery system

Accessory

High-voltage power supplies

Fuser power supply

Formatter

Engine-control system

DC controller

Low-voltage power supply

Laser scanner system

Image-formation system

Pickup, feed, and delivery system

Accessory

High-voltage power supplies

Fuser power supply

DC controller

The DC controller controls the operation of the printer and its components. The DC controller starts the printer operation when the printer power is turned on and the power supply sends DC voltage to the DC controller. After the printer enters the standby period, the DC controller sends out various signals to operate motors, solenoids, and other printer components based on the print command and image data that the host computer sends. 26 Chapter 3 Theory of operations

Figure 3-4 DC controller block diagram

Transfer roller

High-voltage

power supply Fan

Cartridge

LED Fan Motor Clutch Solenoid Switch

Photo interrupter

Formatter

Control panel

Accessory

Laser scanner ass'y Fuser AC input

Low-voltage

power supply

Photo interrupter

Fuser power

supply

DC controller

Switch Sensor Motors The printer has four motors. The motors drive the components in the paper-feed and image-formation systems. The DC controller monitors the fuser motor and the scanner motor to determine if a motor has failed. It notifies the formatter when it encounters the following conditions: ● Startup failure: the motor does not reach a specified speed within a specified time from when the motor starts. ● Rotational failure: the rotational speed of the motor is not in the specified range for a specified time after the motor reaches a specified speed. Motors 27

Table 3-2 Motors

Abbreviation Name Purpose Failure detection

M1 Fuser motor Drives the pressure roller and delivery roller; the pressurization and release of the pressure roller; and the engagement and disengagement of the primary and secondary transfer rollers Yes M2 Pickup motor Drives the Tray 1 pickup roller, the Tray 2 pickup and feed rollers, the registration roller, and the feed roller Yes M3 Scanner motor Drives the scanner mirror Yes M4 Lifter motor Drives the Tray 2 lifter Yes Fans The printer has two fans for preventing the temperature from rising in the printer and for cooling the printed pages. The DC controller determines if there is a fan failure and notifies the formatter if a fan locks for a specified time from when the fan starts. Table 3-3 Fans

Abbreviation Name Cooling area Type Speed

FM1 Main fan Toner cartridge and laser scanner assembly

Intake Full

FM2 Sub fan Around the low-voltage power supply and formatter Intake Full/half

Solenoids

Solenoids are used in printer control. Table 3-4 Solenoids

Component abbreviation Component name

SL1 Tray 2 pickup solenoid SL2 Tray 1 pickup solenoid SL3 Duplex switchback solenoid (duplex models only) Clutches Clutches are used in printer control. Table 3-5 Clutches

Component abbreviation Component name

CL1 Duplex re-pickup clutch (duplex models only) 28 Chapter 3 Theory of operations

Switches Switches are used in printer control. Table 3-6 Switches

Component abbreviation Component name

SW1 Power switch SW2 Cartridge door switch SW4 Tray 2 detection switch SW5 Rear door switch SW201 Test print switch

Photo interrupter sensors

Photo interrupter sensors are used in printer control. Table 3-7 Photo interrupter sensors

Component abbreviation Component name

PS1a Media-width sensor PS2b Duplex feed sensor (duplex models only) PS1 Tray 2 media out sensor PS2 Top-of-page (TOP) sensor PS3 Tray 1 media-out sensor PS4 Output bin media-full sensor PS11 Registration sensor PS12 Media surface sensor PS13 Fuser output sensor
Sensors Sensors are used in printer control. Table 3-8 Sensors

Component abbreviation Component name

TH401 Environment sensor LEDs LEDs indicate if a printer system is correctly functioning. Switches 29
Table 3-9 LEDs

Component abbreviation Component name

LED1 Power supply LED Low-voltage power supply (LVPS) The low-voltage power-supply (LVPS) circuit converts the AC power from the wall receptacle into the DC voltage that the printer components use. Figure 3-5 Low-voltage power-supply circuit (M501)

+3.3VA

AC input +24VA

+3.3VA

+3.3VB +3.3VH

FET FET

FET

+24VB +24VC

+24VD FET +24VE

+24VA +24VBSNS

PWRSW

Low-voltage power supply

Rectifying

circuit

Protection

circuit +3.3V generation circuit Fuse FU101 Fuse FU102 +24V generation circuit Fuser

Power switch

SW1

DC controller

High-voltage power supply

Interlock switch

SW2

Frequency

detection circuit

Fuser control

circuit

Fuser power supply

FREQSNS

30 Chapter 3 Theory of operations

Figure 3-6 Low-voltage power-supply circuit (M506/E50045/M507/E50145, M527/E52545/M528/E52645) AC input +5VA +24VA

+3.3VA

+3.3VB +3.3VH

FET FET

FET

+24VB +24VC

+24VD FET +24VE

+24VA +24VBSNS

FET +5VC +5VA PWRSW

Low-voltage power supply

Rectifying

circuit

Protection

circuit +5V generation circuit Fuse FU101 Fuse FU102 +24V generation circuit Fuser

Power switch

SW1

DC controller

High-voltage power supply

Interlock switch

SW2

Frequency

detection circuit

Fuser control

circuit

Fuser power supply

+3.3V generation circuit

FREQSNS

Low-voltage power supply voltages description

The low-voltage power supply converts the AC power into two DC voltages, which it then subdivides, as described in the following table. Table 3-10 Converted DC voltages

Main DC voltage Sub-voltage Behavior

+24 V +24 VA Constantly supplied

Stopped during inactive OFF

Becomes 4V during active OFF +24 V +24 VB Stopped when cartridge door is opened. (SW2)

Stopped during active OFF or inactive OFF

Low-voltage power supply voltages description 31

Table 3-10 Converted DC voltages (continued)

Main DC voltage Sub-voltage Behavior

+24 V +24 VC Stopped when cartridge door is opened. (SW2)

Stopped during active OFF or inactive OFF

+24 V +24 VD Constantly supplied

Stopped during active OFF or inactive OFF

+24 V +24 VE (M527 only) Constantly supplied

Stopped during active OFF or inactive OFF

+5 V +5 VA Constantly supplied Becomes 3.4V during inactive OFF +5 V +5 VB Constantly supplied

Stopped during active OFF or inactive OFF

+5 V +5 VC Constantly supplied

Stopped during inactive OFF

+3.3 V +3.3 VA Constantly supplied +3.3 V +3.3 VB Constantly supplied

Stopped during inactive OFF

+3.3 V +3.3 VH Constantly supplied

Stopped during inactive OFF

Supplied intermittently during inactive OFF

Over-current/over-voltage protection The low-voltage power supply automatically stops supplying the DC voltage to the printer components whenever it detects excessive current or abnormal voltage. The low-voltage power supply has a protective circuit against over-current and over-voltage to prevent failures in the power supply circuit. CAUTION: If DC voltage is not being supplied from the low-voltage power supply, the protective function might be running. In this case, turn the power switch off and unplug the power cord. Do not turn the power switch on until the root cause is found and corrected. If the protective function is active, the DC controller notifies the formatter of a low-voltage power supply failure. In addition, the low-voltage power supply has two fuses to protect against over-current. If over-current flows into the AC line, the fuse stops the AC power. Sleep mode operation (all except M501) Sleep mode conserves energy by stopping the power to several components when the printer is idle. If the DC controller detects voltage that is too high when the printer is in Sleep mode, it determines that the low-voltage power supply has failed, and it notifies the formatter. 32 Chapter 3 Theory of operations

Low-voltage power supply failure detection (all except M501) The DC controller determines a low-voltage power supply failure and notifies the formatter when the low-voltage power supply does not supply +24 V. Safety Review the following information about the LVPS safety protection. For personal safety, the printer interrupts +24VB and +24VC power when the cartridge door detection switch is turned off (see the "Converted DC voltages" table in the Troubleshooting Manual). This stops DC power supply to the following load: ● High-voltage power supply (HVPS) The remote switch control circuit turns on or off the printer power so that the AC power flows even the power switch is turned

off. Unplug the printer power cord before disassembling the printer.

Low-voltage power supply functions

The printer has the following low-voltage power supply functions: Table 3-11 Low-voltage power supply functions

Function Supported feature

Sleep mode No

Power supply voltage detection No

Automatic power OFF No

Automatic power ON/OFF No

Active OFF Yes

Inactive OFF Yes

Network mode No

Power switch illumination Yes

Low-voltage power supply failure detection Yes

Power save mode No

Fast boot mode Yes

High-voltage power supply

The DC controller controls the high-voltage power supply (HVPS) to generate biases. The high-voltage power supply delivers the high-voltage biases to the following components used to transfer toner during the image-formation process: ● Primary charging roller (in the toner cartridges) ● Developing roller (in the toner cartridges) ● Transfer roller ● Pressure roller Low-voltage power supply failure detection (all except M501) 33

Fuser bias

The printer uses instant-on fusing. The fuser bias is DC positive for improved print quality. The fuser bias circuit is located in the high-voltage power supply.

High-voltage power supply circuits

The high-voltage power supply contains the following separate circuits. Figure 3-7 High-voltage power supply circuits PRI DEV TR

DC controller High-voltage power supply

To primary charge roller

To developing roller

Cartridge

Photosensitive drum

Transfer roller

Primary charging bias circuit

Developing

bias circuit

Transfer bias

circuit Table 3-12 High-voltage power supply circuits

Circuit Description

Primary-charging-bias generation The primary charging bias negatively charges the surface of the photosensitive drum to prepare for image formation. Developing-bias generation The developing bias adheres toner to an electrostatic latent image formed on the photosensitive drum. Transfer-bias generation The primary transfer bias transfers the toner from the photosensitive drum onto the paper.

Fuser control

Learn about fuser control functions. The DC controller and components in the fuser perform the following functions related to fuser operation:

- Control fuser temperature
- Detect fuser failures
- Prevent excessive temperature rise 34 Chapter 3 Theory of operations
- Detect remaining life in the fuser
- Determine if the correct fuser is installed

Fuser circuits

The fuser heater control circuit and the fuser heater safety circuit control the fuser temperature according to commands from the DC controller. The fuser consists of the following major components: Figure 3-8 Fuser components Table 3-13 Fuser components

Type of component Abbreviation Name Function

Heaters H1 Fuser main heater Heats the center of the fuser sleeve

Thermistors

(Contact type) TH1 Main thermistor Detects the center temperature of the fuser heater

Thermal switch

(Contact type) TP1 Thermal switch Prevents an abnormal temperature rise in the fuser heater

Fuser control functions

The diagram below describes the LVPS, DC controller, and fuser systems. Fuser circuits 35

Figure 3-9 Fuser control

Fuser heater

safety circuit

FREQSNS

TP1 Fuser heater (H1) Fuser film ass'y TP1: Thermal switch TH1: Thermistor Fuser

Pressure roller

FSRD RLYD FSRTN +24V

Low-voltage supply supply

Fuser control

RL101

DC controller

Fuser heater

control circuit

Relay control

circuit

Current detection

circuit TH1 The printer has the following fuser control functions. Table 3-14 Fuser control functions

Failure detection function Supported feature

Fuser temperature control Yes

Fuser failure detection Yes

Frequency detection circuit failure detection Yes

Fuser pressure release mechanism failure detection No

Fuser type identification detection Yes

Fuser presence detection No

36 Chapter 3 Theory of operations

Table 3-14 Fuser control functions (continued)

Failure detection function Supported feature

Fuser life detection Yes

Relay failure detection No

Fuser roller cleaning Yes

Fuser heater protection

Fuser heater protection is a feature that detects excessive temperatures in the fuser and interrupts the power supply to the fuser heater. The following three protective components prevent the fuser heater from excessive rising temperature: ● DC controller: When a thermistor or sub-thermistor detects a temperature above a certain threshold, the DC controller interrupts power to the specific heater. ● Fuser-heater safety circuit: The fuser heater safety circuit monitors the detected temperature of the sub thermistors. ● Thermal switch: If the temperature in the heaters is abnormally high, and the temperature in the thermoswitch exceeds a specified value, the contact to the thermoswitch breaks. Fuser unit life detection (all except M501) The fuser life is tracked by fuser rotations, and not by the number of pages printed. This is a more accurate tracking method since the fuser rotates for every print job. There will be variations in fuser life depending on customer usage. Customers who are running one and two page intermittent jobs with long pauses between each job might reach the fuser low message sooner due to the fuser rotating more times per page than it would for larger print jobs. Fuser identification (all except M501) The printer detects the type and presence of the fuser. The DC controller notifies the formatter when it fails to detect the type or presence of the fuser. NOTE: This printer detects if a fuser of the correct voltage for the printer is installed. If a fuser of the incorrect voltage is installed, the DC controller notifies the formatter and an error message is displayed on the control panel. Fuser heater protection 37

Engine laser/scanner system The laser/scanner system forms the latent electrostatic image on the photosensitive drums inside each of the toner cartridges. The DC controller receives instructions from the formatter regarding the image of the page to be printed. The DC controller signals the lasers to emit light, and the laser beams pass through lenses and onto the scanner mirror, which rotates at a constant speed. The mirror reflects the beam onto the photosensitive drum in the pattern required for the image, exposing the surface of the drum so it can receive toner. The main components of the laser/scanner system, which are controlled by signals sent from the DC controller, are: ● Laser assembly ● Scanner motor assembly ● Beam detect (BD) sensor ● Scanner mirror Figure 3-10 Laser/scanner system Laser ass'y

BD sensor

Scanner mirror

Scanner motor ass'y

Photosensitive drum

DC controller

Laser/scanner failure detection The DC controller determines an optical unit failure and notifies the formatter of the error status when any of the following occurs: ● Beam detect (BD) failure detection: The scan control board (SCB) does not detect the laser/scanner and/or the beam detect interval is outside a specified range during printing. 38 Chapter 3 Theory of operations

● Laser/scanner motor startup failure: The scanner motor does not reach a specified rotation frequency within a specified period of time from when the laser/scanner starts up. ● Laser/scanner motor abnormal rotation: The laser/scanner motor does not reach a specified rotational frequency within a specified period of time during a print operation. Safety The laser/scanner assembly has a mechanical laser shutter. For the safety of users and service technicians, the laser shutter interrupts the optical path of the laser/scanner assembly when the top door is opened (SW101).

Image-formation process

The image-formation system creates the printed image on the paper. The system consists of the following components: ● Toner cartridge ● Transfer roller ● Fuser ● Laser/scanner ● High-voltage power supply The DC Controller controls the internal components of the image formation system (according to commands received from the formatter) to form the toner image on the photosensitive drum surface. The toner image is then transferred to the print media and fused. Safety 39

Figure 3-11 Image-formation system Laser scanner ass'y Fuser

Cartridge

Transfer roller

High-voltage power supply

DC controller

The fuser motor (M1) drives the following image formation components: ● Photosensitive drum ● Developing roller ● Primary charging roller (follows the photosensitive drum) ● Transfer roller (follows the photosensitive drum) ● Pressure roller ● Fuser film (follows the pressure roller) NOTE: The primary charging roller and developer roller are located in the toner cartridge. 40 Chapter 3 Theory of operations

Figure 3-12 Fuser motor

DC controller

M1

Abbreviation Component

M1 Fuser motor The following figure shows the location of the toner-level sensor. Image-formation process 41

Figure 3-13 Toner-level sensor

DC controller

Toner level

sensor

Abbreviatio

n

Component

N/A Toner-level sensor The image-formation process consists of seven steps divided into five functional blocks. Figure 3-14 Image-formation process : Paper path : Direction of drum rotation : Functional block : Step

Delivery Pickup

Fusing Transfer

Drum cleaning

Developing

Latent image formation

6. Fusing 5. Separation 7. Drum cleaning 1. Primary charging 2. Laser beam exposure 4. Transfer 3. Developing 42 Chapter 3 Theory of operations

Table 3-15 Image-formation process

Functional block Steps Description

Latent image formation 1. Primary charging 2. Laser-beam exposure

An invisible latent image forms on the surface of

the photosensitive drum. Development 3. Developing Toner adheres to the electrostatic latent image on the photosensitive drum. Transfer 4. Transfer 5. Separation The toner image transfers to the paper. Fusing 6. Fusing The toner fuses to the paper to make a permanent image. Drum cleaning 7. Drum cleaning Residual toner is removed from the drum. Step 1: Primary charging The primary charging roller contacts the

photosensitive drum and charges the drum with negative potential. Figure 3-15 Primary charging

Photosensitive drum

Primary charging roller

Primary charging bias

Step 2: Laser-beam exposure The laser beam strikes the surface of the photosensitive drum in the areas where the image will form. The negative charge neutralizes in those areas, which are then ready to accept toner. Step 1: Primary charging 43

Figure 3-16 Laser-beam exposure

Photosensitive drum

Laser beam

Step 3: Development Toner acquires a negative charge as the developing cylinder contacts the developing blade. Because the negatively charged surface of the photosensitive drums have been neutralized where they have been struck by the laser beam, the toner adheres to those areas on the drums. The latent image becomes visible on the surface of each drum. Figure 3-17 Development

Developer roller

Developing bias

Photosensitive drum

Developer blade

Step 4: Transfer The toner image on the photosensitive drum transfers to the paper. Transfer bias applied to the transfer roller attracts the negatively charged toner to the paper. 44 Chapter 3 Theory of operations

Figure 3-18 Primary transfer

Transfer roller

Paper

Transfer bias

Photosensitive

drum Step 5: Separation The elasticity of the paper and the curvature of the photosensitive drum cause the paper to separate from the photosensitive drum. The static-charge eliminator removes excess charge from the paper to make sure that the toner fuses correctly. Figure 3-19 Separation Paper

Transfer roller

Static charge eliminator

Photosensitive

drum Step 6: Fusing To create the permanent image, the paper passes through heated, pressurized rollers to melt the toner onto the page. Fusing bias is added to the pressure roller to improve the print quality. Step 5: Separation 45

Figure 3-20 Fusing Paper

Fuser film

Fuser heater

Pressure roller

Step 7: Drum cleaning The cleaning blade scrapes the residual toner off the surface of the photosensitive drum, and toner is deposited in the toner-collection portion of the cartridge. Figure 3-21 ITB cleaning

Cleaning blade

Residual toner collection box

Photosensitive

drum

Toner cartridge

The printer has one toner cartridge. Design The toner cartridge is filled with toner and consists of the following components: ● Photosensitive drum ● Developer roller ● Primary-charging roller ● Memory chip 46 Chapter 3 Theory of operations

The DC controller rotates the drum motor to drive the photosensitive drum, developer roller, and the primary-charging roller. Figure 3-22 Toner cartridge system

DC controller

Cartridge

Memory chip

Photosensitive drum

Developer roller

M1

Primary charging

roller

Function Supported feature

Toner cartridge presence detection Yes

Toner level detection Yes

Toner cartridge life detection Yes

Toner cartridge miss-installation detection Not applicable (one cartridge)

Drum discharge No

Transfer cleaning Yes

Memory chip

The memory chip is non-volatile memory that stores information about the usage of the toner cartridge and helps protect the customer from counterfeit cartridges. This chip is also used to detect the presence of a cartridge within the printer. The printer reads and writes the data in the memory chip. NOTE: The memory chip does not store any customer print, scan, copy, fax, or digital send information. The memory chip also supports the printer Jet Intelligence toner cartridge management functions. Memory chip 47

Dynamic security enabled printers in this guide

The printers with dynamic security enabled that are discussed in this guide are the M507, E50145, M528, and E52645 printers.

Dynamic security

Learn about dynamic-security-enabled printers. Certain HP printers use cartridges that have security chips or electronic circuitry. Cartridges using a non-HP chip or modified or non-HP circuitry* might not work. And, those that work today might not work in the future. As is standard in the printing business, HP has a process for authenticating cartridges. HP continues to use security measures to protect the quality of our customer experience, maintain the integrity of our printing systems, and protect our intellectual property. These measures include authentication methods that change periodically and might prevent some third-party supplies from working now or in the future. HP printers and original HP cartridges deliver the best quality, security and reliability. When cartridges are cloned or counterfeited, the customer is exposed to quality and potential security risks, compromising the printing experience. *Non-HP Chips and modified or non-HP electronic circuitry are not produced or validated by HP. HP cannot guarantee that these chips or circuitry will work in your printer now or in the future. If you are using a non-original HP cartridge, please check with your supplier to ensure your cartridge has an original HP security chip or unmodified HP electronic circuitry.

Toner seal

The toner cartridge seal is opened automatically when the toner cartridge is installed into the printer.

Toner level and cartridge life detection

Review the following information about toner cartridge toner level and cartridge life detection. Toner level detection: The DC controller detects the remaining toner in the cartridge by the optical detection method and then notifies the formatter of the remaining toner level. Cartridge life detection: Cartridge life detection is measured in two ways. 1. Toner level: ● For the first 75% of cartridge life, toner level is measure through pixel counting by the DC controller. Once the level reaches 25%, the toner level sensor utilizes electrical properties to determine the amount of toner remaining. 2. Rotations of internal components (OPC and developer): ● The DC controller monitors these two life parameters and reports them to the formatter as percent life remaining. End of cartridge life is determined by the lower value of the two.

JetIntelligence

JetIntelligence supports two features for managing toner cartridges. The authentication feature allows customers to specify the use of only genuine HP toner cartridges in the printer. The anti-theft feature enables locking a cartridge to a specific printer or fleet of printers. 48 Chapter 3 Theory of operations

Authentication

The genuine HP authentication feature allows a customer to specify that only genuine HP supplies can be used in a printer. If a non-HP or used supply is installed, the printer will not print. This feature is disabled by default, and can be enabled or disabled from the control panel or the Embedded Web Server (EWS). If a genuine HP toner cartridge from another printer is moved to a printer with this feature enabled, the toner cartridge will authenticate and print, unless the toner cartridge has passed the low state. If the toner cartridge has passed the low state, an Unauthorized Cartridge message displays on the control panel. If a non-HP toner cartridge is used in a printer with this feature enabled, the message Unauthorized Cartridge appears on the control-panel display. NOTE: If a customer suspects they have a counterfeit cartridge, they should report it by going to www.hp.com/go/anticounterfeit and selecting Report now. The printer still functions normally.

Anti-theft

The toner cartridge anti-theft feature allows a customer to configure the printer to automatically lock genuine HP toner cartridges to a specific printer or fleet of printers when they are installed. A locked toner cartridge will only work in the specified printer or fleet of printers. This feature prevents toner cartridges from being stolen and used in another printer, or from being moved from an authorized printer to an unauthorized printer. This feature is disabled by default, and can be enabled or disabled from the control panel, the Embedded Web Server (EWS), or Web Jetadmin. When the anti-theft feature is enabled, the toner cartridge in a printer will only work in the specified printer or fleet of printers. If a locked toner cartridge is moved to another printer, the cartridge will not print and the message Protected Cartridge appears on the control-panel display. IMPORTANT: When a toner cartridge is locked to a specific printer or fleet of printers, it cannot be unlocked. This is a permanent operation. Authentication 49

Paper handling system

The DC controller controls the paper handling system according to commands from the formatter. The paper handling system uses a series of rollers to move the paper through the printer. The paper handling system consists of the following three functional blocks. The DC controller controls each block to pick up, feed and deliver the paper. ● Pickup-and-feed-block: Controls the movement of the paper from each pickup source to the fuser inlet ● Fuser-and-delivery-block: Controls the movement of the paper from the fuser to the delivery destination ● Duplex block: Controls the movement of the paper from the duplex switchback unit to the duplex re-pickup unit (duplex models only) 50 Chapter 3 Theory of operations

Figure 3-23 Paper handling system

Duplex paper path

Simplex paper path

Duplex block Pickup-and-feed block

Fuser-and-delivery block

Function Supported feature

Tray 2 media size detection No Tray 2 media presence detection Yes Tray 2 media level detection No Tray 2 lift-down control No Paper handling system 51

Function Supported feature

Tray 2 multiple-feed prevention mechanism Yes Tray 1 media presence detection Yes Tray 1 media width detection No Tray 1 last-media detection No

Skew-feed prevention mechanism Yes

Media detection No

OHT detection No

Image leading edge positioning Yes

Media length detection Yes

Media width detection Yes

Pressure roller pressure release control No

Output bin media-full detection Yes

Automatic delivery Yes

Duplex switchback control (duplex models only) Yes Duplex feed control (duplex models only) Yes

Photo sensors and switches

The following figure shows the photo sensors and switches for the paper handling system. Figure 3-24 Photo sensors and switches SW4 PS1a PS2a PS4 PS13 PS2 PS3 PS11 PS1 PS12 52 Chapter 3 Theory of operations

Table 3-16 Photo sensors and switches

Abbreviation Component

PS1a Media width sensor PS2b Duplex feed sensor (duplex models only) PS1 Tray 2 media out sensor PS2 TOP sensor PS3 Tray 1 media out sensor PS4 Output bin media-full sensor PS11 Registration sensor PS12 Media surface sensor PS13 Fuser output sensor SW4 Tray 2 detection switch Motors, clutches, and solenoids The following figure shows the motors, clutches, and solenoids for the paper handling system. Motors, clutches, and solenoids 53

Figure 3-25 Motors, clutches, and solenoids CL1 SL3 M1 SL1 SL2 M2 M4

Abbreviation Component

M1 Fuser motor M2 Pickup motor M4 Lifter motor SL1 Tray 2 pickup solenoid SL2 Tray 1 pickup solenoid SL3 Duplex switchback solenoid (duplex models only) CL1 Duplex re-pickup clutch (duplex models only) Tray 1 (multipurpose) / Tray 2 (base printer) Moving paper from Tray 1 and Tray 2 involves the interaction of multiple components within the printer. The following sections describe these processes. Tray 1 paper pickup and feed The printer picks up one sheet of paper from Tray 1. 54 Chapter 3 Theory of operations

Following are the sequence of steps for the Tray 1 pickup operation. NOTE: Tray 1 and Tray 2 are optimal for paper pickup when using special paper or media other than 20 lb plain paper. For Tray 1, the printer increases the number of attempts to pick up a page, which increases the reliability of successfully picking the page from the tray and decreases the possibility of a mis-pick jam. HP recommends using Tray 1 or Tray 2 if the printer is experiencing excessive or reoccurring jams from trays other than Tray 1 and Tray 2, or for print jobs that require media other than 20 lb plain paper. Always use Tray 1 when printing envelopes or labels. 1. The pickup motor reverses when a print command is received from the formatter. 2. When the DC controller turns on the Tray 1 pickup solenoid, the Tray 1 pickup roller rotates and the lifting plate lifts. 3. As the lifting plate rises, the paper is picked up. 4. The Tray 1 separation pad removes any multiply-fed sheets, and one sheet is fed into the printer. The Tray 1 media-out sensor (SR8) detects whether paper is present in Tray 1. Figure 3-26 Tray 1 pickup mechanism M5 SR8 SL2 Tray 1 pickup solenoid Tray 1 pickup roller Tray 1 separation pad Tray 1 paper pickup and feed 55

Tray 2 paper presence detection The Tray 2 media-out sensor (SR12) detects the presence of paper in Tray 2. The DC controller notifies the formatter when the Tray 2 media-out sensor detects that paper is absent. Tray 2 lift operation The printer keeps the paper stack surface at the correct pickup position. The Tray 2 lift-up operation is performed under the following conditions: ● The printer is turned on ● Tray 2 is installed ● The paper stack surface in Tray 2 lowers The operational sequence of the Tray 2 lift-up is as follows: 1. The lifter motor (M4) rotates and the lifter moves up. 2. When the Tray 2 media stack surface sensor 2 detects the stack surface of media, the lifter motor stops. 3. The lifter motor rotates again to lift the lifter when the Tray 2 media stack surface sensor 1 detects the stack surface and then lowers during printing. When a Tray 2 media stack surface sensors does not detect the stack surface within a specified time period after the lifter motor starts rotating, the DC controller determines a lifter motor failure and notifies the formatter. Tray 2 paper pickup Following are the sequence of steps for the Tray 2 pickup operation. NOTE: Tray 1 and Tray 2 are optimal for paper pickup when using special paper or media other than 20 lb plain paper. For Tray 1, the printer increases the number of attempts to pick up a page, which increases the reliability of successfully picking the page from the tray and decreases the possibility of a miss-pick jam. HP recommends using Tray 1 or Tray 2 if the printer is experiencing excessive or reoccurring jams from trays other than Tray 1 and Tray 2, or for print jobs that require media other than 20 lb plain paper. 1. The printer is

turned on or Tray 2 is inserted. 2. The tray lift-up operation raises the lifting plate so paper can be picked up. 3. The pickup motor rotates when a print command is received from the formatter. 4. The Tray 2 pickup roller and Tray 2 feed roller rotate. 5. The Tray 2 pickup solenoid turns on at a specified time. 6. The Tray 2 pickup cam rotates. 7. As the pickup arm lowers, the Tray 2 pickup roller touches the surface of the paper stack. 8. One sheet of paper feeds into the printer. 56 Chapter 3 Theory of operations

Tray 2 multiple feed prevention The printer uses a separation roller method to prevent multiple sheets of print media from entering the paper path. The separation roller overruns if just one sheet of paper is picked. If two or more sheets are picked, only the top sheet will be fed to registration and the multiply fed sheets will be held at the pickup location by the separation roller. This printer does not have an actively driven separation roller. Figure 3-27 Tray 2 multiple feed prevention Tray 2 separation roller follows the Tray 2 feed roller

Driving force from

the pickup motor

Normal feed

Multiple feed

Paper Tray 2 pickup roller Tray 2 separation roller Tray 1 feed roller Tray 2 separation roller follows the Tray 2 feed roller

Driving force from

the pickup motor

Normal feed

Multiple feed

Paper Tray 2 pickup roller Tray 2 separation roller Tray 1 feed roller Tray 2 presence detection The Tray 2 presence sensor is in the lifter drive unit. The sensor detects the tray-presence sensor flag and determines whether Tray 2 is installed correctly. Tray 2 skew feed prevention The printer can straighten the paper without slowing the feed operation. 1. The leading edge of paper strikes the registration shutter, and the leading edge is aligned with the shutter. 2. As the feed rollers keep pushing the paper, the paper warps. Tray 2 multiple feed prevention 57 3. When the force is great enough, the registration shutter rotates, and the paper passes through straightened. Figure 3-28 Skew-feed prevention

Registration shutter

Paper

Registration shutter

Paper

Feed speed control

The DC controller adjusts the feed speed to improve the print quality depending on the paper type. The paper is fed at a specified speed according to the print mode designated by the formatter.

Feed speeds

● 1/1 speed = 210 mm/s ● Pro speed = 174 mm/s 58 Chapter 3 Theory of operations

● 1/2 speed = 105 mm/s ● 1/3 speed = 70 mm/s Table 3-17 Print mode and feed speed

Print mode Feed speed Media sensor detection

Normal 1/1 Yes Heavy media 1 Pro No Heavy media 2 1/2 Yes Heavy media 3 1/3 Yes Light media 1 1/1 Yes Light media 2 1/1 Yes Light media 3 1/3 Yes Glossy media 1 1/3 Yes Glossy media 2 1/3 Yes Glossy media 3 1/3 Yes Glossy film 1/3 Yes Envelope 1 1/2 No OHT 1/3 Yes Label 1/2 No Designated media 1 1/2 No Designated media 2 1/3 No Designated media 3 Pro No

Duplexing unit

When duplexing, the fuser motor (M1) reverses the paper and feeds it through the paper path to print the second side.

Duplexing reverse and duplex feed control

The duplex reverse control reverses the paper after the first side is printed and feeds it to the duplex re-pickup position to print the second side of the page. 1. At a specified time after the first side of a page is printed, the fuser motor rotates, and the duplex reverse solenoid is turned on. 2. The duplex flapper moves, and the paper feeds to the duplex reverse unit. 3. After a specified period of time, the fuser motor reverses, and the paper feeds to the duplex feed unit. 4. The fuser motor and the pickup motor move the paper to the duplex re-pickup position. 5. The fuser

motor and the duplex feed clutch stop, and the paper feed operation pauses. Duplexing unit 59

6. After a specified period of time, the fuser motor rotates, and the duplex feed clutch is turned on. The paper is then picked up again. Jam detection/prevention The printer uses the following sensors to detect the paper as it moves through the paper path and to report to the DC controller if the paper has jammed. ● Top of page (TOP) sensor (SR2) ● Media width sensor (SR1a) ● Fuser output sensor (SR13) ● Duplex feed sensor (SR2a) ● Registration sensor (SR11; M506/E50045/M507/E50145, M527/E52545/M528/E52645) ● Paper feeder feed sensor (SR432; M506/E50045/M507/E50145, M527/E52545/M528/E52645) Figure 3-29 Jam detection sensors PS432 PS13 PS2a PS2 PS1a PS11

Duplex paper path

Simplex paper path

60 Chapter 3 Theory of operations

The printer determines that a jam has occurred if one of these sensors detects paper at an inappropriate time. The DC controller stops the print operation and notifies the formatter. Table 3-18 Jams that the printer detects

Jam Description

Media input delay jam 1 Media did not reach the registration sensor in time. Media input delay jam 2 Media did not reach the source tray feed sensor in time. Media input delay jam 3 Media did not reach the tray 3 feed sensor in time. Duplex re-feed jam 1 Media did not reach the registration sensor in time. Media input stay jam 1 Media remained at the registration sensor longer than legal-sized media should remain. Fuser delivery delay jam 1 Media did not reach the fuser output sensor in time. Fuser delivery stay jam 1 Media stayed at fuser output sensor longer than it should stay. Media is in duplex path. Fuser delivery stay jam 2 Media stayed at fuser output sensor longer than it should stay. Media is in simplex path. Wrap jam 1 Media is first detected at fuser output sensor and then disappeared from the sensor before it should have disappeared. Door open jam A door is open while paper is moving through the printer. Residual Media in paper path jam 1 Media detected in the paper path. Jam detection/prevention 61

Input accessories

Optional 550-sheet paper feeders are available for this printer. The M506/E50045/M507/E50145 and M527/E52545/M528/E52645 printers support up to three of these paper feeders at a time. The M501 printer supports one paper feeder. Tray 3-5 The 550-sheet paper feeder is installed under the printer. It picks up paper and feeds it into the printer. NOTE: The M501 printer supports one paper feeder. The M506/E50045/M507/E50145 and M527/ E52545/M528/E52645 printers support three paper feeders. 62 Chapter 3 Theory of operations

Figure 3-30 550-sheet paper feeder paper path

Function Supported feature

Cassette lift-up control Yes

Cassette presence detection Yes

Cassette media size detection No

Cassette media stack surface detection Yes

Tray 3-5 63

Function Supported feature

Cassette media out detection Yes

Cassette media level detection No

Multiple-feed prevention Yes

Automatic delivery Yes

Driver PCA

The paper feeder controller controls the operational sequence of the paper feeder. Figure 3-31 Tray 3-5 driver PCA

DC controller Paper feeder controller

Motor 550-sheet paper feeder +24VD

+3.3VB

Solenoid Clutch Switch

Photo interruptor

Electrical components

The 550-sheet paper feeder has one motor for lifting the tray and feeding paper. Table 3-19 550-sheet paper feeder components

Component type Abbreviation Component name

Motor M5 Optional tray lifter motor Solenoid SL4 Cassette pickup solenoid Clutch CL2 Feed clutch Switch SW3 Cassette detection switch
Photo interrupter PS460 Media surface sensor Photo interrupter PS461 Cassette media out sensor Photo interrupter PS432 Feed sensor 64
Chapter 3 Theory of operations

Paper pickup

The paper feeder contains several motors, solenoids, sensors, and switches. Figure 3-32 Paper pickup and feed components SL4 M1 M5 CL2 PS461

PS460 SW3

PS432 Table 3-20 Paper pickup and feed components

Component type Abbreviation Component name

Motors M1 Fuser motor Motors M5 Lifter motor Clutch CL2 Feed clutch Solenoid SL4 Cassette pickup solenoid Switches SW3 Cassette detection switch
Sensors PS460 Media surface sensor Sensors PS461 Cassette media out sensor Sensors PS432 Feed sensor

Multiple feed prevention

The Trays 3-5 multiple feed prevention for the paper feeder is operated in the same way as that of Tray 2.

Tray presence detection

The Trays 3-5 tray presence detection for the paper feeder is the same as that of Tray 2.

Tray lift operation

The 550-sheet paper feeder keeps the paper stack surface at the correct pickup position. The tray lift operation occurs under the following conditions: ● The printer is turned on Paper pickup 65

● The tray is inserted ● The paper stack surface of the tray lowers The sequence occurs as follows: 1. The feeder tray-lifting motor rotates and the lifter moves up. 2. When the paper feeder media-stack-surface sensor detects the stack surface of the paper, the lifting motor stops. 3. The lifting motor rotates again to lift the lifter when the paper feeder media-stack-surface sensor detects the stack surface, and then lowers during printing. If a tray media-stack-surface sensor does not detect a stack surface within a specified period after the lifting motor starts rotating, the paper feeder driver determines that the lifting motor has failed and notifies the formatter through the DC controller. The paper feeder driver notifies the formatter if either of the paper feeder media-stack-surface sensors fails to detect the stack surface within a specified period from when a lift-up operation starts.

Jam detection

The 550-sheet paper feeder uses the paper feeder feed sensor (SR432) to detect the presence of paper and to check whether paper has jammed. 66 Chapter 3 Theory of operations

Figure 3-33 Jam detection (1x550-sheet paper feeder) PS432 PS13 PS2a PS2 PS1a PS11

Duplex paper path

Simplex paper path

The 550-sheet paper feeder detects the following jams: ● Media input delay jam 1 (550-sheet paper feeder): Paper did not reach the registration sensor in time. ● Media input delay jam 2 (550-sheet paper feeder): Paper did not reach the source tray feed sensor in time. ● Media input delay jam 3 (550-sheet paper feeder): Paper did not reach the tray 3 feed sensor in time. ● Pickup stationary jam (550-sheet paper feeder): The feed sensor does not detect the trailing edge of paper within a specified time after the sensor detects the leading edge. ● Residual paper jam (550-sheet paper feeder): The feed sensor detects the presence of paper for a specified time during an automatic delivery operation. ● Right door open jam (550-sheet paper feeder): The right door open is detected during a paper feed operation. Jam detection 67

E52645)

Figures in this section might look slightly different from your printer, but they are correct for the M527/ E52545/M528/E52645 printers. All the information provided in this section is correct for these printers. NOTE: This section is for the M527/E52545/M528/E52645 printers only. The scanner is a carriage-type platen scanner which includes the frame, glass, scan module, and a scan control board (SCB). The scanner has a sensor to detect legal-sized media and a switch to indicate when the document feeder is opened. The document feeder and control panel are attached to the scanner. If the scanner fails, it can be replaced as a whole unit. The scanner replacement part does not include the document feeder, control panel, or SCB. 68 Chapter 3 Theory of operations

Document feeder system (M527/E52545/M528/E52645) Review the following information about the M527/E52545/M528/E52645 document feeder system. NOTE: This section is for the M527/E52545/M528/E52645 printers only.

Document feed system

This section describes the following: The printer supports single-pass electronic duplexing (e-duplex) copy jobs. Two separate scan modules scan the front-side and back-side of an e-duplex copy job page in a single pass through the document feeder. For the WF class, this ADF supports a smart background which auto-crops and adjusts the image extents.

Sensors in the document feeder

The document feeder contains the following sensors: ● ADF paper present sensor: Detects whether a document is present in the document feeder. If paper is present in the document feeder when copies are made, the printer scans the document using the document feeder. If no paper is present when copies are made, the printer scans the document using the scanner glass. ● ADF Y (length) sensor: Detects whether a legal-size original is present in the document feeder. ● ADF jam cover sensor: Detects whether the document feeder cover is open or closed. ● ADF paper path deskew sensor: Detects the top of the page as it enters the deskew rollers. ● ADF paper path pick success sensor: Detects a successful one page feed from the document feeder tray. NOTE: This sensor uses ultrasonic sound to detect a multi-page paper feed. ● Paper path sensor 1: Detects the top of the page as it approaches the front-side scan module (document feeder glass). Document feeder system (M527/E52545/M528/E52645) 69

Figure 3-34 Document feeder sensors 2 5 1 3 4

Item Description

1 ADF Y (length) sensor

2 ADF paper present sensor

3 ADF deskew sensor

4 Paper path sensor 1 NOTE: For an e-duplex copy job, this sensor is used to activate the front-side scan module (in the scanner base) and the front-side background selector (in the document feeder), if needed. 5 ADF jam cover sensor (open the jam access cover and insert a folded piece of paper to activate the flag)

Document feeder paper path

Review the following information about the document feeder paper path. Figure 3-35 Document feeder paper path

Item Description Item Description

1 Input tray 7 Front-side scan module NOTE: This scan module (document feeder glass) is located in the scanner base. 2 Pre-pick roller 8 Back-side scan module 70 Chapter 3 Theory of operations

Item Description Item Description

3 Pick roller 9 ADF pick success transmitter 4 ADF pick success receiver 10 Separator roller 5 Deskew drive roller 11 Exit drive roller 6 Prescan drive roller 12 Lift plate

Document feeder simplex operation

Following is the basic sequence of operation for a document feeder simplex job. 1. The ADF jam cover sensor detects when the cover door is in the closed position. 2. The ADF paper present sensor activates when paper is loaded onto the input tray. 3. The feed motor rotates to raise the lift plate and starts to pick the loaded paper. 4. The ADF multi-pick (ultrasonic) sensor activates when the leading edge of the media is driven past the sensor. The printer firmware registers a successful pick operation. 5. The ADF paper path deskew activates when the leading edge of the paper passes it. The printer firmware registers the leading edge of the paper position. 6. The leading edge of the paper drives into the nip point of the deskew drive roller and the deskew pinch rollers. This creates a buckle of paper by the nip point for pick-skew correction. 7.

The deskew motor rotates the deskew drive roller to pull the paper into the prescan drive roller. 8. The pick motor stops turning and allows both the pick and feed roller to turn freely while the paper is pulled in by the deskew drive roller. 9. The feed motor rotates to drive the paper into the prescan front-side sensor. The firmware registers the leading edge position of the paper as the multi-pick sensor activates. 10. The feed motor continues to rotate and drive the leading edge of the paper through the preset distance from the multi-pick sensor to the front-side scan zone. The scanner begins the scanning and data retrieval process. 11. The ADF multi-pick (ultrasonic) sensor deactivates when the trailing edge of the paper passes the sensor. The firmware registers the trailing edge of the paper position. 12. The feed motor continues to rotate and drive the trailing edge of the paper through the preset distance from the ADF multi-pick (ultrasonic) sensor to the front-side scan zone. The scanner ends the scanning and data retrieval process. 13. The feed motor continues to rotate and ejects the trailing edge of the paper into the output bin. 14. One of the following occurs: ● If the copy job is complete, the ADF paper present sensor deactivates. The feed motor reverses rotation to raise the pick roller. ● If the copy job is not complete, the ADF paper present sensor is active. The printer firmware detects additional pages in the input tray and the process repeats. Document feeder simplex operation 71

Document feeder e-duplex operation

Following is the basic sequence of operation for a document feeder simplex job. NOTE: For an e-duplex copy job, the background scan operation begins immediately after the simplex sequence of operation ends. 1. The feed motor continues to drive the paper until the leading edge activates the prescan back-side sensor. The printer firmware registers the position of the leading edge of the paper. 2. The feed motor continues to rotate to drive the leading edge of the paper through the preset distance from prescan back-side sensor to the back-side background selector scan zone. The back-side background scan module begins scanning and retrieval of the data. 3. The prescan back-side sensor deactivates when the trailing edge of the paper passes it. The printer firmware registers the trailing edge of the paper position. 4. The feed motor continues to rotate to drive the trailing edge of the paper edge past the back-side background selector scan zone. 5. The feed motor continues to rotate and ejects the trailing edge of the paper into the output bin. 6. One of the following occurs: ● If the copy job is complete, the ADF paper present sensor deactivates. The feed motor reverses rotation to raise the pick roller. ● If the copy job is not complete, the ADF paper present sensor is active. The printer firmware detects additional pages in the input tray and the process repeats.

Deskew operation

Sliding side guides on the input tray make sure that the paper stack is correctly aligned at the center of the input tray when paper is loaded in the tray. The correct position of the loaded paper is parallel with the direction of travel into the document feeder paper path. The document feeder further reduces paper skew due to improper loading of paper in the input tray by buckling the paper to create a paper buffer. The document feeder aligns the leading edge of the paper parallel with the deskew drive rollers before it is driven further into the document feeder paper path. NOTE: If the page to be copied is smaller than the minimal sliding guide setting, do not use the document feeder for the copy job. Attempting to copy too small of a page using the document feeder can result in document feeder jams and/or damage to the original page. Instead, use the flatbed glass to copy the page. 72 Chapter 3 Theory of operations

Figure 3-36 Deskew operation

Document feeder hinges

Review the following information about the document feeder hinges. The document feeder hinges allow positioning the assembly vertically above the scanner glass to accommodate the placement of books and other objects up to 25 mm (1.0 in) in height on the scanner glass. The document feeder still closes (the bottom of the ADF is kept parallel to the scanner glass) and allows the printer to operate. The document feeder hinges provide height adjustment of 25 mm (1.0 in) when a maximum downward force of 4.5 kg (10 lb) is applied at the front edge of the assembly, with the fulcrum (such as the spine of a book) centered on the scanner glass and parallel to its long axis. The document feeder will withstand a downward force of about 4.5 kg (10 lb) applied at the front edge center of the assembly—when the fulcrum (such as the spine of a book) is located anywhere on the scanner glass and parallel to its long axis—without breaking, deforming, detaching or experiencing performance degradation. The document feeder hinges support the assembly in the open position and prevent the document feeder from suddenly closing in a damaging or loud manner. The hinges can hold the document feeder static in all positions higher than 100 mm (3.93 in); measured at the front of the assembly. Less than 2.3 kg (5 lb) of force is required to open or close the document feeder. The hinges allow the document feeder to open to an angle of between 60° and 80° from the horizontal position (this angle will not allow the printer to tip over). Document feeder hinges 73

Figure 3-37 Document feeder open (book mode) Figure 3-38 Document feeder open (60° to 80°) 74 Chapter 3 Theory of operations

4 Solve problems

Review the following information about solving printer problems.

HP service and support

Learn about access to additional service and support information.

Additional service and support for channel partners

Channel partners can use the HP Partner Portal or the Channel Services Delivery Platform (CSDP) to access the HP Web-based Interactive Search Engine (WISE). Find information about the following topics. ● Service manuals, service cost data, and service advisories ● The latest control panel message (CPMD) troubleshooting information ● Remove and replace part instructions and videos ● Solutions for printer issues

and emerging issues ● Printer specifications, warranty, and regulatory information ● Install, configure, and how to information View a video of how to use the HP Partner First Portal to access WISE. Access WISE for Channel partners (HP Partner Portal) 1. Sign-in to the HP Partner Portal (click <https://partner.hp.com> to access the sign-in page). 2. Select the Services & Support item (near the top of the screen). 3. Select the Technical Support (WISE) item. 4. Select the Technical Documentation item. 5. Use the WISE portal to search for printer information. Access WISE for Channel partners (CSDP) 1. Sign-in to the CSDP portal (click csdp.hp.com to access the sign-in page). 2. Select the Knowledge and Training item. Solve problems 75

3. Select the HP Technical Documentation item. 4. Use the WISE portal to search for printer information. Additional service and support for HP internal personnel HP internal personnel, go to one of the following Web-based Interactive Search Engine (WISE) sites: View a video of how to access WISE for internal HP users. Americas (AMS) ● WISE - English ● WISE - Spanish ● WISE - Portuguese ● WISE - French Asia Pacific / Japan (APJ) ● WISE - English ● WISE - Japanese ● WISE - Korean ● WISE - Chinese (simplified) ● WISE - Chinese (traditional) ● WISE - Thai Europe / Middle East / Africa (EMEA) ● WISE - English

Additional technical support WISE videos

The videos below provide additional ways to access printer information using WISE. View a video of how to use WISE to find technical support videos (model number search). View a video of how to use WISE to find technical support videos (Product detail page [PDP] search). 76 Chapter 4 Solve problems

Solve problems checklist (M501) Review the following information about the solve problems checklist for the M501 model. Use the solve problems checklist (M501) If the printer is not correctly functioning, complete the steps (in the order given) in the following checklist. If the printer fails a checklist step, follow the corresponding troubleshooting suggestions for that step. If a checklist step resolves the problem, skip the remaining checklist items. 1. Make sure that the printer is set up correctly. a. Press the power button to turn on the printer or to deactivate the Auto-Off mode. b. Check the power-cable connections. c. Make sure that the line voltage is correct for the printer power configuration. See the label that is on the printer for voltage requirements. If you are using a power strip and its voltage is not within specifications, plug the printer directly into the wall. If it is already plugged into the wall, try a different outlet. 2. Check the cable connections. a. Check the cable connection between the printer and the computer. Make sure that the connection is secure. b. Make sure that the cable itself is not faulty, by using a different cable if possible. c. Check the network connection: Make sure that the network light is lit. The network light is next to the network port on the back of the printer. If the printer remains unable to connect to the network, uninstall and then reinstall the printer. If the error persists, contact a network administrator. 3. Check to see if any messages appear on the control panel. 4. Make sure that the paper you are using meets specifications. 5. Make sure that the paper is loaded correctly in the input tray. 6. Make sure that the printer software is installed correctly. 7. Verify that you have installed the printer driver for this printer, and that you are selecting this printer from the list of available printers. 8. Print a configuration page. a. On the printer control panel, press the OK button. b. Open the Reports menu. c. Select Configuration Report. 9. After printing the configuration page, check the following: a. If the page does not print, verify that the input tray contains paper and that the paper is properly loaded. b. Make sure that each toner cartridge is installed correctly. Solve problems checklist (M501) 77

c. If the page jams in the printer, clear the jam. d. If the print quality is unacceptable, complete the following steps: ● Verify that the print settings are correct for the paper you are using. ● Solve print-quality problems. 10. Print a small document from a different program that has printed in the past. If this solution works, then the problem is with the program you are using. If this solution does not work (the document does not print), complete these steps: a. Try printing the job from another computer that has the printer software installed. b. Check the cable connection. Direct the printer to the correct port, or reinstall the software, selecting the connection type you are using. Print the menu map (M501) To more easily navigate individual settings, print a report of the complete printer menus. 1. On the printer control panel, press the OK button. 2. Open the Reports menu. 3. Select Menu Structure. Print the service page (includes the event log) (M501) Printing the service page provides a list of printer settings that might be helpful in the troubleshooting process, as well as the event log, which stores the last 10 error events that the printer experienced. 1. On the printer control panel, press the OK button. 2. Open the Reports menu. 3. Select the Service Page option to print the report. The event log is located in the lower right-hand corner of the service page. 78 Chapter 4 Solve problems

Solve problems checklist (all except M501) Review the following information about the solve problems checklist for the M506/E50045/M507/E50145 and M527/E52545/M528/E52645 models.

Use the solve problems checklist

If the printer is not correctly functioning, complete the steps (in the order given) in the following checklist. If the printer fails a checklist step, follow the corresponding troubleshooting suggestions for that step. If a checklist step resolves the problem, skip the remaining checklist items. 1. If the control panel is blank or black, check the following before proceeding: ● Check to make sure that the printer is not in Sleep mode (tap the touchscreen control panel or press a button on the LCD control panel). ● Check the power cable. ● Check that the power is turned on. ● Make sure that the line voltage is correct for the printer power configuration. (See the label that is on the back of the printer for voltage requirements.) If a power strip is in use, and its voltage is not within specifications, connect the printer directly into the electrical outlet. If it is already connected into the outlet, try a different outlet. If the control panel is not responding to touch (M506x/M507x, E50045/E50145, and M527/E52545/ M528/E52645 models), or if it appears black or blank, try the following: NOTE: The following conditions indicate that the printer has frozen while in Sleep mode. Opening a door, tapping the control panel (touchscreen control panels), or pressing a control panel button (LCD control panels) causes the printer to wake up from Sleep mode. a. Turn the printer power off, and then on again. b. Verify that the control panel is correctly functioning by accessing the control panel diagnostics. See the Control-panel checks section in the printer Troubleshooting Manual. TIP: The LED on the formatter will blink if the control panel is not detected or the cables are not properly seated. If the control panel does not respond to the diagnostic button, try the following: ● Try printing from a host computer. Does the printer print a test page? ● Is the printer HP Embedded Web Server (EWS) accessible? ● Turn the power off. ● Reseat the cable connections on the bottom of the control-panel assembly and the control panel connectors at the formatter. ● Turn the printer power on, and then check for functionality of

the control panel by tapping the touchscreen or pressing a button on the LCD control panel. c. Try upgrading the firmware. If the firmware upgrade fails to resolve the problem, and the printer still freezes while in Sleep mode, elevate the case after collecting the following info: d. If the error persists, replace the control panel assembly. Solve problems checklist (all except M501) 79

2. The control panel should indicate a Ready, Paused, or Sleep mode on status. If an error message displays, resolve the error. ● Try using the Power-on checks section in the printer Troubleshooting Manual to solve the problem. 3. For network connection errors, check the cables. a. Check the network cable connections between the printer and the computer or network port. Make sure that the connections are secure. NOTE: The network LEDs should be illuminated and flashing. b. Make sure that the cables are not faulty by trying different cables, if possible. c. Check the network connection. Verify that the port is active. 4. Print a configuration page. If the printer is connected to a network, an HP Jetdirect page also prints. a. For a printer with a touchscreen control panel, from the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or touch the Reports button (FutureSmart 4). b. Open the following menus: ● Reports (FutureSmart 3) ● Configuration/Status Pages ● Configuration Page c. Touch Configuration Page to select it. TIP: Multiple report pages can be selected, and then printed together. d. Touch the Print button to print the pages. a. For a LCD control panel, from the Home screen on the printer control panel, use the down arrow to scroll to Administration, and then press the OK button (FutureSmart 3); or use the arrow keys to navigate to the Reports, and then press the OK button (FutureSmart 4). b. If necessary, use the down arrow button to scroll to Reports, and then press the OK button (FutureSmart 3). c. If necessary, use the down arrow button to scroll to Configuration/Status Pages, and then press the OK button. d. Use the down arrow button to scroll to Configuration page, and then press the OK button to select it. e. Use the up arrow button to scroll to Print, and then press the OK button to print the pages. TIP: Multiple report pages can be selected, and then printed together. 80 Chapter 4 Solve problems

● If the pages do not print, check that at least one tray contains paper. TIP: Make sure that the selected paper size and type meet HP specifications. Also open the Trays menu on the printer control panel and verify that the tray is configured correctly for the paper type and size. ● If the page jams in the printer, follow the instructions on the control panel to clear the jam. ● If the page does not print correctly, the problem is with the printer hardware. ● If the page prints correctly, the printer hardware is working. The problem is with the host computer, with the print driver, or with the program. 5. Print a supplies status page and then check that the maintenance items below are not at their end-of-life. TIP: If a maintenance item needs to be replaced, order the part number provided below. a. For a printer with a touchscreen control panel, from the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or touch the Reports button (FutureSmart 4). b. Open the following menus: ● Reports (FutureSmart 3) ● Configuration/Status Pages ● Supplies Status Page c. Touch the Print button to print the page, and then check the maintenance items (listed below in this step). TIP: Multiple report pages can be selected, and then printed together. a. For printers with LCD control panels, from the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button (FutureSmart 3); or use the arrow keys to navigate to the Reports, and then press the OK button (FutureSmart 4). b. If necessary, use the down arrow button to scroll to Reports, and then press the OK button (FutureSmart 3). c. If necessary, use the down arrow button to scroll to Configuration/Status Pages, and then press the OK button. d. Use the down arrow button to scroll to Supplies Status Page, and then press the OK button to select it. e. Use the up arrow button to scroll to Print, and then press the OK button to print the page, and then check the maintenance items (listed below in this step). TIP: Multiple report pages can be selected, and then printed together. NOTE: HP long-life consumables and maintenance kit life specifications are estimations. Actual individual life/yield during normal use varies depending on usage, environment, media, and other factors. Estimated life is not an implied guarantee or warrantable. Use the solve problems checklist 81

● Fixing assembly (fuser); estimated life: 150,000 pages

– RM2-5679-000 (110V)

(M501/M506/E50045/M527/E52545)

– RM2-2585-000CN (110V)

(M507/E50145/M528/E52645)

– RM2-5692-000 (220V)

(M501/M506/E50045/M527/E52545)

– RM2-2586-000CN (220V)

(M507/E50145/M528/E52645)

● Document feeder maintenance kit; estimated life: 150,000 pages

– B5L52-67903

6. Verify that the correct print driver for this printer is installed. Check the program to make sure that the print driver for this printer is used. The print driver is on the CD that came with the printer, or can be downloaded from this Web site: <http://www.hp.com/support/ljM501> or <http://www.hp.com/support/ljM506> or <http://www.hp.com/support/ljE50045> or <http://www.hp.com/support/ljM507> or <http://www.hp.com/support/ljE50145> or <http://www.hp.com/support/ljM527mfp> or <http://www.hp.com/support/ljE52545mfp> or <http://www.hp.com/support/ljM528mfp> or <http://www.hp.com/support/E52645mfp>. 7. Print a short document from a different program that has worked in the past. If this solution works, the problem is with the program. If this solution does not work (the document does not print), complete these steps: a. Try printing the job from another computer that has the printer software installed. b. If the printer is connected to the network, connect the printer directly to a host computer with a USB cable. Redirect the printer to the correct port, or reinstall the software (make sure to select the new connection type).

Print the Settings menu map

To more easily navigate individual settings, print a report of the complete Settings menu (FutureSmart 4), or the complete Administration menu (FutureSmart 3). Print the settings menu map from a touchscreen control panel Use the following procedure to print the menu map from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or touch the Reports button (FutureSmart 4). 2. Open the following menus: ● Reports (FutureSmart 3) ● Configuration/Status Pages 82 Chapter 4 Solve problems

3. Select the Administration Menu Map option (FutureSmart 3) or the Settings Menu Map option (FutureSmart 4). 4. Touch the Print button to print the report. TIP: Multiple report pages can be selected, and then printed together. Print the settings menu map from a LCD control panel Use the following procedure to print the menu map from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Open the following menus: ● Reports ● Configuration/Status Pages 3. Use the down arrow button to scroll to Administration Menu Map, and then press the OK button to select it. 4. Use the up arrow button to scroll to Print, and then press the OK button to print the pages. TIP: Multiple report pages can be selected, and then printed together. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Reports, and then select OK. 2. Select Configuration/Status Pages. 3. Select Settings Menu Map, and then select OK. 4. Use the down arrow to scroll down to Cancel, and then use the right arrow to highlight the print icon . Select OK to print the pages. TIP: Multiple report pages can be selected, and then printed together.

Print current settings pages

Printing the current settings page provides a map of the user configurable settings that might be helpful in the troubleshooting process. Print the current settings page from a touchscreen control panel Use the following procedure to print the current settings page from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or touch the Reports button (FutureSmart 4). 2. Open the following menus: ● Reports (FutureSmart 3) Print the settings menu map from a LCD control panel 83

● Configuration/Status Pages 3. Select the Current Settings Page option. 4. Touch the Print button to print the report. TIP: Multiple report pages can be selected, and then printed together. Print the current settings page from a LCD control panel Use the following procedure to print the current settings page from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Open the following menus: ● Reports ● Configuration/Status Pages 3. Use the down arrow button to scroll to Current Settings Page, and then press the OK button to select it. 4. Use the up arrow button to scroll to Print, and then press the OK button to print the pages. TIP: Multiple report pages can be selected, and then printed together. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Reports, and then select OK. 2. Select Configuration/Status Pages. 3. Select Current Settings Page, and then select OK. 4. Use the down arrow to scroll down to Cancel, and then use the right arrow to highlight the print icon . Select OK to print the pages. TIP: Multiple report pages can be selected, and then printed together.

Pre-boot menu options

The Pre-boot menus are available prior to the printer initializing. CAUTION: The Format Disk option (printers with a hard-disk drive only) performs a disk initialization for the entire disk. The operating system, firmware files, and third party files (among other files) will be completely lost. HP does not recommend this action. TIP: The Pre-boot menu can be remotely accessed by using a telnet network protocol to establish an administration connection to the printer. See the Remote Admin (all except M501) topic in the Troubleshooting Manual. 84 Chapter 4 Solve problems

Open the Pre-boot menu from a touchscreen control panel Use the following procedure to open the Pre-boot menu from a touchscreen control panel. 1. Touch the middle of the control-panel display when you see the 1/8 under the logo. Figure 4-1 Open the Pre-boot menu 1 2 2. On the Pre-boot menu screen, use the following buttons to navigate the tests. Figure 4-2 Pre-boot menu

Button Description

Use this button to see more information about a selected item. Use this button to scroll up through menu items. Use this button to select a highlighted menu item. Open the Pre-boot menu from a touchscreen control panel 85

Button Description

Use this button to scroll down through menu items. Use this button to go back to the previous menu. Not used. Use this button to exit a diagnostic test. 3. Use the arrow buttons on the touchscreen to navigate the Pre-boot menu. 4. Touch the OK button to select a menu item.

Open the Pre-boot menu from a LCD control panel

Use the following procedure to open the Pre-boot menu from a LCD control panel. 1. Press the Cancel button when you see the 1/8 under the logo. 2. Use the arrow buttons on the control panel to navigate the Pre-boot menu. 3. Press the OK button to select a menu item. Cold reset using the Pre-boot menu from a touchscreen control panel Use the following procedure to perform a cold reset using the Pre-boot menu from a touchscreen control panel. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost). 1. Touch the middle of the control-panel display when you see the 1/8 under the logo. Figure 4-3 Open the Pre-boot menu 1 2 2. Use the down arrow button to highlight the +3:Administrator item, and then touch the OK button. 86 Chapter 4 Solve problems

3. Use the down arrow button to highlight the +8:Startup Options item, and then touch the OK button. 4. Use the down arrow button to highlight the 2 Cold Reset item, and then touch the OK button to select it. 5. Touch the Home button to return to the main Pre-boot menu and highlight the 1:Continue item, and then touch the OK button. NOTE: The printer will initialize. Cold reset using the Pre-boot menu from a LCD control panel Use the following procedure to perform a cold reset using the Pre-boot menu from a LCD control panel. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost). 1. Press the Cancel button when you see the 1/8 under the logo. 2. Use the down arrow button to highlight the +3:Administrator item, and then press the OK button. 3. Use the down arrow button to highlight the +8:Startup Options item, and then press the OK button. 4. Use the down arrow button to highlight the 2 Cold Reset item, and then press the OK button to select it. 5. Touch the Home button to return to the main Pre-boot menu and highlight the 1:Continue item, and then touch the OK button. NOTE: The printer will initialize. NOTE: Some of the pre-boot options in the following tables are not supported by the current version of the printer firmware and are included for information only. Future versions of firmware will support these options. Table 4-1 Pre-boot menu options (1 of 7) Menu option First level Second level Third level Description Continue Selecting the Continue item exits the Pre-boot menu and continues the normal boot process. If a selection is not made in the initial menu within 30 seconds, the printer returns to a normal boot (the same as selecting Continue). If the user navigates to another menu, the timeout does not apply. Sign In Enter the administrator PIN or service PIN if one is required to open the Pre-boot menu. Administrator This item navigates to the Administrator submenus. If authentication is required (and the user is not already signed in) the Sign In prompt displays. The user is required to sign in. Cold reset using the Pre-boot menu from a LCD control panel 87

Table 4-1 Pre-boot menu options (1 of 7) (continued) Menu option First level Second level Third level Description

Administrator Download Network

USB USB

Thumbdrive

This item initiates a Pre-boot firmware download

process. A USB Thumbdrive option will work on all FutureSmart printers. USB or Network connections are not currently supported. Administrator Format Disk This item reinitializes the disk and cleans all disk partitions. CAUTION: Selecting the Format Disk item removes all data. A delete confirmation prompt is not provided.

The system is not bootable after this action and

a 99.09.67 error displays on the control panel. A firmware download must be performed to return the system to a bootable state. Administrator Partial Clean This item reinitializes the disk (removing all data except the firmware repository where the master firmware bundle is downloaded and saved). CAUTION: Selecting the Partial Clean item removes all data except the firmware repository. A delete confirmation prompt is not provided. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost).

This allows a user to reformat the disk by removing

the firmware image from the active directory without having to download new firmware code (printer remains bootable).

Administrator Change

Password

Select this item to set or change the administrator

password. Administrator Clear Password Select the Clear Password item to remove a password from the Administrator menu. Before the password is actually cleared, a message will be shown asking to confirm that the password should be cleared. Press the OK button to confirm the action. When the confirmation prompt displays, press the OK button to clear the password. Table 4-2 Pre-boot menu options (2 of 7) Menu option First level Second level Third level Description

Administrator

(continued) Manage Disk Clear disk Select the Clear disk item to enable using an external device for job storage. Job storage is normally enabled only for the Boot device. This will be grayed out unless the 99.09.68 error is displayed. 88 Chapter 4 Solve problems

Table 4-2 Pre-boot menu options (2 of 7) (continued) Menu option First level Second level Third level Description Administrator Manage Disk Lock Disk Select the Lock Disk item to lock (mate) a new secure disk to this printer.

The secure disk already locked to this printer will

remain accessible to this printer. Use this function to have more than one encrypted disk accessible by the printer when using them interchangeably.

The data stored on the secure disk locked to this

printer always remains accessible to this printer. Administrator Manage Disk Leave Unlocked Select the Leave Unlocked item to use a new secure disk in an unlocked mode for a single service event. The secure disk that is already locked to this printer will remain accessible to this printer and uses the old disk's encryption password with the new disk. The secure disk that is already locked to this printer remains accessible to this printer. Administrator Manage Disk Clear Disk Pwd Select the Clear Disk Pwd item to continue using the non-secure disk and clear the password associated with the yet-to-be installed secure disk. CAUTION: Data on the missing secure disk will be permanently inaccessible. Administrator Manage Disk Retain Password Select the Retain Password item to use the non secure disk for this session only, and then search for the missing secure disk in future sessions.

Administrator Manage Disk Boot Device Secure

Erase Select the Secure Erase item to erase all of the data on the disk and unlock it if required. This might take a long time. NOTE: The system will be unusable until the system files are reinstalled. The ATA secure-erase command is a one-pass overwrite, which erases the entire disk including firmware. The disk remains an encrypted disk. Administrator Manage Disk Boot Device Erase/ Unlock Select the Erase/Unlock item to cryptographically erase all data on the disk and unlock the disk to allow a user to gain access to it from any printer. NOTE: The system will be unusable until the system files are reinstalled. It erases the encryption key. The encryption key is erased, so the disk becomes a non encrypted disk. Administrator Manage Disk Boot Device Get Status This item provides disk status information if any is available. Table 4-3 Pre-boot menu options (3 of 7) Menu option First level Second level Third level Description

Administrator

(continued)

Manage Disk

(continued) Internal Device Select the Internal Device item to erase the internal device or get a status about the internal device. Cold reset using the Pre-boot menu from a LCD control panel 89

Table 4-3 Pre-boot menu options (3 of 7) (continued) Menu option First level Second level Third level Description

Administrator Manage Disk Internal

Device Secure Erase Select the Secure Erase item to erase all of the data on the disk and unlock it if required. This might take a long time. NOTE: The system will be unusable until the system files are reinstalled. The ATA secure-erase command erases the entire disk, including firmware. The disk remains an encrypted disk.

Administrator Manage Disk Internal

Device Erase/ Unlock Select the Erase/Unlock item to cryptographically erase all of the data on disk and unlock the disk to allow the user to gain access to it from any printer. NOTE: The system will be unusable until the system files are reinstalled. The HP High Performance Secure Hard Disk is erased.

Administrator Manage Disk Internal

Device Get Status This item provides disk status information if any is available.

Administrator Manage Disk External

Device

Select the External Device item to erase the

external device or get status about the external device.

Administrator Manage Disk External

Device Secure Erase Select the Secure Erase item to erase all of the data on the disk and unlock it if required. This might take a long time. NOTE: The system will be unusable until the system files are reinstalled.

The ATA secure-erase command erases the entire

disk, including firmware. The disk remains an encrypted disk.

Administrator Manage Disk External

Device Erase/ Unlock Select the Erase/Unlock item to cryptographically erase all of the data on disk and unlock the disk to allow a user to gain access to it from any printer. NOTE: The system will be unusable until the system files are reinstalled. The encryption key is erased, so the disk becomes a non-encrypted disk.

Administrator Manage Disk External

Device Get Status This item provides disk status information if any is available. Table 4-4 Pre-boot menu options (4 of 7) Menu option First level Second level Third level Description

Administrator

(continued)

Configure LAN

NOTE: This configuration is only active when the Pre-boot menu is open. IP Mode

[DHCP]

The network can be configured to obtain the

network settings from a DHCP server or as static.

Use this item for automatic IP address acquisition

from the DHCP server. 90 Chapter 4 Solve problems

Table 4-4 Pre-boot menu options (4 of 7) (continued) Menu option First level Second level Third level Description

Administrator Configure LAN

NOTE: This configuration is only active when the Pre-boot menu is open. IP Mode

[STATIC]

Use this item to manually assign the network

addresses.

Administrator Configure LAN

NOTE: This configuration is only active when the Pre-boot menu is open. IP Mode

[STATIC]

IP Address Use this item to manually enter the IP addresses.

Administrator Configure LAN

NOTE: This configuration is only active when the Pre-boot menu is open. IP Mode

[STATIC]

Subnet Mask Use this item to manually enter the subnet mask.

Administrator Configure LAN

NOTE: This configuration is only active when the Pre-boot menu is open. IP Mode

[STATIC]

Default Gateway

Use this item to manually enter the default

gateway.

Administrator Configure LAN

NOTE: This configuration is only active when the Pre-boot menu is open. IP Mode

[STATIC]

Save Select the Save item to save the manual settings. Table 4-5 Pre-boot menu options (5 of 7) Menu option First level Second level Third level Description

Administrator

(continued) Startup Options Select the Startup Options item to specify options that can be set for the next time the printer is turned on and initializes to the Ready state. Administrator Startup Options Show Revision Not currently functional: Select the Show Revision item to allow the printer to initialize and show the firmware version when the printer reaches the Ready state. Once the printer power is turned on the next time, the Show Revision item is unchecked so that the firmware revision is not shown. Administrator Startup Options Cold Reset Select the Cold Reset item to clear the IP address and all customer settings. (This item also returns all settings to factory defaults.) NOTE: Items in the Service menu are not reset. Cold reset using the Pre-boot menu from a LCD control panel 91

Table 4-5 Pre-boot menu options (5 of 7) (continued) Menu option First level Second level Third level Description Administrator Startup Options Skip Disk Load Select the Skip Disk Load item to disable installed third-party applications. Administrator Startup Options Skip Cal Select the Skip Cal item to skip the printer calibration for the very next power-initialization cycle only. Administrator Startup Options Lock Service CAUTION: Select the Lock Service item to lock the Service menu access (both in the Pre-boot menu and the Device Maintenance menu).

Service personnel must have the administrator

remove the Lock Service setting before they can open the Service menu. Administrator Startup Options Skip FSK Select the Skip FSK item to disable Chkdisk/ ScanVolume during startup. Administrator Startup Options First Power Not currently functional: This item allows the printer to initialize as if it is the first time it has been turned on. For example, the user is prompted to configure first-time settings like date/time, language, and other settings.

Select this item so that it is enabled for the next

time the printer power is turned on. When the printer power is turned on the next time, this item is unchecked so that the pre-configured settings are used during configuration, and the first-time setting prompt is not used.

Administrator Startup Options Embedded

Jetdirect Off

Select the Embedded Jetdirect Off item to disable

the embedded HP Jetdirect.

By default this item is unchecked so that HP

Jetdirect is always enabled. Administrator Startup Options WiFi Accessory Select the WiFi Accessory item to enable the wireless accessory. Table 4-6 Pre-boot menu options (6 of 7) Menu option First level Second level Third level Description

Administrator

(continued) Diagnostics Diagnostic items are useful to diagnose hardware components and their interface connections. Use these items to troubleshoot specific hardware components, and the interface between them and other components. Administrator Diagnostics Memory Do Not Run Use the Do Not Run item to exclude the Memory diagnostic when executing multiple diagnostics. Administrator Diagnostics Short Use the Short item to select a brief memory test. NOTE: This test requires about four minutes to execute. 92 Chapter 4 Solve problems

Table 4-6 Pre-boot menu options (6 of 7) (continued) Menu option First level Second level Third level Description Administrator Diagnostics Long Use the Long item to select an extended memory test. NOTE: This test requires about twenty minutes to execute. Administrator Diagnostics Disk Do Not Run Use the Do Not Run item to exclude the Disk diagnostic when executing multiple diagnostics. Administrator Diagnostics Short Use the Short item to select a brief firmware self test. NOTE: This test requires about two or three minutes to execute. Administrator Diagnostics Long Use the Long item to select an extended firmware self-test. NOTE: This test requires about sixty minutes to execute. Administrator Diagnostics Optimized Use the Optimized item to select a test that checks the active sectors on the disk. NOTE: This test requires about thirty minutes to execute. Administrator Diagnostics Raw Use the Raw item to select a test that checks every sector on the

disk. NOTE: This test requires about fifty minutes to execute. Administrator Diagnostics Smart Use the Smart item to select a very brief test that checks the drive self-monitoring analysis and reporting technology (SMART) status—the drive detects and reports reliability indicators to help anticipate disk failures (SMART status). Administrator Diagnostics CPB Use the CPB item to verify the integrity of the copy processor board (CPB) and the formatter PCA connections. Administrator Diagnostics Interconnect Use the Interconnect item to verify the integrity of the interconnect PCA (ICB) and its connections. Administrator Diagnostics Run Selected Select the Run Selected item to execute a selected test. NOTE: If more than one test is selected, they are executed in sequence. Cold reset using the Pre-boot menu from a LCD control panel 93

Table 4-7 Pre-boot menu options (7 of 7) Menu option First level Second level Third level Description

Administrator

(continued) Remote Admin Start Telnet The Remote Admin item allows a service technician to access to the printer Pre-boot menu remotely, and to navigate the menu selections from a remote location. IMPORTANT: A Remote Admin connection must be initiated by a person that is physically present at the printer.

This person will also need to provide a randomly

generated PIN to the remote service technician. NOTE: For more information about using the Remote Admin function, see "Remote Admin (M506/ M507/E50145, M527/M528/E52645)" in the Troubleshooting Manual.

Administrator Remote Admin Stop Telnet

Administrator Remote Admin Refresh IP

Administrator System Triage Copy Logs If the device will not boot to the Ready state, or the diagnostic log feature found in the Troubleshooting menu is not accessible, then use the System Triage item to copy the diagnostic logs to a USB flash drive at the next printer start up. The files can then be sent to HP to help diagnose the problem.

Administrator Change Svc

PWD

Use this item to change the Service menu personal

identification number (PIN). Administrator Reset Svc PWD If the Service menu personal identification number (PIN) has been changed. Use this item to reset it to the original PIN. Service Tools Reset Password Use this item to reset the Pre-boot administrator password. Service Tools Subsystems For manufacturing use only. Do not change these values.

Developer Tools Netexec

Remote Admin (all except M501) This section describes the following Remote Admin items. The Remote Admin feature allows remote access the printer Pre-boot menu (BIOS environment). The printer functions as a telnet server which uses the telnet networking protocol to transmit text data. Any computer (with telnet installed and enabled) can function as the telnet client to remotely display and interact with the Pre-boot menu. IMPORTANT: While the Remote Admin function allows remote access the Pre-boot menu, for security reasons the Remote Admin connection must be initiated by a person that is physically present at the printer. 94 Chapter 4 Solve problems

Required software and network connection

Before using the Remote Admin feature, make sure that the telnet network protocol is installed and enabled on the remote telnet client computer. NOTE: This section describes enabling and configuring the telnet feature for computers using a Windows® operating system. HP recommends that the telnet client computer be a Windows-based system; however, there are other operating systems that support the telnet network protocol. For information about enabling and configuring the telnet network protocol for other operating systems, see the owner's manual for that operating system.

Enable the Windows telnet client

All computers using the Windows operating system have the telnet client installed, however, the telnet client function might not be enabled by default. NOTE: The figures and menus in this section are for the Windows 7 Enterprise® operating system. Screens and menu selections might vary slightly for other operating systems. 1. Use the Start menu to open the Control Panel, and then click the Programs item to select it. Figure 4-4 Open the Control Panel Required software and network connection 95

2. Click the Turn Windows features on or off item to select it. Figure 4-5 Turn Windows features on or off 3. In the Windows Features box, scroll down to Telnet Client. If the check box is not checked, click the box to select it, and then click the OK button. TIP: If the check box is already checked then the telnet client function is already enabled. Click the Cancel button. Figure 4-6 Enable the telnet client feature

Network connection

The remote telnet client computer must have direct network access to the printer for the Remote Admin function to operate. This means that the telnet client computer must be on the same network as the printer. The Remote Admin function cannot be accessed through a network firewall or other remote access network security programs. 96 Chapter 4 Solve problems

If a private network is not accessible, ask the network administrator to set up a virtual private network (VPN) connection to the network.

Connect a remote connection

Use the following procedures to connect a remote connection.

Start the telnet server function at the printer

For security reasons the Remote Admin feature must be initiated by a person that is physically present at the printer. The following steps must be performed by a person that is physically present at the printer. NOTE: This person might need to sign in with an administrator or service password depending on how the printer is configured. 1. Turn the printer on. 2. The HP logo displays on the printer control panel. When a 1/8 with an underscore displays, touch the middle of the screen to open the Pre-boot menu. 3. Use the arrow buttons on the touchscreen to scroll down and highlight the +3:Administrator item, and then touch the OK button to select it. Figure 4-7 Select the +3:Administrator item 4. Use the arrow buttons on the touchscreen to scroll down and highlight the +A:Remote Admin item, and then touch the OK button to select it. Figure 4-8 Select the +A:Remote Admin item 5. Use the arrow buttons on the touchscreen to scroll down and highlight the 1:Start Telnet item, and then touch the OK button to select it. Figure 4-9 Select the 1:Start Telnet item Connect a remote connection 97

6. Do one of the following ● If a connecting message displays briefly, go to 7. Figure 4-10 Telnet connecting message ● If an error message displays, use the steps below to identify the problem. Figure 4-11 Telnet error message a. The printer network cable is not correctly connected. b. The BIOS LAN settings are incorrect. – The printer should be configured to use a static IP address, but is configured to use DHCP instead. – The printer is configured to use a static IP address, but the IP address is incorrect. c. The printer is correctly configured to use DHCP, but the DHCP server is not turned on or is malfunctioning. 7. When the printer telnet server function is initialized, the following screen appears. Use the information on this screen to connect the remote telnet client computer to the printer. NOTE: The printer is now ready to receive remote telnet client commands. ● IP: The static or dynamically allocated IP address for the printer. ● Port: The standard telnet port (23). ● Pin: A randomly generated 4-digit personal identification number (PIN). Figure 4-12 Telnet server function initialized 98 Chapter 4 Solve problems

Start the telnet client function at the remote computer The following steps establish a Remote Admin connection from a remote computer to the printer. 1. From the Start menu click Run to open a dialog box, type cmd in the Open field, and then click the OK button to open a Windows command window. Figure 4-13 Open a command window 1 2 2. From any displayed directory, type telnet at the prompt, and then press the Enter key. Figure 4-14 Start a telnet session 3. Type o <IP ADDRESS> at the telnet prompt, and then press the Enter key. NOTE: For <IP ADDRESS>, substitute the IP address that was displayed in step 7 in "Start the telnet server function at the printer" in the Troubleshooting Manual. TIP: If the telnet connection fails to establish a connection, the printer is probably behind a firewall or on a different network that the remote telnet client computer. See "Network connection" in the Troubleshooting Manual. Start the telnet client function at the remote computer 99

Figure 4-15 Establish a telnet connection 4. Type the PIN that was displayed in step 7 in "Start the telnet server function at the printer" in the Troubleshooting Manual at the prompt, and then press the Enter key. IMPORTANT: Make sure to type the PIN correctly. After five incorrect PIN entries, the printer terminates the Remote Admin connection. The Remote Admin feature must be re-initiated at the printer. See "Start the telnet server function at the printer" in the Troubleshooting Manual. Figure 4-16 Enter the PIN 5. The following screen displays when the correct PIN is entered and the Remote Admin connection is successful. For information about the Pre-boot menu and options, see "Pre-boot menu options" in the Troubleshooting Manual. NOTE: Because a Remote Admin connection is an unsecure telnet network protocol connection, the following Pre-boot menu items are disabled for the remote telnet client computer. ● The +3:Administrator menu 4:Change Password item. ● The +3:Administrator menu 5:Clear Password item. ● The +3:Administrator menu 6:Disk Manage item. 100 Chapter 4 Solve problems

Figure 4-17 Remote Admin window

Disconnect a remote connection

The Remote Admin connection can be terminated from the printer control panel or the remote telnet client computer. NOTE: The following procedure describes terminating a Remote Admin connection from the remote telnet client computer. 1. From the Pre-boot main menu, use the arrow buttons on the keyboard to scroll down to the +3:Administrator item, and then press the Enter key. Figure 4-18 Access the administrator menu Disconnect a remote connection 101

2. Use the arrow buttons on the keyboard to scroll down to the +A:Remote Admin item, and then press the Enter key. Figure 4-19 Access the remote admin menu 3. Use the arrow buttons on the keyboard to scroll down to the 2:Stop Telnet item, and then press the Enter key. The Remote Admin connection between the printer and the remote telnet client computer terminates. IMPORTANT: The printer remains in the Pre-boot menu. Have the person that is physically present at the printer do the following: ● Touch the Home button to return to the main Pre-boot menu and highlight the 1:Continue item, and then touch the OK button. The printer will continue to initialize. Figure 4-20 Terminate the telnet connection

Print the event log

Printing the event log might be helpful in the troubleshooting process. For more information, see the interpret control-panel messages and event log entries topics in the printer Troubleshooting Manual. Print the event log from the Administration menu (FutureSmart 3) or Support Tools menu (FutureSmart 4) from a touchscreen control panel Use the following procedure to print the event log from the Administration menu from a touchscreen control panel. 102 Chapter 4 Solve problems

1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Event Log 3. The event log displays on the screen. To print

it, touch the Print button. Print the event log from the Administration menu (FutureSmart 3) or Support Tools menu (FutureSmart 4) from a LCD control panel Use the following procedure to print the event log from the Administration menu (FutureSmart 3) or the Support Tools menu (FutureSmart 4) from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. If necessary, use the down arrow button to scroll to Print Event Log, and then press the OK button to print the page. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then select OK. 2. Select Troubleshooting, and then select OK. 3. Select Event Log. 4. Use the down arrow to scroll down to Cancel, and then use the right arrow to highlight the print icon . Select OK to print the pages. TIP: Multiple report pages can be selected, and then printed together. Print the event log from the Service menu from a touchscreen control panel Use the following procedure to print the event log from the Service menu from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Device Maintenance button (FutureSmart 3), or touch the Support Tools button (FutureSmart 4). 2. Open the Service menu. 3. On the sign-in screen, select the Service Access Code option from the drop-down list. 4. Enter the following service access code for the printer:

● 10050615 (M506/E50045/M507/E50145)

● 11052715 (M527/E52545/M528/E52645)

Print the event log from the Administration menu (FutureSmart 3) or Support Tools menu (FutureSmart 4) from a LCD control panel 103

5. Touch the Print Event Log item. Print the event log from the Service menu from a LCD control panel Use the following procedure to print the event log from the Service menu from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Device Maintenance, and then press the OK button. 2. Use the down arrow button to scroll to Service, and then press the OK button. 3. On the sign-in screen, use the down arrow button to scroll to Service Access Code option and then press the OK button. 4. Enter the following service access code for the printer:

● 10050615 (M506/E50045/M507/E50145)

● 11052715 (M527/E52545/M528/E52645)

5. If necessary, use the down arrow button to scroll to Print Event Log, and then press the OK button to print the page. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then select OK. 2. Select Service, and then select OK. 3. On the sign-in screen, select Service Access Code from the drop-down list. 4. Enter the following service access code for the printer:

● 10050615 (M506/E50045/M507/E50145)

● 11052715 (M527/E52545/M528/E52645)

5. Select Event Log. 6. Use the down arrow to scroll down to Cancel, and then use the right arrow to highlight the print icon . Select OK to print the pages. Clear the event log from a touchscreen control panel Use the following procedure to clear the event log from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Device Maintenance button (FutureSmart 3), or touch the Support Tools button (FutureSmart 4). 2. Open the Service menu. 3. On the sign-in screen, select the Service Access Code option from the drop-down list. 4. Enter the following service access code for the printer:

● 10050615 (M506/E50045/M507/E50145)

104 Chapter 4 Solve problems

● 11052715 (M527/E52545/M528/E52645)

5. Select the Clear Event Log item, and then touch the OK button.

Clear the event log from a LCD control panel

Use the following procedure to clear the event log from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Device Maintenance, and then press the OK button. 2. Use the down arrow button to scroll to Service, and then press the OK button. 3. On the sign-in screen, use the down arrow button to scroll to Service Access Code option and then press the OK button. 4. Enter the following service access code for the printer:

● 10050615 (M506/E50045/M507/E50145)

● 11052715 (M527/E52545/M528/E52645)

5. If necessary, use the down arrow button to scroll to Clear Event Log, and then press the OK button. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then select OK. 2. Select Service, and then select OK. 3. On the sign-in screen, select Service Access Code from the drop-down list. 4. Enter the following service access code for the printer:

● 10050615 (M506/E50045/M507/E50145)

● 11052715 (M527/E52545/M528/E52645)

5. Select Event Log. 6. Use the down arrow to scroll down to Cancel, and then use the right arrow to highlight the trash can icon . Select OK to clear the event log. Clear the event log from a LCD control panel 105

Troubleshooting process

Review the following information about the troubleshooting process.

Determine the problem source

When the printer malfunctions or encounters an unexpected situation, the printer control panel alerts the user to the situation. This section contains a pre-troubleshooting checklist and a troubleshooting flow chart to filter out many possible causes of the problem. ● Use the pre-troubleshooting check list to gather information about the problem from the customer. ● Use the troubleshooting flowchart to pinpoint the root cause of hardware malfunctions. The flowchart guides you to the section of this chapter that contains steps for correcting the malfunction. Before beginning any troubleshooting procedure, check the following issues: ● Are supply items within their rated life? ● Does the configuration page reveal any configuration errors? NOTE: The customer is responsible for checking supplies and for using supplies that are in good condition.

Pre-troubleshooting checklist

The following table includes basic questions to ask the customer to quickly help define the problem(s). Table 4-8 Pre-troubleshooting checklist

General topic Questions

Environment ● Is the printer installed on a solid, level surface (+/- 1°)? ● Is the power-supply voltage within ± 10 volts of the specified power source? ● Is the power-supply plug inserted in the printer and the wall outlet (not a surge protector)? ● Is the operating environment within the specified parameters? ● Is the printer exposed to ammonia gas, such as that produced by diazo copiers or office cleaning materials? NOTE: Diazo copiers produce ammonia gas as part of the copying processes. Ammonia gas (from cleaning supplies or a diazo copier) can have an adverse effect on some printer components (for example, the toner cartridge or cartridges OPC). ● Is the printer exposed to direct sunlight? ● Is the printer exposed to an air conditioning or heating vent that can cause temperature fluctuations? Media ● Does the customer use only supported media? ● Is the media in good condition (no curls, folds, or distortion)? ● Is the media stored correctly and within environmental limits? 106 Chapter 4 Solve problems

Table 4-8 Pre-troubleshooting checklist (continued)

General topic Questions

Input trays ● Is the amount of media in the tray within specifications? ● Is the media correctly placed in the tray? ● Are the paper guides aligned with the stack (no gaps in the stack or excessive pressure causing the stack to bow)? ● Is the tray (or trays) correctly installed in the printer? Toner cartridge ● Is the toner cartridge (or cartridges) installed correctly? Transfer unit and fuser ● Are the transfer unit and fuser installed correctly? NOTE: For printers with an intermediate transfer belt (ITB), is the ITB installed correctly and fully seated. If a replacement ITB was installed, was all of the packing materials removed? Covers ● Is the toner cartridge door closed? Condensation ● Does condensation occur following a temperature change (particularly in winter following cold storage)? If so, wipe affected parts dry or leave the printer on for 10 to 20 minutes. ● Was a toner cartridge (or cartridges) opened soon after being moved from a cold to a warm room? If so, allow the toner cartridge (or cartridges) to sit at room temperature for 1 to 2 hours. Miscellaneous ● Check for and remove any non-HP components (toner cartridges, memory modules, and EIO cards) from the printer. ● Remove the printer from the network and ensure that the failure is associated with the printer before beginning troubleshooting. ● For any color print-quality issues (color printers only), calibrate the printer, and then print a diagnostics page to verify print quality.

Troubleshooting flowchart

This flowchart highlights the general processes to follow to quickly isolate and solve printer hardware problems. Each row depicts a major troubleshooting step. Follow a "yes" answer to a question to proceed to the next major step. A "no" answer indicates that more testing is needed. Go to the appropriate section in this chapter, and follow the instructions there. After completing the instructions, go to the next major

step in this troubleshooting flowchart. Table 4-9 Troubleshooting flowchart

Step Question Action

1 Power on

Is the printer on and

does a readable message display? Yes No Follow the power-on troubleshooting checks. See "Power subsystem" in the Troubleshooting Manual. After the control panel display is functional, see step 2. Troubleshooting flowchart 107

Table 4-9 Troubleshooting flowchart (continued)

Step Question Action

2 Control panel messages

Does the message Ready

display on the control panel? Yes No After the errors have been corrected, go to step 3. 3

Event log

Open the Troubleshooting

menu and print an event log to see the history of errors with this printer. Does the event log print? Yes No If the event log does not print, check for error messages. If paper jams inside the printer, see the jams section of the printer service manual.

If error messages display on the control panel when

trying to print an event log, see the control panel message section of the printer troubleshooting service manual. After successfully printing and evaluating the event log, see step 4. 4

Information

pages

Open the Reports

menu and print the configuration pages to verify that all the accessories are installed.

Are all the accessories

installed? Yes No If accessories that are installed are not listed on the configuration page, remove the accessory and reinstall it. After evaluating the configuration pages, see step 5. 5

Print quality

Does the print quality

meet the customer's requirements? Yes No Compare the images with the sample defects in the image defect tables. See the images defects table in the printer repair service manual. After the print quality is acceptable, see step 6. 6

Interface

Can the customer print

successfully from the host computer? 108 Chapter 4 Solve problems

Table 4-9 Troubleshooting flowchart (continued)

Step Question Action

Yes. This is the end of the troubleshooting process. No Verify that all I/O cables are connected correctly and that a valid IP address is listed on the HP Jetdirect configuration page.

If error messages display on the control panel when

trying to print an event log, see the control-panel message section of the printer troubleshooting service manual. When the customer can print from the host computer, this is the end of the troubleshooting process.

Power subsystem

Review the following information about the power subsystem.

Power-on checks

The basic printer functions should start up when the printer is connected into an electrical outlet and the power switch is pushed to the on position. If the printer does not start, use the information in this section to isolate and solve the problem. If the control panel display remains blank, random patterns display, or asterisks remain on the control panel display, perform power-on checks to find the cause of the problem. NOTE: For touchscreen control panel models, also try the control panel checks in the printer troubleshooting manual.

Power-on troubleshooting overview

During normal operation, cooling fans begins to spin briefly after the printer power is turned on. Place a hand over the vents in the right- and left-side covers. Lean close to the printer to hear the fans operating. If the fan are operating, the dc side of the power supply is functioning. If the fans are operating correctly, the next troubleshooting step is to isolate print engine, formatter, and control panel problems. Perform an engine test. If the formatter is damaged, it might interfere with the engine test. If the engine-test page does not print, try removing the formatter, and then performing the engine test again. If the engine test is then successful, the problem is almost certainly with the formatter, the control panel, or the cable that connects them. NOTE: Depressing, and holding down, the test-page switch causes the printer to continually print test pages. Releasing the switch stops the test page continuous printing.

Troubleshooting power on problems

Use the following procedure to troubleshoot power on problems. 1. Verify that power is available to the printer. If the printer is plugged into a surge protector or uninterruptible power supply (UPS), remove it, and then plug the printer directly into a known operating wall receptacle (make sure that the wall receptacle provides the correct voltage and current for the printer). Unplug any other devices on the same circuit that the printer is using. 2. Try another known operating wall receptacle and a different power cord. Power subsystem 109

3. Listen for startup noises (for example, fans or motors) and look for illuminated lights on the control panel and formatter LEDs. NOTE: Operational fans, motors, and control-panel lights indicate the following: ● AC power is present at the printer. ● The power supply is providing either or both 24 Vdc and 5 Vdc voltages. ● The DC controller microprocessor is functioning. 4. Check the following: a. Turn the printer on, and then listen for startup noises. If normal startup noises are heard, go to step c below. b. If normal startup noises are not heard, turn the printer off, and then remove any installed accessories (for example, envelope feeders, paper feeders, or output accessories). Turn the printer on, and then listen for startup noises. If normal startup noises are heard, the problem might be with one of the accessories. c. Turn the printer off and then remove the power connector from the formatter. Turn the printer on, and then use a small pointed object (like a paper clip) to depress the test-page switch located on the rear side of the printer (near the formatter). NOTE: The test page can only use Tray 2 as the paper source, so make sure that paper is loaded in Tray 2. d. If the engine test page prints, the print engine is operating normally. Replace the formatter. NOTE: M506/E50045/M507/E50145 (all) and M527dn/E52545dn/M528dn/E52645dn models only: If the engine test page does not print, turn the printer off, remove the embedded MultiMedia Card (eMMC), and then try the engine test again. If the page prints, the problem might be the eMMC. 110 Chapter 4 Solve problems

e. If after replacing the formatter normal startup noises are still not heard, replace the DC controller. NOTE: If the error persists after replacing these assemblies, escalate the problem to the Global Business Unit (GBU).

Troubleshooting a blank control panel

A blank control panel display can be caused by one or more of the following: 1. Verify that power is available to the printer. If the printer is plugged into a surge protector or uninterruptible power supply (UPS), remove it, and then plug the printer directly into a known operating wall receptacle (make sure that the wall receptacle provides the correct voltage and current for the printer). 2. Make sure that the power switch is in the on position. 3. Make sure that the fans run briefly, which indicates that the power supply is operational. 4. Make sure that the control-panel display wire harness is connected. 5. Make sure that the formatter connectors are fully seated. Make sure the power switch is in the on position, and then verify that the heartbeat LED is blinking and that the connectivity LED is illuminated. 6. Remove any external solutions, and then try to turn the printer on again. 7. If the control panel display is blank, but the cooling fans run briefly after the printer power is turned on, try printing an engine-test page to determine whether the problem is with the control panel display, formatter, or other printer assemblies. a. Remove the formatter. b. Use a small pointed object to depress the test-page switch located on the rear of the printer. NOTE: The test page can only use Tray 2 as the paper source, so make sure that paper is loaded in Tray 2. Troubleshooting a blank control panel 111

c. If the engine test page prints, the print engine is operating normally (a failed engine test print page does not necessarily indicate that the print engine or DC controller is defective). d. Use the control panel diagnostics to test the control panel. See the control panel checks section in the printer troubleshooting manual. If the error persists, proceed to 8. 8. If the print engine appears to be correctly operating (the engine test page successfully printed) and the control panel is still blank, replace the power supply. 9. If after replacing the power supply normal startup noises and lights are still not present, replace the DC controller. NOTE: If the error persists after replacing these assemblies, escalate the problem to the Global Business Unit (GBU). Control panel checks (M501) The printer includes diagnostic tests for the control panel. This mode allows for troubleshooting issues with the LEDs (2-line control panels), display, and the buttons. 1. Open the secondary service menu. a. From the printer control panel, press the setup button to open the main menu. b. Press and the down arrow button, and then quickly press the Cancel button. c. Press the setup button. The message 2ndary Service appears on the control-panel display. d. Press the OK button to open the 2ndary Service menu. NOTE: Use the arrow buttons to scroll though the 2ndary Service menu. 2. Select one of the following tests: ● LED test: Test the LED lights on 2-line control panels. The touchscreen control panels do not have any LEDs. ● Display Test: Test the control panel display. ● Button Test: Test the control panel buttons. Control panel checks (all except M501) Review the following information about

control panel checks. NOTE: The printer includes a diagnostic test mode for the touchscreen control panels. Diagnostic tests are not available for the LCD control panel. Control panel diagnostics (M506x/M507x and M527/E52545/M528/E52645) Review the following information about the control panel diagnostics. Touchscreen diagnostic mode (M506x, M527/E52545/M528/E52645) Use the diagnostics in this section to test the control panel hardware and embedded firmware. These tests are useful for checking control panel functionality independent of the printer control panel system diagnostics. To test the control panel using the system diagnostics, see "Control panel system diagnostics (M506x/M507x, M527/E52545/M528/E52645)" in the Troubleshooting Manual. 112 Chapter 4 Solve problems

1. M506x only: Locate the diagnostic-tests access button on the back of the control panel. Figure 4-21 Diagnostic-tests access button (M506x) 2. M506x only: Press the diagnostics-access button. Repeatedly pressing the button cycles through the available diagnostics. NOTE: A pen, pencil, or other small blunt object is needed to press the button. TIP: When pressed, the button illuminates green. Figure 4-22 Press the diagnostics-access button (M506x) Touchscreen diagnostic mode (M506x, M527/E52545/M528/E52645) 113

3. M527/E52545/M528/E52645 only: Locate the diagnostic-tests access button on the back of the control panel. Figure 4-23 Diagnostic-tests access button (M527/E52545/M528/E52645) 4. M527/E52545/M528/E52645 only: Press the diagnostics-access button. Repeatedly pressing the button cycles through the available diagnostics. Figure 4-24 Press the diagnostics-access button (M527/E52545/M528/E52645) 5. A yellow screen appears (after the first press of the button) indicating that the control panel firmware is version A (a magenta screen indicates version B firmware). NOTE: If a different color appears on the screen, contact your global business unit (GBU) to determine the firmware version. TIP: After 4 seconds of inactivity, the diagnostic mode times out and is exited. 114 Chapter 4 Solve problems

Figure 4-25 Control panel version A yellow screen 6. Touch the screen to cycle the screen through the following: ● A red screen. ● A green screen. ● A blue screen. ● Five brightness levels of a white screen. Touchscreen diagnostic mode (M506x, M527/E52545/M528/E52645) 115

● A final black screen. TIP: When the final black screen displays, press the Home button to cycle through the diagnostic screens and test its functionality. 7. Pressing the diagnostic-tests button with the black screen displayed exits the diagnostic mode. Figure 4-26 Exit the diagnostic mode Touchscreen diagnostic mode (M507x and E50145dn) Use the diagnostics in this section to test the control panel hardware and embedded firmware. These tests are useful for checking control panel functionality independent of the printer control panel system diagnostics. To test the control panel using the system diagnostics, see "Control panel system diagnostics (M506x/M507x, M527/M528/E52645)" in the Troubleshooting Manual. 116 Chapter 4 Solve problems

1. M507x and E50145dn only: Locate the diagnostic-tests access button on the back of the control panel. Figure 4-27 Diagnostic-tests access button (M507x and E50145dn) 2. Press the diagnostics-access button to enter the diagnostic mode. NOTE: A pen, pencil, or other small blunt object is needed to press the button. TIP: When pressed, the button illuminates green. Figure 4-28 Press the diagnostics-access button (M507x and E50145dn) Repeatedly pressing the button cycles through different brightness levels on the touchscreen control panel. 3. To exit the diagnostic mode, wait approximately five seconds. The diagnostic mode automatically ends, and the touchscreen control panel returns to normal mode. Touchscreen diagnostic mode (M507x and E50145dn) 117

Control panel system diagnostics (M506x/M507x, M527/E52545/M528/E52645) Use the diagnostics in this section to test the control panel hardware and display using the printer firmware system diagnostics.

Open the control panel system diagnostic tests

Use the following procedure to open the control panel system diagnostics tests from a touchscreen control panel. 1. Turn the printer power off, and then on again. 2. Touch the middle of the control panel display when you see the 1/8 under the logo. Figure 4-29 Open the Pre-boot menu 1 2 3. On the Pre-boot menu screen, use the following buttons to navigate the tests. Figure 4-30 Pre-boot menu Table 4-10 Pre-boot menu button descriptions

Button Description

Use this button to see more information about a selected item. 118 Chapter 4 Solve problems

Table 4-10 Pre-boot menu button descriptions (continued)

Button Description

Use this button to scroll up through menu items. Use this button to select a highlighted menu item. Use this button to scroll down through menu items. Use this button to go back to the previous menu. Not used. Use this button to exit a diagnostic test. 4. Use the down arrow button to scroll to +3 Administration, and then press the OK button to select it. Figure 4-31 Access the administration menu Open the control panel system diagnostic tests 119

5. Use the down arrow button to scroll to +E CP Diagnostics, and then press the OK button to select it. NOTE: An administrator password might be required to continue. Figure 4-32 Access the diagnostics menu

Screen test

Use the following procedure to perform a screen test. 1. Open the control panel system diagnostic tests. See "Open the control panel systems diagnostic tests" in the Troubleshooting Manual. 2. With 1 Screen Test highlighted, press the OK button to select it. Figure 4-33 Open the screen test The blue vertical gradient screen appears. 120 Chapter 4 Solve problems

Figure 4-34 Blue vertical gradient screen 3. Touch the touchscreen to scroll through the remaining touchscreen test screens. NOTE: Touch the Home button to exit the test. Table 4-11 Touchscreen test screens

Screen Description

Green vertical gradient

Red vertical gradient

Blue horizontal gradient

Screen test 121

Table 4-11 Touchscreen test screens (continued)

Screen Description

Green horizontal gradient

Red horizontal gradient

Blue with black horizontal interlaced

Green with black horizontal interlaced

Red with black horizontal interlaced

Blue with black vertical interlaced

122 Chapter 4 Solve problems

Table 4-11 Touchscreen test screens (continued)

Screen Description

Green with black vertical interlaced

Red with black vertical interlaced

Black with white center

White with black center

Checkerboard

Multicolor stripes

Screen test 123

Table 4-11 Touchscreen test screens (continued)

Screen Description

Blue green grid meshing

Touch test

Use the following procedure to perform a touch test. 1. Open the control panel system diagnostic tests. See "Open the control panel systems diagnostic tests" in the Troubleshooting Manual. 2. Use the down arrow button to scroll to 2 Touch Test, and then press the OK button to select it. Figure 4-35 Open the touch test 3. Use your finger to touch the white grid on the display. Figure 4-36 Touch the white grid 124 Chapter 4 Solve problems

A mark appears on the grid where it was touched. NOTE: Touch the Home button to exit the test. Figure 4-37 Verify the mark

SoftKey test

Use the following procedure to perform a SoftKey test. 1. Open the control panel system diagnostic tests. See "Open the control panel systems diagnostic tests" in the Troubleshooting Manual. 2. Use the down arrow button to scroll to 3 SoftKey Test, and then press the OK button to select it. Figure 4-38 Open the softkey test SoftKey test 125

3. When prompted, touch the Home button. Figure 4-39 Touch the Home button If the test is successful, the following screen appears on the display. NOTE: Touch the screen to exit the test. Figure 4-40 Successful test

Backlight test

Use the following procedure to perform a backlight test. 1. Open the control panel system diagnostic tests. See "Open the control panel systems diagnostic tests" in the Troubleshooting Manual. 126 Chapter 4 Solve problems

2. Use the down arrow button to scroll to 4 Backlight Test, and then press the OK button to select it. After selecting the 4 Backlight Test, the screen automatically dims, and then returns to full brightness. NOTE: Touch any key to exit the test. Figure 4-41 Open the backlight test

Sound test

Use the following procedure to perform a sound test. 1. Open the control panel system diagnostic tests. See "Open the control panel systems diagnostic tests" in the Troubleshooting Manual. 2. Use the down arrow button to scroll to 5 Sound Test, and then press the OK button to select it. After selecting the 5 Sound Test, the printer emits a series of audible tones. NOTE: Touch any key to exit the test. Sound test 127

Figure 4-42 Open the sound test Keyboard test (M527c/z, E52545c, M528c/z, E52645c only) Use the following procedure to perform a keyboard test on M527c/z, E52545c, M528c/z, and E52645c printers only. IMPORTANT: The control panel system diagnostic tests include a 6 Keyboard Test item. This test is not valid for the M506x, M507x, M527dn/f, E52545dn, M528dn/f, or E52645dn printers even though this option is present in the control panel system diagnostic tests menu. If the 6 Keyboard Test is opened on the M506x, M507x, M527dn/f, E52545dn, M528dn/f, or E52645dn printers, the printer power must be turned off to exit the test. HP does not recommend turning the printer power off during the control panel system diagnostic tests. 1. Open the control panel system diagnostic tests. See the control panel system diagnostics test section in the Troubleshooting Manual. 128 Chapter 4 Solve problems

2. Use the down arrow button to scroll to 6 Keyboard Test, and then press the OK button to select it. Figure 4-43 Do not open the keyboard test except for M527c/z, E52545c, M528c/z, or E52645c printers 3. When prompted, touch the H key on the keyboard or the Home button to exit the test. Version Use the following procedure to view the control panel version information. 1. Open the control panel system diagnostic tests. See "Open the control panel systems diagnostic tests" in the Troubleshooting Manual. 2. Use the down arrow button to scroll to 7 Version, and then press the OK button to select it. NOTE: Touch any key to exit the test. Select 7 Version to view the following types of information: NOTE: The following types of information are for the control panel only, not the printer. ● Panel ID ● Hardware (version) ● Firmware (version) ● KB Hw (version) ● KB Firm (version) ● LCD Vendor ● Touch Controller Version Version 129

Figure 4-44 Open the version information Control panel diagnostic flowcharts (M506x/M507x, M527/E52545/M528/E52645) Use the flowcharts in this section to troubleshoot the following control panel problems. Touchscreen black, white, or dim (no image) To open the diagnostic function, press the button on the back of the control panel. 130 Chapter 4 Solve problems

Figure 4-45 Touchscreen blank, white, or dim (no image) N N Y N N Y Y N Y Y N Y

Touch the display or Home button

to exit sleep mode. Make sure that the product is plugged in and the power is turned on.

Black display

(no backlight or image)

White display

(no image)

Dim display

(no image)

Is the Home button

illuminated (bright white)? Open the diagnostic function.

Does an image

appear on the control panel?

Does the display

turn on? Problem solved.

Are the formatter

LEDs functioning? Turn the product power off.

Remove and reseal the formatter

cables. Do not replace the control panel. Turn the product power off.

Inspect and reseal the

control panel cables.

Remove and reseal the

formatter cables.

Make sure that the

formatter LEDs function. Problem fixed?

Replace the

control panel. Do not replace the control panel. Turn the product power off.

Check connection and cables on the formatter

and at the control panel.

Is the backlight

adjustable? Open the diagnostic function.

Perform the

backlight test.

Is the product in bright

sunlight? If yes, move the product to a different location.

Touchscreen has an unresponsive zone

To open the diagnostic function, press the button on the back of the control panel. Touchscreen has an unresponsive zone 131

Figure 4-46 Touchscreen has an unresponsive zone N Y N Y N Y

Touchscreen

has an unresponsive zone

Is the area of the

touchscreen you are touching greyed out (intentionally deactivated)?

Does the previously

inactive area respond to a touch? Open the diagnostic function.

Perform the

touch test.

Turn the product

power off, and then on again.

Does the previously

inactive area respond to a touch?

Replace the

control panel.

Do not replace the

control panel.

No control panel sound

To open the diagnostic function, press the button on the back of the control panel. 132 Chapter 4 Solve problems

Figure 4-47 No control panel sound N N Y Y

Control panel

has no sound Turn the product power off. Remove the control panel. Check the cables to the speaker. Reseat the cables to the speaker. Turn the product power on. Can sounds be heard?

Can sounds

be heard?

Can sounds

be heard? Replace the control panel. Open the following menus:

Administration

Display Settings

Key Press Sound

Select the following: On Save N

Do not replace the

control panel. Open the diagnostic function. Perform the sound test.

Home button is unresponsive

To open the diagnostic function, press the button on the back of the control panel. Home button is unresponsive 133

Figure 4-48 Home button is unresponsive Y N N Y

Home button

Unresponsive

Note: If illuminated: continue. If not illuminated: verify that the power is turned on, the printer is not in sleep mode, turn the power off, and then on again. Open the diagnostic function.

Perform the Home

button test.

Does the virtual LED

illuminate green when the Home button is pressed? Turn the product power off, and then on again. Open the diagnostic function.

Perform the

Home button test.

Does the virtual LED

illuminate green when the Home button is pressed?

Replace the

control panel.

Do not replace the

control panel. Hardware integration pocket (HIP) is not functioning (control panel functional) Review the following information when the HIP is not functioning, but the control panel is functional. 134 Chapter 4 Solve problems

Figure 4-49 Hardware integration pocket (HIP) is not functioning (control panel functional)

Hardware integration

pocket (HIP) not functional (control panel functional)

Check that all installed hardware integration

pocket devices are within HP specifications. The control panel provides only 0.5 Amps to the port. If an installed device attempts to draw more power than is provided, the port is automatically disabled. Troubleshoot third-party devices with the device manufacturer. Hardware integration pocket (HIP) is not functioning (control panel functional) 135

Tools for troubleshooting

This section describes tools that can help to solve printer problems.

Individual component diagnostics

Review the following information about individual component diagnostics. LED diagnostics (M501) LED, engine, and individual diagnostics can identify and troubleshoot printer problems. Network port LEDs (M501) The formatter has two network port LEDs. When the printer is connected to a properly working network through a network cable, the amber LED indicates network activity, and the green LED indicates the link status. Figure 4-50 Formatter network port LEDs 1 2 1 Link status LED (green) 2 Network activity LED (amber) A blinking amber LED indicates network traffic. If the green LED is off, a link has failed. For link failures, check all of the network cable connections. In addition, you can try to manually configure the network card link speed setting by using the printer control panel. 1. On the printer control panel, press the OK button. 2. Open the following menus: ● Network Setup ● Link Speed 3. Select the appropriate link speed. 136 Chapter 4 Solve problems M501 control panel LEDs The state of the Ready light and Attention light on the printer signal the printer status. The following table outlines the possible control panel light states. Printer state Ready light state Attention light state

Initializing Blinking Blinking

Ready On Off

Receiving data/processing job or cancelling job

Blinking Off

Error message Off Blinking

Fatal error (49 or 79 error)1 On On 1 The printer restarts after one of these errors occurs. LED diagnostics (all except M501) LED, engine, and individual diagnostics can identify and troubleshoot printer problems.

Understand lights on the formatter

Three LEDs on the formatter indicate that the printer is functioning correctly. M501 control panel LEDs 137

Figure 4-51 Formatter LED lights

1 Heartbeat LED

2 HP Jetdirect LEDs

Heartbeat LED

The heartbeat LED indicates that the formatter is functioning correctly. While the printer is initializing after it is turned on, the LED blinks rapidly, and then turns off. When the printer has finished the initialization sequence, the heartbeat LED pulses on and off. The following list describes the heartbeat LED operation while the printer is executing the firmware boot process. NOTE: When the initialization process completes, the heartbeat LED should be illuminated solid green. If after initialization, the heartbeat LED is not solid green, see Table 4-12 Heartbeat LED, printer operational (all except M501) on page 139. ● Green – Blinking: The printer is correctly functioning. – Solid: The FW has not yet booted. ● Amber 138 Chapter 4 Solve problems

– Blinking: The control panel is not detected. – Solid: The FFC between the formatter and DC controller is not connected or is damaged. ● Red – Solid: Valid SPI code; there is a problem releasing ASIC and running the BIOS Solid: BIOS initiates PROSAC to change the LED from red to green. Solid: PROSAC is not running (No ROM or SPI code). – Blinking: Valid SPI code not found. ● Off: No 5V or 3.3V PROSAC power. The following table describes the heartbeat LED operation when the printer completes the firmware boot process and is in the Ready state. Table 4-12 Heartbeat LED, printer operational (all except M501)

LED color Description

Green ● Normal operation – Formatter is operating normally – Firmware is operating normally – Control panel is connected Yellow ● ● Formatter cannot connect to the control panel – Check control panel connections – Verify control panel functionality Red ● Formatter error or failure – Serial peripheral interface (SPI) flash memory boot error – Power on self-test (formatter) failed – Diagnostic (formatter) failed Off TIP: The heartbeat LED is off if the power cable is disconnected, the printer power switch is in the off position, or the printer is in Sleep Mode. ● Firmware or system freeze – Check control panel for an error message – Control panel failure NOTE: This condition is not usually caused by a formatter failure. Turn the power off, and then on again. If the error persists, perform a firmware upgrade.

HP Jetdirect LEDs

The embedded HP Jetdirect print server has two LEDs. The yellow LED indicates network activity, and the green LED indicates the link status. A blinking yellow LED indicates network traffic. If the green LED is off, a link has failed. HP Jetdirect LEDs 139

For link failures, check all the network cable connections. In addition, try to manually configure the link settings on the embedded print server by using the printer control-panel menus. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Settings button (FutureSmart 4). 2. Open the following menus: ● Network Settings (FutureSmart 3) or Networking (FutureSmart 4). ● Embedded Jetdirect Menu ● Link Speed 3. Select the appropriate link speed, and then touch the OK button.

Engine diagnostics

The printer contains extensive internal engine diagnostics that help in troubleshooting print quality, paper path, noise, assembly, and timing issues.

Engine test button

To verify that the printer engine is functioning, print an engine test page. Use a small pointed object to depress the test-page switch located on the rear side of the printer. The test page should have a series of lines that are parallel to the short end of the page. The test page can use only Tray 2 as the paper source, so make sure that paper is loaded in Tray 2. NOTE: Depressing and holding down the test-page switch causes the printer to continually print test pages. Releasing the switch stops the test page continuous printing. Figure 4-52 Engine test button

Defeating interlocks

Different tests can be used to isolate different types of issues. For assembly or noise isolation, run the diagnostic test when the toner cartridge door or rear door is open. 140 Chapter 4 Solve problems

Defeating the door interlocks allows observation of the paper pick operation (and view the page enter registration). WARNING! Be careful when performing printer diagnostics to avoid risk of injury. Only trained service personnel should open and run the diagnostics with a door open. Never touch any of the power supplies when the printer is turned on.

Defeat the rear door interlock

Use the following procedure to defeat the rear door interlock. 1. Open the rear door. 2. Insert a folded piece of paper into the slot. TIP: Fold a stiff piece of paper, for example a business card or index card, into two 10 mm (.375 in) strips, and insert the strips into the slot for the cartridge door logic switches. Figure 4-53 Defeat the rear door interlock

Defeat the toner cartridge door interlock

Use the following procedure to defeat the toner cartridge door interlock. 1. Open the toner cartridge door. 2. Insert a folded piece of paper into the slot. TIP: Fold a stiff piece of paper, for example a business card or index card, into two 10 mm (.375 in) strips, and insert the strips into the slot for the cartridge door logic switches. Defeat the rear door interlock 141

Figure 4-54 Defeat the toner cartridge door interlock Disable cartridge check (all except M501) Use this diagnostic test to print internal pages or send an external job to the printer when the toner cartridge is removed or exchanged. Supply errors are ignored while the printer is in this mode. When the printer is in this mode, access the troubleshooting menus and print internal pages (the print quality pages will be the most useful). This test can be used to isolate problems, such as noise, and to isolate print-quality problems that are related to the toner cartridge. NOTE: Do not remove or exchange the toner cartridge until after beginning the disable cartridge check diagnostic. Disable cartridge check from a touchscreen control panel Use the following procedure to disable cartridge check from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Diagnostic Tests ● Disable Cartridge Check

Disable cartridge check from a LCD control panel

Use the following procedure to disable cartridge check from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 142 Chapter 4 Solve problems

3. Use the down arrow button to scroll to Diagnostics, and then press the OK button. 4. Use the down arrow button to scroll to Disable Cartridge Check, and then press the OK button to select it. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Select Continue to enter Maintenance Mode. 5. Use the down arrow button to scroll to Disable Cartridge Check, and then press the OK button to select it. Paper path and sensor diagnostic tests (all except M501) Review the following information about the paper path and sensor diagnostics test for the M506/M507/ E50145 and M527/M528/E52645 printers. Paper path test (all except M501) This diagnostic test generates one or more test pages. Use these pages to isolate the cause of jams. To isolate a problem, specify which input tray to use, specify whether to use the duplex path (duplex models only), and specify the number of copies to print. Multiple copies can be printed to help isolate intermittent problems. The following options become available after beginning the diagnostic feature: ● Print Test Page: Run the paper-path test from the default settings: Tray 2, no duplex, and one copy. To specify other settings, scroll down the menu, and select the setting, and then scroll back up and select Print Test Page to start the test. ● Source Tray: Select Tray 1, Tray 2, or the optional tray. ● Test Duplex Path: Enable or disable

two-sided printing from designated optional tray (depending on how many optional accessories are installed). ● Number of Copies: Set the numbers of copies to be printed; the choices are 1, 10, 50, 100, or 500. Access the paper path test from a touchscreen control panel Use the following procedure to access the paper path test from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Diagnostic Tests ● Paper Path Test 3. Select the paper path test options for the test. Paper path and sensor diagnostic tests (all except M501) 143

Access the paper path test from a LCD control panel

Use the following procedure to access the paper path test from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Use the down arrow button to scroll to Paper Path Test, and then press the OK button. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Select Continue to enter Maintenance Mode. 5. Use the down arrow button to scroll to Paper Path Test, and then press the OK button. Paper path sensors test (all except M501) This test displays the status of each paper path sensor and allows viewing of sensor status while printing internal pages. Access the paper path sensors test from a touchscreen control panel Use the following procedure to access the paper path sensors test from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Diagnostic Tests ● Paper Path Sensors 3. Touch Start to run the test. Access the paper path sensors test from a LCD control panel Use the following procedure to access the paper path sensors test from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 144 Chapter 4 Solve problems

4. Use the down arrow button to scroll to Paper Path Sensors, and then press the OK button. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Select Continue to enter Maintenance Mode. 5. Use the down arrow button to scroll to Paper Path Sensors, and then press the OK button. NOTE: The menu list of sensors and switches for the Paper path sensors test, Manual Sensor Test, and the Tray/Bin Manual Sensor Test varies depending on which optional accessories are installed. The tables in this section describe the sensor tests available with an optional 1x550-sheet paper feeder installed. For trays other than Tray 1 or Tray 2, the tray number associated with a sensor or switch depends on the number of accessories installed. Table 4-13 Paper path sensors diagnostic tests (all except M501) Sensor name Sensor/Switch number

Replacement part number Descriptions

Tray 3 feed sensor1 PS8008 F2A68-67909 Optional paper feeder Prefeed sensor PS102 NA Not replaceable2 Top-of-Page (TOP) sensor PS215 NA Not replaceable2 Paper width 3 sensor NA NA Not replaceable2 Paper width 2 sensor NA NA Not replaceable2 Paper width 1 sensor NA NA Not replaceable2 Fuser output sensor PS2 RM2-5679-000CN (110 V)

RM2-5692-000CN (220 V)

(M501/M506/E50045/M527/E52545)

Fuser Fuser output sensor PS2 RM2-2585-000CN (110 V)

RM2-2586-000CN (220 V)

(M507/E50145/M528/E52645)

Fuser

Duplexer refeed

sensor NOTE: Duplex models only. PS502 NA Not replaceable2 Access the paper path sensors test from a LCD control panel 145

Table 4-13 Paper path sensors diagnostic tests (all except M501) (continued) Sensor name Sensor/Switch number

Replacement part number Descriptions

Output bin full sensor PS4 NA Not replaceable2 1 Only appears if optional accessories are installed. 2 Require whole unit replacement. Manual sensor tests (all except M501) The table in this section lists the sensors and switches available in the Manual Sensor Test. Access the manual sensor test from a touchscreen control panel The Manual Sensor Test screen shows the sensor number, sensor name (M506x/M507x

and M527/ E52545/M528/E52645 only), sensor state (active or inactive), and the number of times the sensor has been toggled (activated). 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Diagnostic Tests ● Manual Sensor Test 3. Activate the desired sensor, and then check the control panel display to verify the sensor state (active or inactive). NOTE: Only the M506x/M507x and M527/E52545/M528/E52645 control panel displays the State and Toggle descriptor. ● The State virtual LED next to the sensor number and sensor name (M506x/M507x and M527/ E52545/M528/E52645 only) illuminates green when the sensor is active. ● The Toggle virtual LED next to the sensor number and sensor name (M506x/M507x and M527/ E52545/M528/E52645 only) illuminates green after the sensor is activated and increments by one each time the sensor is interrupted (activated or deactivated). For example, opening the front door increments the SW501 Cartridge doorToggled item count two times—once when the door is opened, and once when the door is closed. Access the manual sensor test from a LCD control panel Use the following procedure to access the manual sensor test from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 146 Chapter 4 Solve problems

4. Use the down arrow button to scroll to Manual Sensor Test, and then press the OK button. 5. Activate the desired sensor, and then check the control panel display to verify the sensor state (active or inactive). TIP: Press the return arrow button to reset the sensor or press the Cancel button to exit the test. NOTE: Only the M506x/M507x and M527/E52545/M528/E52645 control panel displays the State and Toggle descriptor. ● The State virtual LED next to the sensor number and sensor name (M506x/M507x and M527/ E52545/M528/E52645 only) illuminates green when the sensor is active. ● The Toggle virtual LED next to the sensor number and sensor name (M506x/M507x and M527/ E52545/M528/E52645 only) illuminates green after the sensor is activated and increments by one each time the sensor is interrupted (activated or deactivated). For example, opening the cartridge door increments the SW501 Front door opening/closing Toggled item count two times—once when the door is opened, and once when the door is closed. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Select Continue to enter Maintenance Mode. 5. Use the down arrow button to scroll to Manual Sensor Test, and then press the OK button. 6. Activate the desired sensor, and then check the control panel display to verify the sensor state (active or inactive). TIP: Press the return arrow button to reset the sensor or press the Cancel button to exit the test. NOTE: Only the M506x/M507x and M527/E52545/M528/E52645 control panel displays the State and Toggle descriptor. ● The State virtual LED next to the sensor number and sensor name illuminates green when the sensor is active. ● The Toggle virtual LED next to the sensor number and sensor name illuminates green after the sensor is activated and increments by one each time the sensor is interrupted (activated or deactivated). Access the manual sensor test from a LCD control panel 147

For example, opening the front door increments the SW3 Front door opening/closing Toggled item count two times—once when the door is opened, and once when the door is closed. NOTE: The following table describes the sensor tests available with an optional 1x550-sheet paper feeder installed. For trays other than Tray 1 or Tray 2, the tray number associated with a sensor or switch depends on the number and type of accessories installed. Table 4-14 Manual sensor diagnostic tests (all except M501) Sensor or switch Replacement Part number Description SW501 Cartridge door sensor RK2-0534-000CN Microswitch (SW2) PS8008 Tray 3 feed sensor1 F2A68-67909 550-sheet paper feeder PS102 Prefeed sensor NA Not replaceable2 PS215 Top-of-Page (TOP) sensor NA Not replaceable2 Paper width 3 sensor NA Not replaceable2 Paper width 2 sensor NA Not replaceable2 Paper width 1 sensor NA Not replaceable2 PS2 Fuser output sensor RM2-5679-000CN (110 V)

RM2-5692-000CN (220 V)

(M501/M506/E50045/M527/E52545)

Fuser PS2 Fuser output sensor RM2-2585-000CN (110 V)

RM2-2586-000CN (220 V)

(M507/E50145/M528/E52645)

Fuser PS502 Duplexer refeed sensor NOTE: Duplex models only. NA Not replaceable2 PS4 Output bin full sensor NA Not replaceable2 1 Only appears if optional accessories are installed. 2 Requires whole unit replacement. Tray/bin manual sensor test (all except M501) The table in this section lists the sensors and switches available in the Tray/Bin Manual Sensor Test. Access the tray/bin manual sensor test from a touchscreen control panel The Tray/Bin Manual Sensor Test screen shows the sensor number, sensor name (M506x/M507x and M527/E52545/M528/E52645 only), sensor state (active or inactive), and the number of times the sensor has been toggled (activated). 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting 148 Chapter 4 Solve problems

● Diagnostic Tests ● Tray/Bin Manual Sensor Test 3. Activate the desired sensor, and then check the control panel display to verify the sensor state (active or inactive). NOTE: Only the M506x/M507x and M527/E52545/M528/E52645 control panel displays the State and Toggle descriptor. ● The State virtual LED next to the sensor number and sensor name (M506x/M507x and M527/ E52545/M528/E52645 only) illuminates green when the sensor is active. ● The Toggle virtual LED next to the sensor number and sensor name illuminates green after the sensor is activated and increments by one each time the sensor is interrupted (activated or deactivated). For example, raising, and then

releasing, the output bin full flag increments the PS4 Output bin full Toggled item count two times—once when the flag is raise, and once when it is released. Access the tray/bin manual sensor test from a LCD control panel Use the following procedure to access the tray/bin manual sensor test from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Use the down arrow button to scroll to Tray/Bin manual Sensor Test, and then press the OK button. 5. Activate the desired sensor, and then check the control panel display to verify the sensor state (active or inactive). TIP: Press the return arrow button to reset the sensor or press the Cancel button to exit the test. NOTE: Only the M506x/M507x and M527/E52545/M528/E52645 control panel displays the State and Toggle descriptor. ● The State virtual LED next to the sensor number and sensor name (M506x/M507x and M527/ E52545/M528/E52645 only) illuminates green when the sensor is active. ● The Toggle virtual LED next to the sensor number and sensor name (M506x/M507x and M527/ E52545/M528/E52645 only) illuminates green after the sensor is activated and increments by one each time the sensor is interrupted (activated or deactivated). For example, raising, and then releasing, the output bin full flag increments the PS4 Output bin full Toggled item count two times—once when the flag is raise, and once when it is released. FutureSmart 4 Access the tray/bin manual sensor test from a LCD control panel 149

1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Select Continue to enter Maintenance Mode. 5. Use the down arrow button to scroll to Tray/Bin manual Sensor Test, and then press the OK button. 6. Activate the desired sensor, and then check the control panel display to verify the sensor state (active or inactive). TIP: Press the return arrow button to reset the sensor or press the Cancel button to exit the test. NOTE: Only the M506x/M507x and M527/E52545/M528/E52645 control panel displays the State and Toggle descriptor. ● The State virtual LED next to the sensor number and sensor name illuminates green when the sensor is active. ● The Toggle virtual LED next to the sensor number and sensor name illuminates green after the sensor is activated and increments by one each time the sensor is interrupted (activated or deactivated). For example, opening the front door increments the SW3 Front door opening/closing Toggled item count two times—once when the door is opened, and once when the door is closed. NOTE: The following table describes the sensor tests available with an optional 1x550-sheet paper feeder installed. For trays other than Tray 1 or Tray 2, the tray number associated with a sensor or switch depends on the number and type of accessories installed. Table 4-15 Tray/bin manual sensors (all except M501) Sensor or switch name Replacement part number Descriptions PS205 Tray 1 paper sensor F2A68-67914 Cartridge door assembly PS3 Tray 2 paper sensor NA Not replaceable1 PS451 Tray 3 paper sensor2 F2A68-67909 Optional paper feeder PS8008 Tray 3 paper feed sensor2 F2A68-67909 Optional paper feeder PS1603 Tray 3 feed sensor2 F2G68-67901 Optional paper feeder PS4 Output bin full sensor NA Not replaceable1 1 Requires whole unit replacement. 2 Only appears if optional accessories are installed. Print/stop test (all except M501) Use this diagnostic test to isolate the cause of problems such as image-formation defects and jams within the engine. 150 Chapter 4 Solve problems

During this test, stop the paper anywhere along the printer paper path. The test can be programmed to stop printing internal pages or an external print job when the paper reaches a certain position. The test can also be programmed to stop from 0 to 60,000 ms. If the timer is set to a value that is greater than the job-print time, the printer can recover in one of two ways. ● After the print job is completed press OK button to return to the Troubleshooting menu before the timer times out. ● After the timer times out, touch the Stop button. Activate the door switch to restart the engine and return it to a normal state. Access the print/stop test from a touchscreen control panel Use the following procedure to access the print/stop test from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Diagnostic Tests ● Print/Stop Test 3. Enter a range, and then touch the OK button. Access the print/stop test from a LCD control panel Use the following procedure to access the print/stop test from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Test, and then press the OK button. 4. Use the down arrow button to scroll to Print/Stop Test, and then press the OK button. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Select Continue to enter Maintenance Mode. 5. Use the down arrow button to scroll to Print/Stop Test, and then press the OK button. Common print stop test timing millisecond (ms) stops Review the following information about common print stop test timing stops. Access the print/stop test from a touchscreen control panel 151

● 600 ms: The page has passed the registration area and the leading edge is just short of entering the fuser. The image can be seen on the paper but has not fused. If the defect is visible then the cause might be the drum, transfer roller, or a roller prior to, or in, the registration area. ● 1200 ms: The leading edge is about 18mm (0.71 in) into the top output bin. The image has gone through the fuser. If the defect was not visible prior to the fuser, and is visible after the fuser, then the fuser it is the likely cause of the print quality defect. Inspect the fuser for damage, debris, or labels stuck to the fuser. Replace the fuser. Discuss media specifications and proper care of the fuser with the customer. Component tests (all except M501) Use the procedure below to test various printer mechanical and electromechanical assemblies. Individual component diagnostics (special-mode test; all except M501) This test activates individual parts independently to isolate problems. NOTE: The following table describes the sensor tests available with an optional 1x550-sheet paper feeder installed. For trays other than Tray 1 or Tray 2, the tray number associated with a sensor or switch depends on the number and type of accessories installed. Each component test can be performed once or repeatedly. If the Repeat option is enabled from the drop-down menu, the test cycles the component on and off. This process continues for two minutes, and then the test terminates. NOTE: The cartridge door interlocks must be defeated to run the component tests. A control panel display prompt appears to indicate removing the toner cartridge, during certain tests. Access the individual component diagnostics from a touchscreen control panel Use the following procedure to access the individual component diagnostics from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Diagnostic Tests ● Component Test 3. Select the component test options for the test. Access the individual component diagnostics from a LCD control panel Use the following procedure to access the individual component diagnostics from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer

control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 152 Chapter 4 Solve problems

3. Use the down arrow button to scroll to Diagnostic Test, and then press the OK button. 4. Use the down arrow button to scroll to Component Test, and then press the OK button. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. Use the down arrow button to scroll to Diagnostic Tests, and then press the OK button. 4. Select Continue to enter Maintenance Mode. 5. Use the down arrow button to scroll to Component Test, and then press the OK button. 6. Select the component and desired options. 7. Select Start to start the test. 8. Select Stop to stop the test. NOTE: The following table describes the sensor tests available with an optional 1x550-sheet paper feeder installed. For trays other than Tray 1 or Tray 2, the tray number associated with a sensor or switch depends on the number and type of accessories installed. Table 4-16 Component test details (all except M501)

Component test Item tested Comments

Fuser motor M1 Activates the specified motor. Pickup motor M2 Activates the specified motor. Tray 1 Pickup Solenoid SL2 Activates the specified solenoid. Tray 2 Pickup Solenoid SL1 Activates the specified solenoid. Tray 3 Pickup Solenoid1 Activates the specified solenoid.

Duplex switchback solenoid

NOTE: Duplex models only. SL3 Activates the specified solenoid. Optional Clutch Drive Activates the specified clutch/drive.

Repick Clutch

NOTE: Duplex models only. CL1 Activates the specified clutch. Laser Scanner Motor M3 Activates the specified motor. Repeat NA Choose Off to execute the test once. Choose On to execute the test continuously. 1 Only appears if an optional 550-sheet paper feeder accessory is installed. Scanner tests (M527/E52545/M528/E52645) Use these diagnostic tests to manually test the document feeder and scanner sensors. Scanner tests (M527/E52545/M528/E52645) 153

Scanner tests (M527/E52545/M528/E52645) This section lists the sensors available in the Scanner Tests. Document feeder and image scanner sensor replacement parts If a document feeder or image scanner sensor fails, replace the following assemblies. ● Document feeder sensors – Document feeder whole unit kit Enterprise printers; 5851-6568 NOTE: Includes the white backing kit. – Document feeder whole unit kit Work Flow printers; 5851-6569 NOTE: Includes the white backing kit. – Scanner sub assembly (SSA) kit; F2A76-67909 NOTE: Includes the white backing kit.

Use the scanner tests

The Scanner Tests screen shows the sensor name, sensor state (active or inactive), and the number of times the sensor has been toggled (activated). 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Diagnostic Tests ● Scanner Tests ● Sensors 3. Touch the sensor name on the Scanner Tests screen to display a sensor location graphic on the control panel display. 4. Activate the desired sensor, and then check the control panel display to verify the sensor state (active or inactive). ● The State virtual LED next to the sensor number and sensor name illuminates green when the sensor is active. ● The Toggle virtual LED next to the sensor number and sensor name illuminates green after the sensor is activated and increments by one each time the sensor is interrupted (activated or deactivated). For example, opening the flatbed cover increments the Flatbed cover Toggle item count two times—once when the door is opened, and once when the door is closed.

Scanner tests sensors

● ADF paper present 154 Chapter 4 Solve problems

● ADF Y (length) ● ADF jam cover ● ADF paper path deskew ● ADF paper path pick success ● Paper path sensor 1 (unreachable) ● Paper path sensor 2 (unreachable) ● Flatbed Y (length) ● Flatbed cover Diagrams Use the diagrams in this section to identify printer components. Diagrams: Block diagrams Use the diagrams in this section to identify printer sensors, switches, and assemblies.

Sensors and switches

Review the following information about the sensors and switches. Figure 4-55 Printer base, sensors and switches block diagram SW4 PS1a PS2a PS4 PS13 PS2 PS3 PS11 PS1 PS12

Sensor Description Sensor Description

SR1a Media width sensor SR4 Output bin media-full sensor SR2b Duplex feed sensor1 SR11 Registration sensor Diagrams 155

Sensor Description Sensor Description

SR1 Cassette media out sensor SR12 Prefeed sensor SR2 Top-of-Page (TOP) sensor SR13 Fuser output sensor SR3 Tray 1 (multipurpose tray) media out and registration sensor SW4 Cassette detection switch 1 Duplex models only. Figure 4-56 1x550-sheet paper feeder, sensors and switches block diagram PS461

PS460 SW3

PS432

Sensor Description

PS460 Media surface sensor PS461 Cassette media out sensor PS432 Feed sensor SW3 Cassette detection switch 156 Chapter 4 Solve problems

Figure 4-57 Printer base, motors, clutches, and solenoids block diagram CL1 SL3 M1 SL1 SL2 M2 M4

Item Description

M1 Fuser motor M2 Pickup motor M4 Lifter motor SL1 Cassette pickup solenoid SL2 Tray 1 (MP) pickup solenoid SL3 Duplex switchback solenoid1 Sensors and switches 157

Item Description

CL1 Duplex re-pickup clutch1 1 Duplex models only. Figure 4-58 1x550-sheet paper feeder, motors, clutches, and solenoids block diagram SL4 M2 M5 CL2 M2 Fuser motor (located in the printer) M5 Lifter motor CL2 Feed clutch SL4 Cassette pickup solenoid

Cross section diagrams

Review the following information about the cross section diagrams. 158 Chapter 4 Solve problems

Figure 4-59 Printer base, cross section diagram 20 19 18 17 16 15 14 13 1 2 3 4 5 6 7 8 9 10 11 12 21

Item Description Item Description

1 Fuser output roller 12 Feed roller 2 Duplex flapper113 Tray 2 feed roller 3 Output roller 14 Tray 2 separation roller 4 Fuser film 15 Tray 2 pickup roller 5 Fuser 16 Duplex re-pickup roller1 6 Photosensitive drum 17 Registration roller 7 Laser/scanner assembly 18 Cassette 8 Toner cartridge 19 Transfer roller 9 Registration shutter 20 Duplex feed roller1 10 Tray 1 (MP) tray pickup roller 21 Pressure roller Cross section diagrams 159

Item Description Item Description

11 Tray 1 (MP) tray separation pad 1 Duplex models only. Figure 4-60 1x550-sheet paper feeder, cross section diagram 1 2 3 4 5

Item Description

1 Cassette

2 Cassette pickup roller

3 Cassette feed roller

4 Feed roller

5 Cassette separation roller

Diagrams: Printed circuit assembly (PCA) connector locations Use the diagrams in this section to identify printer PCA connector locations.

DC controller connections

Each of the connections on the DC controller PCA is indicated in the following figure. 160 Chapter 4 Solve problems

Figure 4-61 DC controller PCA connectors J132 J161 J163

J162 J133 J146 J152 J141

J159 J143 J151 J131

J157 J156 J158 J135 J144 J147

J171 J111 J153 J154

Item Description Item Description

J111 Not used J152 Output bin media-full sensor (PS4) Fuser output sensor (PS13) Thermistor (YH401) J131 Low-voltage power supply (LVPS)

Power LED

J153 Paper feeder J132 Fuser power supply (FPS) J154 Duplex re-pickup clutch (CL1)1 J133 High-voltage power supply (HVPS) J156 Media surface sensor (PS12) J135 Low-voltage power supply (LVPS) J157 Registration sensor (PS11) J141 Duplex switchback solenoid (SL3)1 J158 Cassette detection switch (SW4) J143 Tray 1 (MP) tray pickup solenoid (SL2) J159 Rear door switch (SW5) J144 Lifter motor Cassette pickup solenoid (SL1) J161 Laser/scanner assembly J146 Fuser motor (M1) J162 Laser/scanner motor J147 Sub fan (FM2) J163 Formatter J151 Media width sensor (PS1a) Duplex feed sensor (PS2b)1 Top-of-Page (TOP) sensor (PS2) J171 Pickup motor (M2) 1 Duplex models only. Formatter PCA (M501) Review the following information about the formatter PCA for the M501 printer. Formatter PCA (M501) 161

Figure 4-62 Formatter PCA M501 3 2 1 4 5 6

Item Description Item Description

1 Control panel 4 Local area network (LAN) 2 Engine 5 USB device 3 Power supply 6 USB host Formatter PCA (M506/E50045/M507/E50145) Review the following information about the formatter PCA for the M506/E50045/M507/E50145 printers. 162 Chapter 4 Solve problems

Figure 4-63 Formatter PCA (M506/E50045/M507/E50145) 3 5 6 7 8 9 10 11 2 1 4 14 13 12

Item Description Item Description

1 Power 8 IOD 2 Engine I/F 9 Local area network (LAN) 3 Hardware integration pocket (HIP) 10 USB device 4 Easy-access USB 11 USB host 5 Control panel HDMI 12 Dual in-line memory module (DIMM) 6 Control panel sideband 13 embedded MultiMedia Card (eMMC) 7 Wireless 14 Hard-disk drive (HDD) riser Formatter PCA (M527/E52545/M528/E52645) Review the following information about the formatter PCA for the M527/E52545/M528/E52645 printers. Formatter PCA (M527/E52545/M528/E52645) 163

Figure 4-64 Formatter PCA M527/E52545/M528/E52645 1 2 3 4 5 6 7 8 9 10 12 13 11

Item Description Item Description

1 Power 8 Control panel HDMI 2 Engine and video 9 Island of Data (IOD) 3 Scanner HDMI 10 Local area network (LAN) 4 Hardware integration pocket (HIP) 11 Trusted Platform Module (TPM) 5 embedded MultiMedia Card (eMMC) 12 USB host 6 Walk up USB 13 USB device

7 Control panel side

1x550-sheet paper feeder controller PCA connections Each of the connections on the 1x550-sheet paper feeder controller PCA is indicated in the following figure. 164 Chapter 4 Solve problems

Figure 4-65 1x550-sheet paper feeder controller PCA connectors J611 J610 J620 J612 J625 J624 J623 J622 J621

Item Description Item Description

J610 Printer J622 Lifter motor (M5) J611 Paper feeder J623 Media surface sensor (PS460) J612 Cassette detection switch (SW3) J624 Feed sensor (PS432) J620 Lifter motor (M5) J625 Cassette media out sensor (PS461) J621 Cassette pickup solenoid (SL4) Diagrams: External plug and port locations Use the following figure to locate and identify the printer external port locations. Diagrams: External plug and port locations 165

Figure 4-66 External plug and port locations (M501) 1 2 3 4

Item Description

1 Slot for a cable-type security lock 2 Local area network (LAN) Ethernet (RJ-45) network port 3 Hi-Speed USB 2.0 printing port 4 USB port for job storage feature (requires a 16 GB or larger USB flash drive) 166 Chapter 4 Solve problems

Figure 4-67 External plug and port locations (M506/E50045/M507/E50145) 1 2 3 4

Item Description

1 Slot for a cable-type security lock 2 Local area network (LAN) Ethernet (RJ-45) network port 3 Hi-Speed USB 2.0 printing port 4 USB port for connecting external USB devices (this port might be covered) NOTE: For easy-access USB printing, use the USB port on the top cover. Diagrams: External plug and port locations 167

Figure 4-68 External plug and port locations (M527/E52545/M528/E52645) 1 2 3 4 5

Item Description

1 Slot for a cable-type security lock 2 Fax port (f and z models only) 3 Local area network (LAN) Ethernet (RJ-45) network port 4 USB port for connecting external USB devices (this port might be covered) NOTE: For easy-access USB printing, use the USB port on the top cover. 5 Hi-Speed USB 2.0 printing port Diagrams: Locations of major assemblies Use the figures in this section to identify and locate major components and assemblies in the printer. Main assemblies (printer base) Review the following information about the main assemblies (printer base). 168 Chapter 4 Solve problems

Figure 4-69 Main assemblies 2 1 3 4

Item Description

1 Fuser

2 Registration assembly

3 Feed drive assembly

4 Laser/scanner assembly Rollers and pads (printer base) Review the following information about the rollers and pads (printer base). Rollers and pads (printer base) 169

Figure 4-70 Rollers and pads (printer base) 2 3 1 4 5

Item Description

1 Transfer roller

2 Tray 1 separation pad assembly 3 Tray 2 separation roller assembly 4 Tray 1 pickup roller 5 Tray 2 pickup roller assembly

Motor and fans

Review the following information about the motor and fans. 170 Chapter 4 Solve problems

Figure 4-71 Motor and fans 2 3 1

Item Description

1 Fuser motor (M1) 2 Sub fan (FM2) 3 Main fan (FM1) Printed circuit assemblies (PCAs; printer base) Review the following information about the PCAs (printer base). Printed circuit assemblies (PCAs; printer base) 171

Figure 4-72 PCAs (printer base) 2 3 1 4

Item Description

1 Low-voltage power supply (LVPS) 2 High-voltage power supply (HVPS) 3 Fuser power supply (FPS)

4 DC controller PCA

Diagrams: General timing chart Review the following information about the general timing chart. Figure 4-73 General timing chart Diagrams:

General circuit diagrams Review the following information about the general circuit diagrams. 172 Chapter 4 Solve problems

Figure 4-74 General circuit diagram for the printer base (1 of 2) A 6 5 4 3 2 1 DC B CL SL CL

M M TP

M SL SL M SL M 21 1 2 3 4 J407

PS4 J406

M4 SL4 2 1

J660DH 1 2

J660L

12 J660D

1 2 21 SW4

SOLD4201

SOLD4202

2 1 LED1 21 SW1 PS11 PS2 PS12 PS1a PS2b J7014 J7015 J7013 J7012 J7011 123 J700 J4113 J4111 J4112 J4022 J4021 J4023 J4121 J4122 J4123 12 J860 21 SW5 17 16 15 14 13 12 11 10 9 8 7 6 5 3 2 4 1 SGND SGND SGND

+3.3VA

/VDO1

ENGTYPE

/VCRESET

WAKEVC

WAKEENG

CLEI CLEO /BDO SGND SGND VDO1 /VDO2 VDO2 1 4 2 3 5 6 7 8 9 10 11 12 13 14 15 16 17

LIFTERMT(W)

+24VD(R)

CSTSL +24VD

SGND(V)

CSTSNS(V)

PWSNSAND(GN)

+3.3VB(W)

SGND(R)

TOPSNS(GY)

REFEEDSNS(V)

TOPSNS(GN)

SGND(W)

+3.3VB(PK)

+3.3VB(Y)

CSTPLVLSNS(W)

SGND(GN)

SGND(GN)

PREREGSNS(W)

+3.3VB(BK)

SGND

REARSNS

1 2 3 J157 1 2 3 J156 1 2 3 4 5 J151 1 2 J158 1 2 3 4

J144 17

J163 1 2 J159 17 J300 3456 12 J302 12 TP1

2 1 J293

2 1 H1

2 1 J290

2 1 J291

J4201J4202J4203J4204

12 TH1 Inlet 123

SOLD70SOLD71

PGNDPGNDPGND+5VC+5VC+5VCDCLDCH

AC-HAC-NAC-HAC-N

SGND(BR)FSROUTSNS(GN)NFCS(BL)+3.3VB(GY)MAINTHGND

J243J244J245J246J247J248

J205J206

J201J202J203J204

2 31 J211 1 2 3456 J216

J254J253J252J251

J241 12345678 1234 J410 PWRSWSGND+5VAPWRLEDPWRSVE1INHIBIT24VEON24VAON+3.3VASGND+5VA5VCON 12 11 10982 3 4
5 61 7 J131 12 J143 2 34 1 J135

PGNDPGND+24VA+24VA

4 5 6 3 1 2 J132

FSRDZEROX+24VDSGND+3.3VARLYD LVT

J405 123456

MAINTHFSROUTSNSNFCS

+3.3VB

SGNDFULLSNS

5 6 4 1 2 3 J152 6 7 5 1 2 3 4 4 5 6 7 2 3 1 1234567

J602LH

J602L

J602DH

J602D 2 1

1 2 CL1 J663

/BLMACC/BLMDECBLMFG+24VD+24VDPGNDPGND

1 2 3 4 5 6 7 J712 7 6532 4 1 M1 1 2 3 4 12 3 4 J711 M2

PMOUTB/PMOUTB/PMOUTAPMOUTA

FM2 1 3 2 123 1 2 3

J661DH

J661D J661L 12 SL3 12 SL2 123 12 1234 5

J141 J147 J171 J146

1234567 OPTCL

+3.3VB

+24VD

OPTSCL

OPTSDA

SGND

OPTCSTSNS

1 234567 J153 +24VD

REFEEDCL

12 J154 PGND

LVPSFANON

LVPSFANLOCK

DUPSWBKSL

+24VD MPTSL +24VD

+24VA+24VBSNSSGNDMAINFANONMAINFANLOCK

AIRTEMP

24VCON/TGOFFTGSDATATGRDATATRPWMTRPCLKTRNCLKTRCRNTDEVACPWMDEVACCLKDEVDCSNSPRIACCLKPRIACPWMPRI
DCPWMNRMPTPSNSCSTPSNS+3.3VH 2 1 13121110 9876543 2423222120191817161514
123456789101112131415161718192021222324 24 J133 14 J161

+24VD/SCNACC/SCNDECPCGND

1234 J162 /LDPWM/BDI+3.3VBSGND SGND/VD02VD02SGNDVDO1/VDO1SGNDLDCTRL2LDCTRL1LDCTRL0 87654321 1413121110 9
1234567891011121314 1234 2 3 4 1 J802 M3 14 J851 +24VB

VIN(R)

VSS(W)

+24VA 12 J376

J343J344

+3.3VB(V)

MAINFANON

GND

MAINFANLOCK

MPTPSNS(W)

SGND(GN)

CSTPSNS(W)

SGND(GN)

+3.3VB(R)

123 J374 123 J375 123 J372 24 J341

FT2FT1

2 1 SW2

1 2

J777TAG

FM1 3 2 1

PS3PS1

J4033J4032J4031J4013J4012J4011

2 1 M5 PS461

PS432PS460

J4613J4612J4611J4323J4322J4321J4603J4602J4601

12 1 2 J651D

J651DH

J651L 12 SL6 2 1 1 2 3 4 5 6 7 J604

J604DH

1 2 J650 CL1

CSTSNS(BL)

SGND(BL)

12

+3.3VB(Y)

SGND(BR)

OPTCSTPSNS(GN)

+24VD(BR)

LIFTMT(W)

123 12

PATHSNS(W)

SGND(GN)

+3.3VB(BR)

123

PLEVELSNS(GN)

+3.3VB(V)

SGND(BR)

PICKSL

+24VD 12 123

OPT_CL

+24VD 12

OPTCLOPTSCLOPTSDAOPTCSTSNSSGND+3.3VB+24VD

SGND

+3.3VB

+24VD

OPTCSTSNS

OPTSCL

OPTSDA

OPTCL 1234567 J610

J620 J621 J624J623 J625 J622 J612J611

Formatter

Low-voltage power supply PCA

DC controller

Laser/scanner ass'y

Laser PCA High-voltage power supply PCA

Paper feeder controller PCA Fuser power supply PCA

SOLD6121SOLD6122

2 1 SW3

PS13 21 FU2 21 FU1 Diagrams: General circuit diagrams 173

Figure 4-75 General circuit diagram for the printer base (2 of 2) A 6 5 4 3 2 1 DC B 11 12 13 14 15 10 4 1 2 3 5 6 7 8 99 8 7 6 5 3 2 1 4 15 14 13 12 11 10 15 J4 40 J15 +5VC 1 2 982 3 4 5 61 7 J1 J3 40 J2 17 1816151413121181 2 3 4 5 6 7 9 10 5 J8 5 J6 81 2 3 4 5 6 71 2 3 4 5

J1812S

J1104 J1102

19 J11 17 1816151413121181 2 3 4 5 6 7 9 10 81 2 3 4 5 6 7 9 10 19 J10 J1815 3 4 5 61 2 J1816 PGND PGND +5VC +5VC PGND +24VE

INHIBIT

+24VE +24VA PGND PGND +24VA 8 12345672 3 4 51

J1101 J301

3.3V(V)

SCL(GY)

GND(BL)

SDA(W)

GND(GN)

4 J9 5

J7 J4305

J4304 J4303 J4302 J4301 81 2 3 4 5 6 7 J1103 8 123456781 2 3 4 5 6 7

J1100 J242

Image scanner

4.3 STP

LCD Module

Wireless PCA

USB HIP Memory PCA

Fomatter Fomatter

NFC PCA

Contorl panelLCD PCA

Low-voltagepower supply PCA

Contorl panel SW PCA

Low-voltagepower supply PCA

J1812D

<M506 MF model>

<M506>

<Yellowtail Pro model> <Bluefin>

<M506>

Internal test and information pages

Review the following information about the internal test and information pages. 174 Chapter 4 Solve problems

Print a configuration page

Use the following procedures to print a configuration page. NOTE: Depending on the model, up to three pages print when printing a configuration page. In addition to the main configuration page, the HP embedded Jetdirect configuration and the wireless pages print. Print the configuration page from a LCD control panel (M501) Use the following procedure to print the configuration page from a LCD control panel on an M501 printer. 1. On the printer control panel, press the OK button. 2. Open the Reports menu. 3. Select Configuration Report. Figure 4-76 Configuration report (M501) 5 6 1 2 3 4 7 8

Item Description Item Description

1 Product information 5 Product settings Print a configuration page 175

Item Description Item Description

2 Memory 6 Installed personalities and options 3 Paper settings 7 Print 4 Impressions 8 HP Web services Print the configuration page from a LCD control panel (M506/E50045/M507/E50145) Use the following procedure to print the configuration page from a LCD control panel on an M506/ E50045/M507/E50145 printer. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Open the following menus: ● Reports ● Configuration/Status Pages 3. Use the down arrow button to scroll to Configuration Page, and then press the OK button to select it. 4. Use the up arrow button to scroll to Print, and then press the OK button to print the pages. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Reports, and then press the OK button. 2. Use the down arrow button to scroll to Configuration/Status Pages, and then press the OK button. 3. Use the down arrow button to scroll to Configuration Page, and then press the OK button. 4. Use the down arrow to scroll down to Cancel, and then use the right arrow to highlight the print icon . Select OK to print the pages. Print the configuration page from a touchscreen control panel (M527/E52545/M528/E52645) Use the following procedure to print the configuration page from a touchscreen control panel on an

M527/E52545/M528/E52645 printer. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Reports button (FutureSmart 4). 2. Open the following menus: ● Reports (FutureSmart 3) ● Configuration/Status Pages 3. Touch Configuration Page to select it. 4. Touch the Print button to print the pages. 176 Chapter 4 Solve problems

Figure 4-77 Configuration page (all except M501) 5 6 1 2 3 4 7 HP LaserJet M506

Item Description Item Description

1 Device information 5 Event log 2 Installed personalities and options 6 Security 3 HP Web services 7 Paper trays and options

4 Memory

HP embedded Jetdirect page (all except M501) The second configuration page is the HP embedded Jetdirect page, which contains the following information: Always make sure the status line under the general information line indicates "HP embedded Jetdirect page." HP embedded Jetdirect page (all except M501) 177

Figure 4-78 HP embedded Jetdirect page (all except M501) 5 6 1 2 3 4 HP LaserJet M506

Item Description

1 General Information indicates the printer status, model number, hardware firmware version, port select, port configuration, auto negotiation, manufacturing identification, and manufactured date.

4 Security Settings information

3 Network Statistics indicates the total packets received, unicast packets received, bad packets received, framing errors received, total packets transmitted, unsendable packets, transmit collisions, and transmit late collisions. 4 TCP/IP information, including the IP address 5 IPv4 information 6 IPv6 information Wireless page (all except M501) The third configuration page is the wireless page, which contains the following information: NOTE: Wireless models only. 178 Chapter 4 Solve problems

Always make sure the status line under the general information line indicates "I/O Card Ready." Figure 4-79 Wireless page (all except M501) 5 6 1 2 3 4 8 7 HP LaserJet M506

Item Description

1 General Information indicates the printer status, model number, hardware firmware version, port select, port configuration, auto negotiation, manufacturing identification, and manufactured date. 2 Wireless Direct indicates the wireless function status. 3 Connected Clients indicates the MAC or IP address of connected clients.

4 Security Settings information

5 Network Statistics indicates the total packets received, unicast packets received, bad packets received, framing errors received, total packets transmitted, unsendable packets, transmit collisions, and transmit late collisions. 6 TCP/IP information, including the IP address. 7 IPv4 information. 8 IPv6 information. Wireless page (all except M501) 179

Finding important information on the configuration pages Certain information, such as the firmware date codes, the IP address, and the email gateways, is especially helpful while servicing the printer. This information is on the various configuration pages. Table 4-17 Important information on the configuration pages Type of information Specific information Configuration page

Firmware date codes

When performing a remote firmware

upgrade procedure, all of these firmware components are upgraded. DC controller (all except M501) NOTE: For the M501 printer, the DC controller version is listed as the controller number under product information on the configuration page. Look on the main configuration page, under "Device Information."

Firmware date codes

When performing a remote firmware

upgrade procedure, all of these firmware components are upgraded.

Firmware datecode

NOTE: For the M501 printer, open the main menus, select the Service menu, and then select Firmware Datecode. Look on the main configuration page, under "Product Information" (M501) or "Device Information" (all except M501).

Firmware date codes

When performing a remote firmware

upgrade procedure, all of these firmware components are upgraded.

HP embedded Jetdirect firmware

version (all except M501)

Look on the HP embedded Jetdirect

page, under "General Information."

Accessories and internal storage

All optional devices that are installed on

the printer should be listed on the main configuration page. In addition, separate pages print for the optional paper handling devices and the fax accessory (all except M501). These pages list more-detailed information for those devices. Embedded HP Jetdirect (all except M501) Look on the main configuration page, under "Installed Personalities and Options." Shows model and ID.

Accessories and internal storage

All optional devices that are installed on

the printer should be listed on the main configuration page. In addition, separate pages print for the optional paper handling devices and the fax accessory (all except M501). These pages list more-detailed information for those devices. Total RAM Look on the main configuration page, under "Memory." Additional 550-sheet feeders Additional 550-sheet feeders Look on the main configuration page, under "Paper Trays and Options" (all except M501).

Engine cycles and event logs

Total page counts and maintenance kit

counts are important for ongoing printer maintenance.

The configuration page lists only the three

most recent errors. To see a list of the 50 most recent errors, print an event log from the Diagnostics menu. Engine cycles Look on the main configuration page, under "Impressions" (M501) or "Device Information" (all except M501). Event-log information (all except M501) Event-log information Look on the main configuration page, under "Event Log." 180 Chapter 4 Solve problems

Table 4-17 Important information on the configuration pages (continued) Type of information Specific information Configuration page Error report information (M501) Error-log information Open the 2ndary Service menu, select Service Reports, and then select Error Report. Press the OK button to print the report. NOTE: This report can also be printed using the HP Embedded Web Server (EWS). Print a fuser test page (all except M501) Use the fuser test page to evaluate problems with fuser print quality. Print a fuser test page from a touchscreen control panel Use the following procedure to print a fuser test page from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Troubleshooting ● Print Quality Pages ● Fuser Test Page 3. Touch the Print button.

Print a fuser test page from a LCD control panel

Use the following procedure to print a fuser test page from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to Troubleshooting, and then press the OK button. 3. If necessary, use the down arrow button to scroll to Print Quality Pages, and then press the OK button to start the calibration process. 4. If necessary, use the down arrow button to scroll to Fuser Test Page, and then press the OK button to print the page. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then select OK. 2. Navigate to Maintenance, and then select OK. 3. Navigate to Troubleshooting, and then select OK. 4. Use the down arrow button to scroll to Print Quality Pages, and then press the OK button. Print a fuser test page (all except M501) 181 5. Use the down arrow button to scroll to Fuser Test Page, and then press the OK button. 6. Use the down arrow to scroll down to Cancel, and then use the right arrow to highlight the print icon. Select OK to print the pages. Figure 4-80 Fuser test page (all except M501) Advanced configuration with HP Embedded Web Server (EWS) and HP Device Toolbox (Windows) (M501) Use the HP Embedded Web Server to manage printing functions from your computer instead of the printer control panel. ● View printer status information ● Determine the remaining life for all supplies and order new ones ● View and change tray configurations (paper types and sizes) ● View and print internal pages ● View and change network configuration 182 Chapter 4 Solve problems

The HP Embedded Web Server works when the printer is connected to an IP-based network. The HP Embedded Web Server does not support IPX-based printer connections. You do not have to have Internet access to open and use the HP Embedded Web Server. When the

printer is connected to the network, the HP Embedded Web Server is automatically available. NOTE: HP Device Toolbox is available only if you performed a full installation when you installed the printer. Depending on how the printer is connected, some features might not be available. NOTE: The HP Embedded Web Server is not accessible beyond the network firewall. Open the HP Embedded Web Server (EWS) from the Start menu Use the following procedure to open the HP Embedded Web Server (EWS) from the Start menu. 1. Click the Start button, and then click the Programs item. 2. Click your HP printer group, and then click the HP Device Toolbox item. Open the HP Embedded Web Server (EWS) from a Web browser Use the following procedure to open the HP Embedded Web Server (EWS) from a Web browser. 1. 2-line control panels: On the printer control panel, press the OK button. Open the Network Setup menu, and then select Show IP Address to display the IP address or host name. Touchscreen control panels: From the Home screen on the printer control panel, touch the Connection Information button, and then touch the Network Connected button or the Network Wi-Fi ON button to display the IP address or host name. 2. Open a Web browser, and in the address line, type the IP address or host name exactly as it displays on the printer control panel. Press the Enter key on the computer keyboard. The EWS opens. NOTE: If the Web browser displays a There is a problem with this website's security certificate message when attempting to open the EWS, click Continue to this website (not recommended). Choosing Continue to this website (not recommended) will not harm the computer while navigating within the EWS for the HP printer. Open the HP Embedded Web Server (EWS) from the Start menu 183

Tab or section Description

Home tab Provides printer, status, and configuration information. ● Device Status: Shows the printer status and shows the approximate percent life remaining of HP supplies. ● Supplies Status: Shows the approximate percent life remaining of HP supplies. Actual supply life remaining can vary. Consider having a replacement supply available to install when print quality is no longer acceptable. The supply does not need to be replaced unless the print quality is no longer acceptable. ● Device Configuration: Shows the information found on the printer configuration page. ● Network Summary: Shows the information found on the printer network configuration page. ● Reports: Print the configuration and supplies status pages that the printer generates. ● Event Log: Shows a list of all printer events and errors.

System tab

Provides the ability to configure the

printer from your computer. ● Device Information: Provides basic printer and company information. ● Paper Setup: Change the default paper-handling settings for the printer. ● Print Quality: Change the default print-quality settings for the printer. ● EcoSMART Console: Change the default times for entering Sleep mode or Auto Power Down mode. Configure which events cause the printer to wake. ● Paper Types: Configure print modes that correspond to the paper types that the printer accepts. ● System Setup: Change the system defaults for the printer. ● Service: Perform the cleaning procedure on the printer. ● Save and Restore: Save the current settings for the printer to a file on the computer. Use this file to load the same settings onto another printer or to restore these settings to this printer at a later time. ● Administration: Set or change the printer password. Enable or disable printer features. NOTE: The System tab can be password-protected. If this printer is on a network, always consult with the administrator before changing settings on this tab.

Print tab

Provides the ability to change

default print settings from your computer. ● Printing: Change the default print settings, such as number of copies and paper orientation. These are the same options that are available on the control panel. ● PCL5c: View and change the PCL5c settings. ● PostScript: Turn off or on the Print PS Errors feature. Fax tab (Fax models only) ● Receive Options: Configure how the printer handles incoming faxes. ● Phone Book: Add or delete entries in the fax phone book. ● Junk Fax List: Set fax numbers to block from sending faxes to the printer. ● Fax Activity Log: Review recent fax activity for the printer. 184 Chapter 4 Solve problems

Tab or section Description

Scan tab (MFP models only) Configure the Scan to Network Folder and Scan to E-mail features. ● Network Folder Setup: Configure folders on the network to which the printer can save a scanned file. ● Scan to E-mail Setup: Begin the process to set up the Scan to E-mail feature. ● Outgoing E-mail Profiles: Set an email address that will appear as the "from" address for all emails sent from the printer. Configure the SMTP Server information. ● E-mail Address Book: Add or delete entries in the email address book. ● E-mail Options: Configure a default subject line and body text. Configure default scan settings for emails.

Networking tab

(Network-connected printers only)

Provides the ability to change

network settings from your computer. Network administrators can use this tab to control network-related settings for the printer when it is connected to an IP-based network. It also allows the network administrator to set up Wireless Direct functionality. This tab does not appear if the printer is directly connected to a computer. HP Web Services tab Use this tab to set up and use various Web tools with the printer. Control panel menus (M501) Review the following information about the control panel menus.

Setup menu

To open this menu, either press the OK button (2-line control panels) or touch the Setup button (touchscreen control panels). The following sub menus are available: HP Web Services menu (M501) Review the following information about the HP Web Services menu for the M501 printer. Table 4-18 HP Web Services menu (M501)

Menu item Description

Enable Web Services Use Enable Web Services to set up Web Services on the printer. NOTE: You must be connected to a network to enable HP Web Services. Proxy Settings The Proxy Settings sub-menu includes the following: ● Proxy Server ● Proxy Port ● Username ● Password Reports menu (M501) Review the following information about the Reports menu for the M501 printer. Control panel menus (M501) 185

Table 4-19 Reports menu (M501)

Menu item Description

Demo Page Prints a page that demonstrates print quality. Menu Structure Prints a control-panel menu layout map. Configuration Report Prints a list of the printer settings. Supplies Status Prints the toner cartridge status. Includes the following information: ● Approximate pages remaining ● Supply level ● Serial number ● Number of pages printed ● First install date ● Last used date Network Summary Displays status for: ● Network hardware configuration ● Enabled features ● TCP/IP and SNMP information ● Network statistics Usage Page Displays the number of pages printed by the printer. (Specific items reported are model dependent.) PCL Font List Prints a list of all installed PCL 5 fonts. PS Font List Prints a list of all installed PS fonts. PCL6 Font List Prints a list of all installed PCL 6 fonts. Service Page Prints the service page. The service page includes information about supported paper types, and other settings that are not included on the configuration page. It also includes the event log. Print Quality Page Prints a page that helps solve problems with print quality. Default Info Page Prints a page that shows the default settings for the LaserJet Update feature. System Setup menu (M501) In the following table, items that have an asterisk (*) indicate the factory default setting. Table 4-20 System Setup menu (M501)

Menu item Sub-menu item Sub-menu item Description

Language (List of available control-panel display languages.)

Sets the language in which the control panel

displays messages and printer reports. 186 Chapter 4 Solve problems

Table 4-20 System Setup menu (M501) (continued)

Menu item Sub-menu item Sub-menu item Description

Paper Setup Default Paper

Size Letter A4 Legal

Sets the size for printing internal reports or any

print job that does not specify a size. NOTE: The default setting is determined by the choice of location during the initial printer setup.

Default Paper

Type

Lists available

paper types.

Sets the type for printing internal reports or any

print job that does not specify a type. Tray 1 Paper Type

Paper Size

Configures the size and type for the tray. Tray 2 Paper Type

Paper Size

Configures the size and type for the tray. Tray 3 Paper Type

Paper Size

Configures the size and type for the tray.

Paper Out

Action Wait forever* Cancel Override

Determines how the printer reacts when a

print job requires a paper size or type that is unavailable or when the tray is empty. ● Wait forever: The printer waits until you load the correct paper. ● Override: Print on a different size paper after a specified delay. ● Cancel: Automatically cancel the print job after a specified delay. ● If you select either the Override or Cancel item, the control panel prompts you for the number of seconds to delay.

Print Quality Adjust

Alignment

Print Test Page Shifts the margin alignment to center the image on the page from top to bottom, and left to right. You can also align the image that is printed on the front with the image that is printed on the back. Allows for one-sided and two-sided printing alignment. Print Test Page: Prints a test page that shows the current registration settings. System Setup menu (M501) 187
Table 4-20 System Setup menu (M501) (continued)

Menu item Sub-menu item Sub-menu item Description

Print Quality

(continued) Adjust

Alignment

(continued) Adjust Tray 1 ● X1 Shift: Registration of the image on the paper from side to side, as the paper lies in the tray. For duplex models, this side is the second side (back) of the paper. ● X2 Shift: Registration of the image on the paper from side to side, as the paper lies in the tray, for the first side (front) of a duplexed page. This item appears only on duplex models. NOTE: Set X1 Shift first. ● Y Shift: Registration of the image on the paper from top to bottom as the paper lies in the tray. Adjust Tray 2 ● X1 Shift: Registration of the image on the paper from side to side, as the paper lies in the tray. For duplex models, this side is the second side (back) of the paper. ● X2 Shift: Registration of the image on the paper from side to side, as the paper lies in the tray, for the first side (front) of a duplexed page. This item appears only on duplex models. NOTE: Set X1 Shift first. ● Y Shift: Registration of the image on the paper from top to bottom as the paper lies in the tray. 188
Chapter 4 Solve problems
Table 4-20 System Setup menu (M501) (continued)

Menu item Sub-menu item Sub-menu item Description

Energy Settings Sleep/Auto Off After 5 minutes*

1 Minute

1 Hour

30 Minutes

15 Minutes

Specifies the amount of idle time before the

printer enters sleep mode.

Shut Down

After Never

30 Minutes

1 Hour 2 Hours 4 Hours 8 Hours 24 Hours

Set the amount of elapsed time before the

printer turns itself off.

Delay Shut

Down No Delay

When Ports

Are Active

Select whether or not the printer delays

shutting down after the user presses the power button. No Delay: The printer shuts down immediately. When Ports Are Active: The printer waits until there is no I/O port activity before shutting down. System Setup menu (M501) 189

Table 4-20 System Setup menu (M501) (continued)

Menu item	Sub-menu item	Sub-menu item	Description
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Supply Settings			
-----------------	--	--	--

Black Cartridge Very Low

Setting Stop: The printer stops printing until you replace the print cartridge. Prompt*: The printer stops printing and prompts you to replace the print cartridge. You can acknowledge the prompt and continue printing. Continue: The printer alerts you that the print cartridge is very low, but it continues printing. Low Threshold Enter a percentage for the low threshold setting.

Store Usage

Data Select where to store the printer's usage data, either on the supplies or not on the supplies.

Cartridge

Policy Off

Authorized HP

Use the Cartridge Policy feature to allow only

genuine HP cartridges to be used with this printer. When someone attempts to install a cartridge that is not a genuine HP cartridge, the printer control panel displays a message informing that the cartridge is unauthorized, and it displays information explaining how to proceed.

Cartridge

Protection

Protect

Cartridges

Use the Cartridge Protection feature to

permanently associate toner cartridges with a specific printer so they cannot be used in other printers. When someone attempts to transfer a protected cartridge from the original printer into another printer, that printer will not print.

The printer control panel displays a message

informing that the cartridge is protected, and it displays information explaining how to proceed. After selecting Protect Cartridges, when the printer prompts to confirm, select Continue to enable the feature. NOTE: After enabling cartridge protection for the printer, all subsequent toner cartridges installed in the printer are automatically and permanently protected. To avoid protecting a new cartridge, disable the feature before installing the new cartridge. To disable the feature, select Cancel rather than Continue in this step. 190 Chapter 4 Solve problems

Table 4-20 System Setup menu (M501) (continued)

Menu item	Sub-menu item	Sub-menu item	Description
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Administration Product

Security On Off Set the printer-security feature. If you select the On setting, you must set a password.

Manage Stored

Quick Copy The number of copies sent are printed

immediately, and then that number of copies is printed anytime the stored job is retrieved. ● Job type: temporary ● Job deleted after printing: no ● PIN protection: no Proof and Hold The first copy is printed immediately. Additional copies are printed by using the printer control panel. For example, ten copies are sent to the printer. One copy prints immediately. Nine copies are printed anytime the stored job is retrieved. The job is deleted after all copies are printed. ● Job type: temporary ● Job deleted after printing: yes ● PIN protection: no Personal Job The job is stored in the printer memory until it is printed, and then the job is automatically deleted. ● Job type: temporary ● Job deleted after printing: yes ● PIN protection: yes (optional) Stored Job The job is stored in the printer memory until it is printed, and then the job is manually deleted, or deleted after a predetermined retention time period. ● Job type: standard ● Job deleted after printing: no ● PIN protection: yes (optional)

Inactivity

Timer

Set the amount of time that passes before a

given menu or item will close due to printer inactivity. Courier Font Regular* Dark Set Courier font values. Service menu (M501) In the following table, items that have an asterisk (*) indicate the factory default setting. Service menu (M501) 191

Table 4-21 Service menu (M501)

Menu item Sub-menu item Sub-menu item Description

Cleaning Page Cleans the printer when specks or other marks appear on printed output. The cleaning process removes dust and excess toner from the paper path. When selected, the printer prompts you to load plain Letter or A4 paper in Tray 1. Touch the OK button to begin the cleaning process. Wait until the process completes. Discard the page that prints. USB Speed High* Full

Sets the USB speed for the USB

connection to the computer. For the printer to actually operate at high speed, it must have high speed enabled and be connected to an EHCI host controller that is also operating at high speed. This menu item does not reflect the current operating speed of the printer.

Less Paper Curl On

Off*

When printed pages are consistently

curled, this option sets the printer to a mode that reduces curl.

Archive Print On

Off*

When printing pages that will be stored

for a long time, this option sets the printer to a mode that reduces toner smearing and dusting. Firmware Datecode Displays the current firmware datecode. Restore Defaults Sets all settings to the factory default values. Signature Check Cancel if Invalid*

Prompt if Invalid

Validates HP firmware downloads.

LaserJet Update Check for Updates Now Install Now

Remind Me Later

Skip this Update

Check for printer firmware updates.

Manage Updates Allow Downgrade

Check Automatically

Prompt Before Install

Allow Updates

Manage how the printer handles firmware

updates. SMTP Comm. Report

The SMTP Communication Report

contains the SMTP communication to and from the printer from the last Scan to E mail job. Network Setup menu (M501) In the following table, items that have an asterisk (*) indicate the factory default setting. 192 Chapter 4 Solve problems

Table 4-22 Network Setup menu (M501)

Menu item Sub-menu item Description

Google Cloud Print Print Claim Sheet Press OK to send printer capabilities and model number to Google and print a sheet of registration instructions from Google. Proxy Settings Enter proxy information.

Proxy Server

Proxy Port

Username Password IPv4 Config Method DHCP BOOTP Auto IP Manual The printer automatically configures all the TCP/IP settings via DHCP, BootP or AutoIP. Manual: You can manually configure the IP address, subnet mask, and default gateway. The control panel prompts you to specify values for each address section. As each address is completed, the printer prompts for address confirmation before moving to the next one. After all three addresses are set, the network reinitializes.

Auto Crossover On

Off This item is used when you are connecting the printer directly to a personal computer using an Ethernet cable (you might have to set this to On or Off depending on the computer being used). Network Services IPv4 IPv6 Enable or disable the IPv4 and IPv6 protocols. By default, each protocol is enabled. Link Speed Automatic* 10T Full 10T Half 100TX Full 100TX Half 1000T Full Sets the link speed manually if needed. After setting the link speed, the printer automatically restarts. Security Product Security Enable printer security. If turned on, the printer prompts you to set a password. After it is set, the password will be needed to change printer settings. HTTPS Enforcement An encrypted communication and secure identification of a network web server. Only enabled if the printer is password protected. Firewall Enable, disable, or reset the printer firewall. Access Control List Enable, disable, or reset the network access control list. Reset All Security Reset the security settings to the factory-set default values. Restore Defaults Resets all network configurations to their factory defaults. Quick Forms menu (M501) Review the following information about the Quick Forms menu for the M501 printer. Quick Forms menu (M501) 193

Table 4-23 Quick Forms menu

Menu item Sub-menu item Description

Notebook Paper Narrow Rule

Wide Rule

Child Rule

Prints pages that have preprinted lines. Graph Paper 1/8 inch 5 mm Prints pages that have preprinted graph lines. Checklist 1-Column 2-Column Prints pages that have preprinted lines with check boxes.

Music Paper Portrait

Landscape

Prints pages that have preprinted lines for writing music. Function specific menus (M501) Review the following information about the function specific menus for the M501 printer. Apps (M501) Use the Apps menu to install HP Web Services applications. Jobs (M501) Use the Jobs menu to manage stored jobs. Requires a USB 2.0 storage device with at least 16 GB free space to be installed in the rear host USB port. Supplies Status (M501) Use the Supplies Status menu to print a supplies status page. Press Report. Control panel menus (all except M501) Review the following information about the control panel menus.

Reports menu

Review the following information about the Reports menu. To display: At the printer control panel, select the Reports menu. NOTE: For the M506x/M507x and M527/E52545/M528/E52645 models, use the Print button to print the report. For all other models, use the up arrow button or down arrow button to highlight the report, and then press the OK button. Then use the up arrow button to highlight Print, and then press the OK button to print the report. In the following table, asterisks (*) indicate the factory default setting. 194 Chapter 4 Solve problems

Table 4-24 Reports menu

First level	Second level	Values	Description
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Configuration/Status Pages	Settings Menu	Map	Cancel View Print
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Shows a map of the entire			
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control panel system and the selected values for each setting. Configuration/Status Pages Current Settings Page Cancel View Print			
---	--	--	--

Shows a summary of the			
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current settings for the printer. This might be helpful if you plan to make changes and need a record of the present configuration.			
---	--	--	--

Configuration/Status Pages	Configuration Page	Cancel View Print	
----------------------------	--------------------	-------------------	--

Shows the printer settings			
-----------------------------------	--	--	--

and installed accessories. Configuration/Status Pages How to Connect Page Cancel View Print			
---	--	--	--

Shows the network			
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information typically needed to connect the printer to a network. Configuration/Status Pages Supplies Status Page Cancel View Print			
---	--	--	--

Shows the approximate			
------------------------------	--	--	--

remaining life for the supplies; reports statistics on total number of pages and jobs processed, serial number, page counts, and maintenance information.			
---	--	--	--

HP provides approximations			
-----------------------------------	--	--	--

of the remaining life for the supplies as a customer convenience. The actual remaining supply levels might be different than the approximations provided. Configuration/Status Pages Usage Page Cancel View Print			
---	--	--	--

Shows a count of all paper			
-----------------------------------	--	--	--

sizes that have passed through the printer; lists whether they were simplex or duplex, and reports the page count. Configuration/Status Pages File Directory Page Cancel View Print			
---	--	--	--

Shows the file name and			
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folder name for files that are stored in the printer memory. Configuration/Status Pages Web Services Status Page Cancel View Print			
--	--	--	--

Shows the detected Web			
-------------------------------	--	--	--

Services for the printer.			
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Fax Reports			
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Fax models only			
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Fax Activity Log Cancel			
--------------------------------	--	--	--

View Print			
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Contains a list of the faxes			
-------------------------------------	--	--	--

that have been sent from or received by this printer. Reports menu 195			
--	--	--	--

Table 4-24 Reports menu (continued)

First level	Second level	Values	Description
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Fax Reports			
--------------------	--	--	--

Fax models only			
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Billing Codes Report Cancel			
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View Print			
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Provides a list of billing codes

that have been used for outgoing faxes. This report shows how many sent faxes were billed to each code.

Fax Reports

Fax models only

Blocked Fax List Cancel

View Print

A list of phone numbers that

are blocked from sending faxes to this printer.

Fax Reports

Fax models only

Speed Dial List Cancel

View Print

Shows the speed dials that

have been set up for this printer.

Fax Reports

Fax models only

Fax Call Report Cancel

View Print

A detailed report of the last

fax operation, either sent or received.

Other Pages PCL Font List Cancel

Print Prints the available PCL fonts.

Other Pages PS Font List Cancel

Print Prints the available PS fonts.

Settings menu

You can perform basic printer setup by using the Settings menu. Use the HP Embedded Web Server for more advanced printer setup. To open the HP Embedded Web Server, enter the printer IP address or host name in the address bar of a Web browser. NOTE: Where applicable, the M527/E52545/M528/E52645 printer displays a View button. General menu (all except M501) Review the following information about the Settings menu. To display: At the printer control panel, select the Settings menu, and then select the General menu. In the following table, asterisks (*) indicate the factory default setting. Table 4-25 General menu (M506/E50045/M507/E50145, M527/E52545/M528/E52645)

First level	Second level	Third level	Fourth level	Values	Description	Date/Time Settings	Date/Time Format	Date Format	DD/MMM/YYYY
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MMM/DD/YYYY

YYYY/MMM/DD

Use the Date/Time

Settings menu to

specify the date and time and to configure date/ time settings. 196 Chapter 4 Solve problems

Table 4-25 General menu (M506/E50045/M507/E50145, M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description Date/Time Settings Date/Time Format Time Format 12 hour (AM/PM) 24 hours

Select the format

that the printer uses to show the date and time, for example 12-hour format or 24-hour format. Date/Time Settings Date/Time Time Zone Select the time zone from a list. Date/Time Settings Date/Time Date Select the date from a pop-up calendar. Date/Time Settings Date/Time Time Select the time from a pop-up keypad. Date/Time Settings Date/Time Adjust for Daylight Savings

Checkbox If you are in an

area that uses daylight savings time, select the

Adjust for Daylight

Savings box.

Energy Settings

M527/E52545/

M528/E52645

printer

Sleep Schedule A list of scheduled

events displays. + (Add) Edit Delete

Use to configure

the printer to automatically wake up or go to sleep at specific times on specific days.

Using this feature

saves energy. NOTE: You must configure the date and time settings before you can use this feature.

Energy Settings

M527/E52545/

M528/E52645

printer

Sleep Schedule A list of scheduled

events displays.

Event Type Wake

Sleep

Select whether to

add or edit a Wake event or a Sleep event, and then select the time and the days for the wake or sleep event.

Energy Settings

M527/E52545/

M528/E52645

printer

Sleep Schedule A list of scheduled

events displays.

Event Time

Energy Settings

M527/E52545/

M528/E52645

printer

Sleep Schedule A list of scheduled

events displays.

Event Days Select days of the

week from a list. General menu (all except M501) 197

Table 4-25 General menu (M506/E50045/M507/E50145, M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Energy Settings

M527/E52545/

M528/E52645

printer

Sleep Timer

Settings Sleep Mode/Auto

Off After

Range: 1 to 120 minutes Default = 60 minutes

Set the number

of minutes after which the printer enters Sleep or Auto Off mode.

Use the arrow

buttons on the control panel to increase or decrease the number of minutes.

Energy Settings

M527/E52545/

M528/E52645

printer Wake/Auto On to

These Events

All Events*

Network port

Power button only

Energy Settings

M506/E50045/

M507/E50145

printer

Sleep Timer

Settings Sleep/Auto Off Timer Enabled* Disabled

Enable or disable

the printer sleep or auto off function.

Energy Settings

M506/E50045/

M507/E50145

printer

Sleep Timer

Settings Sleep Mode/Auto

Off After

Range: 1 to 120 minutes Default = 60 minutes

Set the number

of minutes after which the printer enters Sleep or Auto Off mode.

Use the arrow

buttons on the control panel to increase or decrease the number of minutes.

Display Settings Display Brightness A sliding bar

displays with the indicator set in the middle. Use the arrow keys to select the desired brightness and then select Done.

Use to specify the

intensity of the

LCD control panel

display. Display Settings System Sound On* Off

Use to enable or

disable sounds on the printer. Display Settings Language Settings Language Select from a list of languages that the printer supports.

Use to select a

different language for control panel messages and specify the default keyboard layout.

When you select a

new language, the keyboard layout automatically changes to match the factory default for the selected language. 198 Chapter 4 Solve problems

Table 4-25 General menu (M506/E50045/M507/E50145, M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Display Settings Keyboard Layout Each language has

a default keyboard layout. To change it, select from a list of layouts.

Select the default

keyboard layout that matches the language you want to use.

Display Settings Information

Screen

Show connection

information*

Hide connection

information

Use this menu

item to display or hide connection information on the Home screen. Display Settings Inactivity Timeout Range: 10-300 seconds Default = 60 seconds

Specifies the

amount of time that elapses between any activity on the control panel and when the printer resets to the default settings.

When the timeout

expires, the control panel display returns to the Home menu, and any user signed in to the printer is signed out.

Display Settings Clearable

Warnings Display during job*

Display until cleared

Use this feature

to set the period that a clearable warning displays on the control panel. If the

On setting is

selected, clearable warnings appear until the Clearable

Warnings button is

pressed. If the Job setting is selected, clearable warnings stay on the display during the job that generated the warning and disappear from the display when the next job starts. General menu (all except M501) 199

Table 4-25 General menu (M506/E50045/M507/E50145, M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Display Settings Continuable

Events Auto-continue (10 seconds)*

Touch OK to continue

Use this option

to configure the printer behavior when the printer encounters certain errors. If the Auto-continue (10 seconds) option is selected, the job will continue after 10 seconds.

If the Touch OK

to continue option is selected, the job will stop and require the user to touch the

OK button before

continuing.

Enable Device

USB Enabled Disabled*

Enables the printer

to open a file from a USB drive. Quiet Mode Off* On

Automatic transition

Range: 10-300 pages Default = 10 pages

Quiet Mode slows

the printer down to reduce the noise the printer makes when printing.

Automatic

transition allows you to set a predetermined job size of 10 to 999 pages, and print jobs will print in

Quiet Mode for all

print jobs up to the predetermined page size. 200 Chapter 4 Solve problems

Table 4-25 General menu (M506/E50045/M507/E50145, M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description Jam Recovery Automatic* Off On

This printer

provides a jam recovery feature that reprints jammed pages.

Select one of the

following options: Automatic: The printer attempts to reprint jammed pages when sufficient memory is available. This is the default setting. Off: The printer does not attempt to reprint jammed pages. Because no memory is used to store the most recent pages, performance is optimal. NOTE: When using this option, if the printer runs out of paper and the job is being printed on both sides, some pages can be lost. On: The printer always reprints jammed pages.

Additional memory

is allocated to store the last few pages printed. This might cause overall performance to suffer.

Auto Recovery Enabled

Disabled*

If Auto Recovery

is enabled and an unrecoverable error occurs in the device firmware, the device automatically turns off and turns back on to recover from the error. General menu (all except M501) 201

Table 4-25 General menu (M506/E50045/M507/E50145, M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description Enable Auto Send Disabled* Enabled

Browse to the HP

Embedded Web

Server AutoSend

configuration page for advanced set up and the HP online

Privacy Statement

Information.

Use the Enable

Auto Send menu

to enable or disable the AutoSend feature.

The AutoSend

feature enables your product to periodically send product configuration information including serial number, event logs, page usage counts and supplies status information to HP web addresses (URLs), or email addresses.

Information sent

to HP is used to improve products and services, and to monitor the product if you have a relationship with HP that provides you services such as proactive cartridge replacement, pay-per-page contracts, support agreements, or usage tracking. Hold Off Print Job Enabled* Disabled

Enable this feature

if you want to prevent print jobs from starting while a user is initiating a copy job from the control panel. Held print jobs start printing after the copy job is finished, provided that no other copy job is in the print queue. 202 Chapter 4 Solve problems

Table 4-25 General menu (M506/E50045/M507/E50145, M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Reset Factory

Settings Address Book (M527/E52545/ M528/E52645 only) Copy (M527/E52545/ M528/E52645 only) Digital Send (M527/E52545/ M528/E52645 only) E-mail (M527/E52545/ M528/E52645 only) Fax (M527/E52545/ M528/E52645 only) General

(M527/E52545/M528/

E52645 only) Print (M527/E52545/ M528/E52645 only) Security

(M527/E52545/M528/

E52645 only) Cancel (M506/

E50045/M507/E50145

only) Reset (M506/E50045/ M507/E50145 only)

Use to restore

all printer settings to their factory defaults. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) Review the following information about the Print menu for the M506/E50045/M507/E50145) and the Copy/Print menu for the M527/E52545/M528/E52645 printers. To display: At the printer control panel, select the Settings menu, and then select the Print menu (M506/ E50045/M507/E50145) or the Copy/Print menu (M527/E52545/M528/E52645). In the following table, asterisks (*) indicate the factory default setting. Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) First level Second level Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only

Image preview Make optional

Require preview

Disable preview

Use this menu

to configure how Copy behaves. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 203
Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level
Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only Copies 1-9999 Default = 1

Configure the

default options for copy jobs. If the user does not specify the job options when creating the job, the default options are used.

Copy Settings

M527/E52545/

M528/E52645 only Sides Original Sides 1-sided 2-sided

Flip pages up

Use to indicate

whether the original document is printed on one or both sides. For example, select Original Sides 1- sided and Output Sides 2-sided when the original is printed on one side, but you want to make two-sided copies.

Select Flip pages

up to print the pages with long edge of the in the landscape orientation.

Copy Settings

M527/E52545/

M528/E52645 only Sides Output Sides 1-sided 2-sided

Flip pages up

Use to indicate

whether the copies should be printed on one or both sides. For example, select Original Sides 1-sided and Output Sides 2- sided when the original is printed on one side, but you want to make two-sided copies.

Select Flip pages

up to print the pages with long edge of the in the landscape orientation.

Copy Settings

M527/E52545/

M528/E52645 only Watermark Watermark Type None* Text

Use to set a text

watermark. 204 Chapter 4 Solve problems
Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level
Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only

Watermark Watermark Text Draft

Confidential

Secret

Top Secret

Urgent Select a predefined text watermark.

Copy Settings

M527/E52545/

M528/E52645 only Watermark Text Font Letter Gothic*

Antique Olive

New Century

Schoolbook Roman

Garamond Antiqua

Select the font of

the text watermark.

Copy Settings

M527/E52545/

M528/E52645 only Watermark Text Size 30 point 40 point* 60 point

Select the font

point size of the text watermark.

Copy Settings

M527/E52545/

M528/E52645 only Watermark Darkness 1 - (Lighter) 2 3* 4 5 - (Darker)

Select the

darkness of the text watermark.

Copy Settings

M527/E52545/

M528/E52645 only

Stamps Stamp Content User-defined value

None*

IP address

User name

Product information

Page number

Date and time

Use to set a stamp

in any or all the following locations in the document: ● Top Left ● Top Center ● Top Right ● Bottom Left ● Bottom Center ● Bottom Right

Copy Settings

M527/E52545/

M528/E52645 only Stamps Text Font Letter Gothic*

Antique Olive

New Century

Schoolbook Roman

Garamond Antiqua

Select the font of

the stamp. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 205

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only Stamps Text Size 8 point 12 point* 20 point

Select the font

point size of the stamp.

Copy Settings

M527/E52545/

M528/E52645 only

Stamps White

background Disabled* Enabled

Select the check

box to enable a white background for the stamp.

Copy Settings

M527/E52545/

M528/E52645 only Scan Mode Standard document*

Prompt for additional

pages Book 2-sided ID

Use to indicate

the type of original document to be scanned.

Select Prompt for

additional pages to copy or scan an original document that has more pages than the document feeder can accommodate at one time, or to scan originals of different sizes that cannot be scanned together, and then combine these separate scan jobs into a single job.

Copy Settings

M527/E52545/

M528/E52645 only Reduce/Enlarge Automatic*

Include margins

Manual Default=100

Scale the size of

the document up or down.

To reduce the

image, select a scaling percentage that is less than 100. To enlarge the image, select a scaling percentage that is greater than 100.

When Include

margins is enabled, the printer reduces the image slightly to fit the entire scanned image within the printable area on the page. 206 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only Original Size Automatically detect* Letter (8.5x11) Mixed Letter/Legal Legal (8.5x14) Executive (7.25x10.5) Statement (5.5x8.5) Oficio (8.5x13) 4x6 5x7 5x8 A4 (210x297 mm) A5 (148x210 mm) A6 (105x148 mm) RA4 (215x305 mm) B5 (182x257 mm) B6 (128x182 mm) 10x15cm 16K (195x270 mm) 16K (184x260 mm) 16K (197x273 mm)

DPostcard JIS

(148x200 mm) Oficio (216x340 mm)

Specify the size

of the original document. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 207

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only Paper Selection Paper Size Match original size* Letter (8.5x11) Legal (8.5x14) Executive (7.25x10.5) Statement (5.5x8.5) Oficio (8.5x13) 3x5 4x6 5x7 5x8 A4 (210x297 mm) A5 (148x210 mm) A6 (105x148 mm) RA4 (215x305 mm) B5 (182x257 mm) B6 (128x182 mm) 10x15cm 16K (195x270 mm) 16K (184x260 mm) 16K (197x273 mm) Postcard JIS (100x148 mm)

DPostcard JIS

(148x200 mm) Envelope #9 Envelope #10

Envelope Monarch

Envelope #10 Envelope B5 Envelope C5 Envelope C6

Envelope DL

Custom Oficio (216x340 mm)

Select the size of

paper to use when printing or making copies. 208 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only

Paper Selection Paper Type Any Type

Plain* Light 60-74g Intermediate 85-95g Mid-Weight 96-110g Heavy 111-130g Extra Heavy 131-175g Cardstock 176-220g

Mono Transparency

Labels

Letterhead

Envelope

Preprinted

Prepunched

Colored Bond Recycled Rough

HP EcoFFICIENT

Light Bond

Select the type of

paper to use when printing or making copies.

Copy Settings

M527/E52545/

M528/E52645 only Paper Selection Paper Tray Automatically detect*

Manually Feed

Tray 1 Tray 2

Select which tray

to use when printing or making copies.

Copy Settings

M527/E52545/

M528/E52645 only

Booklet Booklet Format Enabled

Disabled*

Select to have

Booklet Format off

or on.

Copy Settings

M527/E52545/

M528/E52645 only

Booklet Booklet Format Borders on each

page Enabled Disabled*

Select to have

borders printed on the page.

Copy Settings

M527/E52545/

M528/E52645 only Content

Orientation

Portrait*

Landscape

Specify the way

the content of the original document is placed on the page. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 209

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only Pages per Sheet One* Two Four

Select how many

pages to print on one sheet.

Copy Settings

M527/E52545/

M528/E52645 only Page Order Right, then down Down, then right

Select to print the

pages in rows or columns.

Copy Settings

M527/E52545/

M528/E52645 only

Add page borders Enabled

Disabled*

Select the Add

page borders checkbox to add borders to the pages.

Copy Settings

M527/E52545/

M528/E52645 only Image Adjustment Darkness 1 - (Lighter) 2 3 4 5* 6 7 8 9 - (Darker)

Adjust to increase

or decrease the amount of white and black in the colors.

Copy Settings

M527/E52545/

M528/E52645 only Image Adjustment Contrast 1 - (Less) 2 3 4 5* 6 7 8 9 - (More)

Adjust to increase

or decrease the difference between the lightest and darkest color on the page. 210 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only

Image Adjustment Background

Cleanup 1 - (Normal) 2 3* 4 5* 6 7 8 9 - (Cleaner)

Adjust if you

are having trouble copying a faint image.

Copy Settings

M527/E52545/

M528/E52645 only Image Adjustment Sharpness 1 - (Less) 2 3* 4 5 - (More)

Adjust to sharpen

or soften the image.

Copy Settings

M527/E52545/

M528/E52645 only Optimize Text/ Picture Text Mixed*

Printed picture

Photograph

Select to optimize

for the output of a particular type of content.

Copy Settings

M527/E52545/

M528/E52645 only

Edge-to-Edge Normal

(recommended)

Edge-to-Edge output

When the Edge to-Edge feature is enabled, the product minimizes margins and prints as close to the edge of the paper as possible.

Copy Settings

M527/E52545/

M528/E52645 only Erase Edges Use inches Enabled* Disabled

Use the Erase

Edges feature to

remove blemishes, such as dark borders or staple marks, by cleaning theedges of the scanned image.

Disabling Use

inches changes the measurements to millimeters. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 211

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Copy Settings

M527/E52545/

M528/E52645 only

Erase Edges Front Side Specify a different

width for each edge

Apply same width to

all edges*

All edges value

Default = 0.00

Sets the width of

the edge to clean.

Copy Settings

M527/E52545/

M528/E52645 only

Erase Edges Back Side Specify a different

width for each edge

Apply same width to

all edges Mirror front side*

Sets the width of

the edge to clean.

Copy Settings

M527/E52545/

M528/E52645 only Collate Collate on*

Collate off

When Collate on is

selected, each set of copied pages are assembled in the same order as the original document.

Copy Settings

M527/E52545/

M528/E52645 only

Multi-feed

Detection

Disabled Enabled*

Use this feature

to detect when multiple pages are fed simultaneously into the ADF (automatic document feeder) or sheet-feed scanner.

Copy Settings

M527/E52545/

M528/E52645 only

Enable Print from

USB Drive

Enabled Disabled*

Enables the printer

to open a file from a USB drive.

Manage Stored

Jobs Job Sort Order Job Name* Date*

This option

allows you list the jobs either alphabetically or chronologically.

Manage Stored

Jobs

Retain Temporary

Jobs

Do not retain

Personal jobs only

All temporary jobs

Sets which

temporary jobs will be retained in the event of a printer reboot. 212 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level
Third level Fourth Level Values Description

Manage Stored

Jobs

Temporary Job

Storage Limit

1-300 Default = 32

Configure global

settings for jobs that are stored in the printer memory.

The Temporary

Job Storage Limit

feature specifies the number of temporary jobs that can be stored on the printer. The maximum allowed value is 300.

Manage Stored

Jobs

Temporary Job

Storage Retention

Off 30 minutes 1 hour 4 hours 1 day 1 week 4 weeks 3 days

Configure global

settings for temporary jobs that are stored in the printer memory.

The Temporary

Stored Job

Retention feature

specifies the amount of time temporary jobs can be stored on the printer.

Manage Stored

Jobs

Standard Stored

Job Retention

Off 30 minutes 1 hour 4 hours 1 day 1 week 4 weeks 3 days

Configure global

settings for jobs that are stored in the printer memory.

The Standard

Stored Job

Retention feature

specifies the amount of time jobs can be stored on the printer.

Default Print

Options Number of Copies Range: 1-32000 Default = 1

Sets the default

number of copies for a copy job. This default applies when the

Copy function or

the Quick Copy function is initiated from the printer Home screen.

Default Print

Options

Paper Selection Paper size

Paper type

Paper tray

Select from a list of

sizes that the printer supports.

Configures the

default paper size, type and tray used for print jobs. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 213

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Default Print

Options

Default Custom

Paper Size

X Dimension Range: 3.00-8.50 inches Default = 8.5 inches

Configures the

default paper size that is used when the user selects

Custom as the

paper size for a print job.

Default Print

Options

Default Custom

Paper Size

Y Dimension Range: 5.00-14.00 inches Default = 14 inches

Configures the

default paper size that is used when the user selects

Custom as the

paper size for a print job.

Default Print

Options

Default Custom

Paper Size

Use Inches Enabled* Disabled

Default Print

Options Output Sides 1-sided* 2-sided

Use to indicate

whether the original document is printed on one or both sides.

Default Print

Options

Edge-to-Edge Normal

(recommended)*

Edge-to-Edge output

Use to avoid

shadows that can appear along the edges of copies when the original document is printed close to the edges.

Default Print

Options Resolution FastRes 1200* ProRes 1200

Use to select the

level of desired print quality.

Default Print

Options

Resolution Economode Enabled

Disabled*

Text is printed

using less toner.

This setting is

useful when you are printing drafts.

You can turn

on this option independently of other print quality settings. PCL and

Postscript

Settings Courier Font Regular* Dark

Select which

version of the

Courier font you

want to use. The factory default setting is Regular, which uses an average stroke width. The Dark setting can be used if a heavier

Courier font is

needed. 214 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description PCL and

Postscript

Settings Wide A4 Enabled Disabled*

Changes the

printable area of A4-size paper. If you enable this option, eighty 10- pitch characters can be printed on a single line of A4 paper. PCL and

Postscript

Settings

Print PS Errors Enabled

Disabled*

Use this feature

to select whether a PostScript (PS) error page is printed when the printer encounters a PS error. PCL and

Postscript

Settings

Print PDF Errors Enabled

Disabled*

Selects whether a

PDF error page is

printed when the printer encounters a PDF error. PCL and

Postscript

Settings Personality Automatic* PCL PS PDF

Configures the

default print language or personality for the printer. Normally you should not change the printer language. If you change the setting to a specific printer language, the printer does not automatically switch from one language to another unless specific software commands are sent to it. PCL and

Postscript

Settings

PCL Font Settings Font Source Internal

Disk resident*

Selects the font

source for the user-soft default font. The list of available options varies depending on the installed printer options. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 215

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description PCL and

Postscript

Settings PCL Font Settings Font Number Range: 0-110 Default = 0

Specifies the font

number for the user-soft default font using the source that is specified in the Font Source menu.

The printer assigns

a number to each font and lists it on the PCL font list.

The font number

displays in the Font # column of the printout. PCL and

Postscript

Settings PCL Font Settings Font Pitch Range: 0.44-99.99 Default = 10

If the Font Source

option and the Font

Number setting

indicate a contour font, then use this feature to select a default pitch (for a fixed-spaced font). PCL and

Postscript

Settings PCL PCL Settings Form Length Range: 5-128 lines Default = 60

Controls the PCL

print-command options. PCL is a set of printer commands that

HP developed to

provide access to printer features.

Use the Form

Length feature to

select the user soft -default vertical form length. PCL and

Postscript

Settings PCL PCL Settings Orientation Portrait*

Landscape

Select the

orientation that is most often used for copy or scan originals. Select the Portrait option if the short edge is at the top or select the Landscape option if the long edge is at the top. 216 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description PCL and

Postscript

Settings

PCL PCL Settings Symbol Set Select from a list of

symbol sets.

Select any one of

several available symbol sets from the control panel.

A symbol set

is a unique grouping of all the characters in a font. The factory default value for this option is PC-8. Either PC-8 or PC-850 are recommended for line-draw characters. PCL and

Postscript

Settings

PCL PCL Settings Append CR to LF Enabled

Disabled* When enabled, this option appends a carriage return to each line feed encountered in backwards compatible PCL jobs. PCL and

Postscript

Settings

PCL PCL Settings Suppress Blank

Pages Enabled Disabled*

This option is for

users who are generating their own PCL, which could include extra form feeds that would cause blank pages to be printed. When the

On option is

selected, form feeds are ignored if the page is blank. PCL and

Postscript

Settings

PCL PCL Settings Media Source

Mapping Standard* Classic

Use to select

and maintain input trays by number when you are not using the printer driver, or when the software program has no option for tray selection. The following options are available: Standard: Tray numbering is based on newer HP LaserJet models. Classic: Tray numbering is based on HP LaserJet 4 and older models. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 217

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description Print Quality General Toner Density A sliding bar appears with the indicator set in the middle between Less and More.

Lighten or darken

the print on the page by changing the toner density setting.

Print Quality General REt Disabled

Enabled*

Use this setting

to enable or disable Resolution

Enhancement

technology (RET), which produces smoother angles, curves, and edges.

Print Quality Image

Registration

Tray Tray 1 Tray 2 Tray 3 Tray 4 Tray 5

Depends upon the

number of trays installed

Specify tray to be

adjusted 218 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Print Quality Image

Registration

Front-side

Horizontal Shift

Front-side

Vertical Shift

Back-side

Horizontal Shift

Back-side

Vertical Shift

-5.00 mm to 5.00 mm Shift the margin alignment to center the image on the page from top to bottom and from left to right. You can also align the image on the front with the image printed on the back.

The direction that

is perpendicular to the way the paper passes through the printer is referred to as X. This is also known as the scan direction. X1 is the scan direction for a single-sided page or for the second side of a two-sided page. X2 is the scan direction for the first side of a two-sided page.

The direction that

the paper feeds through the printer is referred to as Y. Y1 is the feed direction for a single-sided page or for the second side of a two-sided page. Y2 is the feed direction for the first side of a two sided page.

Use the Adjust Tray

<X> menu to adjust the registration settings for each tray. Before adjusting these values, print a registration test page. It provides alignment guides in the X and

Y directions so

you can determine which adjustments are necessary. You can adjust values for X1 Shift, X2 Shift, Y1 Shift, and Y2 Shift. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 219

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Print Quality Image

Registration

Print Test Page Use the Print Test

Page option to

print a page to test the image registration. It provides alignment guides in the X and Y directions so you can determine which adjustments are necessary.

Print Quality Auto Sense

Behavior Tray 1 Sense every page

Sense first page

Sense transparency

only

Print Quality Auto Sense

Behavior

All Other Trays Sense first page

Sense transparency

only

Print Quality Adjust Paper

types

Select from a list

of paper types that the printer supports. The available options are the same for each paper type.

Print mode Select from a list of

paper types

Reset Paper Types

Changing the Print

mode setting is usually the first first thing to try to resolve print quality problems.

Problems can

include toner not sticking well to the page, a faint image of the page repeated on the same or following page, incorrect gloss level, and so on.

Print Quality Adjust Paper

types

Select from a list

of paper types that the printer supports. The available options are the same for each paper type. Resistance mode Normal* Up 1 Up 2

Use this setting

to correct print quality problems in low-humidity environments and highly resistive paper.

Use the Up options

to solve print quality problems that are related to faded images or scattered toner on certain paper types.

The Up options

raise the secondary transfer bias. 220 Chapter 4 Solve problems

Print Quality Adjust Paper

types

Select from a list

of paper types that the printer supports. The available options are the same for each paper type. Paper curl mode Normal* Reduced

Use this setting to

reduce paper curl in print jobs. Print Quality Optimize Line Detail Normal* Off

Alternate

Use this setting if

you have scattered lines in printed pages. Print Quality Optimize Moisture Control Normal*

Alternate

Print Quality Optimize Envelope control Normal

Reduced Temp

Print Quality Optimize Tray 1 Normal

Alternate

Print Quality Optimize Background Normal

Alternate

Print Quality Optimize Uniformity

Control Normal Alternate 1 Alternate 2 Alternate 3 Print Quality Optimize Cac03 Off On

Print Quality Optimize Best Normal Off

On

Print Quality Optimize Tracking Control Off

On

Print Quality Optimize Registration Normal

Alternate

Print Quality Optimize Transfer Control Normal

Alternate 1

Print Quality Optimize Moisture Control Normal

Alternate

Print Quality Optimize Reset Optimize Reset

Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 221

Print Quality Edge Control Off

Light Normal* Maximum

Manage Trays Use Requested

Tray Exclusively*

When available

Controls how the

printer handles jobs that have specified a specific input tray.

Two options are

available: Exclusively: The printer never selects a different tray when the user has indicated that a specific tray should be used, even if that tray is empty. When available:

The printer pulls

from another tray if the specified tray is empty, even though the specific tray was indicated for the job.

Manage Trays Manually Feed

Prompt Always prompt*

Prompt on mismatch

Indicate whether

a prompt should appear when the type or size for a job does not match the specified tray and the printer pulls from the multipurpose tray instead.

Two options are

available: Always: A prompt always displays before using the multipurpose tray.

Prompt on

mismatch: A prompt displays only if the size or type do not match or the tray is empty. 222 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description Manage Trays Size/Type Prompt Display*

Do not display

Controls whether

the tray configuration message displays whenever a tray is closed.

Two options are

available: Display: Shows the tray configuration message when a tray is closed. The user is able to configure the tray settings directly from this message. Do not display:

Prevents the

tray configuration message from automatically appearing. Manage Trays Use Another Tray Allow*

Do not allow

Use to turn on

or off the control panel prompt to select another tray when the specified tray is empty.

Two options are

available: Allow: When this option is selected the user is prompted to either add paper to the selected tray or to choose a different tray. This is the factory default. Do not allow: When this option is selected, the user is not given the option of selecting a different tray. The printer prompts the user to add paper to the tray that was initially selected. Print menu (M506/E50045/M507/E50145) and Copy/Print menu (M527/E52545/M528/E52645) 223

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description

Manage Trays Alternative

Letterhead Mode

Off* On

Use to load

letterhead or preprinted paper into the tray the same way for all print jobs, whether you are printing to one side of the sheet or to both sides of the sheet.

When this option is

selected, load the paper as you would for printing on both sides. See the user documentation that came with the printer for instructions about loading letterhead for printing on both sides. When this option is selected, the printer speed slows to the speed required for printing on both sides.

Manage Trays Duplex Blank

Pages Automatic* Always

Controls how the

printer handles two-sided jobs (duplexing). Two options are available: Automatic: Choose this option to skip printing blank sides during a two sided print job. The printer can print jobs faster when blank sides are skipped. Always: Choose this option to print all sides of a two sided job, even if one side is blank. This might be preferable for certain jobs that use paper types such as letterhead or prepunched paper. 224 Chapter 4 Solve problems

Table 4-26 Print menu (M506/E50045/M507/E50145) or Copy/Print menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth Level Values Description Manage Trays Override A4/ Letter Yes* No

Prints on letter-size

paper when an A4 job is sent but no A4-size paper is loaded in the printer (or to print on A4 paper when a letter-size job is sent but no letter-size paper is loaded). This option will also override A3 with ledger-size paper and ledger with A3-size paper. Scan/Digital Send Settings menu (M527/E52545/M528/E52645) Review the following information about the Scan/Digital Send Settings menu for M527/E52545/M528/E52645 printers. To display: At the printer control panel, select the Settings menu, and then select the Scan/Digital Send Settings menu. In the following table, asterisks (*) indicate the factory default setting. Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) First level Second level Third level Fourth level Values Description

Scan to Email

Settings

Scan to Network

Folder Settings

Scan to USB Drive

Settings NOTE: The same options are available for each of these features, except where noted.

E-mail Setup

NOTE: Email

Settings only

E-mail Setup

Wizard

Use to configure

settings that apply to sending documents through email or saving documents to a folder on the network or on a USB multi-drive.

The E-mail

Setup Wizard

feature configures the printer to send scanned images as email attachments. To open the printer

HP Embedded Web

Server and set

up the email notification server, enter the printer IP address into a Web browser. Scan/Digital Send Settings menu (M527/E52545/M528/E52645) 225

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Default Job

Options Image Preview Make optional*

Require preview

Disable preview

Defines the default

job options for each function. If you do not specify the job options when creating the job, the default options are used. For complete setup, go to the

HP Embedded Web

Server by typing

the IP address of the printer into a Web browser.

Use the Image

Preview feature to

scan a document and display a preview before completing the job.

Select whether

this feature is available on the printer. Make optional: The feature is optional, depending on the user who is signed in.

Require preview

Previews are

required for all users. Disable preview:

Previews are

disabled for all users. Default File Name [Untitled]* The printer is shipped with a factory default file name of [Untitled] for any scanned files that are sent or saved. Use this feature to specify a different default file name. If you are saving a file to a network folder or USB storage device and a file with the default file name already exists, a number is appended to the file name, for example, [Untitled]001. 226 Chapter 4 Solve problems

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Document File

Type

Select from a list

of file types.

PDF provides the

best overall image and text quality.

JPEG is a good

choice for most graphics. Most computers have a browser that can view .JPEG files. This file type produces one file per page.

TIFF is a standard

file format that many graphics programs support.

This file type

produces one file per page. MTIFF: stands for multi-page TIFF.

This file type saves

multiple scanned pages in a single file. XPS (XML Paper Specification) creates an XAML file that preserves the original formatting of the document and supports color graphics and embedded fonts. PDF/A (Archivable): provides archivable text and image quality. NOTE: OCR file types are not supported on this printer unless attached to DSS. Scan/Digital Send Settings menu (M527/E52545/M528/E52645) 227

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level
Values Description Optimize Text/ Picture Mixed* Text

Printed picture

Photograph

Use to optimize

the output for a particular type of content. You can optimize the output for text, printed pictures, or a mixture. Mixed: Use to optimize the setting for text and for pictures. Text: Use to optimize the text portion of the copy when text and/or pictures are on the original. Printed picture:

Use for line

drawings and preprinted images, such as magazine clippings or pages from books. Photograph: Best suited for making copies of printed pictures.

Quality and File

Size High (large file) Medium* Low (small file)

Use to select

the quality for the output. Higher quality images require a larger file size than lower quality images.

Larger files take

more time to send, and some recipients might have trouble receiving larger files. 228 Chapter 4 Solve problems

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level
Values Description Original Sides 1-sided 2-sided

Use to describe

the layout for each side of the original document. First select whether the original document is printed on one side or both sides. Then touch the Orientation setting to indicate whether the original has portrait or landscape orientation. If it is printed on both sides, also select the 2-sided format that matches the original document.

Orientation Automatically

detect Portrait*

Landscape

For some features

to work correctly, you must specify the way the content of the original document is placed on the page. Portrait orientation means the short edge of the page is along the top. Landscape orientation means the long edge of the page is along the top. In the Orientation area, select whether the original document has a portrait or landscape orientation. Scan/Digital Send Settings menu (M527/E52545/M528/E52645) 229

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level
Values Description Resolution 600 dpi 400 dpi 300 dpi 200 dpi 150 dpi* 75 dpi Sets the resolution for sent documents. Higher resolution images have more dots per inch (dpi), so they show more detail. Lower resolution images have fewer dots per inch and show less detail, but the file size is smaller. Some file types, for example a file that will be processed with OCR, require a specific resolution.

When these file

types are selected, the Resolution setting might automatically change to a valid value. Content

Orientation

Orientation Auto Detect

Portrait*

Landscape

For some features

to work correctly, you must specify the way the content of the original document is placed on the page. Portrait orientation means the short edge of the page is along the top. Landscape orientation means the long edge of the page is along the top. In the Orientation area, select whether the original document has a portrait or landscape orientation. 230 Chapter 4 Solve problems

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level
Values Description Color/Black Automatically detect color or black*

Automatically

detect color or gray Color Black Black/Gray

Use to enable

or disable color scanning.

Automatically

detect color or black: When pages without color are detected, the printer creates an image of the page in 1-bit black if other settings allow. If the other settings don't allow (File Type, for example), the image is in grayscale.

Automatically

detect color or gray: When pages without color are detected, the printer creates an image of the page in grayscale.

Select this option

for the best image quality for non color pages. Color: Scans documents in color. Black/Gray: Scans documents in grayscale. Black: Scans documents in black and white with a compressed file size.

Original Size Select from a list

of supported sizes.

Use to describe

the page size of the original document. Scan/Digital Send Settings menu (M527/E52545/M528/E52645) 231

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level
Values Description Notification Do not notify*

Notify when job

completes

Notify only if job

fails

Configure

to receive notification about the status of a sent document. Do not notify: Turns off this feature.

Notify when

job completes:

Select to receive

notification for this job only. Print E-mail

Notify only

if job fails:

Select to receive

notification only if the job is not sent successfully. Print: Select to print the notification at this printer. Include Thumbnail NOTE: When sending an analog fax, select Include

Thumbnail to

receive a thumbnail image of the first page of the fax in your notification.

Notification E-mail

address E-mail: Select to receive the notification in an email. Touch the text box following Email Address, and then type the email address for the notification.

Image Adjustment Darkness Use to improve the

overall quality of the copy.

Adjust the

Darkness setting

to increase or decrease the amount of white and black in the colors. 232 Chapter 4 Solve problems

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level
Values Description

Contrast Adjust the

Contrast setting

to increase or decrease the difference between the lightest and darkest color on the page.

Background

Cleanup

Adjust the

Background

Cleanup setting if

you are having trouble copying a faint image.

Sharpness Adjust the

Sharpness setting

to clarify or soften the image. For example, increasing the sharpness could make text appear crisper, but decreasing it could make photographs appear smoother.

Automatic Tone

Default Select this to make

the selected Image

Adjustment setting

the default value. Cropping Options Do not crop*

Crop to content

Crop to paper

Use this menu item

to automatically crop the scan for digital sending.

Use the Crop to

content option to scan the smallest possible area that has detectable content. Scan/Digital Send Settings menu (M527/E52545/M528/E52645) 233

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level
Values Description

Erase Edges Use Inches

Back side erase

Front side erase

Use this menu

item to remove blemishes, such as dark borders or staple marks, by cleaning the specified edges of the scanned image. In each of the text boxes enter the measurements, in millimeters or inches, for how much of the top edge, bottom edge, left edge, and right edge to clean.

Multi-feed

Detection

On* Off

This setting stops

the scanning process if it detects multiple page feeds through the document feeder.

Blank Page

Suppression

Disabled* Enabled

Use to prevent

blank pages in the original document from being included in the output document.

Digital Send

Service Setup

Allow Usage of

Digital Sending

Software (DSS) Server Enabled* Disabled

Configure how the

printer interacts with the HP

Digital Sending

Software (DSS) server. HP DSS handles digital sending tasks, such as faxing, emailing, and sending scanned documents to a network folder or USB storage device.

Use the Allow

Usage of Digital

Sending Software

(DSS) Server option to configure the printer to use

HP DSS.

234 Chapter 4 Solve problems

Table 4-27 Scan/Digital Send Settings menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level
Values Description

Allow Transfer

to New Digital

Sending Software

(DSS) Server Enabled* Disabled

Use the Allow

Transfer to New

Digital Sending

Software (DSS)

Server option to

specify whether

DSS management

of a printer is transferable to a different DSS. Fax menu (M527/E52545/M528/E52645) Review the following information about the Fax menu for the M527/E52545/M528/E52645 printers. To display: At the printer control panel, select the Settings menu, and then select the Fax menu. In the following table, asterisks (*) indicate the factory default setting. Table 4-28 Fax menu (M527/E52545/M528/E52645) First level Second level Third level Fourth level Values Description

Internal Fax

Modem Setup

Configure settings

for sending faxes from the printer.

Use the Internal

Fax Modem Setup

feature to set up options for faxing. NOTE: To set up

LAN fax or Internet

fax, use the HP

Embedded Web

Server. To open the

HP Embedded Web

Server, type the printer network address into a Web browser. To configure the fax features, select the Fax tab.

Fax Send Settings Fax Send Setup Fax Dialing

Settings

Fax Dial Volume Off

Low* High

These settings

control how the fax modem dials the outgoing fax number when faxes are sent.

Fax Send Settings Fax Send Setup Fax Dialing

Settings Dialing Mode Tone* Pulse

Fax Send Settings Fax Send Setup Fax Dialing

Settings Redial Interval 1 – 5 Minutes Default = 5 minutes Fax menu (M527/E52545/M528/E52645) 235

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Send Settings Fax Send Setup Fax Dialing

Settings Fax Send Speed Fast* Medium Slow

Fax Send Settings Fax Send Setup Fax Dialing

Settings

Dialing Prefix

Fax Send Settings Fax Send Setup Fax Dialing

Settings

Detect Dial Tone

Fax Send Settings Fax Send Setup Fax Dialing

Settings Redial on Error Range: 0 – 9 Default = 2

Fax Send Settings Fax Send Setup Fax Dialing

Settings

Redial on No

Answer Range: 0 – 2 Default = 0

Fax Send Settings Fax Send Setup Fax Dialing

Settings Redial on Busy Range: 0 – 9 Default = 3

Fax Send Settings Fax Send Setup General Fax Send

Settings

Fax Number

Confirmation

Enabled Disabled*

If this feature

is enabled, you must enter the fax number twice.

Fax Send Settings Fax Send Setup General Fax Send

Settings PC Fax Send Enabled* Disabled

Enables users who

have the correct driver installed to send faxes through the printer from their computers.

Fax Send Settings Fax Send Setup General Fax Send

Settings JBIG Compression Enabled* Disabled The JBIG compression reduces fax transmission time, which can result in lower phone charges. However, using

JBIG compression

sometimes causes compatibility problems with older fax machines. If this occurs, turn off the JBIG compression.

Fax Send Settings Fax Send Setup General Fax Send

Settings

Error Correction

Mode Enabled* Disabled When error correction mode is enabled and an error occurs during fax transmission, the printer sends or receives the error portion again. 236 Chapter 4 Solve problems

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Send Settings Fax Send Setup General Fax Send

Settings Fax Header Prepend* Overlay

Use to prepend

or overlay the fax header page. Fax Send Settings Fax Send Setup Billing Codes Enable Billing Codes Off On*

When billing codes

are enabled, a prompt displays that asks the user to enter the billing code for an outgoing fax. This prompt does not appear if the Allow users to edit billing codes check box is not checked.

You can also

use the billing codes report in the

Reports menu to

view the list of the billing codes that have been used for faxes that have been sent from the printer. The list is grouped by billing code and also shows fax details.

This feature can be

used for billing or usage tracking. Fax Send Settings Fax Send Setup Billing Codes Default Billing Code

Specify a default

billing code for faxing. If you specify a default billing code, this code displays in the Billing Code field when the user sends an outgoing fax. If this field is blank, no default billing code is provided for the user. Fax Send Settings Fax Send Setup Billing Codes Minimum Length Range: 1 – 16 Default = 1

Specify the

required length of the billing code.

Billing codes can

be between 1 and 16 characters long. Fax Send Settings Fax Send Setup Billing Codes Allow users to edit billing codes Off On* Fax menu (M527/E52545/M528/E52645) 237

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Send Settings Default Job

Options Image Preview Make optional*

Require preview

Disable preview

Use the Image

Preview feature to

scan a document and display a preview before completing the job.

Select whether

this feature is available on the printer. Make optional: The feature is optional, depending on the user who is signed in.

Require preview

Previews are

required for all users. Disable preview:

Previews are

disabled for all users.

Fax Send Settings Default Job

Options Resolution Standard (100 x 200dpi)* Fine (200 x 200dpi) Superfine (300 x 300dpi)

Select the

resolution for outgoing faxes. If you increase the resolution, faxes might be clearer but they could transmit more slowly. Some file types, for example a file that will be processed with OCR, require a specific resolution.

When these file

types are selected, the Resolution setting might be automatically changed to a valid value. 238 Chapter 4 Solve problems

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Send Settings Default Job

Options Original Sides 1-sided* 2-sided

Pages flip up

Use to describe

the layout for each side of the original document. First select whether the original document is printed on one side or both sides. Then touch the Orientation setting to indicate whether the original has portrait or landscape orientation. If it is printed on both sides, also select the 2-sided format that matches the original document.

If the Pages flip up

option is selected, the back side of the page is printed upside-down. This option is for print jobs that are bound along the top edge.

Fax Send Settings Default Job

Options Original Sides Orientation Portrait*

Landscape

For some features

to work correctly, you must specify the way the content of the original document is placed on the page. Portrait: This setting means the short edge of the page is along the top. Landscape: This setting means the long edge of the page is along the top. Fax menu (M527/E52545/M528/E52645) 239

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Send Settings Default Job

Options Notification Do not notify*

Notify when job

completes

Notify only if job

fails

Use to receive

notification about the status of a sent document. Do not notify: Turns off this feature.

Notify when

job completes:

Select to receive

notification for this job only.

Notify only

if job fails:

Select to receive

notification only if the job is not sent successfully.

Fax Send Settings Default Job

Options

Notification Print

E-mail Print: Select to print the notification at this printer. E-mail: Select to receive the notification in an email. Touch the text box following Email Address, and then enter the email address for the notification.

Fax Send Settings Default Job

Options

Notification Include Thumbnail When sending an

analog fax, select

Include Thumbnail

to receive a thumbnail image of the first page of the fax in your notification.

Fax Send Settings Default Job

Options

Notification Notification E-mail

address

Provide the

email address that will receive notifications. 240 Chapter 4 Solve problems

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Send Settings Default Job

Options Content

Orientation

Orientation Portrait*

Landscape

For some features

to work correctly, you must specify the way the content of the original document is placed on the page. Portrait: This setting means the short edge of the page is along the top. Landscape: This setting means the long edge of the page is along the top.

Fax Send Settings Default Job

Options

Original Size Select from a list

of sizes that the printer supports.

Use to describe

the page size of the original document.

Fax Send Settings Default Job

Options

Image Adjustment Darkness Use to improve the

overall quality of the copy.

Adjust the

Darkness setting

to increase or decrease the amount of white and black in the colors.

Fax Send Settings Default Job

Options

Image Adjustment Contrast Adjust the

Contrast setting

to increase or decrease the difference between the lightest and darkest color on the page.

Fax Send Settings Default Job

Options

Image Adjustment Background

Cleanup

Adjust the

Background

Cleanup setting if

you are having trouble copying a faint image. Fax menu (M527/E52545/M528/E52645) 241

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Send Settings Default Job

Options

Image Adjustment Sharpness Adjust the

Sharpness setting

to clarify or soften the image. For example, increasing the sharpness could make text appear crisper, but decreasing it could make photographs appear smoother.

Fax Send Settings Default Job

Options

Image Adjustment Automatic Tone The printer

automatically adjusts the Darkness, Contrast, and

Background

Cleanup settings

to the most appropriate for the scanned document.

Fax Send Settings Default Job

Options Optimize Text/ Picture Manually adjust* Optimize For Text

Printed picture

Photograph

Optimizes the

output for a particular type of content. You can optimize the output for text, printed pictures, or a mixture. Manually adjust:

Use to manually

optimize the setting for text or for pictures. Text: Use to optimize the text portion of the copy where text and/or pictures are on the original. Printed picture:

Use for line

drawings and preprinted images, such as magazine clippings or pages from books. Photograph: Best suited for making copies of printed pictures. 242 Chapter 4 Solve problems

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Send Settings Default Job

Options

Multi-feed

Detection

Disabled Enabled*

This setting stops

the scanning process if it detects multiple page feeds through the document feeder.

Fax Send Settings Default Job

Options

Blank Page

Suppression

Disabled* Enabled

Prevents blank

pages in the original document from being included in the output document.

Fax Receive

Settings

Fax Receive Setup Ringer Volume Off

Low* High

Use to configure

settings for receiving faxes.

Fax Receive

Settings Fax Receive Setup Rings To Answer Range: 1–6 Default = 1

Fax Receive

Settings Fax Receive Setup Fax Send Speed Fast* Medium Slow

Fax Receive

Settings Fax Receive Setup Ring Interval Range: 220–600 ms Default = 600 ms

Fax Receive

Settings Fax Receive Setup Ring Frequency Range: 1–200 Default = 68hz

Fax Receive

Settings

Fax Printing

Schedule

Always store faxes

Always print faxes*

Use schedule

If you have

concerns about the security of private faxes, use this feature to store faxes rather than having them automatically print.

Select Incoming

Fax Options, and then you can choose to always store faxes, always print them, or you can set up a schedule for each day of the week. Fax menu (M527/E52545/M528/E52645) 243

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Receive

Settings

Fax Printing

Schedule Schedule

Touch this to

set up a fax printing schedule if you selected the

Use Fax Printing

Schedule option. Add Edit Delete

Print incoming

faxes

Store incoming

faxes Time

Event Days

If you are using

a fax printing schedule, use this menu to configure when to print faxes.

Fax Receive

Settings

Blocked Fax

Numbers

Fax Number to

Block

The blocked fax

list can contain up to 30 numbers.

When the printer

receives a call from one of the blocked fax numbers, it deletes the incoming fax.

It also logs

the blocked fax in the activity log along with job-accounting information.

Add blocked

numbers: Enter a fax number into the Fax Number to Block field, and then touch the arrow button to add a new number to the blocked fax list.

To remove blocked

numbers: Select a number and touch the Delete button to delete it from the blocked fax list.

To clear all blocked

numbers: Touch the Delete All button to clear all of the numbers from the blocked fax list.

You can also use

the Blocked Fax

List report in the

Information menu

to view the list of the fax numbers that have been blocked on this printer. 244 Chapter 4 Solve problems

Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Receive

Settings

Default Job

Options Notification Do not notify*

Notify when job

completes

Notify only if job

fails

Configure

to receive notification about the status of a sent document. Do not notify: Turns off this feature.

Notify when

job completes:

Select to receive

notification for this job only.

Notify only

if job fails:

Select to receive

notification only if the job is not sent successfully.

Fax Receive

Settings

Default Job

Options Print E-mail* E-mail: Select to receive the notification in an email. Touch the text box following Email Address, and then enter the email address for the notification.

Fax Receive

Settings

Default Job

Options Include Thumbnail NOTE: When sending an analog fax, select Include

Thumbnail to

receive a thumbnail image of the first page of the fax in your notification.

Fax Receive

Settings

Default Job

Options

Notification E-mail

address

Fax Receive

Settings

Default Job

Options

Stamp Received

Faxes Enabled Disabled*

Use this option to

add the date, time, sender's phone number, and page number to each page of the faxes that this printer receives. Fax menu (M527/E52545/M528/E52645) 245
Table 4-28 Fax menu (M527/E52545/M528/E52645) (continued) First level Second level Third level Fourth level Values Description

Fax Receive

Settings

Default Job

Options Fit to Page Enabled* Disabled

Use to shrink faxes

that are larger than Letter-size or A4-size so that they can fit onto a Letter-size or A4-size page. If this feature set to Disabled, faxes larger than Letter or A4 will flow across multiple pages.

Fax Receive

Settings

Default Job

Options Paper Selection Automatic*

Select from a list

of the trays.

Fax Receive

Settings

Default Job

Options Output sides 1-sided* 2-sided

Pages flip up

Use to describe

the layout for each side of the original document. First select whether the original document is printed on one side or both sides. Then touch the Orientation setting to indicate whether the original has portrait or landscape orientation. If it is printed on both sides, also select the 2-sided format that matches the original document.

Fax Forwarding Enable Fax

Forwarding

Disabled* Enabled

Use to forward

received faxes to another fax machine.

Fax Forwarding Type of Fax Job to

Forward

All faxes

Sent faxes

Received faxes

Fax Forwarding Fax Forwarding

Number

Enter the fax

fowarding number.

Clear Fax Activity

Log Cancel Clear

Clears all events

from the Fax Activity Log list.

Manage Supplies menu

Review the following information about the Manage Supplies menu. 246 Chapter 4 Solve problems

To display: At the printer control panel, select the Settings menu, and then select the Manage Supplies menu. In the following table, asterisks (*) indicate the factory default setting. Table 4-29 Manage Supplies menu First level Second level Third level Fourth level Values Description

Low Warning

Thresholds

Black Cartridge 1-100% Default = 4%

Set the estimated

percentage at which the printer notifies you when a toner cartridge is very low.

Low Warning

Thresholds

Document Feeder

Kit

M527/E52545/

M528/E52645 only 1-100% Default = 10%

Set the estimated

percentage at which the printer notifies you when the document feeder kit is very low.

Low Warning

Thresholds

Maintenance Kit

M506/E50045/

M507/E50145 only 1-100% Default = 10%

Set the estimated

percentage at which the printer notifies you when the maintenance kit is very low.

Low Warning

Threshold

Message On* Off

Displays a

message on the control panel when a cartridge is very low. Low Behavior Continue* Stop

Use one of these

options to specify the action for the device to take when a supply reaches a low condition. Stop: Stops the current print job. Continue:

Completes the

current print job. Manage Supplies menu 247

Table 4-29 Manage Supplies menu (continued) First level Second level Third level Fourth level Values Description

Very Low Behavior Black Cartridge Stop

Continue*

Prompt to

continue

Use one or more

of these options to specify the action for the device to take when a supply reaches a very low condition. Stop: Stops the current print job. Continue:

Completes the

current print job. Prompt to continue:

Allows the user

to decide whether or not to finish printing.

Very Low Behavior Document Feeder

Kit

M527/E52545/

M528/E52645 only Stop Continue

Prompt to

continue*

Use one or more

of these options to specify the action for the device to take when a supply reaches a very low condition. Stop: Stops the current print job. Continue:

Completes the

current print job. Prompt to continue:

Allows the user

to decide whether or not to finish printing.

Very Low Behavior Maintenance Kit

M506/E50045/

M507/E50145 only Stop Continue

Prompt to

continue*

Use one or more

of these options to specify the action for the device to take when a supply reaches a very low condition. Stop: Stops the current print job. Continue:

Completes the

current print job. Prompt to continue:

Allows the user

to decide whether or not to finish printing. 248 Chapter 4 Solve problems

Table 4-29 Manage Supplies menu (continued) First level Second level Third level Fourth level Values Description Store Usage Data On supplies*

Not on supplies

Store Usage Data

provides a way to suppress the toner cartridges from storing most of the information gathered exclusively for the purpose of understanding the usage of the printer. Select the

On supplies setting

to store the data on the toner cartridge memory chip. Select the Not on supplies setting to suppress the information from being stored on the memory chip.

Cartridge

Protection

Off*

Protect cartridges

Select to

permanently protect cartridges so that they can be used only in this product or fleet of products. Cartridge Policy Off*

Authorized HP

Set Authorized

HP to allow

only genuine HP cartridges to be used in this printer.

Reset Supplies Document Feeder

Kit (M527/E52545/

M528/E52645

only)

Maintenance

Kit (M506/E50045/ M507/E50145 only) Reset Cancel

Select this option

if you have installed a new document feeder kit (M527/E52545/

M528/E52645)

or maintenance kit (M506/E50045/

M507/E50145).

Networking menu

Review the following information about the Networking menu. To display: At the printer control panel, select the Settings menu, and then select the Networking menu. In the following table, asterisks (*) indicate the factory default setting. Networking menu 249

Table 4-30 Networking menu First level Second level Third level Fourth level Values Description

Ethernet Information Print Security

Report Yes No* Yes: Prints a page that contains the current security settings on the

HP Jetdirect print

server. No: A security settings page is not printed. Ethernet TCP/IP Host Name Use the arrow buttons to edit the host name.

NPXXXXXX*

An alphanumeric

string, up to 32 characters, used to identify the printer. This name is listed on the HP Jetdirect configuration page. The default host name is NPIxxxxxx, where xxxxxx is the last six digits of the LAN hardware (MAC) address. 250 Chapter 4 Solve problems

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Ethernet TCP/IP IPV4 Settings
Config Method Bootp DHCP* Auto IP Manual

Specifies the

method that TCP/ IPv4 parameters will be configured on the HP Jetdirect print server. Bootp (Bootstrap Protocol): Use for automatic configuration from a BootP server. DHCP (Dynamic

Host Configuration

Protocol): Use for automatic configuration from a DHCPv4 server.

If selected and

a DHCP lease exists, the DHCP

Release menu and

the DHCP Renew menu are available to set DHCP lease options. Auto IP: Use for automatic link-local IPv4 addressing. An address in the form 169.254.x.x is assigned automatically.

If you set this

option to the Manual setting, use the Manual

Settings menu to

configure TCP/IPv4 parameters. Networking menu 251

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Ethernet TCP/IP IPV4 Settings
Default IP Auto IP* Legacy

Specify the IP

address to default to when the print server is unable to obtain an IP address from the network during a forced TCP/IP reconfiguration (for example, when manually configured to use BootP or DHCP). NOTE: This feature assigns a static IP address that might interfere with a managed network. Auto IP: A link local IP address 169.254.x.x is set. Legacy: The address 192.0.0.192 is set, consistent with older HP Jetdirect printers. Ethernet TCP/IP IPV4 Settings DHCP Release Yes No* Ethernet TCP/IP IPV4 Settings DHCP Renew Yes No* Ethernet TCP/IP IPV4 Settings Primary DNS Range: 0-255 Default = xxx.xxx.xx.xx

Specify the IP

address (n.n.n.n) of a Primary Domain

Name System

(DNS) Server. Ethernet TCP/IP IPV4 Settings Secondary DNS Range: 0-255 Default = 0.0.0.0

Specify the IP

address (n.n.n.n) of a Secondary DNS Server. Ethernet TCP/IP IPV6 Settings Enable Off On*

Use this item to

enable or disable IPv6 operation on the print server. Off: IPv6 is disabled. On: IPv6 is enabled. 252 Chapter 4 Solve problems

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Ethernet TCP/IP IPV6 Settings
Address Manual Settings Enable Address

Use this item

to enable and manually configure a TCP/IPv6 address. Ethernet TCP/IP IPV6 Settings DHCPv6 Policy Router Specified Router Unavailable* Always Router Specified:

The stateful

auto-configuration method to be used by the print server is determined by a router. The router specifies whether the print server obtains its address, its configuration information, or both from a DHCPv6 server. Router Unavailable: If a router is not available, the print server should attempt to obtain its stateful configuration from a DHCPv6 server. Always: Whether a router is available, the print server always attempts to obtain its stateful configuration from a DHCPv6 server. Ethernet TCP/IP IPV6 Settings Primary DNS Range: 0-255 Default = xxx.xxx.xx.xx

Specify the IP

address (n.n.n.n) of a Primary Domain

Name System

(DNS) Server. Ethernet TCP/IP IPV6 Settings Secondary DNS Range: 0-255 Default = 0.0.0.0

Specify the IP

address (n.n.n.n) of a Secondary DNS Server. Networking menu 253

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Ethernet TCP/IP Proxy Server Select from a provided list.

Specifies the proxy

server to be used by embedded applications in the printer. A proxy server is typically used by network clients for Internet access. It caches Web pages, and provides a degree of Internet security for those clients.

To specify a

proxy server, enter its IPv4 address or fully-qualified domain name. The name can be up to 255 octets. For some networks, you might need to contact your

Internet Service

Provider (ISP) for the proxy server address. Ethernet TCP/IP Proxy Port Range: 1-65535 Default = 00080

Enter the port

number used by the proxy server for client support.

The port number

identifies the port reserved for proxy activity on your network, and can be a value from 0 to 65535. Ethernet TCP/IP Idle Timeout Range: 1-3600 Default = 0270 The time period, in seconds, after which an idle

TCP print data

connection is closed (default is 270 seconds, 0 disables the timeout). Wi-Fi Direct Status On* Off

Wi-Fi Direct Wi-Fi Direct Name DIRECT-xx-HP

[printer model]* xx represents two unique characters to help identify the printer. Displays the Wi Fi Direct printer name. 254 Chapter 4 Solve problems

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Wi-Fi Direct Channel 1 2* 3 4 5 6 7 8 9 10 11 36 40 44 48 149 153 157 161 165

Displays the

available channels.

Wi-Fi Direct Connection

Method Automatic* Manual Automatic: A mobile device can automatically connect to the printer and print a document. Manual: You must acknowledge the connection by touching the printer display when prompted or by entering a PIN code, provided by the printer, into the mobile device.

Wi-Fi Direct Wi-Fi Direct

Password 12345678* Displays the Wi-Fi

Direct password

for the printer. Networking menu 255

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description

Wireless Information Print Security

Report Yes No* Yes: Prints a page that contains the current security settings on the

HP Jetdirect print

server. No: A security settings page is not printed.

Wireless Wireless Station Status

Wireless Wireless Station SSID

Wireless Wireless Station Authentication

Wireless Wireless Station Reset 802.11

Wireless Wireless Station Wi-Fi Protected

Setup Wireless TCP/IP Host Name NPIXXXXXX*

Edit the host name

as needed.

An alphanumeric

string, up to 32 characters, used to identify the printer. This name is listed on the HP Jetdirect configuration page. The default host name is NPIxxxxxx, where xxxxxx is the last six digits of the LAN hardware (MAC) address. 256 Chapter 4 Solve problems

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Wireless TCP/IP IPV4 Settings
Config Method Bootp DHCP* Auto IP Manual

Specifies the

method that TCP/ IPv4 parameters will be configured on the HP Jetdirect print server. Bootp (Bootstrap Protocol): Use for automatic configuration from a BootP server. DHCP (Dynamic

Host Configuration

Protocol): Use for automatic configuration from a DHCPv4 server.

If selected and

a DHCP lease exists, the DHCP

Release menu and

the DHCP Renew menu are available to set DHCP lease options. Auto IP: Use for automatic link-local IPv4 addressing. An address in the form 169.254.x.x is assigned automatically.

If you set this

option to the Manual setting, use the Manual

Settings menu to

configure TCP/IPv4 parameters. Networking menu 257

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Wireless TCP/IP IPV4 Settings
Default IP Auto IP* Legacy

Specify the IP

address to default to when the print server is unable to obtain an IP address from the network during a forced TCP/IP reconfiguration (for example, when manually configured to use BootP or DHCP). NOTE: This feature assigns a static IP address that might interfere with a managed network. Auto IP: A link local IP address 169.254.x.x is set. Legacy: The address 192.0.0.192 is set, consistent with older HP Jetdirect printers. Wireless TCP/IP IPV4 Settings DHCP Release Yes No* Wireless TCP/IP IPV4 Settings DHCP Renew Yes No* Wireless TCP/IP IPV4 Settings Primary DNS Range: 0-255 Default = xxx.xxx.xx.xx

Specify the IP

address (n.n.n.n) of a Primary Domain

Name System

(DNS) Server. Wireless TCP/IP IPV4 Settings Secondary DNS Range: 0-255 Default = 0.0.0.0

Specify the IP

address (n.n.n.n) of a Secondary DNS Server. Wireless TCP/IP IPV6 Settings Enable Off On*

Use this item to

enable or disable IPv6 operation on the print server. Off: IPv6 is disabled. On: IPv6 is enabled. 258 Chapter 4 Solve problems

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Wireless TCP/IP IPV6 Settings Address Manual Settings Enable Address

Use this item

to enable and manually configure a TCP/IPv6 address. Wireless TCP/IP IPV6 Settings DHCPv6 Policy Router Specified Router Unavailable* Always Router Specified:

The stateful

auto-configuration method to be used by the print server is determined by a router. The router specifies whether the print server obtains its address, its configuration information, or both from a DHCPv6 server. Router Unavailable: If a router is not available, the print server should attempt to obtain its stateful configuration from a DHCPv6 server. Always: Whether a router is available, the print server always attempts to obtain its stateful configuration from a DHCPv6 server. Wireless TCP/IP IPV6 Settings Primary DNS Range: 0-255 Default = xxx.xxx.xx.xx

Specify the IP

address (n.n.n.n) of a Primary Domain

Name System

(DNS) Server. Wireless TCP/IP IPV6 Settings Secondary DNS Range: 0-255 Default = 0.0.0.0

Specify the IP

address (n.n.n.n) of a Secondary DNS Server. Networking menu 259

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description Wireless TCP/IP Proxy Server Select from a provided list.

Specifies the proxy

server to be used by embedded applications in the printer. A proxy server is typically used by network clients for Internet access. It caches Web pages, and provides a degree of Internet security for those clients.

To specify a

proxy server, enter its IPv4 address or fully-qualified domain name. The name can be up to 255 octets. For some networks, you might need to contact your

Internet Service

Provider (ISP) for the proxy server address. Wireless TCP/IP Proxy Port Range: 1-65535 Default = 80

Enter the port

number used by the proxy server for client support.

The port number

identifies the port reserved for proxy activity on your network, and can be a value from 0 to 65535. Wireless TCP/IP Idle Timeout Range: 1-3600 Default = 270 The time period, in seconds, after which an idle

TCP print data

connection is closed (default is 270 seconds, 0 disables the timeout). Wireless Security Secure Web HTTPS Required*

HTTPS Optional

Wireless Security IPSEC Keep

Disable* Wireless Security 802.1X Reset Keep* 260 Chapter 4 Solve problems

Table 4-30 Networking menu (continued) First level Second level Third level Fourth level Values Description

Wireless Security Announcement

Agent Enable* Disable

Wireless Security Reset Security Yes

No* I/O Timeout Range: 5 – 300 sec Default = 10

Use to set the

I/O timeout period in seconds. I/O timeout refers to the elapsed time before a print job fails. If the stream of data that the printer receives for a print job gets interrupted, this setting indicates how long the printer will wait before it reports that the job has failed.

Support Tools menu

The menus under the Support Tools menu contain the options for maintaining the printer and troubleshooting printer problems. Learn about these menus and settings. Maintenance menu (all except M501) Review the following information about the Maintenance menu for the M506/E50045/M507/E50145 and M527/E52545/M528/E52645 printers. Backup/Restore menu (all except M501) Review the following information about the Backup/Restore menu for the M506/E50045/M507/E50145 and M527/E52545/M528/E52645 printers. To display: At the printer control panel, select the Support Tools menu, select the Maintenance menu, and then select the Backup/Restore menu. In the following table, asterisks (*) indicate the factory default setting. Table 4-31 Backup/Restore menu (all except M501) First level Second level Third level Values Description

Back up Data Enable Scheduled

Backups

Backup Time Enter a time

Days Between

Backups

Enter the number of

days

Back up Now

Export Last Backup

Support Tools menu 261

Table 4-31 Backup/Restore menu (all except M501) (continued) First level Second level Third level Values Description

Restore Data Insert a USB drive that

contains the backup file. Calibration/Cleaning menu (all except M501) Review the following information about the Calibration/Cleaning menu for the M506/E50045/M507/E50145 and M527/E52545/M528/E52645 printers. To display: At the printer control panel, select the Support Tools menu, select the Maintenance menu, and then select the Calibration/Cleaning menu. In the following table, asterisks (*) indicate the factory default setting. Table 4-32 Calibration/Cleaning menu (all except M501)

First level Values Description

Cleaning Page Automatic Cleaning Interval

Off* 1000 pages 2000 pages 5000 pages 10000 pages 20000 pages

Use this menu to configure

the settings for the product to automatically print a cleaning page.

Select Off to disable

automatic cleaning.

Select a cleaning interval to

set the number of pages to be printed before an automatic cleaning page is printed.

Touch Print to manually print a

cleaning page. NOTE: Cleaning pages print on the default paper size configured for the printer.

Calibrate Scanner

(M527/E52545/M528/E52645

only) Done Start

Touch Start to calibrate the

device scanner. Messages on the control panel display will lead you through the calibration process.

Clean Document Feeder

Rollers

(M527/E52545/M528/E52645

only)

Set cleaning reminder

interval Range: 0 – 100% Default = 10% Done

Clean Now

Configure cleaning settings

for the document feeder rollers. USB Firmware Upgrade menu (all except M501) Review the following information about the USB Firmware Upgrade menu for the M506/E50045/M507/ E50145 and M527/E52545/M528/E52645 printers. 262 Chapter 4 Solve problems
To display: At the printer control panel, select the Support Tools menu, select the Maintenance menu, and then select the USB Firmware Upgrade menu. Insert a USB storage device with a firmware upgrade bundle into the USB port, and follow the on-screen instructions. Service menu (all except M501) Review the following information about the Service menu for the M506/E50045/M507/E50145 and M527/ E52545/M528/E52645 printers. To display: At the printer control panel, select the Support Tools menu, and then select the Service menu. The Service menu is locked and requires a PIN for access. This menu is intended for use by authorized service personnel. See the Service mode function section in the printer troubleshooting manual.

Troubleshooting menu

Review the following information about the Troubleshooting menu. To display: At the printer control panel, select the Support Tools menu, and then select the Troubleshooting menu. In the following table, asterisks (*) indicate the factory default setting. Table 4-33 Troubleshooting menu First level Second level Third level Fourth level Values Description Reports Configuration/

Status Pages

Settings Menu

Map

Current Settings

Page

Configuration

Page

Supplies Status

Page

Usage Page

Paper Path Page

[File Directory Page](#)

[Web Services](#)

[Status Page](#)

[PCL Font List](#)

[PS Font List](#)

[Event Log](#)

[Warning Log Page](#)

[T.30 Protocol Trace PQ](#)

[Troubleshooting](#)

[Pages](#)

[Paper Path Test](#)

[Select the desired](#)

[report and then select the print icon to print the page. Service menu \(all except M501\) 263](#)

[Table 4-33 Troubleshooting menu \(continued\) First level Second level Third level Fourth level Values Description](#)

[Reports Fax Reports Fax Activity Log](#)

[Billing Codes](#)

[Report](#)

[Blocked Fax List](#)

[Fax Call Report](#)

[Select the desired](#)

[report and then select the print icon to print the page.](#)

[Reports Other Pages PCL Font List](#)

[PS Font List](#)

[Select the desired](#)

[report and then select the print icon to print the page. Fax Tools Fax T.30 Trace Report Never automatically print*](#)

[Print after every](#)

[fax](#)

[Print only after fax](#)

[send jobs](#)

[Print only after](#)

[received faxes](#)

[Print only after fax](#)

[send errors](#)

[Print only after fax](#)

receive errors

Print after any fax

error

Use to print

or configure the fax T.30 trace report. T.30 is the standard that specifies handshaking, protocols, and error correction between fax machines.
Fax Tools (Fax models only) Fax V.34 Enable* Disable

Use to disable

V.34 modulations if several fax failures have occurred or if phone line conditions require it. Fax Tools (Fax models only)

JBIG Compression On

Off* Fax Tools (Fax models only) Fax Speaker Mode Normal*

Diagnostic

Used by a

technician to evaluate and diagnose fax issues by listening to the sounds of fax modulations 264 Chapter 4 Solve problems

Table 4-33 Troubleshooting menu (continued) First level Second level Third level Fourth level Values Description Fax Tools (Fax models only)

Fax Service Log Print The standard fax

log includes basic information such as the time and whether the fax was successful.

The detailed

fax log shows the intermediate results of the redial process not shown in the standard fax log.

Print Quality Pages PQ

Troubleshooting

Pages

Print Use to print pages

that help you resolve problems with print quality. Event Log Prints the 50 most recent events in the Event Log. For each event, the printed log shows the error number, page count, error code, and description or personality.

Paper Path Page Print Shows how many

pages were printed from each tray.

Diagnostic Tests Disable Cartridge

Check

Continue Use this diagnostic

test to print internal pages or send an external job to the printer when the toner cartridge is removed or exchanged. Supply errors are ignored while the printer is in this mode. Diagnostic Tests Paper Path Test Source Tray Select from a list of the available trays.

Generates a test

page for testing paper handling features. You can define the path that is used for the test in order to test specific paper paths Troubleshooting menu 265

Table 4-33 Troubleshooting menu (continued) First level Second level Third level Fourth level Values Description Diagnostic Tests Paper Path Test Number of Copies Range: 1–500 Default = 1

Sets the default

number of copies for a copy job. This default applies when the Copy or Quick Copy function is initiated from the printer Home screen. The factory default setting is 1.

Diagnostic Tests Paper Path Test Test Duplex Path

(Duplex models only) Enable Disable*

Prints on both

sides of the paper. Diagnostic Tests Print/Stop Test Specify the length of time in milliseconds (0-60,000).

Diagnostic Tests Paper Path Sensor

Test

Select from a

list of the printer sensors.

Initiates a test of

the paper path sensors.

Diagnostic Tests Manual Sensor

Test

Select from a

list of available components Reset

Test the printer

sensors and switches for correct operation.

Each sensor is

displayed on the control-panel screen, along with its status. Manually trip each sensor and watch for it to change on the screen. Press the

Stop button to

abort the test. Diagnostic Tests Tray/Bin Manual

Sensor Test

Select from a

list of available components Reset

Test the printer

sensors and switches for correct operation.

Each sensor is

displayed on the control-panel screen, along with its status. Manually trip each sensor and watch for it to change on the screen. Press the

Stop button to

abort the test. 266 Chapter 4 Solve problems

Table 4-33 Troubleshooting menu (continued) First level Second level Third level Fourth level Values Description

Diagnostic Tests Component Test Select from a

list of available components.

Use to exercise

individual parts independently to isolate noise, leaking, or other issues. To start the test, select one of the components.

The test will run the

number of times specified by the Repeat option. You might be prompted to remove parts from the printer during the test.

Press the Stop

button to abort the test.

Diagnostic Tests Continuous Scan

(M527/E52545/

M528/E52645

only) 2-sided

Diagnostic Tests Scanner Tests

(M527/E52545/

M528/E52645

only) Sensors

Generate Debug

Data Start Retrieve

Diagnostic Data

Create device data

file

Create zipped

debug information file

Include crash

dump files

Clean up debug

information

Send to E-mail

Export to USB

Create files

that contain information about the printer that can help identify the cause of problems. Troubleshooting menu 267
Table 4-33 Troubleshooting menu (continued) First level Second level Third level Fourth level Values Description

Retrieve Fax

Diagnostic Data

Create device data

file

Create zipped

debug information file

Include crash

dump files

Clean up debug

information

Send to E-mail

Export to USB

Create files

that contain information about the printer that can help identify the cause of problems. Error-code and control-panel-message troubleshooting overview This section provides information on error codes and searching for information. Error codes (types and structure) Error codes and control-panel messages display on the printer control panel to indicate the current printer status or situations that might require action. Error codes are numerical, or alphanumeric, and have a set structure with six characters (example: 13.WX.YZ). ● The first two characters are numeric and represent the system component that is causing the error. For example, in error code 10.22.15, 10 = Supplies for HP LaserJets. ● The remaining four characters (W, X, Y, and Z values) further define the error. HP LaserJet and HP PageWide Enterprise error codes are documented in the control panel message document (CPMD) for each printer. The CPMD is a comprehensive list of error codes, diagnostic and troubleshooting steps to clear or resolve the error, and other helpful information such as service mode pins and part numbers. The CPMD is continually updated and republished with the latest information for the following error codes. Table 4-34 ERROR CODES: The first two characters Error code System Component System Error Description 10.WX.YZ Supplies (LaserJet) Supply error or supply memory error. 11.WX.YZ Real-time clock Internal error with the clock on the formatter. 13.WX.YZ Jam (LaserJet) Paper jam or open door jam error. 15.WX.YZ Jam (PageWide) Paper jam or open door jam error. 17.WX.YZ Supplies (PageWide) Supply error or supply memory error. 20.WX.YZ Printer memory Insufficient memory or buffering error. 21.WX.YZ Page Page complexity causing a decompression error when trying to process job. 268 Chapter 4 Solve problems

Table 4-34 ERROR CODES: The first two characters (continued) Error code System Component System Error Description 30.WX.YZ Scanner Flatbed scanner error occurring inside the unit. 31.WX.YZ Document feeder Document feeder, scanner, or jam error. 32.WX.YZ Backup, restore, or reset Backup, restore, or reset notification or error. 33.WX.YZ Security Backup, Disk, EFI BIOS, Firmware integrity (SureStart), or Trusted Platform Module (TPM) notification or error. 40.WX.YZ Input/Output (I/O) Partition manager, secure erase, or USB accessory error. 41.WX.YZ Fuser, Laser scanner, or Paper path Miscellaneous error including general and misprint or mismatch errors typically involving (but not limited to) the fuser, the laser scanner, or the paper path. 42.WX.YZ Firmware Firmware failure involving the Event Log, Shell, System Manager, or other component. 44.WX.YZ Firmware Firmware failure involving a digital sending component. 45.WX.YZ OXPd/Web Kit (PageWide) Informational notifications involving the OXPd Web Kit (communications log). 46.WX.YZ Engine (LaserJet) Engine communication error. 47.WX.YZ Firmware Job parser or printer calibration error. 48.WX.YZ Firmware PJA job accounting, job management, or job pipeline error. 49.WX.YZ Firmware Firmware communication error. 50.WX.YZ Fuser (LaserJet) Fuser error. 51.WX.YZ Laser scanner (LaserJet) Laser scanner beam error. 52.WX.YZ Laser scanner (LaserJet) Laser scanner startup error. 54.WX.YZ Sensor Sensor error (not jam related). 55.WX.YZ DC controller (LaserJet) DC controller communication error. 56.WX.YZ Paper handling (LaserJet) Paper input/output or accessory error. 57.WX.YZ Fan Fan error. 58.WX.YZ Sensor Engine sensor failure. 59.WX.YZ Motor (LaserJet) Motor error. 60.WX.YZ Tray motor error (LaserJet) Tray lifting or pick up error. 61.WX.YZ Engine (PageWide) Print engine error with the 8-bit data package. 62.WX.YZ System (LaserJet) Print bar (PageWide) LaserJet internal system error. PageWide print system error. 63.WX.YZ Engine (LaserJet) General engine error (electrical, communication, etc.). 65.WX.YZ Connector Output accessory connection error. Error codes (types and structure) 269

Table 4-34 ERROR CODES: The first two characters (continued) Error code System Component System Error Description 66.WX.YZ Output accessory Output accessory error. 67.WX.YZ Input accessory Input accessory connection error 69.WX.YZ Duplexer Duplexer error. 70.WX.YZ DC controller (LaserJet) DC controller firmware error. 76.WX.YZ DC controller (LaserJet) Power supply fluctuations causes formatter and

DC controller communication issue and results

them not to be in synchronization. 80.WX.YZ Managed device Embedded Jetdirect error. 81.WX.YZ Near Field Communication (NFC) Wireless, Bluetooth or internal EIO error. 82.WX.YZ Memory (hard disk, EMMC, etc.) Disk hardware error. 90.WX.YZ Internal diagnostics Internal test of systems (i.e. disk, CPB, display) or interconnection error. 98.WX.YZ Hard disk Hard disk partition error. 99.WX.YZ Firmware installer Remote Firmware Upgrade (RFU), firmware install (engine or accessory), or disk error.

How to search for printer documentation

The CPMD, error codes, and other support documentation for each printer is available on the HP portals. For HP internal access to WISE, go to WISE. For HP channel partners access WISE, see Accessing WISE for HP channel partners (HP Partner Portal) on page 277 or Accessing WISE for HP channel partners (CSDP) on page 276. WISE is available to call agents, service technicians, and HP internal users. The level of detail available depends on your access credentials. To learn how to find support content in WISE, watch the video here. How to search WISE for printer information and documentation These instructions are for HP internal use by call agents, service technicians, and other internal users. To learn how to find support content in WISE, watch the video here: Finding Error Code and Control Panel Message Document in WISE. TIP: The Product Detail Page (PDP) is available from the WISE home page. Enter, and then select a product name or number in the Go to a Product Detail Page area (callout 1), or select the ...or select your product from a list item (callout 2) on the WISE home page. 1 2 270 Chapter 4 Solve problems

Many types of printer documentation and information are available on WISE. This section details methods for finding error code descriptions and solutions. TIP: To view a list of control panel message documents (CPMD) per printer, search for the following topic in WISE: HP LaserJet, OfficeJet, PageWide, ScanJet Enterprise - Control panel message document (CPMD) list or click the link below. ● HP LaserJet, HP OfficeJet, HP PageWide, HP ScanJet - Control panel messages document (CPMD), Service manual, and Service cost data list Use one of the following methods to search for CPMD error code information. ● Method 1: Use the WISE home page Error Code Lookup tool on page 271 ● Method 2: Search for error codes using the WISE search function on page 272 ● Method 3: Search for error codes using the Product Detail Page on page 274 Method 1: Use the WISE home page Error Code Lookup tool Use the WISE Error Code Lookup tool to find error code troubleshooting procedures using the following steps. 1. On the WISE home page, click the Error Code Lookup tool icon. Figure 4-81 WISE Error Code Lookup tool icon 2. Enter a printer model number (for example, M606) in the Error Code Lookup dialog box. Figure 4-82 WISE Error Code Lookup tool dialog box How to search WISE for printer information and documentation 271

3. A drop-down list appears. Click on the full printer series name (for example, HP LaserJet Enterprise M606 series) in the list. NOTE: Selecting the series item from the drop-down list is recommended for the most accurate search return. Figure 4-83 WISE Error Code Lookup tool drop-down list M606 4. Type the error code (for example, 13.b2.d2) in the Error Code Lookup keyword field, and then select the search icon. Figure 4-84 WISE Error Code Lookup tool search 1 2 5. The error code troubleshooting content displays in the All Search Results area. Figure 4-85 WISE Search results Method 2: Search for error codes using the WISE search function Use the WISE search function to find error code troubleshooting procedures using the following steps. 272 Chapter 4 Solve problems

1. On the WISE home page, enter a printer model number (for example, M606) in the Search dialog box. Figure 4-86 WISE Search dialog box 2. A drop-down list appears. Click on the full printer series name (for example, HP LaserJet Enterprise M606 series) in the list. NOTE: Selecting the series item from the drop-down list is recommended for the most accurate search return. Figure 4-87 WISE Search drop-down list 3. Type the error code (for example, 13.E1.D3) in the Search keyword field, and then select the search icon. Figure 4-88 WISE Search 1 2 4. The error code troubleshooting content displays in the All Search Results area. How to search WISE for printer information and documentation 273

Figure 4-89 WISE Search results Method 3: Search for error codes using the Product Detail Page Use the WISE Product Detail Page (PDP) to find error code troubleshooting procedures using the following steps. 1. On the WISE home page, enter a printer model number (for example, M606) in the Go to a Product Detail page dialog box. TIP: To find a PDP from a list of products by product type, select the ...or select your product from a list item just below the dialog box. Figure 4-90 WISE PDP dialog box 2. A drop-down list appears. Click on the full printer series name (for example, HP LaserJet Enterprise M606 series) in the list. NOTE: Selecting the series item from the drop-down list is recommended for the most accurate search return. 274 Chapter 4 Solve problems

Figure 4-91 WISE PDP drop-down list 3. Type the error code (for example, 13.E1.D3) in the Error Code Lookup field, and then select the search icon. Figure 4-92 WISE PDP search 1 2 4. The error code troubleshooting content displays in the All Search Results area. Figure 4-93 WISE PDP search results How to search WISE for printer information and documentation 275

Accessing WISE for HP channel partners (CSDP) Learn how to access WISE by accessing the HP Channel Services Delivery Platform (CSDP) if you are an HP Channel Partner. IMPORTANT: A Partner Admin must requested access, via the CSDP portal, to create a new user before you can sign into the CDSP portal. 1. Sign-in to the CSDP portal (click csdp.hp.com to access the sign-in page). Figure 4-94 CSDP sign-in page 2. On the CSDP home page, click the Knowledge and Training item. TIP: Use the Knowledge and Training item at the top of the CSDP home page (callout 1), or the Knowledge and Training action icon (callout 2). Figure 4-95 CSDP portal home page 1 2 276 Chapter 4 Solve problems

3. Select HP Technical Documentation on the Knowledge and Training page to open the WISE portal home page. Figure 4-96 CSDP portal HP Knowledge and Training page 4. Use the WISE portal to search for printer information. Figure 4-97 WISE portal home page Accessing WISE for HP channel partners (HP Partner Portal) Learn how to access WISE by accessing the HP Partner Portal if you are an HP Channel Partner. NOTE: If this is your first visit to the HP Partner Portal, you will be asked to create an account. Follow the setup directions using your HP Partner credentials. View a video of how to access WISE for HP channel partners 1. Sign-in to the HP Partner Portal (click <https://partner.hp.com> to access the sign-in page). Figure 4-98 HP Partner Portal sign-in page Accessing WISE for HP channel partners (HP Partner Portal) 277

2. On the HP Partner Portal home page, click the Services & Support item. Figure 4-99 HP Partner Portal home page 3. Select the Technical Support (WISE) item. Figure 4-100 HP Partner Portal Service & Support page 278 Chapter 4 Solve problems

4. Select the Technical Documentation item. Figure 4-101 HP Partner Portal Technical Support (WISE)page 5. Use the WISE portal to search for printer information. Figure 4-102 WISE portal home page Accessing WISE for HP channel partners (HP Partner Portal) 279

Resolving mono print quality problems

Use the information below to troubleshoot print-quality and image-quality problems on mono (black and white) HP LaserJet Enterprise printers.

Introduction

Use the information below to troubleshoot print-quality problems, including issues with image quality. The troubleshooting steps can help resolve the following defects: ● Blank pages ● Black pages ● Curled paper ● Dark or light bands ● Dark or light streaks ● Fuzzy print ● Gray background or dark print ● Light print ● Loose toner ● Missing toner ● Scattered dots of toner ● Skewed images ● Smears ● Streaks

Troubleshoot print quality

To resolve print-quality issues, try these steps in the order presented. To troubleshoot specific image defects, see Troubleshoot image defects.

Print from a different software program

Try printing from a different software program. If the page prints correctly, the problem is with the software program from which you were printing.

Check the paper-type setting for the print job

Check the paper type setting when printing from a software program and the printed pages have smears, fuzzy or dark print, curled paper, scattered dots of toner, loose toner, or small areas of missing toner.

Check the paper type setting on the control panel

Check the paper type setting on the printer control panel, and change the setting as needed. 280 Chapter 4 Solve problems

1. Open and close the paper tray. 2. Follow the control panel prompts to confirm or modify the paper type and size settings for the tray. 3. Make sure the paper loaded meets specifications. 4. Adjust the humidity and resistance setting on the control panel to match the environment.
- a. Open the following menus: i. Settings ii. Copy/Print or Print iii. Print Quality iv. Adjust Paper Type b. Select the paper type that matches the type loaded in the tray. c. Use the arrows to increase or decrease the humidity and resistance setting. 5. Make sure the driver settings match the control panel settings. NOTE: The print driver settings will override any control panel settings. Check the paper type setting (Windows) Check the paper type setting for Windows, and change the setting as needed. NOTE: The print driver settings will override any control panel settings. 1. From the software program, select the Print option. 2. Select the printer, and then click the Properties or Preferences button. 3. Click the Paper/Quality tab. 4. From the Paper Type drop-down list, click the More... option. 5. Expand the list of Type is: options. 6. Expand the category of paper types that best describes your paper. 7. Select the option for the type of paper you are using, and click the OK button. 8. Click the OK button to close the Document Properties dialog box. In the Print dialog box, click the OK button to print the job. Check the paper type setting (macOS) Check the paper type setting for macOS, and change the setting as needed. 1. Click the File menu, and then click the Print option. 2. In the Printer menu, select the printer. 3. Click Show Details or Copies & Pages. 4. Open the menus drop-down list, and then click the Paper/Quality menu. Check the paper type setting (Windows) 281
5. Select a type from the Media Type drop-down list. 6. Click the Print button.

Check toner-cartridge status

Follow these steps to check the estimated life remaining in the toner cartridges and if applicable, the status of other replaceable maintenance parts. Step one: Print the Supplies Status Page The supplies status page indicates the cartridge status. 1. From the Home screen on the printer control panel, select the Reports menu. 2. Select the Configuration/Status Pages menu. 3. Select Supplies Status Page, and then select Print to print the page. Step two: Check supplies status Check the supplies status report as follows. 1. Look at the supplies status report to check the percent of life remaining for the toner cartridges and, if applicable, the status of other replaceable maintenance parts. Print quality problems can occur when using a toner cartridge that is at its estimated end of life. The supplies status page indicates when a supply level is very low. After an HP supply has reached the very low threshold, HP's premium protection warranty on that supply has ended. The toner cartridge does not need to be replaced now unless the print quality is no longer acceptable. Consider having a replacement available to install when print quality is no longer acceptable. If you determine that you need to replace a toner cartridge or other replaceable maintenance parts, the supplies status page lists the genuine HP part numbers. 2. Check to see if you are using a genuine HP cartridge. A genuine HP toner cartridge has the word "HP" on it, or has the HP logo on it. For more information on identifying HP cartridges go to www.hp.com/go/learnaboutsupplies.

Print a cleaning page

During the printing process paper, toner, and dust particles can accumulate inside the printer and can cause print-quality issues such as toner specks or spatter, smears, streaks, lines, or repeating marks. Follow these steps to clean the printer paper path: 1. From the Home screen on the printer control panel, select the Support Tools menu. 2. Select the following menus: a. Maintenance b. Calibration/Cleaning c. Cleaning Page 282 Chapter 4 Solve problems

3. Select Print to print the page. A Cleaning... message displays on the printer control panel. The cleaning process takes a few minutes to complete. Do not turn the printer off until the cleaning process has finished. When it is finished, discard the printed page.

Visually inspect the toner cartridge or cartridges

Follow these steps to inspect each toner cartridge: 1. Remove the toner cartridge from the printer, and verify that the sealing tape has been removed. 2. Check the memory chip for damage. 3. Examine the surface of the green imaging drum. CAUTION: Do not touch the imaging drum. Fingerprints on the imaging drum can cause print quality problems. 4. If you see any scratches, fingerprints, or other damage on the imaging drum, replace the toner cartridge. 5. Reinstall the toner cartridge and print a few pages to see if the problem has resolved.

Check paper and the printing environment

Use the following information to check the paper selection and the printing environment. Step one: Use paper that meets HP specifications Some print-quality problems arise from using paper that does not meet HP specifications. ● Always use a paper type and weight that this printer supports. ● Use paper that is of good quality and free of cuts, nicks, tears, spots, loose particles, dust, wrinkles, voids, staples, and curled or bent edges. ● Use paper that has not been previously printed on. ● Use paper that does not contain metallic material, such as glitter. ● Use paper that is designed for use in laser printers. Do not use paper that is designed only for use in Inkjet printers. ● Use paper that is not too rough. Using smoother paper generally results in better print quality. Step two: Check the environment The environment can directly

affect print quality and is a common cause of print-quality or paper feeding issues. Try the following solutions: ● Move the printer away from drafty locations, such as open windows or doors, or air-conditioning vents. ● Make sure the printer is not exposed to temperatures or humidity outside of printer specifications. ● Do not place the printer in a confined space, such as a cabinet. ● Place the printer on a sturdy, level surface. Visually inspect the toner cartridge or cartridges 283

● Remove anything that is blocking the vents on the printer. The printer requires good air flow on all sides, including the top. ● Protect the printer from airborne debris, dust, steam, grease, or other elements that can leave residue inside the printer. Step three: Set the individual tray alignment Follow these steps when text or images are not centered or aligned correctly on the printed page when printing from specific trays. 1. From the Home screen on the printer control panel, select the Settings menu. 2. Select the following menus: a. Copy/Print or Print b. Print Quality c. Image Registration 3. Select Tray, and then select the tray to adjust. 4. Select Print Test Page, and then follow the instructions on the printed pages. 5. Select Print Test Page again to verify the results, and then make further adjustments if necessary. 6. Select Done to save the new settings.

Try a different print driver

Try a different print driver when printing from a software program and the printed pages have unexpected lines in graphics, missing text, missing graphics, incorrect formatting, or substituted fonts. Download a different print driver from the HP support Web site:

<http://www.hp.com/support/ljM501> or <http://www.hp.com/support/ljM506> or <http://www.hp.com/support/ljE50045> or <http://www.hp.com/support/ljM507> or <http://www.hp.com/support/ljE50145> or <http://www.hp.com/support/ljM527mfp> or <http://www.hp.com/support/ljE52545mfp> or <http://www.hp.com/support/ljM528mfp> or <http://www.hp.com/support/E52645mfp>.

Troubleshoot image defects

Review examples of image defects and steps to resolve these defects. 284 Chapter 4 Solve problems

Table 4-35 Image defects table quick reference

Sample Description Possible solutions

Table 4-42 Light print on page 290 Table 4-40 Gray background or dark print on page 289 Table 4-38 Blank page — No print on page 287 Table 4-37 Black page on page 287 Table 4-36 Banding defects on page 286 Table 4-44 Streak defects on page 291 Table 4-39 Fixing/fuser defects on page 288 Table 4-41 Image placement defects on page 289 Table 4-43 Output defects on page 290 Troubleshoot image defects 285

Image defects, no matter the cause, can often be resolved using the same steps. Use the following steps as a starting point for solving image defect issues. 1. Reprint the document. Print quality defects can be intermittent in nature or can go away completely with continued printing. 2. Check the condition of the cartridge or cartridges. If a cartridge is in a Very Low state (it has passed the rated life), replace the cartridge. 3. Make sure that the driver and tray print mode settings match the media that is loaded in the tray. Try using a different ream of media or a different tray. Try using a different print mode. 4. Make sure that the printer is within the supported operating temperature/humidity range. 5. Make sure that the paper type, size, and weight are supported by the printer. See support.hp.com for a list of the supported paper sizes and types for the printer. NOTE: The term “fusing” refers to the part of the printing process where toner is affixed to paper. The following examples depict letter-size paper that has passed through the printer short edge first. Table 4-36 Banding defects

Sample Description Possible solutions

Dark or light lines which repeat down

the length of the page, and are wide pitch and/or impulse bands. They might be sharp or soft in nature. The defect displays only in areas of fill, not in text or sections with no printed content. 1. Reprint the document. 2. Try printing from another tray. 3. Replace the cartridge. 4. Use a different paper type. 5. Enterprise models only: From the

Home screen on the printer control

panel, go to the Adjust Paper Types menu, and then choose a print mode that is designed for a slightly heavier media than what you are using. This slows the print speed and might improve the print quality. 6. If the issue persists, go to support.hp.com. 286 Chapter 4 Solve problems

Table 4-37 Black page

Sample Description Possible solutions

The entire printed page is black. 1. Visually inspect the cartridge to check for damage. 2. Make sure that the cartridge is installed correctly. 3. Replace the cartridge. 4. If the issue persists, go to support.hp.com. Table 4-38 Blank page — No print

Sample Description Possible solutions

The page is completely blank and

contains no printed content. 1. Make sure that the cartridges are genuine HP cartridges. 2. Make sure that the cartridge is installed correctly. 3. Print with a different cartridge. 4. Check the paper type in the paper tray and adjust the printer settings to match. If necessary, select a

lighter paper type. 5. If the issue persists, go to support.hp.com. Troubleshoot image defects 287

Table 4-39 Fixing/fuser defects

Sample Description Possible solutions

Slight shadows or offsets of the image

are repeated down the page. The repeated image might fade with each recurrence. 1. Reprint the document. 2. Check the paper type in the paper tray and adjust the printer settings to match. If necessary, select a lighter weight paper type. 3. If the issue persists, go to support.hp.com.

Toner rubs off along either edge of the

page. This defect is more common at the edges of high-coverage jobs, and on light media types, but can occur anywhere on the page. 1. Reprint the document. 2. Check the paper type in the paper tray and adjust the printer settings to match. If necessary, select a heavier paper type. 3. Enterprise models only: From the printer control panel, go to the

Edge-to-Edge menu and then select

Normal. Reprint the document. 4. Enterprise models only: From the printer control panel, select Auto

Include Margins and then reprint

the document. 5. If the issue persists, go to support.hp.com. 288 Chapter 4 Solve problems

Table 4-40 Gray background or dark print

Sample Description Possible solutions

The image or text is darker than

expected and/or the background is gray. 1. Make sure that the paper in the trays has not already been run through the printer. 2. Use a different paper type. 3. Reprint the document. 4. Mono models only: From the Home screen on the printer control panel, go to the Adjust Toner Density menu, and then adjust the toner density to a lower level. 5. Make sure that the printer is within the supported operating temperature and humidity range. 6. Replace the cartridge. 7. If the issue persists, go to support.hp.com. Table 4-41 Image placement defects

Sample Description Possible solutions

The image is not centered, or is skewed on the page. The defect occurs when the paper is not positioned properly as it is pulled from the tray and moves through the paper path. 1. Reprint the document. 2. Remove the paper and then reload the tray. Make sure that all the paper edges are even on all sides. 3. Make sure that the top of the paper stack is below the tray-full indicator. Do not overfill the tray. 4. Make sure that the paper guides are adjusted to the correct size for the paper. Do not adjust the paper guides tightly against the paper stack. Adjust them to the indentations or markings in the tray. 5. If the issue persists, go to support.hp.com. Troubleshoot image defects 289

Table 4-42 Light print

Sample Description Possible solutions

The printed content is light or faded on

the entire page. 1. Reprint the document. 2. Remove the cartridge, and then shake it to redistribute the toner.

Reinsert the toner cartridges into

the printer and close the cover. For a graphical representation of this procedure, see Replace the toner cartridges. 3. Mono models only: Make sure that the EconoMode setting is disabled, both at the printer control panel and in the print driver. 4. Make sure that the cartridge is installed correctly. 5. Print a Supplies Status Page and check the life and usage of the cartridge. 6. Replace the cartridge. 7. If the issue persists, go to support.hp.com. Table 4-43 Output defects

Sample Description Possible solutions

Printed pages have curled edges. The curled edge can be along the short or long side of the paper. Two types of curl are possible: ● Positive curl: The paper curls toward the printed side. The defect occurs in dry environments or when printing high-coverage pages. ● Negative curl: The paper curls away from the printed side. The defect occurs in high-humidity environments or when printing low coverage pages. 1. Reprint the document. 2. Positive curl: From the printer control panel, select a heavier paper type. The heavier paper type creates a higher temperature for printing. Negative curl: From the printer control panel, select a lighter paper type. The lighter paper type creates a lower temperature for printing.

Try storing the paper in a dry

environment prior to use, or use freshly opened paper. 3. Print in duplex mode. 4. If the issue persists, go to support.hp.com. 290 Chapter 4 Solve problems

Table 4-43 Output defects (continued)

Sample Description Possible solutions

The paper does not stack well in the

output tray. The stack might be uneven, skewed, or the pages might be pushed out of the tray and onto the floor. Any of the following conditions can cause this defect: ● Extreme paper curl ● The paper in the tray is wrinkled or deformed ● The paper is a non-standard paper type, such as envelopes ● The output tray is too full 1. Reprint the document. 2. Extend the output bin extension. 3. If the defect is caused by extreme paper curl, refer to the troubleshooting steps for "Output curl." 4. Use a different paper type. 5. Use freshly opened paper. 6. Remove the paper from the output tray before the tray gets too full. 7. If the issue persists, go to support.hp.com. Table 4-44 Streak defects

Sample Description Possible solutions

Light vertical streaks that usually span

the length of the page. The defect displays only in areas of fill, not in text or sections with no printed content. 1. Reprint the document. 2. Remove the cartridge, and then shake it to redistribute the toner.

Reinsert the toner cartridges into

the printer and close the cover. For a graphical representation of this procedure, see Replace the toner cartridges. 3. If the issue persists, go to support.hp.com. NOTE: Both light and dark vertical streaks can occur when the printing environment is outside the specified range for temperature or humidity. Refer to your printer's environmental specifications for allowable temperature and humidity levels. Troubleshoot image defects 291

Table 4-44 Streak defects (continued)

Sample Description Possible solutions

Dark vertical lines which occur down the

length of the page. The defect might occur anywhere on the page, in areas of fill or in sections with no printed content. 1. Reprint the document. 2. Remove the cartridge, and then shake it to redistribute the toner.

Reinsert the toner cartridges into

the printer and close the cover. For a graphical representation of this procedure, see Replace the toner cartridges. 3. Print a cleaning page. 4. Check the toner level in the cartridge. 5. If the issue persists, go to support.hp.com. 292 Chapter 4 Solve problems

Print quality troubleshooting guide

Review the following information about print quality troubleshooting.

Image defects table

Image defects, no matter what their cause, can often be resolved using the same steps. Use the following steps as a starting point for solving image defect issues. 1. Reprint the document. Print quality defects can be intermittent in nature or can go away completely with continued printing. 2. Check the condition of your cartridge(s). If it is in a Very Low state (it has passed the rated life), replace your cartridge(s). 3. Check that the driver/tray print mode settings match the media that is loaded in a given tray. Try using a different ream of media or a different tray. Try using a different print mode. 4. Verify that your printer is in within its supported operating temperature/humidity range. 5. Verify the paper type, size and weight are supported by the printer. See the printer support page at www.hp.com/support for a list of the supported paper sizes and types for your printer. NOTE: The term "fusing" refers to the part of the printing process where toner is affixed to paper. The following examples depict letter-size paper that has passed through the printer short-edge first. Table 4-45 Image defect examples

Description Sample Possible solutions

Banding

Repetitive wide-pitch banding and

impulse bands: Dark or light lines which repeat down the length of the page. They might be sharp or soft in nature. The defect displays only in areas of fill, not in text or sections with no printed content. 1. Reprint the document. 2. Try printing from another tray. 3. Replace the cartridge. 4. Use a different paper type. 5. Enterprise models only: From the

Home screen on the printer control

panel, go to the Adjust Paper Types menu, and then choose a print mode that is designed for a slightly heavier media than what you are using. This slows the print speed and might improve the print quality. 6. If the problem persists, go to www.hp.com/support. Streaks Print quality troubleshooting guide 293

Table 4-45 Image defect examples (continued)

Description Sample Possible solutions

Light vertical streaks: Light streaks that usually span the length of the page. The defect displays only in areas of fill, not in text or sections with no printed content. 1. Reprint the document. 2. Remove the cartridge, and then shake it to redistribute the toner. 3. If the problem persists, go to www.hp.com/support. NOTE: Both light and dark vertical streaks can occur when the printing environment is outside the specified range for temperature or humidity. Refer to your printer's environmental specifications for allowable temperature and humidity levels.

Dark vertical streaks and ITB cleaning

streaks (color models only): Dark lines which occur down the length of the page.

The defect might occur anywhere on the

page, in areas of fill or in sections with no printed content. 1. Reprint the document. 2. Remove the cartridge, and then shake it to redistribute the toner. 3. Print a cleaning page. 4. Check the toner level in the toner cartridge(s). 5. If the problem persists, go to www.hp.com/support. Fixing/fuser Hot fuser offset (shadow): Slight shadows, or offsets, of the image repeated down the page. The repeated image might fade with each recurrence. 1. Reprint the document. 2. Check the paper type in the paper tray and adjust the printer settings to match. If necessary, select a lighter paper type. 3. If the problem persists, go to www.hp.com/support. 294 Chapter 4 Solve problems

Table 4-45 Image defect examples (continued)

Description Sample Possible solutions

Poor fusing: Toner rubs off along either edge of page. This defect is more common at the edges of high-coverage jobs and on light media types but can occur anywhere on the page. 1. Reprint the document. 2. Check the paper type in the paper tray and adjust the printer settings to match. If necessary, select a heavier paper type. 3. Enterprise models only: From the printer control panel, go to the

Edge-to-Edge menu and then select

Normal. Reprint the document. 4. Enterprise models only: From the printer control panel, select Auto

Include Margins and then reprint

the document. 5. If the problem persists, go to www.hp.com/support.

Image placement

Margins and skew

The image is not centered, or is skewed on the page. The defect occurs when the paper is not positioned properly as it is pulled from the tray and moves through the paper path. 1. Reprint the document. 2. Remove the paper and then reload the tray. Make sure that all the paper edges are even on all sides. 3. Make sure that the top of the paper stack is below the tray full indicator. Do not overfill the tray. 4. Make sure that the paper guides are adjusted to the correct size for the paper. Do not adjust the paper guides tightly against the paper stack. Adjust them to the indentations or markings in the tray. 5. If the problem persists, go to www.hp.com/support.

Color plane registrations

Color plane registrations (color models only)

One or more color of the planes is not

aligned with the other color planes. The primary error will typically occur with yellow. 1. Reprint the document. 2. From the printer control panel, calibrate the printer. 3. If a cartridge has reached a Very

Low state or the printed output

is severely faded, replace the cartridge. 4. From the printer control panel use the Restore Calibration feature to reset the printer's calibration settings to the factory defaults. 5. If the problem persists, go to www.hp.com/support. Output Image defects table 295

Table 4-45 Image defect examples (continued)

Description Sample Possible solutions

Output curl

Printed paper has curled edges. The curled edge can be along the short or long side of the paper. Two types of curl are possible: ● Positive curl: The paper curls toward the printed side. The defect occurs in dry environments or when printing high coverage pages. ● Negative curl: The paper curls away from the printed side. The defect occurs in high-humidity environments or when printing low coverage pages. 1. Reprint the document. 2. Positive curl: From the printer control panel, select a heavier paper type. The heavier paper type creates a higher temperature for printing. Negative curl: From the printer control panel, select a lighter paper type. The lighter paper type creates a lower temperature for printing.

Try storing the paper in a dry

environment prior or use freshly opened paper. 3. Print in duplex mode. 4. If the problem persists, go to www.hp.com/support.

Output stacking

The paper does not stack well in the

output tray. The stack might be uneven, skewed, or the pages might be pushed out of the tray and onto the floor. Any of the following conditions can cause this defect: ● Extreme paper curl ● The paper in the tray is wrinkled or deformed ● The paper is a non-standard paper type, such as envelopes ● The output tray is too full 1. Reprint the document. 2. Flip out the output bin extension. 3. If the defect is caused by extreme paper curl, complete the troubleshooting steps for Output curl. 4. Use a different paper type. 5. Use freshly opened paper. 6. Remove the paper from the output tray before the tray gets too full. 7. If the problem persists, go to www.hp.com/support.

Product specific image defects

Review the following information about product specific image defects.

Repetitive image defect ruler

Review the following information about a repetitive image defect ruler. When troubleshooting the source of some print image defects, one solution is to identify if it is a repetitive defect (does the print quality defect appear multiple times on the printed page?). If this is the case, use a ruler to measure occurrences of repetitive image defects to help solve image-quality problems. For more information, see Use a ruler to measure between repetitive defects on page 297. 296 Chapter 4 Solve problems

Use a ruler to measure occurrences of repetitive image defects to help solve image-quality problems. Place the ruler next to the first occurrence of the defect on the page. Find the distance between identical defects and use the table below to identify the component that is causing the defect. IMPORTANT: Do not use solvents or oils to clean rollers. Instead, rub the roller with a lint-free cloth. If dirt is difficult to remove, rub the roller with a lint-free cloth that has been dampened with water. NOTE: Defects on the tray pickup rollers or the Tray 1 pickup roller do not cause a repetitive defect. Defects on these rollers cause a defect to appear only on the leading edge of the image. TIP: To make a printer specific repetitive defect ruler, use a metric ruler to transfer the measurements in Table 4-46 Repetitive defects on page 297 to a transparency or the edge of a piece of paper—clearly label each ruler mark with the associated defective assembly. Table 4-46 Repetitive defects

Assembly Distance between defects

Developer roller (toner cartridge) 1 27.8 mm (1.09 in) Charge roller (toner cartridge) 1 28.3 mm (1.11 in) Face down roller 36.9 mm (1.45 in) Transfer roller 39 mm (1.53 in) Duplex feed roller 40.3 mm (1.58 in)

Registration roller

Middle feed roller (cassette) Middle feed roller (optional input feeder) 43.3 mm (1.70 in) Switchback roller 43.4 mm (1.70 in) Feed roller cassette) Feed roller (optional input feeder) 47.4 mm (1.87 in) Pick roller (cassette) Pickup roller (optional input feeder) 49.7 mm (1.95 in) Pressure roller² (fuser) 63.5 mm (2.50 in) Photosensitive drum¹ (OPC drum; toner cartridge) Fuser sleeve² (fuser) 75.4 mm (2.96 in) Pick roller (Tray 1) 78.7 mm (3.09 in) 1 The primary charging roller and photosensitive drum cannot be cleaned because they are internal assemblies in the toner cartridge. If one of these assemblies is causing the defect, replace the toner cartridge. 2 The primary fuser sleeve unit or pressure roller cannot be cleaned because they are internal assemblies in the fuser. If one of these assemblies is causing the defect, replace the fuser.

Use a ruler to measure between repetitive defects

The figures in this section show color repetitive defect pages. However, the process for measuring repetitive defects is valid for mono pages. Use a ruler to measure between repetitive defects 297

1. Identify a repetitive defect on the page. TIP: Print a cleaning page to see if that resolves the defect. NOTE: Some printers allow loading Letter and A4 media in short-edge-first or long-edge-first orientation in the paper trays. When measuring repetitive defects, make sure to place the ruler at the leading edge of the page. This is the edge of the page that feeds into the printer first. The example pages below show the following types of repetitive defects. ● Lines (callout 1) ● Smudges (callout 2) ● Dots or spots (callout 3) Figure 4-103 Examples of repetitive defects 1 2 3 NOTE: These are examples only, other types of repetitive defects might appear on a page. 298 Chapter 4 Solve problems
2. Position a metric ruler on the page with the “zero” ruler mark at one occurrence of the defect (callout 1). Figure 4-104 Place the ruler on the page 1 3. Locate the next occurrence of the defect (callout 1). Figure 4-105 Locate the next repetitive defect 1 Use a ruler to measure between repetitive defects 299

4. Measure the distance (in millimeters) between the two occurrences (callout 1), and then use the Repetitive defects table to determine the defective assembly. TIP: Always measure from and to the same point on the defects. For example, if the ruler is “zeroed” at the top edge of a defect, measure to the top edge of the next occurrence of that defect. Figure 4-106 Determine the defective assembly 1

Print-quality troubleshooting

Review the following information about print-quality troubleshooting. NOTE: The image defects described in this section are printer-specific image defects.

Image defect events

Review the following information about image defect events. NOTE: The image defects described in this section are defects that occurred during development and that could not be fully eliminated. Customers might observe these defects.

Graininess

Review the following information about graininess image defect events. Graininess/fixing mottle Review the following information about the graininess/fixing mottle image defect event. 300 Chapter 4 Solve problems

Figure 4-107 Graininess/fixing mottle

Description of the defect

This defect is large areas of fill that appear blotchy, grainy, non-uniform, or mottled (marked with spots or smears of color). The graininess/fixing mottle defect is caused by how the toner (dot) is pressed or melted and the severity of the defect depends on the paper roughness. This leads to the density variance or uneven gloss. Graininess/fixing mottle is caused by inconsistently melted toner.

Conditions that can cause the defect

The defect appears in large areas of fill and is more severe in low temperature and humidity environments.

Solutions for the defect

Use the following steps to solve the defect. ● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. Description of the defect 301

– Paper Type 2. Select Intermediate 85–95g. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray – Modify 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray <X> NOTE: Select the tray you want to configure. – Type 2. Select Intermediate 85–95g. ● Try using a smoother paper.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Toner cartridge ● Fuser Streaks Review the following information about streak image defect events. Vertical streaks - high temperature/humidity Review the following information about the vertical streaks - high temperature/humidity image defect event. 302 Chapter 4 Solve problems

Figure 4-108 Vertical streaks - high temperature/humidity

Description of the defect

Light streaks can appear and they usually span the length of the page. This problem is caused by toner clumping at the developer blade which reduces the amount of toner available for transfer at that page location. There is a stirring shaft inside the toner cartridge to reduce the severity of this defect. This defect appears in areas of fill, not in text.

Conditions that can cause the defect

This defect primarily occurs in high temperature and high humidity environments.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. Description of the defect 303

● If the defect does not improve within two to three print jobs, remove the toner cartridge and gently rock it back and forth from side to side (this distributes the toner evenly in the toner cartridge). ● If the toner cartridge has reached the Very Low state, replace it. ● If the defect persists, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge Banding Review the following information about banding image defect events. Dark streaks (early in toner cartridge life) Review the following information about the

dark streaks early in toner cartridge life image defect event. Figure 4-109 Dark streaks (early in toner cartridge life) 304 Chapter 4 Solve problems

Description of the defect

Smearred dark streaks come and go down the page, and are more common at the right and left edges of the page. This issue is caused by grease from toner cartridge stirring system contaminating the supply hopper which then causes the toner to cluster and results in dark streaks on the page. Countermeasures include optimizing grease: amount, position, and stickiness. This defect appears in areas of fill, not in text.

Conditions that can cause the defect

This defect occurs when using a new toner cartridge and usually decreases over toner cartridge life.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. ● If the defect does not improve within two to three print jobs, remove the toner cartridge and gently rock it back and forth from side to side (this distributes the toner evenly in the toner cartridge). ● If the defect persists, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge

Fine-pitch banding

Review the following information about the fine-pitch banding image defect event. Description of the defect 305

Figure 4-110 Fine-pitch banding

Description of the defect

This defect appears as alternating light and dark (evenly spaced) repetitive horizontal lines. The 0.93 mm (0.03 in) fine-pitch banding is caused by the main drive motor gear tooth engagement. This defect appears in areas of fill, not in text.

Conditions that can cause the defect

The alignment of the motor gear and drum drive gear is the important factor for this banding and specifically the accuracy of the motor installed surface and accuracy of the bearing shaft position at the both sides of the drum drive gear shaft.

Solutions for the defect

Use the following steps to solve the defect. ● Try using a smoother paper. 306 Chapter 4 Solve problems

● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. – Paper Type 2. Select Intermediate 85–95g. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray – Modify – Next 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray <X> NOTE: Select the tray you want to configure. – Type 2. Select Intermediate 85–95g.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge ● Printer engine (whole unit replacement) Wide-pitch banding 27.8 mm (1.09 in) Review the following information about the wide-pitch banding (27.8 mm (1.09 in)) image defect event. Parts related to the defect 307

Figure 4-111 Wide-pitch banding 27.8 mm (1.09 in)

Description of the defect

This defect appears as soft, gradual bands can be seen over a constant density area. It appears as slight gradients which repeat at approximately a 27.8 mm (1.09 in) pitch. This defect appears in areas of fill, not in text.

Conditions that can cause the defect

This defect is a density variation which appears on the PQ due to the gap variation between the OPC and the developer sleeve. This gap varies due to the accuracy of the OPC, developer sleeve, and the spacer placed between the OPC and the developer sleeve.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. ● If the defect persists, replace the toner cartridge. 308 Chapter 4 Solve problems

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge AC banding 4.6 mm (0.18 in) Review the following information about the AC banding (4.6 mm (0.18 in)) image defect event. Figure 4-112 AC banding

Description of the defect

This defect appears as alternating light and dark bands that repeat at 4.6 mm (0.18 in) pitch might be visible starting midway down the page in high-temperature and high-humidity environments. This defect is caused by the transfer current getting pulled into the fusing area through the paper from the fuser heater voltage supply. The transfer banding occurs at the power supply voltage frequency. A capacitor was added to the transfer front guide to improve the level as a countermeasure during development. Parts related to the defect 309

Conditions that can cause the defect

This defect usually occurs when printing in Plain print mode in high-humidity environments on fully acclimated (low resistivity) paper, which makes the transfer current flow to the fusing area more easily. The defect occurs more often with 220V units, which have higher impressed voltage to the fuser heater. This defect appears in areas of fill, not in text.

Solutions for the defect

Use the following steps to solve the defect. ● Try printing the job on a newly opened supply of paper. ● Cool the printer environment, and then resend the print job. ● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. – Paper Type 2. Select Intermediate 85–95g. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray – Modify – Next 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray <X> NOTE: Select the tray you want to configure. 310 Chapter 4 Solve problems – Tray 2. Select Intermediate 85–95g.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Fuser OPC sharp bands at 75mm (2.95 in) pitch (version 1) Review the following information about the OPC sharp bands at 75mm (2.95 in) pitch (version 1) image defect event. Figure 4-113 OPC sharp bands (version 1) Parts related to the defect 311

Description of the defect

This defect appears as sharp bands which repeat at a 75mm (2.95 in) pitch down the page as a light/dark line. Waste products (for example, toner or paper) accumulate at the contact position between the C-blade and the OPC. When the OPC stops, the waste material can form a light attachment to the OPC. When the OPC begins to turn again, the material removal can affect the OPC charge for several rotations. This defect appears in areas of fill, not in text. NOTE: This defect is also called C-blade blur and is common to many color and mono products. This defect looks similar to OPC sharp bands version 2, but appears to be light/dark.

Conditions that can cause the defect

This defect might occur any time the toner cartridge is idle for long periods of time.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. The defect should fade with subsequent printed pages. ● If the defect persists, replace the toner cartridge

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge OPC sharp bands at 75 mm (2.95 in) pitch (version 2) Review the following information about the OPC sharp bands at 75mm (2.95 in) pitch (version 2) image defect event. 312 Chapter 4 Solve problems

Figure 4-114 OPC sharp bands (version 2)

Description of the defect

This defect appears as sharp bands which repeat at a 75 mm (2.95 in) pitch down the page as a dark line. The OPC drum is rubbed by the cleaning blade or charge roller due to vibration or impact. The rubbing effects the OPC charge but the defect fades with usage and time. This defect appears in areas of fill, not in text. NOTE: This defect looks similar to OPC sharp bands version 1, but appears to be dark.

Conditions that can cause the defect

This defect can occur at the beginning of toner cartridge use, or if the cartridge is subject to vibration (for example, transportation).

Solutions for the defect

Use the following steps to solve the defect. Description of the defect 313

- Resend the print job. The defect should fade with subsequent printed pages.
- If the defect persists, replace the toner cartridge

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge Impulse band 49-50 mm (1.92-1.96 in) from the leading edge Review the following information about the impulse band 49-50 mm (1.92-1.96 in) from the leading edge image defect event. Figure 4-115 Impulse band (leading edge)

Description of the defect

This defect appears as a dark and usually sharp band and occurs 49-50 mm (1.92-1.96 in) from the leading edge of the page. 314 Chapter 4 Solve problems

This impulse band is caused by the trailing edge of the previous page touching the OPC as it leaves the transfer area, which causes a memory charge disturbance. This defect appears in areas of fill, not in text.

Conditions that can cause the defect

This defect only occurs during continuous printing, on the second and subsequent pages of a job.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. Impulse bands can be variable. ● M501: If the defect appears during continuous printing, from the printer control panel enable the Less Paper Curl print mode. NOTE: This mode adds a rotation of the OPC drum to the inter-page gap timing that conditions the drum and minimizes the defect. 1. Open the following menus: – Setup – Service – Less Paper Curl 2. Select On. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode to Paper Curl Mode. NOTE: This mode adds a rotation of the OPC drum to the inter-page gap timing that conditions the drum and minimizes the defect. 1. Open the following menus: – Administration – General Settings – Print Quality – Adjust Paper Types 2. Select Paper Curl Mode. ● M507/E50145 and M528/E52645: change the print mode to Paper Curl Mode. NOTE: This mode adds a rotation of the OPC drum to the inter-page gap timing that conditions the drum and minimizes the defect. 1. Open the following menus: – Settings – Copy/Print – Print Quality Conditions that can cause the defect 315

– Adjust Paper Types 2. Select the appropriate paper type for the paper used. 3. Select Paper Curl Mode.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Toner cartridge Impulse band 75-82 mm (2.95-3.22 in) from the trailing edge Review the following information about the impulse band 75-82 mm (2.95-3.22 in) from the trailing edge image defect event. Figure 4-116 Impulse band (trailing edge) 316 Chapter 4 Solve problems

Description of the defect

This defect appears as a dark band and occurs 75-82 mm (2.95-3.22 in) from the trailing edge of the page. This impulse band is a development blur which occurs when the paper trailing edge goes through the registration roller. This defect appears in areas of fill, not in text.

Conditions that can cause the defect

During the print job, the transfer top guide might be vibrating.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. Impulse bands can be variable. ● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. – Paper Type 2. Select Intermediate 85–95g. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray – Modify – Next 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays Description of the defect 317

– Tray <X> NOTE: Select the tray you want to configure. – Type 2. Select Intermediate 85–95g.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Registration assembly Impulse band 102 mm (4.01 in) from the trailing edge Review the following information about the impulse band 102 mm (4.01 in) from the trailing edge image defect event. 318 Chapter 4 Solve problems

Figure 4-117 Impulse band (trailing edge) 102 mm (4.01 in)

Description of the defect

This defect appears as a soft dark band and occurs 102 mm (4.01 in) from the trailing edge of the page. The cause of this defect is poor sliding performance of the transfer bearing (TR bearing) and transfer roller shaft.

Conditions that can cause the defect

This defect is more likely to occur in low temperature/low humidity environments and at cold start conditions.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. Impulse bands can be variable. ● Try using a different media type.

Description of the defect 319

● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. – Paper Type 2. Select Intermediate 85–95g. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray – Modify – Next 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray <X> NOTE: Select the tray you want to configure. – Type 2. Select Intermediate 85–95g.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Transfer roller 320 Chapter 4 Solve problems

Image defect (developmental) events Review the following information about image defect (developmental) events. NOTE: The image defects described in this section are defects that occurred during printer development, and HP implemented a solution to eliminate them. Customers should not observe these defects. These defects are included for the unlikely event that they reoccur.

Cleaning defect events

Review the following information about cleaning image defect (developmental) events.

Rain-toner attached to the OPC

Review the following information about the rain-toner attached to the OPC cleaning defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Image defect (developmental) events 321

Figure 4-118 Rain-toner attached to the OPC

Description of the defect

This defect appears as very small white voids which appear in solid black areas. This issue occurs when toner attached to the OPC cannot be removed by the cleaning blade. This defect appears in areas of solid fill, not in text.

Conditions that can cause the defect

This defect is likely to occur in the later stages of toner cartridge life.

Solutions for the defect

Use the following step to solve the defect. ● Verify that the toner is within the expected life. If not, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. 322 Chapter 4 Solve problems

● Toner cartridge Developer defect repeats at 27.8 mm (1.09 in) pitch Review the following information about the developer defect repeats at 27.8 mm (1.09 in) pitch cleaning defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-119 Developer defect

Description of the defect

This defect appears as a soft white spot which repeats at 27.8 mm (1.09 in) (developer roller) pitch. This defect appears in areas of solid fill, not in text.

Conditions that can cause the defect

This defect is caused by a cartridge cleaning performance issue caused by contamination. Developer defect repeats at 27.8 mm (1.09 in) pitch 323

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job (the defect should fade with subsequent printed pages). ● If the defect persists, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge

Random missing toner

Review the following information about the random missing toner cleaning defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-120 Random missing toner 324 Chapter 4 Solve problems

Description of the defect

This defect appears as white spots which appear in solid black areas (sometimes with a tail). This defect appears in areas of solid fill, not in text.

Conditions that can cause the defect

This defect is caused by contaminants attaching to the OPC. If the contaminant is large, the size of missing toner will be large.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. This defect is highly variable.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge Part/assembly defect events Review the following information about part/assembly image defect (developmental) events.

OPC gear slip

Review the following information about the OPC gear slip part/assembly defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Description of the defect 325

Figure 4-121 OPC gear slip

Description of the defect

This defect appears as severe banding. This defect is caused by OPC gear slip (a failed connection between the OPC gear and the drum).

Conditions that can cause the defect

This defect is not dependent on environmental conditions.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. ● If the defect persists, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge 326 Chapter 4 Solve problems

OPC wide-pitch banding

Review the following information about the OPC wide-pitch banding part/assembly defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-122 OPC wide-pitch banding

Description of the defect

This defect appears as overlapping sets of wide-pitch bands (repeating at 75 mm (2.95 in)).

Conditions that can cause the defect

This defect is caused by OPC deformation during the assembly process.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job (the defect should fade with subsequent printed pages). OPC wide-pitch banding 327

- If the defect persists, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge

Transfer bias defect events

Review the following information about the transfer bias image defect (developmental) events.

Leading edge - mid-page toner scatter

Review the following information about the leading edge - mid-page toner scatter transfer bias defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-123 Leading edge - mid-page toner scatter 328 Chapter 4 Solve problems

Description of the defect

This defect appears as a non-uniform or smear that might appear in the middle of the page at the top of a page. This occurs when the toner scatters at transfer in the area where the media rubs the pick roller and the middle roller. This defect is more likely to occur when media with high resistivity is used. The static charge eliminator (metal brush) on the transfer guide keeps this defect from occurring.

Conditions that can cause the defect

This defect might occur in low temperature/low humidity environments and in cold start conditions.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job (the defect should fade with subsequent printed pages as the printer warms up). ● Try using a different media type.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Printer engine (whole unit replacement) Density change at 75 mm (2.95 in) from the leading edge Review the following information about the density change at 75 mm (2.95 in) from the leading edge transfer bias defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Description of the defect 329

Figure 4-124 Density change

Description of the defect

This defect appears as a light image in the first 75 mm (2.95 in) of the page. This defect is created by a charge ghost due to the lack of charge current.

Conditions that can cause the defect

This defect happens when there is high print coverage across the longitudinal direction and charging is not able to reach the target potential due to the lack of current that charges OPC in the area and the density becomes high at the next OPC rotation. This shows up in areas of fill, not in text.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. 330 Chapter 4 Solve problems

- No parts are related to this defect.

Transfer issue - random voids

Review the following information about the transfer issue - random voids transfer bias defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-125 Transfer issue - random voids

Description of the defect

This defect appears as very small sections of missing toner in random locations on the page. The transfer bias was too strong and caused the missing toner.

Conditions that can cause the defect

This defect is not dependent on environmental conditions. Transfer issue - random voids 331

Solutions for the defect

Use the following steps to solve the defect. ● All except M501: If the defect is persistent, reprint your document with a lower transfer bias setting. 1. Open the following menus: – Administration – General Settings – Print Quality – Adjust Paper Types 2. Select Resistance Mode. ● M507/E50145 and M528/E52645: If the defect is persistent, reprint your document with a lower transfer bias setting. 1. Open the following menus: – Settings – Copy/Print – Print Quality – Adjust Paper Types 2. Select the appropriate paper type for the paper used. 3. Select Resistance Mode.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● No parts are related to this defect.

Toner leak defect events

Review the following information about the toner leak image defect (developmental) events.

Right to left fade and banding

Review the following information about the right to left fade and banding toner leak defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. 332 Chapter 4 Solve problems

Figure 4-126 Right to left fade and banding

Description of the defect

This defect appears as fading and banding at an edge and occurs when the cartridge seals leak allowing toner into the gap between the OPC drum and the SD roller/sleeve. This appears in areas of fill, not in text

Conditions that can cause the defect

This defect is likely to occur in the later stages of toner cartridge life.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. ● Verify that the toner is within the expected life, if not replace the toner cartridge. ● If the defect persists, replace the toner cartridge. Description of the defect 333

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge Cartridge fine pitch (1.5 mm (0.05 in)) banding Review the following information about the cartridge fine pitch (1.5 mm (0.05 in)) banding toner leak defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-127 Cartridge fine pitch banding 334 Chapter 4 Solve problems

Description of the defect

This defect appears as alternating light and dark, evenly spaced, repetitive horizontal lines at a 1.5 mm (0.05 in) pitch. This defect is caused by toner leaking in the drive gear area of the toner cartridge. This appears in areas of fill, not in text.

Conditions that can cause the defect

This defect is likely to occur in the later stages of toner cartridge life.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. ● Verify that the toner is within the expected life, if not replace the toner cartridge. ● If the defect persists, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge 382 mm (15.03 in) + IPG repeating defect Review the following information about the 382 mm (15.03 in) + IPG repeating toner leak defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Description of the defect 335

Figure 4-128 IPG repeating defect 336 Chapter 4 Solve problems

Description of the defect

This defect appears as a dark crescent moon shape defect and it appears in non-print areas of the page. The defect occurs when the toner is thrown outside the toner cartridge at the stirring pitch due to the waviness of the blow out seal. The waviness creates a gap between sleeve and blow out seal. The cause of the blow out seal waviness is that the edge of blow out seal contacts with the edge of the sleeve/drum spacer.

Conditions that can cause the defect

This defect is not dependent on environmental conditions.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. ● If the defect does not improve within two to three print jobs, remove the toner cartridge and gently rock it back and forth from side to side (this distributes the toner evenly in the toner cartridge). ● If the defect persists, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge

Paper path impulse defect events

Review the following information about paper path impulse image defect (developmental) events. Impulse band 15 mm (0.59 in) from the leading edge Review the following information about the impulse band 15 mm (0.59 in) from the leading edge paper path impulse defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Description of the defect 337

Figure 4-129 Impulse band leading edge

Description of the defect

This defect appears as a dark and usually sharp band occurs 15 mm (0.59 in) from the leading edge of the page. This band shows up in areas of fill, not in text.

Conditions that can cause the defect

The defect occurs when the paper leading edge goes into the transfer area.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. Impulse bands can be variable. ● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: 338 Chapter 4 Solve problems – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. – Paper Type 2. Select Intermediate 85–95g. ● Try using a different media type. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray – Modify – Next 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray <X> NOTE: Select the tray you want to configure. – Type 2. Select Intermediate 85–95g.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Registration assembly Parts related to the defect 339

Toner in the leading edge margin (fuser slap) Review the following information about the toner in the leading edge margin (fuser slap) paper path impulse defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-130 Toner in the leading edge margin (fuser slap)

Description of the defect

This defect appears as scattered toner which occurs at the leading edge of the page. This defect occurs when the page enters the fuser and the page touches the fuser film which rubs the image on the page.

Conditions that can cause the defect

This defect is not dependent on environmental conditions, but is more likely to occur on pages with small top margins or when using curled media. 340 Chapter 4 Solve problems

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. This defect can be variable. ● Try using a different media type. ● M501: From the printer control panel, increase the leading edge margin. Before beginning, use the Print Test Page option to print a registration test page. This page provides alignment guides in the X and Y directions. 1. Open the following menus: – Setup – System Setup – Print Quality – Adjust Alignment – Adjust Tray <X> NOTE: Select the tray you want to configure. 2. Adjust the Y Shift setting. NOTE: Adjust the “Y” value in .25 mm increments. ● M506/E50045 and M527/E52545: From the printer control panel, increase the leading edge margin. Before beginning, use the Print Test Page option to print a registration test page. This page provides alignment guides in the X and Y directions. 1. Open the following menus: – Administration – General Settings – Print Quality – Image Registration – Adjust Tray <X> NOTE: Select the tray you want to configure. 2. Adjust the Y1 Shift (simplex pages) and/or Y2 Shift (duplex pages) settings. NOTE: Adjust the “Y” values in .25 mm increments. ● M507/E50145 and M528/E52645: From the printer control panel, increase the leading edge margin. Before beginning, use the Print Test Page option to print a registration test page. 1. Open the following menus: Solutions for the defect 341

– Settings – Copy/Print – Print Quality – Image Registration 2. Select the tray you want to configure. 3. Adjust the Front-side Vertical Shift (simplex pages) and/or Back-side Vertical Shift (duplex pages). NOTE: Adjust the Vertical Shift values in .25 mm increments.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Fuser 6-7 mm (0.23-0.27 in) wide-pitch banding Review the following information about the 6-7 mm (0.23-0.27 in) wide-pitch banding paper path impulse defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. 342 Chapter 4 Solve problems

Figure 4-131 Wide-pitch banding 6-7 mm (.23-.27 in)

Description of the defect

This defect appears as soft bands that repeat at 6-7 mm (0.23-0.27 in). This banding occurs when the transfer top guide vibrates, which is conveyed to the page as well as the page at the transfer area, causing the uneven density. The left and right edges of the back side of transfer top guide contact with the positioning guide at the transfer frame by the spring at the right edge of the pivot shaft. Since the spring is used only at the right edge, the left side of the alignment feature receives smaller pressing force. When the pressing force is small, the sheet metal vibrates easily. This band shows up in areas of fill, not in text.

Conditions that can cause the defect

During the print job, the transfer top guide might be vibrating.

Solutions for the defect

Use the following steps to solve the defect. Description of the defect 343

● Try using a different media type. ● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. – Paper Type 2. Select Intermediate 85–95g. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray – Modify – Next 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray <X> NOTE: Select the tray you want to configure. – Type 2. Select Intermediate 85–95g.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Transfer assembly 344 Chapter 4 Solve problems

Fuser/fixing defect events Review the following information about the fuser/fixing image defect (developmental) events.

Fuser blisters

Review the following information about the fuser blisters fuser/fixing defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-132 Fuser blisters

Description of the defect

This defect appears as repeating print defects visible in the image at the fuser pitch of 75.4 mm (2.96 in). These defects were caused by contamination between the films on the fuser. Fuser/fixing defect events 345

Figure 4-133 Contaminated fuser film

Conditions that can cause the defect

A defect similar in appearance could occur if a page with sharp objects (for example, staples) is put through the printer.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. ● If the defect persists, replace the fuser.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Fuser

Hot fuser offset

Review the following information about the hot fuser offset fuser/fixing defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. 346 Chapter 4 Solve problems

Figure 4-134 Hot fuser offset

Description of the defect

This defect appears as slight offset of the image that repeats down the page spaced approximately 75.0 mm (2.95 in) apart. This defect occurred when the fuser thermistor detection temperature did not follow the target temperature.

Conditions that can cause the defect

This defect is likely to occur in the later stages of fuser life.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job and use a lighter fuser mode. ● If the defect persists, replace the fuser. Description of the defect 347

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Fuser

Poor edge fixing - within the image assurance area

Review the following information about the poor edge fixing - within the image assurance area fuser/fixing defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-135 Poor edge fixing - within the image assurance area 348 Chapter 4 Solve problems

Description of the defect

This defect appears as missing toner inside the 6 mm (0.23 in) image assurance area. Toner rubs off the page. The fuser temperature is too cold. NOTE: The image assurance area is 5 mm (0.19 in) from the top and bottom edge of the page and 5 mm (0.19 in) from the edge on each side of the page.

Conditions that can cause the defect

This defect is not dependent on environmental conditions, but is more common at the edges of high coverage print jobs.

Solutions for the defect

Use the following steps to solve the defect. ● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. – Paper Type 2. Select Intermediate 85–95g. ● M506/M507/E50145 and M527/M528/E52645: Confirm that the correct print mode is selected for the media being used. Use a print mode for a heavier media type. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray – Modify – Next 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays Description of the defect 349

– Tray <X> NOTE: Select the tray you want to configure. – Type 2. Select Intermediate 85–95g.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Fuser

Poor edge fixing - outside the image assurance area

Review the following information about the poor edge fixing - outside the image assurance area fuser/ fixing defect event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. 350 Chapter 4 Solve problems

Figure 4-136 Poor edge fixing - outside the image assurance area

Description of the defect

This defect appears as missing toner outside the 6 mm (0.23 in) image assurance area. Toner rubs off the page. Poor edge fixing might occur when the printer is using Edge to Edge print mode and the image extends past the 5 mm (0.19 in) image assurance area. NOTE: The image assurance area is 5 mm (0.19 in) from the top and bottom edge of the page and 5 mm (0.19 in) from the edge on each side of the page.

Conditions that can cause the defect

This defect is not dependent on environmental conditions, but is more common at the edges of high coverage print jobs.

Solutions for the defect

Use the following steps to solve the defect. Description of the defect 351

● M506/E50045 and M527/E52545: Resend the print job with Edge to Edge mode set to Normal. From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Administration – Default print options – Edge to Edge 2. Select Normal. ● M507/E50145 and M528/E52645: Resend the print job with Edge to Edge mode set to Normal. 1. Open the following menus: – Settings – Copy/Print – Default Print Options – Edge-to-Edge 2. Select Normal. ● M501: Resend the print job with increased side margins. From the printer control panel, adjust the page margins. Before beginning, use the Print Test Page option to print a registration test page. This page provides alignment guides in the X and Y directions. 1. Open the following menus: – Setup – System Setup – Print Quality – Adjust Alignment – Adjust Tray <X> NOTE: Select the tray you want to configure. 2. Adjust the X1 Shift and X2 Shift settings. NOTE: Adjust the “X” values in .25 mm increments. ● M506/E50045 and M527/E52545: Resend the print job with increased side margins. From the printer control panel, adjust the page margins. Before beginning, use the Print Test Page option to print a registration test page. This page provides alignment guides in the horizontal and vertical directions. 1. Open the following menus: 352 Chapter 4 Solve problems – Administration – General Settings – Print Quality – Image Registration – Adjust Tray <X> NOTE: Select the tray you want to configure. 2. Adjust the X1 Shift (simplex pages) and/or X2 Shift (duplex pages). NOTE: Adjust the “X” values in .25 mm increments. ● M507/E50145 and M528/E52645: Resend the print job with increased side margins. From the printer control panel, adjust the page margins. Before beginning, use the Print Test Page option to print a registration test page. This page provides alignment guides in the horizontal and vertical directions. 1. Open the following menus: – Settings – Copy/Print – Print Quality – Image Registration 2. Select the tray you want to configure. 3. Adjust the Front-side Horizontal Shift (simplex pages) and/or Back-side Horizontal Shift (duplex pages). NOTE: Adjust the Horizontal Shift values in .25 mm increments. ● M501: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Setup – System Setup – Paper Setup – Tray <X> NOTE: Select the tray you want to configure. – Paper Type 2. Select Intermediate 85–95g ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). Solutions for the defect 353

1. Open the following menus: – Trays – Tray – Modify – Next 2. Select Intermediate 85–95g. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode from plain to intermediate (this slows the print speed). 1. Open the following menus: – Trays – Tray <X> NOTE: Select the tray you want to configure. – Type 2. Select Intermediate 85–95g.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Fuser

Miscellaneous defect events

Review the following information about miscellaneous image defect (developmental) events.

Uneven density - across the page

Review the following information about the uneven density - across the page image defect (developmental) event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. 354 Chapter 4 Solve problems

Figure 4-137 Uneven Density - across the page

Description of the defect

This defect appears as areas of varying (light/dark) density across the page. This is caused by variation in the position of the stirring system at initial cartridge assembly resulting in toner material attached to the D-blade. Uneven attachment of toner component to D-blade is caused by uneven toner coating on the sleeve during initial sequence. This banding shows up in areas of fill, not in text.

Conditions that can cause the defect

This defect is likely to occur in the later stages of toner cartridge life.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. ● If the defect does not improve within two to three print jobs, remove the toner cartridge and gently rock it back and forth from side to side (this distributes the toner evenly in the toner cartridge). Description of the defect 355

● If the defect persists, replace the toner cartridge.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Toner cartridge Water drop (condensation) Review the following information about the water drop (condensation) image defect (developmental) event. NOTE: This defect occurred at some point during printer development. HP has implemented a solution to the problem. This defect is included in the image defects section for the unlikely event that it reoccurs. Figure 4-138 Water drop (condensation) 356 Chapter 4 Solve problems

Description of the defect

This defect appears as short, soft, light streaks at the top of a page. Condensation collects on the fuser exit guide ribs and transfers to the trailing (unprinted side) of the first pass page of a duplex job. When the duplexed page is printed, the condensation affects the transfer of the toner and causes the defect.

Conditions that can cause the defect

This defect might occur in low temperature high humidity environments and in cold start conditions.

Solutions for the defect

Use the following steps to solve the defect. ● Resend the print job. This defect can be variable. ● M506/E50045 and M527/E52545: If the defect does not improve within two to three print jobs, change the print mode Moisture Control from Normal to Alternate. 1. Open the following menus: – Administration – General Settings – Print Quality – Optimize – Moisture Control 2. Select Alternate. ● M507/E50145 and M528/E52645: If the defect does not improve within two to three print jobs, change the print mode Moisture Control from Normal to Alternate. 1. Open the following menus: – Settings – Copy/Print – Print Quality – Optimize – Moisture Control 2. Select Alternate.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Fuser Description of the defect 357

Other events

Review the following information about other print-quality events. NOTE: The examples in this section describe other printer events that can occur. Customers might observe these problems.

Fuser contamination

Review the following information about the fuser contamination event. Figure 4-139 Fuser contamination

Description of the defect

This defect appears as dark globs of toner observed on the front or back of the page. Toner builds up on the fuser sleeve and pressure roller and comes off on subsequent pages. This defect can be either repeating at 75 mm (2.95 in) or does not repeat. 358 Chapter 4 Solve problems

Conditions that can cause the defect

This defect might occur in the following conditions: ● Inappropriate media type print mode is selected (fuser too cold) ● Fuser jams were observed on the previous jobs ● Narrow media was printed for extended periods of time

Solutions for the defect

Use the following steps to solve the defect. ● M501: From the printer control panel, process a cleaning page to remove the contamination in the fuser. NOTE: If a repeating defect does not disappear after processing multiple cleaning pages, a permanent defect might be present on the fuser sleeve. – Open the following menus: ■ Setup ■ Service ■ Cleaning Page NOTE: Touch the OK button to begin the cleaning process. Wait until the process completes. Discard the page that prints. ● M506/E50045 and M527/E52545: From the printer control panel, process a cleaning page to remove the contamination in the fuser. NOTE: If a repeating defect does not disappear after processing multiple cleaning pages, a permanent defect might be present on the fuser sleeve. 1. Open the following menus: – Device Maintenance – Calibration/Cleaning – Cleaning Page 2. Select Print. NOTE: The process takes up to 1.5 minutes. ● M507/E50145 and M528/E52645: From the printer control panel, process a cleaning page to remove the contamination in the fuser. NOTE: If a repeating defect does not disappear after processing multiple cleaning pages, a permanent defect might be present on the fuser sleeve. 1. Open the following menus: – Support

Tools Conditions that can cause the defect 359

– Maintenance – Calibration/Cleaning – Cleaning Page 2. Select Print. NOTE: The process takes up to 1.5 minutes.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Fuser

Image placement - margins and skew

Review the following information about the image placement - margins and skew event. 360 Chapter 4 Solve problems

Figure 4-140 Image placement - margins and skew

Description of the defect

This defect appears as image that is not centered or skewed on the page. The media is not positioned properly as it is pulled from the tray and goes through the paper path.

Conditions that can cause the defect

This defect might occur if the IPA is off in any environment and from any tray.

Solutions for the defect

Use the following steps to solve the defect. ● Before loading paper in the tray, hold the stack of paper, and then tap the bottom edge on a flat surface so that the stack is even on all sides. ● Make sure that the paper guides are adjusted to the correct size for the selected paper and do not overfill the tray. Verify that the top of the stack is below the tray full indicator. Description of the defect 361

● Do not adjust the paper guides tightly against the paper stack. Adjust them to the indentations or markings in the tray. ● M501: Before beginning, use the Print Test Page option to print a registration test page. This page provides alignment guides in the X and Y directions. 1. Open the following menus: – Setup – System Setup – Print Quality – Adjust Alignment – Adjust Tray <X> NOTE: Select the tray you want to configure. 2. Adjust the appropriate margin settings. NOTE: Adjust the “X” and “Y” values in .25 mm increments. – X1 Shift NOTE: Registration of the image on the paper from side to side, as the paper lies in the tray. For duplex models, this side is the second side (back) of the paper. – X2 Shift NOTE: Registration of the image on the paper from side to side, as the paper lies in the tray, for the first side (front) of a duplexed page. This item appears only on duplex models. – Y Shift NOTE: Registration of the image on the paper from top to bottom as the paper lies in the tray. ● M506/E50045 and M527/E52545: For image placement defects do the following. Resend the print job. If the error persists, change the image placement settings for the tray that the defect occurs. From the printer control panel, process a cleaning page to remove the contamination in the fuser. Before beginning, use the Print Test Page option to print a registration test page. This page provides alignment guides in the X and Y directions. 1. Open the following menus: – Administration – General Settings – Print Quality – Registration 362 Chapter 4 Solve problems

– Adjust Tray <X> NOTE: Select the tray you want to configure. 2. Adjust the appropriate margin settings. NOTE: Adjust the “X” and “Y” values in .25 mm increments. ● X1 (horizontal; simplex) ● X2 (vertical; simplex) ● Y1 (horizontal; duplex) ● Y2 (vertical; duplex) ● M507/E50145 and M528/E52645: For image placement defects do the following. Resend the print job. If the error persists, change the image placement settings for the tray that the defect occurs. From the printer control panel, process a cleaning page to remove the contamination in the fuser. Before beginning, use the Print Test Page option to print a registration test page. This page provides alignment guides in the X and Y directions. 1. Open the following menus: – Settings – Copy/Print – Print Quality – Image Registration 2. Select the tray you want to configure. 3. Adjust the appropriate margin settings. Front-side settings are for simplex pages, and back-side settings are for duplex pages. NOTE: Adjust the Vertical and Horizontal Shift values in .25 mm increments.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Tray guides ● Registration assembly ● Feed rollers

Output curl

Review the following information about the output curl event. Parts related to the defect 363

Figure 4-141 Output curl

Description of the defect

This defect appears as printed pages are curled. Two types of curl are possible: ● Positive curl: paper curls toward the printed side ● Negative curl: paper curls away from printed side This defect is caused by uneven drying conditions for the two sides of the paper and fuser location close to the paper output.

Conditions that can cause the defect

Positive curl occurs in dry environments or when printing high coverage pages. Negative curl occurs in high humidity environments when printing low coverage pages.

Solutions for the defect

Use the following steps to solve the defect.

General curl defect solutions

364 Chapter 4 Solve problems

● Resend the print job. Use the duplex option in the printer driver. ● Do not adjust the paper guides tightly against the paper stack. Adjust them to the indentations or markings in the tray. ● M501: From the printer control panel, enable the Paper Curl print mode option. 1. Open the following menus: – Setup – Service – Less Paper Curl 2. Select On. ● M506/E50045 and M527/E52545: From the printer control panel, change the print mode Paper Curl from Normal to Reduced. 1. Open the following menus: – Administration – General Settings – Print Quality – Optimize – Paper Curl 2. Select Reduced. ● M507/E50145 and M528/E52645: From the printer control panel, change the print mode Paper Curl from Normal to Reduced. 1. Open the following menus: – Settings – Copy/Print – Print Quality – Adjust Paper Types 2. Select the appropriate paper type for the paper used. 3. Select Paper Curl Mode. 4. Select Reduced.

Specific curl defect solutions

● Positive curl: Use a higher temperature media type (heavier weight) mode which will increase the negative curl tendency reducing the positive curl. Also, more fusing could be selected in the extended print modes for the selected paper type. Solutions for the defect 365
● Negative curl: Use a lower temperature media type (lighter weight) mode or select a reduced paper curl or lower fusing temperature setting in extended print modes. If possible, try storing media in a dry environment prior to printing or use freshly opened paper (non-acclimated).

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Fuser

Sticky output

Review the following information about the sticky output event. Figure 4-142 Sticky output

Description of the defect

This defect appears as output pages that stick together when printing heavy coverage images in duplex mode. When printing large jobs the output stack retains enough heat that the toner on the top page does not cool sufficiently and can be tacky in heavy coverage areas. When the next page is output if there is heavy coverage that lands on the top page in the stack, the toner-to-toner contact might stick.

Conditions that can cause the defect

This defect might occur in the following conditions: ● Large jobs greater than 25 pages (50 Images) ● Plain fuser mode ● Back-to-back images with heavy (greater than 80% density) coverage ● More severe on smooth heavy media

Solutions for the defect

Use the following steps to solve the defect. 366 Chapter 4 Solve problems

● Confirm that the correct print mode is selected for the paper being used. ● Flex the output stack before separating pages. ● Print smaller jobs and remove the output before sending the next job. ● Use the Intermediate Fuser mode which slows the printing and allows more time for cooling. ● Try printing in Quiet mode which lowers the fuser temperature and slows the output. M501: From the printer control panel, enable the Quiet Print Mode option. 1. Open the following menus: – Setup – System Setup – Quiet Print Mode 2. Select On.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Fuser

Output stacking

Review the following information about the output stacking event. Parts related to the defect 367

Figure 4-143 Output stacking

Description of the defect

This defect appears as output pages that do not stack properly in the output bin. The stack may be uneven or skewed or pages could be pushed out of the tray and onto the floor.

Conditions that can cause the defect

Poor stacking can occur when there is significant paper curl present, if the paper being used is wrinkled or deformed, if using non-standard media types such as envelopes, or if the output bin gets too full. Causes for this defect vary depending on the type of media being used. If paper curl is present, the cause can be attributed to uneven drying of the media in the fuser.

Solutions for the defect

Use the following steps to solve the defect. ● Use the output bin extension. ● If applicable, use the solutions in "Output curl" in the Troubleshooting Manual. ● Use different paper that meets HP specifications for this printer. 368 Chapter 4 Solve problems

- Use freshly opened paper (non-acclimated).
- Remove paper from the output tray before the stack becomes too large.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Fuser

Paper handling - misprints

Review the following information about the paper handling - misprints event.

Description of the defect

This defect appears as media in the paper path that does not behave as the printer user expects. The control panel displays a 41.XX.XX error. A sensor is triggered at an unexpected timing and displays an error message.

Conditions that can cause the defect

Misprints occur in any environment.

Solutions for the defect

Use the following steps to solve the defect. ● Make sure that the size of the paper loaded in a specified tray matches the control panel setting. ● Make sure that the paper guides in the tray are adjusted correctly for the size of paper and the rear door is fully closed. ● Make sure that the input tray and output bin are not overfilled. ● Try using a new supply of paper or a different type of paper.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Tray guides ● Registration assembly ● Feed rollers

Paper handling - multifeeds

Review the following information about the paper handling - multifeeds event.

Description of the defect

This defect appears as one (or many) pages feed through the paper path together. Sheets of paper stick together (physically or statically) and the separation pressure from the pick roller is unable to separate the sheets. Parts related to the defect 369

Conditions that can cause the defect

The multifeed defect can occur in any environment, but occurs more frequently in HH and NN (as opposed to LL), on glossy media (as opposed to plain media) and with freshly opened (as opposed to fully acclimated) paper.

Solutions for the defect

Use the following steps to solve the defect. ● Make sure that the input tray is not overfilled. ● Make sure that the paper guides in the tray are adjusted correctly for the size of paper. ● Remove the stack of paper from the tray, flex it, rotate it 180 degrees and then flip it over. Insert the media in a new orientation. ● Load a smaller stack of media in the input tray. ● Fan the media. ● Try a new supply of paper or a different type of paper.

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Tray guides ● Registration assembly ● Feed rollers

Paper handling - jams

Review the following information about the paper handling - jams event.

Description of the defect

This defect appears as media stuck in the paper path. The control panel displays a 13.XX.XX error. Media does not enter, pass through, or exit the paper path correctly.

Conditions that can cause the defect

Jams occur in any environment.

Solutions for the defect

Use the following steps to solve the defect. ● Clear the jam (input tray, rear door, bottom duplex tray, or output bin). ● Make sure that the paper guides in the tray are adjusted correctly for the size of paper and the rear door is fully closed. ● Make sure that the input tray and output bin are not overfilled. ● Load a smaller stack of media in the input tray. ● Try a new supply of paper or a different type of paper. 370 Chapter 4 Solve problems

Parts related to the defect

The following item(s) might need to be repaired or replaced to prevent this defect from occurring. ● Media ● Tray guides ● Registration assembly ● Feed rollers Parts related to the defect 371

Clean the printer

Review the following information about cleaning the printer. NOTE: To clean the printer exterior, use a soft, water-moistened cloth.

Clean the paper path

Over time, particles of toner and paper accumulate inside the printer. This can cause print-quality problems during printing. Cleaning the paper path eliminates or reduces these problems. Clean the paper path and toner-cartridge areas every time that the toner cartridge is changed or whenever print-quality problems occur. As much as possible, keep the printer free from dust and debris.

Print a cleaning page

Use the following procedures to print a cleaning page. Print the cleaning page from a LCD control panel (M501) Use the following procedure to print the cleaning page from a LCD control panel on an M501 printer. 1. From the printer control panel, press the OK button. 2. Open the Service menu. 3. Use the arrow keys to select the Cleaning mode option, and then press the OK button. The printer prints the first side and then prompts you to remove the page from the output bin and reload it in Tray 1, keeping the same orientation. Wait until the process is complete. Discard the page that prints. Print the cleaning page from a LCD control panel (M506/E50045/M507/E50145) Use the following procedure to print the cleaning page from a LCD control panel on an M506/E50045/ M507/E50145 printer. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Device Maintenance, and then press the OK button. 2. Use the down arrow button to scroll to Calibration/Cleaning, and then press the OK button. 3. If necessary, use the down arrow button to scroll to Cleaning, and then press the OK button to print the page. 4. Follow the instruction on the printed cleaning page to finish the cleaning page process. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then press the OK button. 2. Use the down arrow button to scroll to Maintenance, and then press the OK button. 3. Use the down arrow button to scroll to Calibration/Cleaning, and then press the OK button. 4. Select Continue to enter Maintenance Mode. 372 Chapter 4 Solve problems 5. Use the down arrow button to scroll to Cleaning page, and then press the OK button to print the page. 6. Follow the instruction on the printed cleaning page to finish the cleaning page process. 7. A Cleaning... message displays on the printer control panel. The cleaning process takes a few minutes to complete. Do not turn the printer off until the cleaning process has finished. When it is finished, discard the printed page. Print the cleaning page from a touchscreen control panel (M527/E52545/M528/E52645) Use the following procedure to print the cleaning page from a touchscreen control panel (M527/E52545/

M528/E52645).

1. From the Home screen on the printer control panel, scroll to and touch the Device Maintenance button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Maintenance (FutureSmart 4) ● Calibration/Cleaning 3. Touch Cleaning Page, and then press the OK button (FutureSmart 3) or the Print button (FutureSmart 4) to print the page. 4. The cleaning process can take several minutes. When it is finished, discard the printed page. Enable and configure auto cleaning (M527/E52545/M528/E52645) Use the procedure in this section to enable and configure the automatic cleaning function from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Device Maintenance button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the following menus: ● Maintenance (FutureSmart 4) ● Calibration/Cleaning ● Auto Cleaning (FutureSmart 3) or Cleaning Page (FutureSmart 4). 3. FutureSmart 3: Select the Enable item, and then touch the Save button. 4. FutureSmart 4: Under Automatic Cleaning Interval, select the desired page interval when the printer should automatically print a cleaning page, and then touch the Done button. Check the scanner glass for dirt and smudges (M527/E52545/M528/E52645) Over time, specks of debris might collect on the scanner glass and white plastic backing, which can affect performance. Use the following procedure to clean the scanner if the printed pages have streaks, unwanted lines, black dots, poor print quality, or unclear text. Print the cleaning page from a touchscreen control panel (M527/E52545/M528/E52645) 373

1. . Press the power button to turn the printer off, and then disconnect the power cable from the electrical outlet. 2. . Open the scanner lid. 3. . Clean the scanner glass, the document feeder strips, and the white plastic backing with a soft cloth or sponge that has been moistened with nonabrasive glass cleaner. CAUTION: Do not use abrasives, acetone, benzene, ammonia, ethyl alcohol, or carbon tetrachloride on any part of the printer; these can damage the printer. Do not place liquids directly on the glass or platen. They might seep and damage the printer. NOTE: If you are having trouble with streaks on copies when you are using the document feeder, be sure to clean the small strip of glass on the left side of the scanner. 4. . Dry the glass and white plastic parts with a chamois or a cellulose sponge to prevent spotting. 374 Chapter 4 Solve

problems

5. . Connect the power cable to an outlet, and then press the power button to turn the printer on. Clean the pickup rollers and separation pad in the document feeder (M527/

E52545/M528/E52645)

Use the following procedure to clean the pickup rollers and separation pad in the document feeder for the M527/E52545/M528/E52645 printer.

1. . Lift the document-feeder latch. 2. . Open the document-feeder cover. Clean the pickup rollers and separation pad in the document feeder (M527/E52545/M528/E52645) 375

3. . Remove any visible lint or dust from each of the feed rollers and the separation pad using compressed air or a clean lint-free cloth moistened with warm water. NOTE: Lift up the roller assembly to access and clean the second roller. 4. . Close the document-feeder cover. If the error persists, install a document feeder maintenance kit (B5L52-67903). Clean the Tray 1 roller and separation pad Use the following procedure to clean the Tray 1 roller and separation pad. NOTE: The figures in this section show the M506x. However, the procedure is correct for all M506/ E50045/M507/E50145 models and the M501 and M527/E52545/M528/E52645 printers. Step 1: Remove the roller Use the following procedure to remove the roller. 1. Open the toner-cartridge door. Figure 4-144 Open the toner-cartridge door 376 Chapter 4 Solve problems

2. Release two tabs between the roller collar and roller (callout 1), and then rotate the top of the roller out and away from the printer (callout 2). TIP: Pushing down on the top of the roller might make it easier to release the tabs. Figure 4-145 Release two tabs 1 1 2 3. Remove the roller. Figure 4-146 Remove the roller Step 2: Remove the separation pad assembly Use the following procedure to remove the separation pad assembly. Step 2: Remove the separation pad assembly 377

1. Open the toner-cartridge door. Figure 4-147 Open the toner-cartridge door 2. Release two tabs between the roller collar and roller (callout 1), and then rotate the top of the roller out and away from the printer (callout 2). TIP: Pushing down on the top of the roller might make it easier to release the tabs. Figure 4-148 Release two tabs 1 1 2 378 Chapter 4 Solve problems

3. Remove the roller. Figure 4-149 Remove the roller Step 3: Clean the Tray 1 roller and separation pad Use the following procedure to clean the Tray 1 roller and separation pad. ■ Use a damp, lint-free cloth to gently clean the rollers. CAUTION: When handling the roller and pad, avoid touching the spongy surfaces. Skin oils and fingerprints on a roller surface can cause print-quality problems. Figure 4-150 Clean the rollers and pad Step 4: Install the separation pad assembly Use the following procedure to install the separation pad assembly. CAUTION: Do not touch the black rubber portion of the pad assembly. Skin oils on the pad can cause paper handling problems. HP recommends washing your hands before handling the assembly. Step 3: Clean the Tray 1 roller and separation pad 379

1. Install the spring (callout 1) from the discarded separation pad assembly (or use the one supplied in this kit) on to the replacement assembly (callout 2). Figure 4-151 Install the spring 1 2 2. Align the slots in the assembly (callout 1) with the rails on the holder (callout 2). Figure 4-152 Align the assembly with the holder 1 2 3. Install the pad assembly into the holder. Reinstallation tip: The separation pad should freely move up and down on the spring when correctly installed. If the pad does not freely move, remove it, check the spring, and then reinstall it. 380 Chapter 4 Solve problems

Figure 4-153 Install the pad assembly 4. Slide the roller carriage back to the center of the shaft (the locking tab snaps into place). NOTE: Slightly depress the separation pad so that the roller carriage can slide over it. Figure 4-154 Slide the roller carriage to the left Step 5: Install the roller Use the following procedure to install the roller. CAUTION: Do not touch the gray spongy portion of the replacement roller. Skin oils on the roller can cause paper handling problems. HP recommends washing your hands before handling the assembly. Step 5: Install the roller 381

1. Place the pins on the under-side of the pickup roller in the slots on the holder, and then rotate the top of the roller up and toward the printer. NOTE: In this step, you start with the roller upside down, and then rotate it up and into its final installed position. Figure 4-155 Install the roller 2. Continue to rotate the top of the roller toward the printer until two tabs snap into place. Figure 4-156 Engage two tabs 2 1 1 3. Close the toner-cartridge door. NOTE: If the toner cartridge was removed, do not forget to reinstall it. 382 Chapter 4 Solve problems

Figure 4-157 Close the toner-cartridge door Clean the Tray 2-X rollers Use the following procedures to clean the Tray 2-X rollers. NOTE: The figures in this section show the M506x and M527. However, the procedure is correct for all M506/E50045/M507/E50145 models, and the M501 and the M527/E52545/M528/E52645 printers. Step 1: Remove the tray Use the following procedure to remove the tray. 1. Pull the tray straight out of the printer until it stops. Figure 4-158 Pull the tray out until it stops Clean the Tray 2-X rollers 383

2. Lift the front of the tray, and then pull it out of the printer to remove it. Figure 4-159 Release and remove the tray Step 2: Remove the roller assembly Use the following procedure to remove the roller assembly. CAUTION: When handling the roller assembly, do not touch the gray spongy portion of the rollers. Skin oils on the rollers can cause paper handling problems. HP recommends washing your hands before handling the assembly. NOTE: The figures in this section show the M506 and M527 Tray 2. However, the procedure is correct for replacing the Tray 2 pickup and feed roller and separation roller assemblies in the M501, as well as the optional Tray 3, Tray 4, and Tray 5 accessories.

1. Look up into the tray cavity to locate the roller assembly. Figure 4-160 Locate the roller assembly 384 Chapter 4 Solve problems

2. Grasp the white tab on the roller holder (callout 1), and then slide the roller assembly to the left to compress the spring loaded shaft (callout 2). Figure 4-161 Compress the spring-loaded shaft 1 3. With the spring loaded shaft depressed, rotate the right side of the roller assembly down and towards you to release it. Figure 4-162 Remove the roller assembly Step 3: Clean the Tray 2-X rollers Use the following procedure to clean the Tray 2-X rollers. ■ Use a damp, lint-free cloth to gently clean the rollers. CAUTION: When handling the rollers, avoid touching the spongy surfaces. Skin oils and fingerprints on a roller surface can cause print-quality problems. Step 3: Clean the Tray 2-X rollers 385

Figure 4-163 Clean the rollers Step 4: Install the roller assembly Use the following procedure to install the roller assembly. CAUTION: When handling the roller assembly, do not touch the gray spongy portion of the rollers. Skin oils on the rollers can cause paper handling problems. HP recommends washing your hands before handling the assembly. NOTE: The figures in this section show the M506 and M527 Tray 2.

However, the procedure is correct for replacing the Tray 2 pickup and feed roller and separation roller assemblies in the M501, as well as the optional Tray 3, Tray 4, and Tray 5 accessories. 1. When the assembly is installed, the actuator pin (callout 1) on the roller assembly must be installed in the slot (callout 2) in the actuator arm. Figure 4-164 Check the pin on the assembly 2 1 386 Chapter 4 Solve problems

2. Position the left end of the assembly on the spring loaded shaft and push it left to compress the shaft (callout 1), keep the roller holder parallel to the underside of the printer, and then rotate the right end up and into the printer (callout 2). Figure 4-165 Install the roller assembly 3. Slowly release the depressed spring loaded shaft to allow the right end of the roller assembly to engage with the right-side drive shaft. Make sure that the right side of the assembly is fully engaged with the right-side drive shaft. Figure 4-166 Decompress the spring-loaded shaft 1 4. To check the installation, repeatedly push up and release, the tray actuator (callout 1) and observe the roller assembly. If the assembly is correctly installed, it moves up and down (callout 2). NOTE: If the assembly does not properly move, remove it and then reinstall it, making sure that the pin on the assembly is installed in the slot in the actuator arm. See Figure 4-164 Check the pin on the assembly on page 386. Step 4: Install the roller assembly 387

Figure 4-167 Check the installation 1 2 Step 5: Install the tray Use the following procedure to install the tray. 1. With the tray at a slight angle, align the sides of the tray with the rails in the accessory, and then partially slide the tray into the printer. Figure 4-168 Install the tray 388 Chapter 4 Solve problems

2. Push the tray straight into the accessory to close it. Figure 4-169 Close the tray Step 5: Install the tray 389

Solve paper handling problems

Review the following information to solve paper handling problems.

Printer feeds incorrect page size

Review the following information when the printer feeds an incorrect page size. NOTE: To print using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus). Table 4-47 Printer feeds incorrect page size

Cause Solution

The correct size paper is not loaded in the tray. Load the correct size paper in the tray. The correct size paper is not selected in the software program or printer driver. Confirm that the settings in the software program and printer driver are correct, because the software program settings override the printer driver and control panel settings, and the printer driver settings override the control panel settings. The correct size paper for the tray is not selected in the printer control panel. From the control panel, select the correct size paper for the tray. The paper size is not configured correctly for the tray. Print a configuration page to determine the paper size for which the tray is configured. The guides in the tray are not against the paper. Verify that the paper guides are touching the paper.

Printer pulls from incorrect tray

Review the following information when the printer pulls from an incorrect tray. NOTE: To print using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus). Table 4-48 Printer pulls from incorrect tray

Cause Solution

A driver for a different printer is in use. Use a driver for this printer. The specified tray is empty. Load paper in the specified tray. The paper size is not configured correctly for the input tray. Print a configuration page or use the control panel to determine the paper size for which the tray is configured. The guides in the tray are not against the paper. Verify that the guides are touching the paper. Printer will not duplex or duplexes incorrectly (duplex models) Review the following information when the printer will not duplex or duplexes incorrectly. 390 Chapter 4 Solve problems

Table 4-49 Printer will not duplex (print 2-sided jobs) or duplexes incorrectly

Cause Solution

The duplex job is trying to use unsupported paper. Verify that the paper is supported for duplex printing. The printer driver is not set up for duplex printing. Set up the printer driver to enable duplex printing. The first page is printing on the back of preprinted forms or letterhead. Load preprinted forms and letterhead in Tray 1 with the letterhead or printed side down, with the top of the page leading into the printer. For Tray 2-X, load the paper printed side up with the top of the page toward the right of the printer. The printer model does not support automatic 2-sided printing. The printer model does not support automatic 2-sided printing. Paper does not feed from Tray 2-X Review the following information when the paper does not feed from Tray 2-X. NOTE: To print using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus). Table 4-50 Paper does not feed from Tray 2-X

Cause Solution

The correct size paper is not loaded. Load the correct size paper. The input tray is empty. Load paper in the input tray. The correct paper type for the input tray is not selected in the printer control panel. From the printer control panel, select the correct paper type for the input tray. Trays configured for a paper type with a specific weight range will not match a print job that specifies an exact weight, even if the specified

weight is within the weight range. Paper from a previous jam has not been completely removed. Open the printer and remove any paper in the paper path. Closely inspect the fuser area for jams. None of the optional trays appear as input tray options. The optional trays only display as available if they are installed. Verify that any optional trays are correctly installed. Verify that the printer driver has been configured to recognize the optional trays. An optional tray is incorrectly installed. Print a configuration page to confirm that the optional tray is installed. If not, verify that the tray is correctly attached to the printer. The paper size is not configured correctly for the input tray. Print a configuration page or use the control panel to determine the paper size for which the tray is configured. The guides in the tray are not against the paper. Verify that the guides are touching the paper.

Output is curled or wrinkled

Review the following information when the output is curled or wrinkled. Paper does not feed from Tray 2-X 391

Table 4-51 Output is curled or wrinkled

Cause Solution

Paper does not meet the specifications for this printer. Use only paper that meets the HP paper specifications for this printer. Non-recycled, 75 g/m² (20 lb) paper is optimal for office use. The correct paper type for the input tray is not selected in the printer control panel. From the printer control panel, select the correct paper type for the input tray. Trays configured for a paper type with a specific weight range will not match a print job that specifies an exact weight, even if the specified weight is within the weight range. Paper is damaged or in poor condition. Remove paper from the input tray and load paper that is in good condition. The printer is operating in an excessively humid environment. Verify that the printing environment is within humidity specifications. The print job consist of large, solid-filled areas. Large, solid-filled areas can cause excessive curl. Try using a different pattern. Paper used was not stored correctly and might have absorbed moisture. Remove paper and replace it with paper from a fresh, unopened package. Store paper in a plastic bag to protect it from humidity. Paper has poorly cut edges. Remove paper, flex it, rotate it 180 degrees or turn it over, and then reload it into the input tray. Do not fan paper. If the problem persists, replace the paper. The specific paper type was not configured for the tray or selected in the software. Configure the software for the paper (see the software documentation). Configure the tray for the paper. The paper has previously been used for a print job. Do not re-use paper.

Printer does not pick up paper or misfeeds

Use the following procedures if the printer will not pick up paper or misfeeds paper.

The printer does not pick up paper

If the printer does not pick up paper from the tray, try these solutions. NOTE: Tray 1 and Tray 2 are optimal for paper pickup when using special paper or media other than 20lb plain paper. For Tray 1 and Tray 2 the printer increases the number of attempts to pick up a page, which increases the reliability of successfully picking the page from the tray and decreases the possibility of a mispick jam. HP recommends using Tray 1 or Tray 2 if the printer is experiencing excessive or reoccurring jams from trays other than Tray 1 and Tray 2, or for print jobs that require media other than 20lb plain paper. 1. Open the printer and remove any jammed sheets of paper. 2. Load the tray with the correct size of paper for the job. 3. Make sure the paper size and type are set correctly on the printer control panel. 4. Make sure the paper guides in the tray are adjusted correctly for the size of paper. Adjust the guides to the appropriate indentation in the tray. 392 Chapter 4 Solve problems 5. Check the printer control panel to see if the printer is waiting for an acknowledgment to the feed the paper manually prompt. Load paper, and continue. 6. The rollers above the tray might be contaminated. Clean the rollers with a lint-free cloth dampened with warm water.

The printer picks up multiple sheets of paper

If the printer picks up multiple sheets of paper from the tray, try these solutions. 1. Remove the stack of paper from the tray and flex it, rotate it 180 degrees, and flip it over. Do not fan the paper. Return the stack of paper to the tray. 2. Use only paper that meets HP specifications for this printer. 3. Use paper that is not wrinkled, folded, or damaged. If necessary, use paper from a different package. 4. Make sure the tray is not overfilled. If it is, remove the entire stack of paper from the tray, straighten the stack, and then return some of the paper to the tray. 5. Make sure the paper guides in the tray are adjusted correctly for the size of paper. Adjust the guides to the appropriate indentation in the tray. 6. Make sure the printing environment is within recommended specifications. 7. The tray pick and/or feed rollers might be contaminated. Clean the rollers with a lint-free cloth dampened with warm water. The document feeder jams, skews, or picks up multiple sheets of paper (MFP) Learn about MFP document feeder paper handling problems. Review the following information when the document feeder jams, skews, or picks up multiple sheets of paper. ● Check to see if there are areas on the page that might have had staples removed. This can cause jams and/or mispicks. ● The original might have something on it, such as staples or self-adhesive notes that must be removed. ● Check that all rollers are in place and correctly installed. ● Make sure that the top document-feeder cover is closed. ● The pages might not be placed correctly. Straighten the pages and adjust the paper guides to center the stack. ● The paper guides must be touching the sides of the paper stack to work correctly. Make sure that the paper stack is straight, and the guides are against the paper stack. ● The document feeder input tray or output bin might contain more than the maximum number of pages. Make sure the paper stack fits below the guides in the input tray and remove pages from the output bin. ● Verify that there are no pieces of paper, staples, paper clips, or other debris in the paper path. The printer picks up multiple sheets of paper 393

● Clean the document-feeder rollers and the separation pad. Use compressed air or a clean, lint-free cloth moistened with warm water. If misfeeds still occur, replace the rollers. ● Use the control panel menus to check the status of the document-feeder kit and replace it if necessary.

Paper does not feed automatically

Review the following information when the paper does not feed automatically. NOTE: To print using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus). Table 4-52 Paper does not feed automatically

Cause Solution

Manual feed is selected in the software program. Load Tray 1 with paper, or, if the paper is loaded, press the OK button. The correct size paper is not loaded. Load the correct size paper. The input tray is empty. Load paper into the input tray. Paper from a previous jam has not been completely removed. Open the printer and remove any paper in the paper path. The paper size is not configured correctly for the input tray. Print a configuration page or use the control panel to determine the paper size for which the tray is configured. The guides in the tray are not against the paper. Verify that the rear and width paper guides are touching the paper. 394 Chapter 4 Solve problems

Clear paper jams

Use the following information and procedures to clear paper jams.

Paper path jam sensor locations

Use the figures below to identify the locations of sensors where reoccurring jams are found. Figure 4-170 Jam sensors PS432 PS13 PS2a PS2 PS1a PS11

Duplex paper path

Simplex paper path

Table 4-53 Jam sensors

Sensor Description Sensor Description

SR2 TOP sensor SR2a Duplex feed sensor1 SR1a Media-width sensor SR11 Registration sensor Clear paper jams 395

Table 4-53 Jam sensors (continued)

Sensor Description Sensor Description

SR13 Fuser output sensor SR432 Paper feeder feed sensor2 1 Duplex models only. 2 550-sheet paper feeder accessory.

Auto-navigation for clearing jams

The auto-navigation feature assists you in clearing jams by providing step-by-step instructions on the control panel. When you complete a step, the product displays instructions for the next step until you have completed all steps in the procedure. Experiencing frequent or recurring paper jams? To reduce the number of paper jams, try these solutions. 1. Use only paper that meets HP specifications for this product. 2. Use paper that is not wrinkled, folded, or damaged. If necessary, use paper from a different package. 3. Use paper that has not previously been printed or copied on. 4. Make sure the tray is not overfilled. If it is, remove the entire stack of paper from the tray, straighten the stack, and then return some of the paper to the tray. 5. Make sure the paper guides in the tray are adjusted correctly for the size of paper. Adjust the guides so they are touching the paper stack without bending it. 6. Make sure that the tray is fully inserted in the product. 7. If you are printing on heavy, embossed, or perforated paper, use the manual feed feature and feed sheets one at a time. 8. Open the Trays menu on the product control panel. Verify that the tray is configured correctly for the paper type and size. 9. Make sure the printing environment is within recommended specifications.

Clear paper jams in the document feeder

The following information describes how to clear paper jams in the document feeder. When a jam occurs, the control panel displays an animation that assists in clearing the jam. 396 Chapter 4 Solve problems

1. . Lift the latch to release the document-feeder cover. 2. . Open the document-feeder cover. Clear paper jams in the document feeder 397
3. . Remove any jammed paper. 4. . Lift the document-feeder input tray. 398 Chapter 4 Solve problems
5. . Remove any jammed paper. 6. . Lower the document-feeder input tray. Clear paper jams in the document feeder 399

7. . Close the document-feeder cover. NOTE: Verify that the latch on the top of the document-feeder cover is completely closed. NOTE: To avoid jams, make sure the guides in the document-feeder input tray are adjusted against the document, without bending the document. To copy narrow documents, use the flatbed scanner. Remove all staples and paper clips from original documents. NOTE: Original documents that are printed on heavy, glossy paper can jam more frequently than originals that are printed on plain paper. Clear paper jams in Tray 1 Use the following procedure to clear paper jams in Tray 1. When a jam occurs, the control panel displays an animation that assists in clearing the jam. 1. . If most of the sheet of paper is visible in the tray, slowly pull the jammed paper out of the printer. Make sure that the entire sheet is removed. If it tears, continue with the following steps to find the remnants.

If most of the sheet of paper has been pulled

inside the printer, continue with the following steps. 400 Chapter 4 Solve problems

2. . Press the top-cover-release button on the left side of the printer. 3. . Open the front door. 4. . Remove the toner cartridge. Clear paper jams in Tray 1 401

5. . Lift the jam-access cover and slowly remove any jammed paper. Be careful not to tear the paper. 6. . Reinsert the toner cartridge. 402

Chapter 4 Solve problems

7. . Close the front door. Clear paper jams in Tray 2 and the 550-sheet trays Use the following procedure to check for paper in all possible jam locations related to Tray 2 and the 550-sheet trays. When a jam occurs, the control panel displays an animation that assists in clearing the jam. 1. . Remove the tray from the printer. 2. . Remove the paper from the tray, and discard any damaged paper. Clear paper jams in Tray 2 and the 550-sheet trays 403

3. . Verify that the rear paper guide is adjusted to the indentation for the correct paper size. If necessary, pinch the release and move the rear paper guide to the correct position. It should click into place. B5

11 LTR

14 LGL

4. . Load the paper into the tray. Make sure that the stack is flat at all four corners and that the top of the stack is below the maximum-height indicators. 5. . Reinsert and close the tray. 404 Chapter 4 Solve problems

6. . Press the top-cover-release button on the left side of the printer. 7. . Open the front door. 8. . Remove the toner cartridge. Clear paper jams in Tray 2 and the 550-sheet trays 405

9. . Lift the jam-access cover and slowly remove any jammed paper. Be careful not to tear the paper. 10. . Reinsert the toner cartridge. 406

Chapter 4 Solve problems

11. . Close the front door.

Clear paper jams in the toner-cartridge area

Use the following procedure to check for paper in all possible jam locations in the toner-cartridge area. When a jam occurs, the control panel displays an animation that assists in clearing the jam. 1. . Press the top-cover-release button on the left side of the printer. 2. . Open the front door. Clear paper jams in the toner-cartridge area 407

3. . Remove the toner cartridge. 4. . Lift the jam-access cover and slowly remove any jammed paper. Be careful not to tear the paper. 408

Chapter 4 Solve problems

5. . Reinsert the toner cartridge. 6. . Close the front door. Clear paper jams in the rear door and the fuser area Use the following procedure to clear paper jams in the rear door and fuser area. When a jam occurs, the control panel displays an animation that assists in clearing the jam. NOTE: The fuser is hot while the printer is in use. Wait for the fuser to cool before clearing jams. Clear paper jams in the rear door and the fuser area 409

1. . Open the rear door. 2. . Remove any jammed paper. 3. . Close the rear door 410 Chapter 4 Solve problems

Clear paper jams in the output bin

Use the following procedure to check for paper in all possible jam locations in the output bin. When a jam occurs, the control panel displays an animation that assists in clearing the jam. 1. . If paper is visible in the output bin, grasp the leading edge and remove it.

Clear paper jams in the duplexer

Use the following procedure to check for paper in all possible jam locations in the automatic duplexer. When a jam occurs, the control panel displays an animation that assists in clearing the jam. 1. . Remove Tray 2 from the printer. Clear paper jams in the output bin 411

2. . Pull forward on the green tab inside the tray area to release the duplex pan. 3. . Remove any jammed paper. 4. . Close the duplex pan. 412 Chapter 4 Solve problems

5. . Reinsert and close the tray. 6. . Open the rear door. 7. . Remove any jammed paper. Clear paper jams in the duplexer 413

8. . Close the rear door 9. . Press the top-cover-release button on the left side of the printer. 10. . Open the front door. 414 Chapter 4 Solve problems

11. . Remove the toner cartridge. 12. . Lift the jam-access cover and slowly remove any jammed paper. Be careful not to tear the paper. Clear paper jams in the duplexer 415

13. . Reinsert the toner cartridge. 14. . Close the front door. Change jam recovery (all except M501) This printer provides a jam recovery feature that reprints jammed pages. Change jam recovery from a touchscreen control panel Use the following procedure to change jam recovery from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Settings button (FutureSmart 4). 2. Open the General Settings menu (FutureSmart 3) or the Settings menu (FutureSmart 4), and then open the Jam Recovery menu. 3. Select one of the following options: ● Automatic — The printer attempts to reprint jammed pages when sufficient memory is available. This is the default setting. 416 Chapter 4 Solve problems

● Off — The printer does not attempt to reprint jammed pages. Because no memory is used to store the most recent pages, performance is optimal. NOTE: Some pages can be lost if the printer runs out of paper while printing a duplex job with Jam Recovery set to Off. ● On — The

printer always reprints jammed pages. Additional memory is allocated to store the last few pages printed.

Change jam recovery from a LCD control panel

Use the following procedure to change jam recovery from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to General Settings, and then press the OK button. 3. Use the down arrow button to scroll to Jam Recovery, and then press the OK button. 4. Use the down arrow button to scroll to an option, and then press the OK button to select it. ● Automatic — The printer attempts to reprint jammed pages when sufficient memory is available. This is the default setting. ● Off — The printer does not attempt to reprint jammed pages. Because no memory is used to store the most recent pages, performance is optimal. NOTE: Some pages can be lost if the printer runs out of paper while printing a duplex job with Jam Recovery set to Off. ● On — The printer always reprints jammed pages. Additional memory is allocated to store the last few pages printed. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Settings, and then press the OK button. 2. Use the down arrow button to scroll to General, and then press the OK button. 3. Use the down arrow button to scroll to Jam Recovery, and then press the OK button. 4. Use the down arrow button to scroll to an option, and then press the OK button to select it. ● Auto — The printer attempts to reprint jammed pages when sufficient memory is available. This is the default setting. ● Off — The printer does not attempt to reprint jammed pages. Because no memory is used to store the most recent pages, performance is optimal. NOTE: Some pages can be lost if the printer runs out of paper while printing a duplex job with Jam Recovery set to Off. ● On — The printer always reprints jammed pages. Additional memory is allocated to store the last few pages printed. Change jam recovery from a LCD control panel 417

Solve performance problems

Review the following information and procedures to solve performance problems. NOTE: Tray 1 and Tray 2 are optimal for paper pickup when using special paper or media other than 75-80gsm (20lb) plain paper. For Tray 1 and Tray 2 the printer increases the number of attempts to pick up a page, which increases the reliability of successfully picking the page from the tray and decreases the possibility of a mis-pick jam. HP recommends using Tray 1 or Tray 2 if the printer is experiencing excessive or reoccurring jams from trays other than Tray 1 and Tray 2, or for print jobs that require media other than 75-80gsm (20lb) plain paper.

Factors affecting print performance

Review the following information about factors affecting print performance. NOTE: To print using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus). Table 4-54 Solve performance problems

Problem Cause Solution

Pages print but are totally blank. The document might contain blank pages.

Check the original document to see if

content is present on all of the pages. Pages print but are totally blank. The printer might be malfunctioning. To check the printer, print a Configuration page. Pages print but are totally blank. Make sure that the printer is not feeding multiple pages (especially if very thin paper is used).

Make sure that the paper meets HP

specifications for this printer. For a complete list of specific HP brand paper that this printer supports, go to <http://www.hp.com/support/ljM501> or <http://www.hp.com/support/ljM506> or <http://www.hp.com/support/ljE50045> or <http://www.hp.com/support/ljM507> or <http://www.hp.com/support/ljE50145> or <http://www.hp.com/support/ljM527mfp> or <http://www.hp.com/support/ljE52545mfp> or <http://www.hp.com/support/ljM528mfp> or <http://www.hp.com/support/E52645mfp>. Pages print very slowly. NOTE: Some software programs process print jobs slowly.

Heavier paper types can slow the print

job. Print on a different type of paper. Pages print very slowly. NOTE: Some software programs process print jobs slowly. Complex pages can print slowly. Proper fusing might require a slower print speed to ensure the best print quality. Pages print very slowly. NOTE: Some software programs process print jobs slowly. Large batches, narrow paper, and special paper such as gloss, transparency, cardstock, and HP Tough Paper can slow the print job. Print in smaller batches, on a different type of paper, or on a different size of paper. 418 Chapter 4 Solve problems

Table 4-54 Solve performance problems (continued)

Problem Cause Solution

Pages did not print. The printer might not be pulling paper correctly.

Make sure paper is loaded in the tray

correctly. Pages did not print. The paper is jamming in the printer. Clear the jam. Pages did not print. The USB cable might be defective or incorrectly connected. ● Disconnect the USB cable at both ends and reconnect it. ● Try printing a job that has printed in the past. ● Try using a different USB cable. Pages did not print. Other devices are running on the host computer. The printer might not share a USB port.

If an external hard drive or network

switchbox is connected to the same port as the printer, the other device might be interfering with the printer. To connect and use the printer, disconnect the other device or use two USB ports on the host computer. Pages did not print. The print job might not have arrived at the printer. Check the printer status queue. Also, the

Printing message should appear on the

control panel display.

Print speeds

Print speed is the number of pages that print in one minute. Print speed depends on different engine process speeds or operational pauses between printed pages during normal printer operation. Factors that determine the print speed of the printer include the following: ● Page formatting time The printer must pause for each page to be formatted before it prints. Complex pages take more time to format, resulting in reduced print speed. However, most jobs print at full engine speed. ● Media size Legal-size media reduces print speed because it is longer than the standard Letter- or A4-size media. A reduce print speed is used when printing on narrow media to prevent the edges of the fuser from overheating. ● Media mode Some media types require a reduced print speed to achieve maximum print quality on that media. For example, glossy, heavy, and specialty media (for example, envelopes or photos) require a reduced print speed. To maximize the print speed for special media types, make sure that the correct media type in the print driver is selected. ● Printer temperature To prevent printer damage, print speed is reduced if the printer reaches a specific internal temperature (thermal slow down). The starting temperature of the printer, ambient environment temperature, and the print job size effect the number of pages that can be printed before the printer reduces the print speed. Thermal slow down reduces print speed by printing four pages and then pausing for an amount of time before printing continues. Print speeds 419

● Other print speed reduction factors Other factors (especially during large print jobs) that can cause reduced print speeds include: – Density control sequence; occurs every 150 pages and takes about 120 seconds

The printer does not print or it prints slowly

Use the following procedures when the printer does not print or it prints slowly.

The printer does not print

If the printer does not print at all, try the following solutions. NOTE: To print using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus). 1. Make sure the printer is turned on and that the control panel indicates it is ready. ● If the control panel does not indicate the printer is ready, turn the printer off and then on again. ● If the control panel indicates the printer is ready, try sending the job again. 2. If the control panel indicates the printer has an error, resolve the error and then try sending the job again. 3. Make sure the cables are all connected correctly. If the printer is connected to a network, check the following items: ● Check the bottom LED next to the network connection on the printer. If the network is active, the light is green. ● Make sure that a network cable and not a phone cord is used to connect to the network. ● Make sure the network router, hub, or switch is turned on and that it is working correctly. 4. Install the HP software for the printer. Using generic printer drivers can cause delays clearing jobs from the print queue. 5. From the list of printers on your computer, right-click the name of this product, click Properties, and open the Ports tab. ● If a network cable is used to connect to the network, make sure the printer name listed on the Ports tab matches the one on the printer configuration page. ● If a USB cable is used, and the printer is connected to a wireless network, make sure the box is checked next to Virtual printer port for USB. 6. If a personal firewall system on the computer is used, it might be blocking communication with the printer. Try temporarily disabling the firewall to see if it is the source of the problem. 7. If the host computer or the printer is connected to a wireless network, low signal quality or interference might be delaying print jobs.

The printer prints slowly

If the printer prints, but it seems slow, try the following solutions. 420 Chapter 4 Solve problems

1. Make sure the computer meets the minimum specifications for this printer. For a list of specifications, go to this Web site: <http://www.hp.com/support/ljM501> or <http://www.hp.com/support/ljM506> or <http://www.hp.com/support/ljE50045> or <http://www.hp.com/support/ljM507> or <http://www.hp.com/support/ljE50145> or <http://www.hp.com/support/ljM527mfp> or <http://www.hp.com/support/ljE52545mfp> or <http://www.hp.com/support/ljM528mfp> or <http://www.hp.com/support/E52645mfp>. 2. When the printer is configured to print on some paper types, such as heavy paper, the printer prints more slowly so it can correctly fuse the toner to the paper. If the paper type setting is not correct for the type of paper you are using, change the setting to the correct paper type. 3. If the host computer or the printer is connected to a wireless network, low signal quality or interference might be delaying print jobs. The printer prints slowly 421

Solve connectivity problems

Review the following information and procedures to solve connectivity problems.

Solve USB connection problems

If you have connected the product directly to a computer, check the cable. ● Verify that the cable is connected to the computer and to the product. ● Verify that the cable is not longer than 2 m (6.65 ft). Try using a shorter cable. ● Verify that the cable is working correctly by connecting it to another product. Replace the cable if necessary.

Solve wired network problems

Review the following information and procedures to solve wired network problems.

Introduction

Learn about solving wired network problems. Certain types of problems can indicate there is a network communication problem. These problems include the following issues: ● The periodic loss of ability to communicate with the printer ● The printer cannot be found during driver installation ● A periodic failure to print Check the items in this topic to verify that the printer is communicating with the network. Before beginning, print a configuration page from the printer control panel and locate the printer IP address that is listed on this page. NOTE: To print using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus).

Poor physical connection

Use the following procedure when the printer has a poor physical connection. 1. Verify that the printer is attached to the correct network port using a cable of the correct length. 2. Verify that cable connections are secure. 3. Look at the network port connection on the back of the printer, and verify that the amber activity light and the green link-status light are lit. 4. If the problem continues, try a different cable or port on the hub. The computer is using the incorrect IP address for the printer Use the following procedure when the computer is using the incorrect IP address for the printer. 422 Chapter 4 Solve problems

1. Open the printer properties and click the Ports tab. Verify that the current IP address for the printer is selected. The printer IP address is listed on the printer configuration page. NOTE: To print a configuration page using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus). 2. If you installed the printer using the HP standard TCP/IP port, select the box labeled Always print to this printer, even if its IP address changes. 3. If you installed the printer using a Microsoft standard TCP/IP port, use the hostname instead of the IP address. 4. If the IP address is correct, delete the printer and then add it again. The computer is unable to communicate with the printer Use the following procedure when the computer is unable to communicate with the printer. 1. Test network communication by pinging the network. a. Open a command-line prompt on your computer. ● For Windows, click Start, click Run, type cmd, and then press Enter. ● For macOS, go to Applications, then Utilities, and open Terminal. b. Type ping followed by the IP address for your printer. c. If the window displays round-trip times, the network is working. 2. If the ping command failed, verify that the network hubs are on, and then verify that the network settings, the printer, and the computer are all configured for the same network. The printer is using incorrect link and duplex settings for the network HP recommends leaving these settings in automatic mode (the default setting). If you change these settings, you must also change them for your network. New software programs might be causing compatibility problems Verify that any new software programs are correctly installed and that they use the correct print driver. The computer or workstation might be set up incorrectly Use the following procedure when the computer or workstation might be set up incorrectly. 1. Check the network drivers, print drivers, and the network redirection settings. 2. Verify that the operating system is configured correctly. The printer is disabled, or other network settings are incorrect Use the following procedure when the printer is disabled, or other network settings are incorrect. The computer is unable to communicate with the printer 423

1. Review the configuration page to check the status of the network protocol. Enable it if necessary. NOTE: To print a configuration page using an HP Plus printer, the printer must have genuine original HP toner cartridges installed, must have an ongoing internet connection and an active HP Plus account (for more information, go to hp.com/plus). 2. Reconfigure the network settings if necessary. 424 Chapter 4 Solve problems

Service mode functions (M501) Review the following information about service mode functions on an M501 printer. Service menu (M501) Use the control panel Service menu to troubleshoot printer problems.

Service menu settings

The Service menu is used to adjust print settings, restore factory default settings, and clean the print paper path. The following Service menu items are available: ● Cleaning Page—used to remove dust and toner from the print paper path. ● USB Speed—used to set the USB speed to high or full. ● Less Paper Curl—used to put the printer into a mode that reduces paper curl. ● Archive Print—used to put the printer into a mode that produces output less susceptible to toner smearing and dusting for preservation and archival. ● Firmware Date—used to display the product's firmware date code. ● Restore Defaults—used to reset all customer-accessible menu settings back to the factory default settings (except language) via the control panel or software. NOTE: This Service menu item does not reset factory-settable settings, including formatter number, page counts, factory paper settings, language, and so on. ● Signature Check—used to configure how the printer proceeds when a firmware upgrade file does not have a valid signature. ● LaserJet Update—used to manually update the firmware or to set up automatic firmware updates.

Restore the factory-set defaults

Use the following procedure to restore the factory-set defaults. Restore the factory-set defaults from a LCD control panel (M501) Use the following procedure to restore the factory-set defaults from a LCD control panel on an M501 printer. 1. On the printer control panel, press the

OK button to open the menus. 2. Open the following menus: ● Service ● Restore Defaults The printer automatically restarts. Service mode functions (M501) 425

Secondary service menu (M501) Use the secondary service menu to print service-related reports and to run special tests. Customers do not have access to this menu. Open the secondary service menu (M501) Use the following procedure to open the secondary service menu on an M501 printer. Open the secondary service menu from a LCD control panel (M501) Use the following procedure to open the secondary service menu from a LCD control panel on an M501 printer. 1. Make sure the printer is in the Ready state. 2. From the printer control panel, press the setup button to open the main menu. 3. Press and the down arrow button, and then quickly press the Cancel button. 4. Press the setup button. The message 2ndary Service appears on the control panel display. 5. Press the OK button to open the 2ndary Service menu. NOTE: Use the arrow buttons to scroll though the 2ndary Service menu. Secondary service menu structure (M501) Review the following information about the secondary service menu structure on an M501 printer. Table 4-55 Secondary Service menu (M501)

Menu item Sub-menu item Description

Service Reports Cont-Self Test Print a continuous configuration page. Service Reports Error Report Print an error report. Service Reports Extended key map 10-key pad values for job storage. Speed Use this item to toggle between high and low. Default settings Taylor system defaults (IT managed or self-managed).

Location A list of available locations

appears This item sets certain printer parameters that are dependent on the location, such as the default paper size and the symbol set. Scroll to the appropriate location and select Yes to set the location. The printer automatically restarts after you change the location. Line Frequency This item allows the refresh rate of the touchscreen control panel to be changed between 50 and 60 Hz. LED Test This item allows the testing of the different LED Displays 426 Chapter 4 Solve problems

Table 4-55 Secondary Service menu (M501) (continued)

Menu item Sub-menu item Description

Display Test Use this test to verify that the LEDs and characters on the control panel display function correctly. At the beginning of the test, each of the LEDs is turned on one-at time. Press the OK button to continue to the next LED. After the LED test is complete, the character test begins by testing the pixels on each line. Then, each of the 255 characters is displayed in groups of 16. Press the OK button to continue to the next group of 16 characters. You can cancel the test at any time by pressing the Cancel button. Button Test Use this test to verify that the control-panel buttons function correctly. The display prompts you to press each button. CP FW Version This item shows the current date code of the firmware. NAND Reset This item allows the NAND to be reset. Permanent Config Allow or prevent the action to be performed. Printer resets (M501) Review the following information about printer resets on an M501 printer. NVRAM initialization (M501) Use the following procedure to perform an NVRAM initialization on an M501 printer. CAUTION: All HP Jetdirect settings are also reset. Be sure to print a configuration page before performing an NVRAM initialization. Make note of the IP address that is listed on the Jetdirect configuration page. You need to restore the IP address after performing an NVRAM initialization. An NVRAM initialization erases all data stored in the unprotected NVRAM sections. Performing an NVRAM initialization resets the following settings and information: ● All menu settings are reset to factory default values. ● All localization settings, including language and country/region, are reset. After performing an NVRAM initialization, reconfigure any computers that print to this printer so that the computers can recognize the printer. 1. Turn the printer off. 2. Simultaneously press and hold the up arrow button and the Cancel button. Keep these buttons depressed as you turn the printer on. 3. When the Permanent Storage Init. message appears on the display, release the buttons. 4. When the printer has finished the NVRAM initialization, it returns to the Ready state. Printer resets (M501) 427

Service mode functions

Learn about the printer service mode functions.

Service menu

Learn about the printer service menu items. Service menu access is restricted by using a personal identification number (PIN). Only authorized service people should access the Service menu. When selecting Service from the list of menus, the printer prompts the user to enter an eight-digit PIN. NOTE: The printer automatically exits the Service menu after about one minute if no items are selected or changed. Open the service menu from a touchscreen control panel Use the following procedure to open the service menu from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Device Maintenance button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). 2. Open the Service menu. 3. On the sign-in screen, select Service Access Code from the drop-down list. 4. Enter the following service access code for the printer:

● 10050615 (M506/E50045)

● 11052715 (M527/E52545)

● 03050719 (M507, E50145)

● 03052819 (M528/E52645)

Open the service menu from a LCD control panel

Use the following procedure to open the service menu from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Device Maintenance, and then press the OK button. 2. Use the down arrow button to scroll to Service, and then press the OK button to select it. 3. Use the down arrow button to scroll to Service Access Code, and then press the OK button to select it. 4. Enter the following service access code for the printer: NOTE: After entering the PIN, press the OK button.

● 10050615 (M506/E50045/M507/E50145)

● 11052715 (M527/E52545/M528/E52645)

FutureSmart 4 428 Chapter 4 Solve problems

1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then select OK. 2. Navigate to Maintenance, and then select OK. 3. Navigate to Service, and then select OK. 4. Enter the following service access code for the printer: NOTE: After entering the PIN, select Sign In.

● 10050615 (M506/E50045/M507/E50145)

● 11052715 (M527/E52545/M528/E52645)

Service menu items

The following menu items appear in the Service menu. Table 4-56 Service menu (all except M501)

First level Second level Value Description

Event Log Print Print or view the event log.

Clear Event Log Clear Use this item to clear the

printer event log. Cycle Counts Total Engine Cycles Set the page count that was stored in NVRAM prior to installing a new formatter. Cycle Counts Refurbish Cycle Count Use this item to record the page count when the printer was refurbished. Cycle Counts Document (M527/E52545/ M528/E52645 only)

Total number of pages since

the document feeder kit was replaced.

Cycle Counts Document Feeder Kit Interval

(M527/E52545/M528/E52645

only)

Use this item to

set the interval that causes the printer to prompt the customer to replace document feeder maintenance kit. Cycle Counts Clean Rollers Count (M527/ E52545/M528/E52645 only)

Total number of pages since

the document feeder rollers were cleaned. Cycle Counts Clean Rollers Interval (M527/ E52545/M528/E52645 only)

Use this item to set

the interval that causes the printer to prompt the customer to clean the document feeder rollers and separation pad. Cycle Counts ADF Count (M527/E52545/ M528/E52645 only)

Set the total pages fed

through the document feeder. Cycle Counts Flatbed Count (M527/E52545/ M528/E52645 only)

Set the total pages scanned

from the flatbed. Service menu items 429

Table 4-56 Service menu (all except M501) (continued)

First level Second level Value Description

Cycle Counts ADF Simplex Count (M527/ E52545/M528/E52645 only)

Set the total single-sided

pages fed through the document feeder. Cycle Counts ADF Duplex Count (M527/ E52545/M528/E52645 only)

Set the total two-sided pages

fed through the document feeder. Cycle Counts Copy Scan Count (M527/ E52545/M528/E52645 only)

Set the total copy pages that

have been scanned. Cycle Counts Send Scan Count (M527/ E52545/M528/E52645 only)

Set the number of scanned

pages sent to email.

Cycle Counts Fax Scan Count

NOTE: M527/E52545/M528/

E52645 fax models only.

Set the number of scanned

pages that have been faxed. Cycle Counts Copy Pages Count (M527/ E52545/M528/E52645 only)

Set the number of scanned

pages that have been printed. Scanner Settings (M527/E52545/ M528/E52645 only)

ADF Settings Leading edge front

Leading edge back

Trailing edge front

Trailing edge back

Left side front

Left side back

Set the calibration values. WARNING! Do not change these values unless instructed to do so. Scanner Settings (M527/E52545/ M528/E52645 only)

Glass Settings Leading edge glass

Left Side Glass

Serial Number Set the serial number.

Service ID Use this item to show the

date that the printer was first used on the control panel.

This eliminates the need for

users to keep paper receipts for proof of warranty.

Cold Reset Paper When you perform a cold

reset, the paper size that is stored in NVRAM is reset to the default factory setting. If you replace a formatter board in a country/region that uses A4 as the standard paper size, use this menu to reset the default paper size to A4. LETTER and A4 are the only available values.

Low Alerts Enable

Disable Turn on (or off) low alerts (for supplies). 430 Chapter 4 Solve problems

Table 4-56 Service menu (all except M501) (continued)

First level Second level Value Description

Reset Low Alerts ● Reset to level 1 ● Reset to level 2 ● Reset to level 3 ● Set to non-HP managed mode

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only.

Test the internal modem for

the analog fax accessory.

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only.

Hook Operations Off Hook

On Hook

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only.

Generate Random Data Select a value from

the list.

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only.

Generate DTMF Tone Burst Select a value from

the list.

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only.

Generate DTMF Continuous

Tone

Select a value from

the list.

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only.

Generate Pulse Burst Select a value from

the list.

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only. Generate Tone Dial Number Enter dial number.

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only. Generate Pulse Dial Number Enter dial number.

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only. Generate Single Modem Tone Range: 1100–2100 Hz Default = 2100 Hz

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only.

Line Measurements

PTT Test Mode

NOTE: M527/E52545/M528/E52645

fax models only.

Fax Transmit Signal Loss

Service menu items 431

Table 4-56 Service menu (all except M501) (continued)

First level Second level Value Description

Test Support Continuous Scan (M527/ E52545/M528/E52645 only) 2-sided

Save to Disk

Test Support Continuous Copy (M527/ E52545/M528/E52645 only) 2-sided

Save to Disk

Test Support Raw Scan (M527/E52545/ M528/E52645 only) 2-sided

Mechanical

Calibration

Test Support Continuous Print from USB

Test Support Automatic Calibrations (M527/ E52545/M528/E52645 only) Disabled Enabled* Test Support Runtime Configuration (M527/ E52545/M528/E52645 only) Standard

StandardEIC

Workflow

WorkflowEIC

Reconfigure

Printer resets (all except M501) Review the following information and procedures about printer resets. Restore factory-set defaults (all except M501) Use the following procedures to restore the factory-set defaults. NOTE: The printer restarts automatically after the reset operation completes. Restore factory-set defaults from a touchscreen control panel Use the following procedure to restore factory-set defaults from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Settings button (FutureSmart 4). 2. Open the following menus: ● General Settings (FutureSmart 3) or General (FutureSmart 4). ● Restore Factory Settings (FutureSmart 3) or Reset Factory Settings (FutureSmart 4). 3. A verification message advises that completing the reset function might result in loss of data. Touch the Reset button to complete the process. Restore factory-set defaults from a LCD control panel Use the following procedure to restore factory-set defaults from a LCD control panel. 432 Chapter 4 Solve problems

FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Use the down arrow button to scroll to General Settings, and then press the OK button to select it. 3. Use the down arrow button to scroll to Restore Factory Settings, and then press the OK button to select it. 4. Use the down arrow button to scroll to Restore, and then press the OK button to select it. 5. A verification message advises that completing the reset function might result in loss of data. Touch the Reset button to complete the process. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Settings, and then select OK. 2. Use the down arrow button to scroll to General, and then press the OK button. 3. Use the down arrow button to scroll to Reset Factory Settings, and then press the OK button to select it. 4. A verification message advises that completing the reset function might result in loss of data. Touch the Reset button to complete the process. Restore the service ID (all except M501) Use the following procedure to restore the service ID.

Restore the service ID

When replacing the formatter, the date is lost. Use this menu item to reset the date to the original date that the printer was first used. The date format is YYDDD. Use the following formula to calculate the dates: 1. To calculate YY, subtract 1990 from the calendar year. For instance, if the printer was first used in 2002, calculate YY as follows: $2002 - 1990 = 12$. $YY = 12$. 2. Subtract 1 from 10 (October is the tenth month of the year): $10 - 1 = 9$. ● Multiply 9 by 30: $9 \times 30 = 270$ or add 17 to 270: $270 + 17 = 287$. Thus, $DDD = 287$.

Convert the service ID to an actual date

Use the printer Service ID number to determine whether the printer is still under warranty. Use the following formula to convert the Service ID into the installation date as follows: 1. Add 1990 to YY to get the actual year that the printer was installed. 2. Divide DDD by 30. If there is a remainder, add 1 to the result. This is the month. 3. The remainder from the calculation in step 2 is the date. Using the Service ID 12287 as an example, the date conversion is as follows: 1. $12 + 1990 = 2002$, so the year is 2002. Restore the service ID (all except M501) 433

2. 287 divided by 30 = 9 with a remainder of 17. Because there is a remainder, add 1 to 9 to get 10, which represents October. 3. The remainder in step 2 is 17, so that is the date. The complete date is 17-October-2002. NOTE: A six-day grace period is built into the date system. Printer cold reset (all except M501) Use the following procedures to perform a printer cold reset. Cold reset using the Pre-boot menu from a touchscreen control panel Use the following procedure to perform a cold reset using the Pre-boot menu from a touchscreen control panel. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost). 1. Touch the middle of the control panel display when you see the 1/8 under the logo. Figure 4-171 Open the Pre-boot menu 1 2 2. Use the down arrow button to highlight the +3:Administrator item, and then touch the OK button. 3. Use the down arrow button to highlight the +8:Startup Options item, and then touch the OK button. 4. Use the down arrow button to highlight the 2 Cold Reset item, and then touch the OK button to select it. 5. Touch the Home button to return to the main Pre-boot menu and highlight the 1:Continue item, and then touch the OK button. NOTE: The printer will initialize. Cold reset using the Pre-boot menu from a LCD control panel Use the following procedure to perform a cold reset using the Pre-boot menu from a LCD control panel. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost). 1. Press the Cancel button when you see the 1/8 under the logo. 2. Use the down arrow button to highlight the +3:Administrator item, and then press the OK button. 434 Chapter 4 Solve problems

3. Use the down arrow button to highlight the +8:Startup Options item, and then press the OK button. 4. Use the down arrow button to highlight the 2 Cold Reset item, and then press the OK button to select it. 5. Touch the Home button to return to the main Pre-boot menu and highlight the 1:Continue item, and then touch the OK button. NOTE: The printer will initialize. Format Disk and Partial Clean functions (all except M501) Review the following information about the Format Disk and Partial Clean functions. NOTE: Only for printers with an optional

hard-disk drive (HDD) installed).

Active and repository firmware locations

The firmware bundle consists of multiple parts. The main components are the Windows CE Operating System and the printer/peripheral firmware files. There are two locations/partitions on the hard drive where the firmware components are stored: ● The Active, where the operating system and firmware currently are executing. ● The Repository, the recovery location. If the Active location is damaged, or a Partial Clean was performed, the printer automatically copies over the OS and firmware files from the Repository location and the printer recovers. If both the Active and Repository locations are damaged, or a Format Disk was performed, then both locations are gone and the error message 99.09.67 displays on the control-panel display. The user must upload the firmware to the printer in order for it to function again. CAUTION: The Format Disk option performs a disk initialization for the entire disk. The operating system, firmware files, and third party files (among other files) will be completely lost. HP does not recommend this action.

Partial Clean

The Partial Clean option erases all partitions and data on the disk drive, except for the firmware repository where a backup copy of the firmware file is stored. This allows the disk drive to be reformatted without having to download a firmware upgrade file to return the printer to a bootable state.

Characteristics of a Partial Clean

- Customer-defined settings, third-party solutions, firmware files, and the operating system are deleted. ● Rebooting the printer restores the firmware files from the Repository location, but does not restore any customer-defined settings. Format Disk and Partial Clean functions (all except M501) 435
- For previous HP printers, a Hard Disk Initialization is similar to executing the Partial Clean function for this printer. CAUTION: HP recommends backing-up printer configuration data before executing a Partial Clean to retain customer-defined settings (if needed). See the Backup/Restore item in the Device Maintenance menu.

Reasons for performing Partial Clean

- The printer continually boots up in an error state. NOTE: Try clearing the error prior to executing a Partial Clean. ● The printer will not respond to commands from the control panel. ● Executing the Partial Clean function is helpful for troubleshooting hard disk problems. ● To reset the printer by deleting all solutions and customer-defined settings. ● The printer default settings are not properly working.

Execute a Partial Clean

Use the following procedures to execute a Partial Clean on the printer. Execute a Partial Clean from a touchscreen control panel Use the following procedure to execute a Partial Clean from a touchscreen control panel. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost). 1. Touch the middle of the control-panel display when you see the 1/8 under the logo. Figure 4-172 Open the Pre-boot menu 1 2 2. Use the down arrow button to highlight the +3:Administrator item, and then touch the OK button. 3. Use the down arrow button to highlight Partial Clean and then touch the OK button. 4. Touch the OK button again. 5. Touch the Home button to highlight Continue, and then touch the OK button. NOTE: The printer initializes. 436 Chapter 4 Solve problems

Execute a Partial Clean from a LCD control panel

Use the following procedure to execute a Partial Clean from a LCD control panel. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost). 1. Press the Cancel button when you see the 1/8 under the logo. 2. Use the down arrow button to highlight the +3:Administrator item, and then press the OK button. 3. Use the down arrow button to highlight Partial Clean and then press the OK button. 4. Press the OK button again. 5. Press the Home button to highlight Continue, and then press the OK button. NOTE: The printer initializes.

Format Disk

The Format Disk option erases the entire disk drive. CAUTION: After executing a Format Disk option, the printer is not bootable.

Characteristics of a Format Disk

- Customer-defined settings, third-party solutions, firmware files, and the operating system are deleted. NOTE: Rebooting the printer does not restore the firmware files. ● Rebooting the printer restores the firmware files from the Repository location, but does not restore any customer-defined settings. ● After executing the Format Disk function, the message 99.09.67 displays on the control panel. ● After executing the Format Disk function, the printer firmware must be reloaded. CAUTION: HP recommends not using the Format Disk option unless an error occurs and the solution in the printer service manual recommends this solution. After executing the Format Disk function, the printer is unusable. HP recommends backing-up printer configuration data before executing a Format Disk to retain customer-defined settings (if needed). See the Backup/Restore item in the Device Maintenance menu.

Reasons for performing Format Disk

● The printer continually boots up in an error state. NOTE: Try clearing the error prior to executing a Format Disk. ● The printer will not respond to commands from the control panel. ● Executing the Format Disk function is helpful for troubleshooting hard disk problems. ● To reset the printer by deleting all solutions and customer-defined settings. Execute a Partial Clean from a LCD control panel 437

Execute a Format Disk

Use the following procedures to execute a Format Disk on the printer. Execute a Format Disk from a touchscreen control panel Use the following procedure to execute a Format Disk from a touchscreen control panel. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost). 1. Touch the middle of the control-panel display when you see the 1/8 under the logo. Figure 4-173 Open the Pre-boot menu 1 2 2. Use the down arrow button to highlight the +3:Administrator item, and then touch the OK button. 3. Use the down arrow button to highlight Format Disk, and then touch the OK button. 4. Touch the OK button again. NOTE: When the Format Disk operation is complete, reload the printer firmware.

Execute a Format Disk from a LCD control panel

Use the following procedure to execute a Format Disk from a LCD control panel. CAUTION: This procedure resets all printer configurations and settings to factory defaults (customer configurations and settings are lost). 1. Press the Cancel button when you see the 1/8 under the logo. 2. Use the down arrow button to highlight the +3:Administrator item, and then press the OK button. 3. Use the down arrow button to highlight Format Disk, and then press the OK button. 4. Press the OK button again. NOTE: When the Format Disk operation is complete, reload the printer firmware. 438 Chapter 4 Solve problems

Firmware upgrades (M501) HP offers periodic printer updates, new Web Services apps, and new features to existing Web Services apps. Follow these steps to update the firmware for a single printer. When you update the firmware, Web Service apps will update automatically. There are two supported methods to perform a firmware update on this printer. Use only one of the following methods to update the printer firmware. Determine the installed revision of firmware (M501) Use the following procedure to determine the installed revision of firmware on an M501 printer. NOTE: Print a configuration page to determine the installed revision of firmware. 1. On the printer control panel, press the OK button. 2. Open the Reports menu. 3. Select Configuration Report. Figure 4-174 Configuration page firmware date code location (M501)

Firmware Datecode

Method one: Update the firmware using the control panel (M501) Use these steps to load the firmware from the control panel (for network-connected printers only), and/or set the printer to automatically load future firmware updates. For USB-connected printers, use method two. Firmware upgrades (M501) 439

1. Make sure the printer is connected to a wired (Ethernet) or wireless network with an active Internet connection. NOTE: The printer must be connected to the internet to update the firmware via a network connection. 2. From the Home screen on the printer control panel, open the Setup menu. ● For standard control panels, press the left or right arrow button. 3. Scroll to and open the Service menu, and then open the LaserJet Update menu. NOTE: If the LaserJet Update option is not listed, use method two. 4. Check for updates. ● For standard control panels, select Check for Update. NOTE: The printer automatically checks for an update, and if a newer version is detected, the update process automatically starts. 5. Set the printer to automatically update the firmware when updates become available. From the Home screen on the printer control panel, open the Setup menu. ● For standard control panels, press the left or right arrow button. Scroll to and open the Service menu, open the LaserJet Update menu, and then select the Manage Updates menu. Set the printer to automatically update the firmware. ● For standard control panels, set the Allow Updates option to YES, and then set the Automatic Check option to ON. Method two: Update the firmware using the Firmware Update Utility (M501) Use these steps to manually download and install the Firmware Update Utility from HP.com. NOTE: This method is the only firmware update option available for printers connected to the computer via a USB cable. It also works for printers connected to a network. 1. Go to www.hp.com/go/support, click the Drivers & Software link, type the printer name in the search field, press the ENTER button, and then select the printer from the list of search results. 2. Select the operating system. 3. Under the Firmware section, locate the Firmware Update Utility. 4. Click Download, click Run, and then click Run again. 5. When the utility launches, select the printer from the drop-down list, and then click Send Firmware. NOTE: To print a configuration page to verify the installed firmware version before or after the update process, click Print Config. 6. Follow the on-screen instructions to complete the installation, and then click the Exit button to close the utility. 440 Chapter 4 Solve problems

Firmware upgrades (all except M501) Review the following information about downloading firmware upgrades. To download the most recent firmware upgrade for the printer, go to: ● In the US, go to <http://www.hp.com/support/ljM501> or <http://www.hp.com/support/ljM506> or <http://www.hp.com/support/ljE50045> or <http://www.hp.com/support/ljM507> or <http://www.hp.com/support/ljE50145> or <http://www.hp.com/support/ljM527mfp> or <http://www.hp.com/support/ljE52545mfp> or <http://www.hp.com/support/ljM528mfp> or <http://www.hp.com/support/E52645mfp>. a. Select Get drivers, Software, and Firmware, and then select the appropriate product by name. NOTE: More than one printer model might be listed. Make sure to select the correct model so that the upgraded firmware supports all of the printer functions. b. Select the driver language and operating system. c. Locate the firmware download, and then select Download. ● Outside the U.S., go to www.hp.com/support. a. Select your country/region. b. Select Drivers & Downloads. c. Enter the product name in the Find my product dialogue box, and then select Go. TIP: Click on the How do I find my product name/number? link to see a short video on identifying the printer name and number. d. Select the appropriate product by name. NOTE: More than one printer model might be listed. Make sure to select the correct model so that the upgraded firmware supports all of the printer functions. e. Select the driver language and operating system. f. Locate the firmware download, and then select Download. Determine the installed revision of firmware (all except M501) Use the following procedures to determine the installed revision of firmware. NOTE: Print a configuration page to determine the installed revision of firmware. Print the configuration page from a touchscreen control panel Use the following procedure to print the configuration page from a touchscreen control panel. 1. From the Home screen on the printer control panel, scroll to and touch the Administration button (FutureSmart 3), or scroll to and touch the Reports button (FutureSmart 4). 2. Open the following menus: ● Reports (FutureSmart 3) Firmware upgrades (all

except M501) 441

● Configuration/Status Pages 3. Touch Configuration Page to select it. 4. Touch the Print button to print the pages. Print the configuration page from a LCD control panel Use the following procedure to print the configuration page from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Administration, and then press the OK button. 2. Open the following menus: ● Reports ● Configuration/Status Pages 3. Use the down arrow button to scroll to Configuration Page, and then press the OK button to select it. 4. Use the up arrow button to scroll to Print, and then press the OK button to print the pages. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Reports, and then press the OK button. 2. Use the down arrow button to scroll to Configuration/Status Pages, and then press the OK button. 3. Use the down arrow button to scroll to Configuration Page, and then press the OK button. 4. Use the down arrow to scroll down to Cancel, and then use the right arrow to highlight the print icon . Select OK to print the pages. 442 Chapter 4 Solve problems

Figure 4-175 Configuration page firmware version (all except M501) HP LaserJet M506

Firmware Revision

Perform a firmware upgrade (all except M501) The firmware bundle is a .bdl file. This file requires an interactive upgrade method. The traditional FTP, LPR, or Port 9100 methods of upgrading are not available. Use one of the following methods to upgrade the firmware for this printer. HP Embedded Web Server (all except M501) Use the following procedure to perform a firmware upgrade using the HP Embedded Web Server. NOTE: The printer should be at the Ready state. The firmware update might take 10 minutes or longer based on the input/output (I/O) transfer rates and the time it takes for the printer to reinitialize. 1. Open an Internet browser window. 2. Enter the printer IP address in the URL line. 3. Select the Firmware Upgrade link from the General tab or from the Troubleshooting tab. 4. Browse to the location that the firmware upgrade file was downloaded to, and then select the firmware file. The file has a .bdl file extension. Select the Install button to perform the upgrade. NOTE: Do not close the browser window OR interrupt communication until the HP Embedded Web Server (EWS) displays the confirmation page. Perform a firmware upgrade (all except M501) 443

5. After the printer reinitializes, print a configuration page and verify that the latest firmware version has been installed. USB flash drive (Pre-boot menu) (all except M501) Use the following procedures to perform a firmware upgrade using a USB flash drive and the Pre-boot menu. IMPORTANT: Only use this method of performing a firmware upgrade if the printer cannot initialize to the Ready state. USB flash drive firmware (Pre-boot menu) update from a touchscreen control panel Use the following procedure to perform a USB flash drive firmware (Pre-boot menu) update from a touchscreen control panel. 1. Copy the .bdl file to a portable USB flash drive. 2. Touch the middle of the control panel display when you see the 1/8 under the logo. Figure 4-176 Open the Pre-boot menu 1 2 3. Touch the down arrow button to highlight +3 Administrator, and then touch the OK button. 4. If necessary, touch the down arrow button to highlight +1 Download, and then touch the OK button. 5. Insert the USB flash drive with the .bdl file on it into the USB port on the printer. NOTE: If the error message No USB Thumbdrive Files Found displays on the control panel display, try using a different portable storage device. 6. Touch the down arrow button to highlight USB Thumbdrive, and then touch the OK button. 7. Touch the down arrow button to highlight the .bdl file, and then touch the OK button. NOTE: The upgrade process can take 10 minutes or longer to complete. TIP: If there is more than one .bdl file on the storage device, make sure to select the correct file for this printer. 8. When the message Complete displays on the control panel display, touch the down arrow button several times until the message Continue displays. 9. Touch the OK button to begin the upgrade. When the upgrade is complete, the printer will initialize to the Ready state. 444 Chapter 4 Solve problems

10. When the upgrade process is complete, print a configuration page and verify that the upgrade firmware version was installed. USB flash drive firmware (Pre-boot menu) update from a LCD control panel Use the following procedure to perform a USB flash drive firmware (Pre-boot menu) update from a LCD control panel. 1. Press the Cancel button when you see the 1/8 under the logo. 2. Press the down arrow button to highlight +3 Administrator, and then press the OK button. 3. If necessary, press the down arrow button to highlight +1 Download, and then press the OK button. 4. Insert the USB flash drive with the .bdl file on it into the USB port on the printer. NOTE: If the error message No USB Thumbdrive Files Found displays on the control panel display, try using a different portable storage device. 5. Press the down arrow button to highlight USB Thumbdrive, and then press the OK button. 6. Press the down arrow button to highlight the .bdl file, and then press the OK button. NOTE: The upgrade process can take 10 minutes or longer to complete. TIP: If there is more than one .bdl file on the storage device, make sure to select the correct file for this printer. 7. When the message Complete displays on the control panel display, turn the printer power off, and then on again. 8. When the upgrade process is complete, print a configuration page and verify that the upgrade firmware version was installed. USB flash drive (control panel menu) (all except M501) Use the following procedures to perform a firmware upgrade using a USB flash drive and the control panel menu. NOTE: USB flash drives that are not using a FAT32 format, or do not have a CD formatted partition, might not be recognized by the printer. If the printer does not recognize a USB flash drive, try using a different USB flash drive. TIP: The USB port on the printer must be enabled. If it is disabled, use the Enable Retrieve from USB item in the General Settings menu to enable it. USB flash drive firmware (control panel menu) update from a touchscreen control panel Use the following procedure to perform a USB flash drive firmware (control panel menu) update from a touchscreen control panel. 1. Copy the .bdl file to a portable USB flash drive. 2. Turn the printer on, and then wait until it reaches the Ready state. 3. From the Home screen on the printer control panel, scroll to and touch the Device Maintenance button (FutureSmart 3), or scroll to and touch the Support Tools button (FutureSmart 4). USB flash drive firmware (Pre-boot menu) update from a LCD control panel 445

4. Open the Maintenance menu (FutureSmart 4). 5. Touch the USB Firmware Upgrade button. 6. Insert the USB flash drive with the .bdl file on it into the USB port on the printer. 7. Touch the .bdl file, and then touch the Upgrade button. TIP: If there is more than one .bdl file on the storage device, make sure to select the correct file for this printer. 8. Select one of the following options: ● Upgrade NOTE: The upgrade process can take 10 minutes or longer to complete. ● Re-install ● Downgrade 9. When the upgrade is complete, the printer will initialize to the Ready state. Print a configuration page and verify that the upgrade firmware version was installed. USB flash drive firmware (control panel menu) update from a LCD control panel Use the following procedure to perform a USB flash drive firmware (control panel menu) update from a LCD control panel. FutureSmart 3 1. From the Home screen on the printer control panel, use the down arrow button to scroll to Device Maintenance, and then press the OK button. 2. Open the following menus: ● USB Firmware Upgrade 3. Insert the USB flash drive with the

.bdl file on it into the USB port on the printer. 4. If necessary, use the down arrow button to scroll to the appropriate .bdl file, and then press the OK button to select. 5. Use the down arrow button to scroll to Upgrade, and then press the OK button to start the firmware upgrade. NOTE: If the firmware in the .bdl file on the portable USB storage device matches the installed firmware on the printer, Upgrade is replaced by Re-Install. FutureSmart 4 1. From the Home screen on the printer control panel, use the arrow keys to navigate to Support Tools, and then select OK. 2. Navigate to Maintenance, and then select OK. 3. Navigate to USB Firmware Upgrade, and then select OK. 4. Insert the USB flash drive with the .bdl file on it into the USB port on the printer. 446 Chapter 4 Solve problems

5. If necessary, use the down arrow button to scroll to the appropriate .bdl file, and then press the OK button to select.

Solve fax or email problems

Learn about solving fax and email problems. Fax or email troubleshooting information is not provided in this service manual. The most current information is available in WISE. Search using model number then use "fax troubleshooting" as the search term. For HP Channel partners, open the HP Partner First Portal located at <https://partner.hp.com>, and then do the following: 1. Select the Services & Support tab, and then select Technical Support. 2. Select Technical Documentation. 3. You will be taken to the WISE portal. For HP service personnel, go to one of the following Web-based Interactive Search Engines (WISE) sites: Americas (AMS) ● WISE - English ● WISE - Spanish ● WISE - Portuguese ● WISE - French Asia Pacific / Japan (APJ) ● WISE - English ● WISE - Japanese ● WISE - Korean ● WISE - Chinese (simplified) ● WISE - Chinese (traditional) ● WISE - Thai Europe / Middle East / Africa (EMEA) ● WISE - English Solve fax or email problems

447

5 Removal and replacement

When servicing the printer, several items must be taken into account to ensure a successful repair and to avoid damage to the printer or injury. Learn about these considerations and find detailed instructions for removing and replacing printer parts.

Conventions used in this guide

Learn about the conventions used in this publication. TIP: Tips provide helpful hints or shortcuts. NOTE: Notes provide important information to explain a concept or to complete a task. CAUTION: Cautions indicate procedures that you should follow to avoid losing data or damaging the product. WARNING! Warnings alert you to specific procedures that you should follow to avoid personal injury, catastrophic loss of data, or extensive damage to the product.

HP service and support

Learn about access to additional service and support information.

Additional service and support for channel partners

Channel partners can use the HP Partner Portal or the Channel Services Delivery Platform (CSDP) to access the HP Web-based Interactive Search Engine (WISE). Find information about the following topics. ● Service manuals, service cost data, and service advisories ● The latest control panel message (CPMD) troubleshooting information ● Remove and replace part instructions and videos ● Solutions for printer issues and emerging issues ● Printer specifications, warranty, and regulatory information ● Install, configure, and how to information View a video of how to use the HP Partner First Portal to access WISE. 448 Chapter 5 Removal and replacement

Access WISE for Channel partners (HP Partner Portal) 1. Sign-in to the HP Partner Portal (click <https://partner.hp.com> to access the sign-in page). 2. Select the Services & Support item (near the top of the screen). 3. Select the Technical Support (WISE) item. 4. Select the Technical Documentation item. 5. Use the WISE portal to search for printer information. Access WISE for Channel partners (CSDP) 1. Sign-in to the CSDP portal (click csdp.hp.com to access the sign-in page). 2. Select the Knowledge and Training item. 3. Select the HP Technical Documentation item. 4. Use the WISE portal to search for printer information. Additional service and support for HP internal personnel HP internal personnel, go to one of the following Web-based Interactive Search Engine (WISE) sites: View a video of how to access WISE for internal HP users. Americas (AMS) ● WISE - English ● WISE - Spanish ● WISE - Portuguese ● WISE - French Asia Pacific / Japan (APJ) ● WISE - English ● WISE - Japanese ● WISE - Korean ● WISE - Chinese (simplified) ● WISE - Chinese (traditional) ● WISE - Thai Europe / Middle East / Africa (EMEA) ● WISE - English

Additional technical support WISE videos

The videos below provide additional ways to access printer information using WISE. View a video of how to use WISE to find technical support videos (model number search). HP service and support 449

View a video of how to use WISE to find technical support videos (Product detail page [PDP] search). 450 Chapter 5 Removal and replacement

Removal and replacement strategy

The printer uses a field repair strategy. Defective parts are diagnosed and replaced at the Field Replaceable Unit (FRU) assembly level. Follow some basic safety precautions to avoid injury or printer damage. Learn about the tools required to service the printer and the types of fasteners used.

Introduction

Learn the overall process for troubleshooting failures and replacing parts, as well as items of note related to installing and tracking printer supplies. Repair normally begins by using the printer internal diagnostics and the following two-step process: 1. Isolate the problem to the major system (for example, the network, the server, or the printer). 2. Identify the cause of failures according to the troubleshooting section in the printer service manual, and follow the disassembly procedures to replace the defective parts or the consumable parts. After locating a faulty part, the printer can usually be repaired at the assembly level by replacing the field replaceable unit (FRUs). HP does not support replacement of components on the printed circuit assemblies (PCAs). The user replaces supplies (cartridges) as they are depleted. Additional instructions about other user replaceable parts are provided in this section. The printer tracks the amount of use on the customer-replaceable supplies by keeping a page count. The printer prompts the user to replace certain items when a supply is depleted or a specific number of pages have been printed. Swapping supplies between products might be necessary in some test scenarios. However, this might cause a misrepresentation of supply life values, and is not recommended.

Considerations during removal and replacement

Understand the items to take into account when removing and replacing parts. This chapter describes the FRUs. Reinstalling FRUs is generally the reverse of removal. Notes are included to provide directions for difficult or critical replacement procedures. Removal and replacement strategy 451

HP does not support repairing individual subassemblies or troubleshooting at the PCA component level. **WARNING!** Never operate or service the printer with the protective cover removed from the laser scanner assembly. The reflected beam, although invisible, can cause damage to the eyes. The sheet-metal parts can have sharp edges. Be careful when handling sheet-metal parts. Turn the printer off, wait five seconds, and then remove the power cord before attempting to service the printer. If this warning is not followed, severe injury can result as well as damage to the printer. The power must be on for certain functional checks during troubleshooting. However, the power cord must be disconnected during parts removal. AC voltage is still present inside the printer when the power switch is in the off position. The power cord must be disconnected before servicing the printer. **CAUTION:** Incorrectly routed or loose wire harnesses or flat flexible cables (FFCs) might interfere with other internal components or assemblies and can be damaged, pinched, or frayed. Make sure that wire harnesses are correctly routed and retained when installing assemblies. Do not bend or fold the FFCs during removal or installation. Do not expose the toner cartridges or imaging drums to strong light even for a short time. **IMPORTANT:** When an assembly is removed that includes a rating plate, tag, or a printer code label, make sure to transfer the plate, tag, or code label to the replacement assembly. Do not replace the formatter, DC controller, memory PCA (island of data), eMMC PCA, or hard-disk drive (HDD) simultaneously during a single printer servicing. Doing so might cause the printer to become unstable or inoperable. When replacing the formatter, make sure that all accessories installed on the formatter, such as the fax PCA, hard disk drive, trusted platform module, and DIMM, are removed from the defective formatter and installed on the replacement formatter. Make sure that the formatter is a new service part that has not been installed in a different device. **NOTE:** During assembly removal and replacement, or if the printer is moved, remove the toner cartridges, the imaging drums, and the toner collection unit. Toner is a non-poisonous substance composed of plastic and a small number of colored components. If toner gets on the skin or clothing, wipe it off with dry tissue paper and wash in cold water. Hot water sets toner and it might be difficult, or impossible, to remove. Toner easily breaks down vinyl materials, so avoid letting toner contact vinyl. **TIP:** Some figures might show assemblies removed or installed that have not yet been removed or installed at that specific step. However, the procedures are correct for this printer and the target assembly. Always thoroughly read the instructions that accompany each figure.

Electrostatic discharge

Learn how to protect sensitive parts against damage from electrostatic discharge. **CAUTION:** Some parts are sensitive to electrostatic discharge (ESD). Look for the ESD reminder when removing printer parts. Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD-sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. 452 Chapter 5 Removal and replacement

Required tools

Find information about the tools required to remove and replace parts on the printer. ● #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length ● Small flat-blade screwdriver ● Needle-nose pliers ● ESD strap (if one is available) ● Penlight **CAUTION:** Always use a #2 Phillips screwdriver (callout 1). Do not use a Pozidriv screwdriver (callout 2) or any motorized screwdriver. These can damage screws or screw threads. Figure 5-1 Screwdrivers

Fasteners used in this printer

Learn about the different types of screws used in the printer and the importance of installing the correct type of screw in the correct location when replacing a part. **WARNING!** Make sure that assemblies are replaced with the correct screw type. Using the incorrect screw (for example, substituting a long screw for the correct shorter screw) can cause damage to the printer or interfere with printer operation. Do not intermix screws that are removed from one assembly with the screws that are removed from another assembly. **NOTE:** To install a self-tapping screw, first turn it counterclockwise to align it with the existing thread pattern, and then carefully turn it clockwise to tighten. Do not overtighten. If a self-tapping screw hole becomes stripped, repair the screw hole or replace the affected assembly. Always take note of the length, diameter, color, type, and location of each removed screw. Make sure that screws are installed in their original location during reinstallation.

Types of screws used in the printer

● Screw, BH M3x10 ● Screw, W/Washer, M3x8 ● Screw, D, M3x8 ● Screw, Machine, Truss Head, M3x6 Required tools 453

Figure 5-2 Screw size chart 454 Chapter 5 Removal and replacement

Service approach

Follow these steps before and after performing service on the printer to prevent damage to the printer and ensure that the repair was successful.

Before performing service

Follow these steps before performing any type of service on the printer. **WARNING!** Turn the printer off, wait five seconds, and then remove the power cord before attempting to service the printer. If this warning is not followed, severe injury can result as well as damage to the printer. The power must be on for certain functional checks during troubleshooting. However, the power cord must be disconnected during parts removal. AC voltage is still present inside the printer when the power switch is in the off position. The power cord must be disconnected before servicing the printer. 1. Turn the printer off using the power button. 2. Unplug the power cable and interface cable or cables. 3. Place the product on an ESD workstation or mat, or use an ESD strap (if one is available). If an ESD workstation, mat or strap is not available, ground yourself by touching the sheet-metal chassis before touching an ESD-sensitive part. 4. Remove the toner cartridge(s) and/or imaging drum(s). Place a sheet of paper over the cartridges or imaging drums to protect them from excessive light exposure. 5. Remove all paper from the trays. 6. Remove the tray(s).

After performing service

Follow these steps after performing service on the printer. 1. Verify that you have completed the necessary reassembly steps. 2. Reinstall the tray(s). 3. Return all paper to the trays. 4. Reinstall the toner cartridge(s) and/or imaging drum(s). 5. Plug in the power cable and interface cable or cables. 6. Turn the printer on using the power button. 7. Verify that the expected start up sounds occur.

Print quality test

Ensure that the print quality is acceptable after performing printer repairs. Use the printer internal print quality pages to help diagnose and solve print-quality problems. For more information, see the troubleshooting section in the service manual. **NOTE:** To get further assistance in print quality troubleshooting, go to <http://www.hp.com/support/colorljM751> or <http://www.hp.com/support/colorljE75245>. Select Troubleshooting, select Print, and then select Print Quality. Service approach 455

Removal and replacement procedures

Learn how to remove and replace selected parts on the printer. Customer self-repair (CSR) A parts and accessories Customer self-repair (CSR) A parts are parts that a customer replaces without assistance from a field technician. These parts can be replaced without the use of any tools. Removal and replacement: Toner cartridge Learn how to remove and replace the toner cartridge. To view a short video of this procedure, click the link below. View a video of how to remove and replace the toner cartridge. Mean time to repair: 5 minutes Service level: CSR A

Before performing service

Turn the printer power off

■ Disconnect the power cable. **WARNING!** To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Use the table below to identify the correct part number for your printer. To order the part, go to www.hp.com/buy/parts. Table 5-1 Part number

Part number Part description

CF287A

CF287-67901 (service) HP 872A Black Original LaserJet Toner Cartridge (M501/M506/E50045/M527/ E52545 only.)

Standard-capacity replacement black toner cartridge

CF287X

CF287-67902 (service) HP 872X High Yield Black Original LaserJet Toner Cartridge (M501/M506/ E50045/M527/E52545 only.)

High-capacity replacement black toner cartridge

CF289A

CF289-67901 (service) HP 89A Black Original LaserJet Toner Cartridge (M507/M528/E52545/E52645 only)

Standard-capacity replacement black toner cartridge

CF289X

CF289-67902 (service) HP 89X High Yield Black Original LaserJet Toner Cartridge (M507/M528/ E52545/E52645 only)

High-capacity replacement black toner cartridge

CF289Y

CF289-67903 (service) HP 89Y Extra High Yield Black Original LaserJet Toner Cartridge (M507/M528/ E52545/E52645 only) Extra High-capacity replacement black toner cartridge 456 Chapter 5 Removal and replacement
Table 5-1 Part number (continued)

Part number Part description

CF289YC

CF289-67904 (service) HP Black contract LaserJet Toner Cartridge (M507/M528/E52545/E52645 only)

CF289A

CF289-67905 (service) HP Black managed LaserJet Toner Cartridge (M507/M528/E52545/E52645 only)

Required tools

No special tools are required to remove or install this assembly.

After performing service

Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

Post service test

Make sure that the printer initializes to a Ready state. Print a configuration page to make sure that the printer is functioning correctly. 1.

Remove the toner cartridge. An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Press the cartridge-door-release button. Figure 5-3 Press the cartridge-door-release button Removal and replacement: Toner cartridge 457

b. Open the toner-cartridge door. Figure 5-4 Open the toner-cartridge door c. Remove the toner cartridge from the printer. Figure 5-5 Remove the toner cartridge 2. Unpack the replacement toner cartridge. a. Dispose of the old part. NOTE: HP recommends responsible disposal of the defective part. b. Unpack the replacement part from the packaging.

<http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> 458 Chapter 5 Removal and replacement

Figure 5-6 Recycle and unpack 1 2 1 3 2 NOTE: When installing a replacement toner cartridge, hold both ends of the toner cartridge and rock it back and forth 5-6 times. 3. Install the toner cartridge. An MFP printer is shown in this section. However, the procedure is correct for all printer models. Removal and replacement: Toner cartridge 459

a. Align the toner cartridge with the tracks inside the printer, and then install the toner cartridge until it is firmly seated. Figure 5-7 Install the toner cartridge b. Close the toner-cartridge door. Figure 5-8 Close the toner-cartridge door Removal and replacement: Staple cartridge (MFP) Learn how to remove and replace the staple cartridge. To view a short video of this procedure, click the link below. View a video of how to remove and replace the staple cartridge. Mean time to repair: 5 minutes Service level: CSR A

Before performing service

Turn the printer power off

460 Chapter 5 Removal and replacement

■ Disconnect the power cable. **WARNING!** To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Use the table below to identify the correct part number for your printer. To order the part, go to www.hp.com/buy/parts. Table 5-2 Part information

Part number Part description

Q7432A HP staple cartridge pack (new) Q7432-67901 HP staple cartridge pack (service)

Required tools

No special tools are required to remove this part. No special tools are required to remove or install this assembly.

After performing service

Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

Post service test

Make sure that the printer initializes to a Ready state. Print a configuration page to make sure that the printer is functioning correctly. 1.

Remove the staple cartridge. **NOTE:** Stapler models only Removal and replacement: Staple cartridge (MFP) 461

a. Open the stapler door. Figure 5-9 Open the stapler door b. Pull the staple cartridge straight out of the printer to remove it. Figure 5-10 Remove the staple cartridge 2. Unpack the replacement assembly. a. Dispose of the defective part. **NOTE:** HP recommends responsible disposal of the defective part. <http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> 462 Chapter 5 Removal and replacement

b. Unpack the replacement part from the packaging. **CAUTION:** Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. **IMPORTANT:** Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. **NOTE:** When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. 3. Install the staple cartridge. **NOTE:** Stapler models only a. Push the staple cartridge straight into the printer to install it. Figure 5-11 Install the staple cartridge Removal and replacement: Staple cartridge (MFP) 463

b. Close the stapler door. Figure 5-12 Close the stapler door Removal and replacement: embedded Multi-Media Card (eMMC) (all except M501) Learn how to remove and replace the embedded Multi-Media Card (eMMC) To view a short video of this procedure, click the link below. View a video of how to remove and replace the eMMC. Mean time to repair: 5 minutes Service level: CSR A

Before performing service

Turn the printer power off

■ Disconnect the power cable. **WARNING!** To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. **CAUTION:** This part contains components that are electrostatic discharge (ESD) sensitive. To reduce the possibility of ESD damage, always touch the sheet-metal chassis to ground yourself before touching an ESD sensitive part. **IMPORTANT:** Do not replace the formatter, DC controller, and eMMC PCA simultaneously during a single printer servicing. Doing so might cause the printer to become unstable or inoperable. **NOTE:** Data stored on the eMMC is not secure and cannot be encrypted. Use the table below to identify the correct part number for your printer. To order the part, go to www.hp.com/buy/parts. 464 Chapter 5 Removal and replacement

Table 5-3 Part information

Part number Part description

5851-6436 (M527dn/E52545dn/M528dn/ E52645dn) embedded Multi-Media Card (eMMC) with instruction guide B5L32-67901

(M506/E50045/M507/E50145) embedded Multi-Media Card (eMMC) with instruction guide

Required tools

No special tools are required to install this part.

After performing service

Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

Post service test

Make sure that the printer initializes to a Ready state. Print a configuration page to make sure that the printer is functioning correctly. 1. Remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). Figure 5-13 Remove the formatter cover (M501) 2 1 ● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. Removal and replacement: embedded Multi-Media Card (eMMC) (all except M501) 465

Figure 5-14 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

E52645)

2. Remove the embedded Multi-Media Card (eMMC models). CAUTION: ESD sensitive part. ■ Locate the eMMC component on the formatter, and then pull it straight off of the formatter to remove it. Figure 5-15 Remove the eMMC 3. Unpack the replacement assembly. 466 Chapter 5 Removal and replacement

a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part.

<http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. 4. Install the eMMC (eMMC models). CAUTION: ESD sensitive part. ■ Align the connector on the replacement eMMC (callout 1) with the connector on the formatter, and then push the eMMC onto the formatter to install it. IMPORTANT: Push on the eMMC at the location shown by the arrow. Ignore any "Push here" label on the eMMC itself. Make sure that the eMMC is fully seated in the connector . NOTE: The eMMC can only be installed in one direction on the formatter. Removal and replacement: embedded Multi-Media Card (eMMC) (all except M501) 467

Figure 5-16 Install the eMMC 1 5. Install the formatter cover. ■ Do one of the following: ● M501: Slide it toward the front of the printer to install it (callout 1), and then install one screw (callout 2). Figure 5-17 Install the formatter cover (M501) 1 2 ●

M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide it toward the front of the printer to install it. 468 Chapter 5 Removal and replacement

Figure 5-18 Install the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

E52645)

6. Reverse the removal steps to install the assembly. Also use the following special installation instructions.

Reinstall the printer firmware

a. Go to www.hp.com/go/futuresmart. b. Select Upgrade now. c. Find your Enterprise printer model. d. Select the link to open the firmware download page. e. Select OS Independent from the list of operating systems. f. Under the Firmware section, find the file for multiple operating systems. g. Select Download. Reinstallation tip: To view installation instructions, go to <http://www.hp.com/go/futuresmart>. Select Upgrade now, and then select How to perform a firmware update. Removal and replacement: Hard-disk drive (HDD) Learn how to remove and replace the hard-disk drive (HDD). To view a short video of this procedure, click the link below. View a video of how to remove and replace the HDD. Mean time to repair: 5 minutes Service level: CSR A

Before performing service

Turn the printer power off

Removal and replacement: Hard-disk drive (HDD) 469

■ Disconnect the power cable. WARNING! To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. CAUTION: This part contains components that are electrostatic discharge (ESD) sensitive. To reduce the possibility of ESD damage, always touch the sheet-metal chassis to ground yourself before touching an ESD sensitive part. Use the table below to identify the correct part number for your printer. To order the part, go to www.hp.com/buy/parts. Table 5-4 Part information

Part number Part description

B5L29-67907 Hard-disk drive (HDD) accessory (M506/E50045/M527/E52545/M528/E52645) with instruction guide 5EL03A Hard-disk drive (HDD) (government only)

Required tools

No special tools are required to remove or install this assembly.

After performing service

Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

Post service test

Make sure that the printer initializes to a Ready state. Print a configuration page to make sure that the printer is functioning correctly. 1.

Remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). 470 Chapter 5 Removal and replacement

Figure 5-19 Remove the formatter cover (M501) 2 1 ● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. Figure 5-20 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

E52645)

2. Remove the hard-disk drive (HDD models). CAUTION: ESD sensitive part. Removal and replacement: Hard-disk drive (HDD) 471

a. Release the locking connector (callout 1), and then pinch the retainer (callout 2) to release it. Figure 5-21 Release the HDD 1 2 b. Rotate the connector end of the HDD out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-22 Remove the HDD 1 2 3. Unpack the replacement assembly. a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part. <http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> 472 Chapter 5 Removal and replacement

b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. 4. Install the HDD (HDD models). CAUTION: ESD sensitive part. a. Before proceeding, take note of the location of the slot (callout 1) in the sheet-metal where the HDD cradle mounting tab (callout 2) must be installed. CAUTION: This portion of the sheet-metal is a knock-out plate (callout 1). Do not push on it with sufficient force to detach it when installing the HDD. Figure 5-23 Locate the slot in the sheet-metal 1 2 Removal and replacement: Hard-disk drive (HDD) 473